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QatarEnergy– IS LIFTING OPERATIONS GUIDELINE

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1.0 PURPOSE

The main purpose of this guidelines is to provide guidance and information to personnel involved with lifting and cargo handling operations. It should prove useful for the selection, purchasing, repair, ongoing maintenance and safe use of lifting equipment and gear. Having identified various areas and practices which have proved problematic in the past, a number of procedures are incorporated in the document to eliminate these problems in the future.

The procedures and guidance contained within this document have been compiled with the following objectives in mind:

- 1) The prevention of equipment failure due to poor quality (Systems of approval of manufacturers / vendors / products)
- 2) The prevention of equipment failure due to misuse (Training of personnel - issue of "safe use" guidance)
- 3) The prevention of equipment failure due to inadequate / poor maintenance (Standardization of Repair Procedures and PMR's)
- 4) The prevention of equipment failure due to improved Inspection techniques (Standardization of Inspection Procedures)

The document also emphasizes the pre-use examination of equipment as a number of incidents / accidents if not caused by misuse, are caused by using gear which is faulty or has been previously damaged i.e.:

- 1) Lifting equipment can be damaged in transit
- 2) Lifting equipment can be misused and returned to the rigging loft
- 3) Repairs can sometimes be of poor quality
- 4) Lifting equipment in use can suffer damage from outside sources

Note: It has also been known for new equipment to contain faults.

QatarEnergy-IS present philosophy for their worksites is to:

- Employ a contractor to supply on hire, mobile cranes.
- Ensure that contractors employ an approved 3rd Party to carry out regular inspection of fixed & loose lifting equipment

2.0 SCOPE AND APPLICABILITY

This Guideline covers all **QatarEnergy-IS** operated worksites. It applies to all employees of **QatarEnergy-IS** including employees of any contractor or outside agency that work at any facilities owned, operated and / or managed by **QatarEnergy-IS**.

3.0 ILLUSTRATED TECHNICAL REFERENCE MANUAL ON LIFTING EQUIPMENT

3.1 Introduction

This section of the manual is intended to impart to the engineer, the relevant technical information on the various standard types of lifting equipment available in the market place to enable them to select the equipment most suitable for the job in hand. In the interest of safety, any restrictions in the use of the equipment have also been highlighted. Wherever possible, standard equipment should be used but for special



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cases it is possible to obtain modified equipment to allow an awkward job to be carried out. Manufacturers should always be consulted regarding any proposed modification and under no circumstances should this be carried out without their written consent. Further information as to the safe use of the equipment can be found in the NSL International Rigging and Lifting Handbook.

3.2 Design Notes

Generally, items of lifting equipment are manufactured to a standard or code but where neither British nor International Standards exist to cover the design and / or manufacture the following criteria shall be met:

- 1) The equipment will have a factor of safety commensurate with the type of usage it is liable to be subjected to. This factor of safety will be 5:1 for normal lifting purposes and 10:1 for man-riding purposes.
- 2) All materials used in the manufacture shall conform to a recognized national standard.
- 3) All welding shall conform to either recognized national standards or welding procedures approved by **QatarEnergy-IS**.
- 4) All welds shall be of adequate strength suitable for the designed rated capacity of the equipment.
- 5) Each unit shall be manufactured to the highest specified quality and standards of workmanship.
- 6) The surface of each unit shall be prepared correctly prior to the application of a protective coating and if specified, this coating shall be suitable for use in a marine environment.

British and International Standards have been quoted throughout this section but the dates of issue have purposely been omitted to eliminate the need for revising the document each time a standard is re-issued. It is to be understood that the latest issue / revision complete with any amendments will apply to equipment supplied to or used by **QatarEnergy-IS**.

3.3 Overhead Cranes, Runway Beams and Lifting Beams

3.3.1 Overhead Cranes

Overhead cranes shall be so designed that the stresses do not exceed the allowable as specified in BS 2573 / BS EN 13001. Should the cranes be powered, they must also meet with the requirements of BS 466. Generally, the construction of the building or module in which the crane is to be installed will dictate the cranes maximum capacity. Overhead cranes are available with two basic bridge designs i.e. single girder and double girder. With single girder construction, the hoist unit is suspended from a trolley mounted on the lower flange to give "cross travel" (**see Fig 1**).

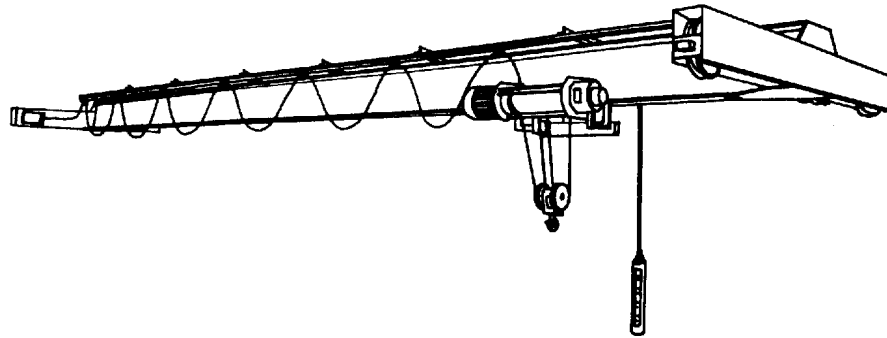


Fig 1

In the case of double girder construction, the crab unit travels on rails mounted on top of the girders or between them with the crab wheels running on the lower inside flanges (see Fig 2).

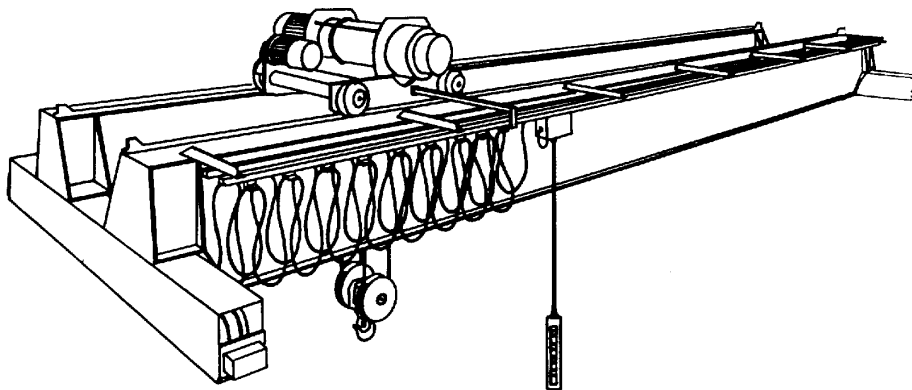


Fig 2

The main advantage of double girder cranes is the reduced headroom and increased height of lift due to the decreased beam depths and the ability to raise the hook up between the beams. In either case, long travel (also known as 'down shop travel') is obtained by means of carriages fitted at each end of and at right angles to the cross girder(s).

The hoisting unit can either be a standard under-slung trolley hoist (fitted with either wire rope or chain) or a crab unit mounted between or on top of the bridge girders (for wire rope only). The bridge would be compatible with the width of the building or module.

The crane operation both hoisting and traveling can either be powered (electric, pneumatic or hydraulic), manual (operated by hand chains) or a combination of both i.e. they are supplied to suit specific requirements.

Most powered models have single speed or dual speed options but all are fitted with overhoist limits as a safety feature. Additional safety features available are cross and long travel brakes / limits etc. and for certain applications, anti-collision equipment.

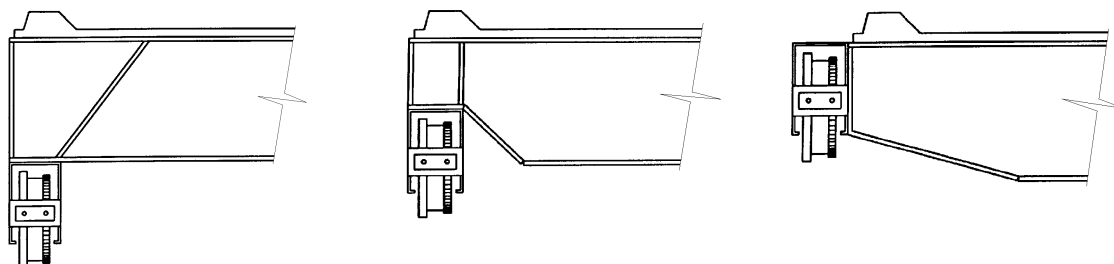
Where headroom is restricted, the jointing of the bridge to the end carriages can be either cut down or even made flush (see Fig 3).



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Standard

Cut down

Flush

Fig 3

For special applications, the cranes can be supplied with the following specifications:

- 1) Spark proof
- 2) Explosion proof
- 3) Coated for use in a marine environment

The following information would be required from the crane supplier to allow the engineering department to carry out the required stress calculations:

- 1) Total weight of crane bridge including end carriages and crab / trolley hoist unit
- 2) Maximum individual wheel loads including horizontal loads
- 3) End carriage trolley wheel centres
- 4) Track width of trolley wheels
- 5) Recommended minimum section of down shop support beams / rails

3.3.2 Runway Beams

Runway beams shall be designed in accordance with BS 2853 and shall be of proper size and adequate strength for the application. They shall be adequately supported or suspended and be capable of being properly maintained. Materials used shall comply with the relevant British Standard.

Stress calculations shall take account of the load to be lifted, weight of beam, weight of lifting appliance and all other superimposed loads on the structure. Allowance must be made for dynamic efforts due to swinging loads and the kinetic action of acceleration and retardation. This may be in the form of increasing the maximum wheel loads by 25% for power-operated appliances and 10% for hand-operated appliances.

These allowances are applicable only if the load is applied in the plane of the web. If this is not the situation, I-section beams should be replaced with beams specially designed to resist induced torsional stresses. If the beam is to be used in particularly hazardous areas, higher factors of safety should be used with proportionately lower working stresses. The guidance given to BS 2573 / BS EN 13001 for permissible stress values should be followed. In addition, it should be borne in mind that the runway beam will have to withstand a 125% proof load prior to being put into use for the first time.

Runway beams shall be supported in such a way that stresses due to torsional bending moments are negligible. They shall also be capable of withstanding the maximum



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vertical bending moment and shearing force induced by travelling loads. Connections to the supporting structure shall be of adequate strength and stiffness to withstand safely, the maximum loads applied.

Joints in runways shall be of adequate strength to withstand the maximum loads occurring at the joints, and to ensure the maintenance of true alignment. Effective end stops shall be provided on the runway to prevent the trolley either falling from the beams or fouling the structure of the module in which the runway is installed.

The maximum measured deflection of a runway beam under the safe working load shall not exceed $1/500^{\text{th}}$ of the span. For cantilever beams the maximum measured deflection under the Safe Working Load shall not exceed $1/250^{\text{th}}$.

Runway beams shall be certified by a competent person before being taken into use for the first time after erection, re-erection or having undergone any substantial alteration or repair. A certificate shall be submitted only after the competent person is satisfied that the maximum allowable stresses, and calculated deflections, are not exceeded.

Runway beams shall be subjected to a Proof Load Test equal to the weight of the heaviest lifting appliance supported by the runway, plus 25 per cent in excess of the Safe Working Load lifted by this appliance. If the runway supports more than one lifting appliance, an allowance will be made for the permissible proximity of any other lifting appliance. The beam shall be inspected by a competent person before, during and after the load test.

The Safe Working Load (SWL) and identification number shall be marked clearly on the beam such that they are clearly visible to the operator. The Safe Working Load does not refer to the lifting appliance or associated equipment.

Runway beams shall be examined by a competent person in accordance with the Written Scheme of Examination. This person shall have access to the original Test Certificate. An examination report shall be issued by the competent person on satisfactory conclusion of the examination. During any examination, if any defect is identified, the report must be referred to the appropriate **QatarEnergy-IS Representative** before any repairs are undertaken.

Any load supporting arrangement which cannot be used safely, and is accessible such that casual use could occur, shall be identified and tagged, **'NOT SUITABLE FOR USE'**, and any associated equipment removed or isolated to prevent normal usage.

3.3.3 Runway Beams

Spreader Beams and Lifting Frames are primarily designed to remove the compressive forces generated by slinging arrangements with a single point lift. For this reason the structural members are generally in compression and hence can be quite slender in section. A common oversight is the requirement for additional bracing to ensure that the main load bearing members remain in compression and are not subjected to excessive bending moments due to damage or distortion sustained in general day-to-day use (see Fig 5).

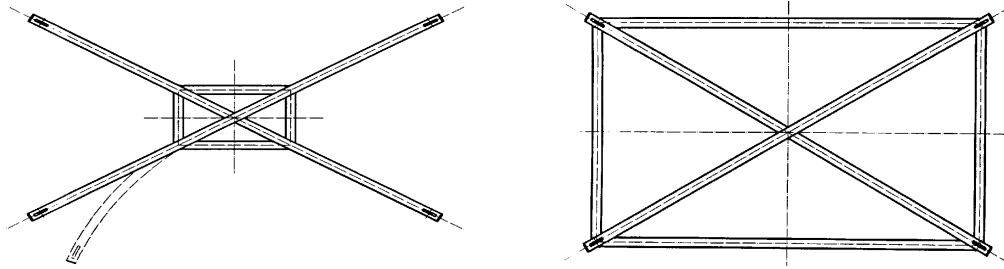


Fig 5

Impact Damage subsequent loss of residual strength use and rough handling

Note: In view of the different types and styles of such structures, it is considered practicable to state only the basic design and manufacturing requirements.

The most popular and efficient method of fixing pad eyes to a spreader beam or frame is to manufacture the top and bottom padeye as one plate and slot it into the tubular or position between the channels etc. depending on the design (**refer to Fig 6**).

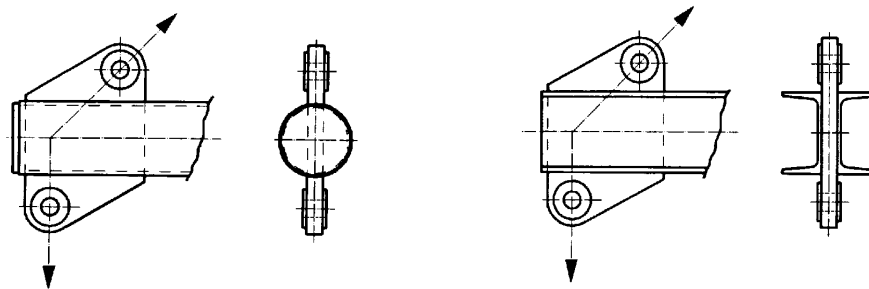


Fig 6

Unless the design is supplied by **QatarEnergy-IS**, the manufacturer / supplier of the lifting fabrication will be responsible for providing a suitable design for the intended application. Any drawings and calculations etc. must be approved by **QatarEnergy-IS** since it is **QatarEnergy-IS** ultimate responsibility. The testing and certification of the unit however will be the responsibility of the fabricator. These tests may require to be witnessed. Where applicable, a copy of the manufacturer's design calculations and final approved drawings shall accompany the Certificate of Test and Examination.

Note: Fabrications that do not have supporting design criteria but have been successfully proof-load tested and NDT'd will be acceptable but only if the workmanship is of a good standard. However, they will have to be regularly monitored in case fatigue is an issue.

Each integral padeye or lifting point shall be orientated in such a direction as to reduce out-of-plane loading of the padeye or lifting point and associated shackle. Padeyes and lifting points should be designed with the main connections in shear, rather than in tension. Wherever possible, to avoid tension loads in the thickness direction, padeye plates should be slotted through horizontal flanges and welded directly to vertical web plates.

Note: For further details on padeye design, refer to section 2.0 (lifting points).



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Each shackle and hook supplied as an integral part of the assembly shall conform to the appropriate standards. The material used in fabrications shall conform to BS 7668, Specification for Weldable Structural Steel. Alternatively, if other materials are used, evidence must be available to confirm, to the satisfaction of **QatarEnergy-IS**, that it is essentially of the same quality as the preferred material.

Welding shall comply with BS EN 1011, Metal-arc Welding of Carbon and Carbon Manganese Steel.

Each unit must be produced to the highest standards of quality and workmanship, and be free from defects. The structure of each unit will be properly cleaned to remove all contaminants prior to the application of a high quality protective coating.

The test program will be the responsibility of the manufacturer. An independent competent person nominated by the manufacturer shall supervise and monitor all aspects of the test program. Each unit shall be subject to a Proof Load Test (**See Table 1**).

Table 1 - Proof Load Test for Spreader Beams and Lifting Frames

	Minimum Proof Load
≤ 3 tonnes	SWL + 100%
$> 3 \leq 6$ tonnes	SWL + 3 tonnes
$> 6 \leq 20$ tonnes	SWL + 50%
$> 20 \leq 40$ tonnes	SWL + 10 tonnes
> 40 tonnes	SWL + 25 tonnes

Immediately prior to, and after the application of, the proof load test, all material and welds associated with the critical lifting points shall be 100% non-destructively tested. The test load shall be a live load applied to the complete structure. Loading of the individual lifting points separately is acceptable only in exceptional circumstances.

Upon completion of the test, only structure, lifting points and associated lifting gear which do not show any permanent deformation shall be put into use.

A Certificate of Test and Examination shall be supplied for each spreader beam or lifting frame. This test certificate will refer to the test certificates for the associated equipment, e.g. those covering the shackles, hooks etc., and the Non Destructive Examination (NDE) reports. It shall also be accompanied by a copy of the approved fabrication drawings and stress calculations.

Each spreader beam or lifting frame shall be inspected by a competent person prior to each occasion it is used. In addition, each unit shall be thoroughly examined at six-monthly intervals by a competent person who is independent of **QatarEnergy-IS**.

When a deviation from a certified load supporting arrangement is required, such a deviation must be fully evaluated by a competent person.

Any load supporting arrangement which cannot be used safely, and is accessible such that casual use could occur, shall be identified and tagged, **'NOT SUITABLE FOR USE'**, and any associated equipment removed or isolated to prevent normal usage.



3.4 Lifting Points

This section provides guidance on the design, use and examination of lifting points located on drilling rigs and offshore installations. At present there are neither National nor International Standards pertinent to the full range of fittings considered as lifting points.

Where reference is made in this document to the use and selection of associated lifting gear it will be in accordance with the appropriate **QatarEnergy-IS** standards. This guidance is not intended to cover lifting points used in the general construction phase of the main topside modules and those lifting points which are an integral part of the structure of cargo containers.

Definitions for terms used in this Section, are in accordance with the standard definitions for Lifting Gear and Lifting Appliances, as contained in Glossary of Terms of this Manual. The exceptions are:

- 1) Lifting Point: generic term for the calculated and designed point(s) or attachment(s) on an item of plant, by which it can be lifted safely using certified lifting gear and appliances of adequate capacity. The term also applies to points or attachments fixed to structural members and from which a load can be suspended. It includes such common names as lifting lugs, lifting eyes, trunnions, fabricated lifting plates, padeyes and jacking points
- 2) Rated Load: the maximum vertical load which can be accepted by a lifting point
- 3) Plant: the term includes machines, sub-assemblies and structure

The following is a selection of different types of lifting points. The type of lifting points depends on the design of the plant to be lifted but the first two illustrated are the most popular fabricated type:

- 1) Welded-on lifting points with holes suitable for shackles or hooks, e.g. padeye (see **Fig 1 below and QatarEnergy-IS pad eye table at appendix 11**)

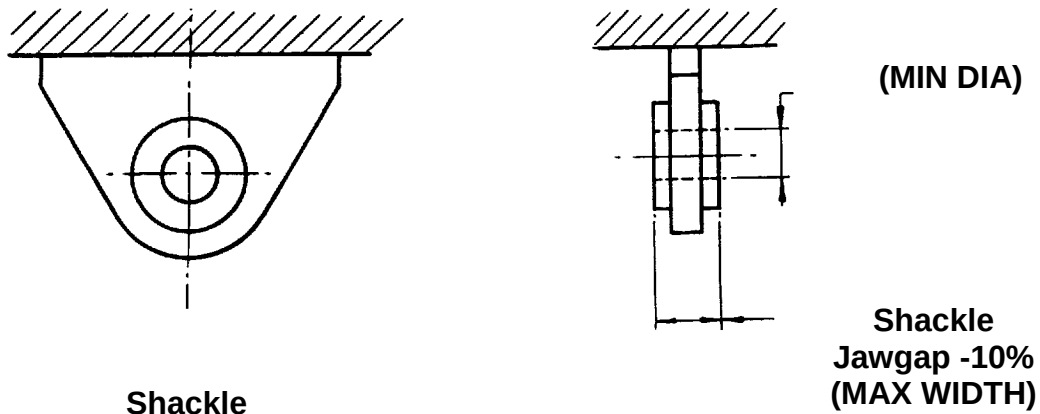


Fig 1



- 2) Bolted-on lifting points with holes suitable for shackles or hooks (see Fig 2)

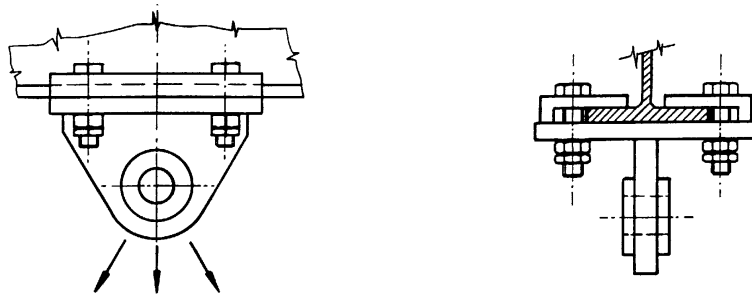
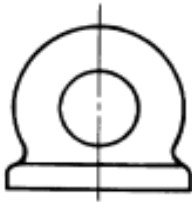


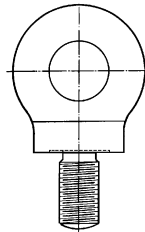
Fig 2

As an alternative to the fabricated type, there are standard manufactured items i.e.:

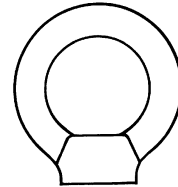
- 3) Eyeplates, Eyebolts and Eyenuts (see Fig 3)



Weld On Eyeplates



Collar Eyebolt



Screw on Eyenut

Fig 3

The remainder are as follows:

- 1) Integral lugs with holes suitable for shackles or hooks.
- 2) Holes inherent in the design of the plant which may be designated as lifting points.
- 3) Trunnions. – are not acceptable for dynamic lifts.
- 4) Links welded on to reinforcing members in concrete structure.
- 5) Welded-on or integral pads for jacking points.

When practicable, the number of lifting points incorporated in the design should be sufficient to ensure stability, and limit stress in the structure to within design limits. They should also be positioned so that each point supports an equal proportion of the total load.

Fabricated lifting points should be designed to withstand transverse bending of the plates and welds.

Each lifting point should be designed to have a factor of safety of not less than 4:1, based on the rated load applied vertically, and the ultimate tensile stress of material used.

When calculating the rated load factors, such as weight allowance and dynamic amplification factor, the location of the centre of gravity should be considered.

Lifting points should be orientated in such a direction as to minimise out-of-plane loading.

Consideration should be given to the number of lifting points which will bear the load at any one time.



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In practice it is unlikely that the total load will be evenly distributed amongst all the lifting points. The actual load distribution will depend on a number of factors including the construction of the plant, e.g. flexibility of the structure, sling length, tolerances etc. It is recommended to assume that in a four point lift situation, 75% of the load is distributed between two diagonally opposite legs, the third leg supports 25% of the load and the fourth leg is redundant. This equates to a possible increase in load on each lifting point of 50%. Therefore each lifting point shall have a rated load of not less than 1.5 times the share of the load which it is intended to take when the load is applied vertically. As a general rule however, the following formula should be adopted:

Load x 1.5

Rated load at 0° = No of lifting points

For rigid loads the majority of the load may be taken on only two of the four lifting points and the rated load for each lifting point should be considered as 0.5 times the total load. This is an extreme condition.

Each lifting point should be rated with a Safe Working Load (SWL) for lifting at 0° to 45° to the vertical. This equates to the rating of 2, 3 and 4-point lifts using the uniform load method 0° to 90° where the maximum included angle between the legs is 90°.

Note: When lifting points are designed for use with shackles, the dimensions will be such as to accept the pin of the shackle and suit the shackle jaw gap. As a guideline, the hole diameter should equate to the shackle pin diameter plus 10% and the plate width (including cheek plates if required) equate to the jaw gap minus 10%. Any tighter tolerances will cause problems especially after the application of a coating to an offshore specification. (Information regarding lift points on major items, e.g. modules can be found in the Engineering Reference Document "Guidelines for Lifting Points and Heavy Lift Criteria".) Lifting points will be designed in accordance with recognised design codes such as BS 449-2 Specification for the use of Structural Steel in Buildings or BS 5950.

Welding carried out in the fabrication of lifting points and attachments shall be performed by welders qualified to ASME IX and in accordance with BS EN 1011 or other standards approved by **QatarEnergy-IS**.

Each lifting point shall be clearly and permanently marked with:

- 1) Lifting equipment identification code number
- 2) Safe Working Load at 0° to 90°

Tapped holes, when used to locate a lifting point (e.g. eyebolt), shall have the thread form and diameter permanently marked adjacent to each hole. All plant provided with lifting points, shall have its weight clearly marked on a permanently attached plate.

When plant forms part of a larger assembly, a list of the assembled parts, their individual weights, and the total assembled weight shall be given, e.g.

Assembly Plate

Pump	2.6t
Motor	1.8t
Base Plate	0.9t
Total Assembly	5.3t

The total weight of all contents shall be included in the displayed weight, and the plate shall be located so as to be clearly visible at all stages of lifting, assembly and disassembly.

Each lifting point shall be subject to a Proof Load Test by one of the following methods:

- 1) Applying a vertical load test equal to twice the rated load of the lifting point at 0°
- 2) Applying a live load which is equal to twice the total weight of the item

All lifting points shall be subjected to a proof load test and examination prior to being used for the first time. If the design of the lifting point included welding in its fabrication,



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the examination will include a Non Destructive Examination (NDE) (Magnetic Particle Inspection (MPI) in accordance with ASME V) and a thorough visual examination. A Certificate of Test and Examination of Lifting Gear and Lifting Appliances, shall be supplied, together with a full NDE test report. The test certificate will be qualified by a design drawing and corresponding calculations. The only exception to this requirement shall be when the lifting point is a recognized standard design conforming to a national standard.

The certificate of test shall include the following information:

- 1) The name of the manufacturer
- 2) Description of the lifting point including dimensions
- 3) The rated load and the factor of safety used
- 4) The quality and grade of material used in the fabrication
- 5) The national standard to which the lifting point was manufactured (if appropriate)
- 6) The lifting equipment identification code number
- 7) Proof load applied and method of application

If the lifting points are designed for a particular method of rigging arrangement, then this must be stated on the design drawing and calculations.

Each lifting point shall be inspected by the user immediately prior to each use. In addition, each lifting point shall be examined after substantial alteration or repair and by a competent person every six months.

Should it be necessary to perform a proof load test, then this will be followed by a Non Destructive Examination (NDE) where appropriate (MPI to ASME V) and a thorough visual examination.

All lifting points should be proof load tested prior to first use. The only exceptions to this requirement are:

- 1) Lifting points which are an integral part of a piece of equipment or its casing. Such lifting points may be of inadequate design, and therefore the application of a Proof Load, may cause damage to the equipment to an extent which may render the equipment unserviceable. This must be avoided, and therefore an acceptable alternative should be to establish, by calculation, the integrity of the lifting point.
- 2) Lifting points which have been identified as being used infrequently and labelled **'NOT TO BE USED UNTIL TESTED'**.

When a deviation from a certified load supporting arrangement is required, such a deviation must be fully evaluated by a competent person.

Any load supporting arrangement which cannot be used safely, and is accessible such that casual use could occur, shall be identified and tagged, **'NOT SUITABLE FOR USE'**, and any associated equipment removed or isolated to prevent normal usage.

3.5 Manually Operated Chain Hoists

All hoists used by **QatarEnergy-IS** shall be designed and constructed in accordance with BS 3243 or an equivalent international standard.

A manual chain hoist or chain block is a lifting machine which, through the use of a gearbox and brake mechanism, allows the user to raise heavy loads (up to 50 tonne) by applying a force of approximately 35kgf to the operating chain (hand chain). They can be procured in various capacities i.e. WLL 500kg, 1te, 3te, 5te, 10te, 15te, 25te and 50te. (Certain manufacturers supply capacities between those stated above).

The standard range of lift (**see Fig 1**) is 3 metres but this can be increased to suit any



specific requirement. Should this be a substantial increase, the manufacturer should be consulted as it may be necessary to down-rate the hoist due to the excess weight of load chain.

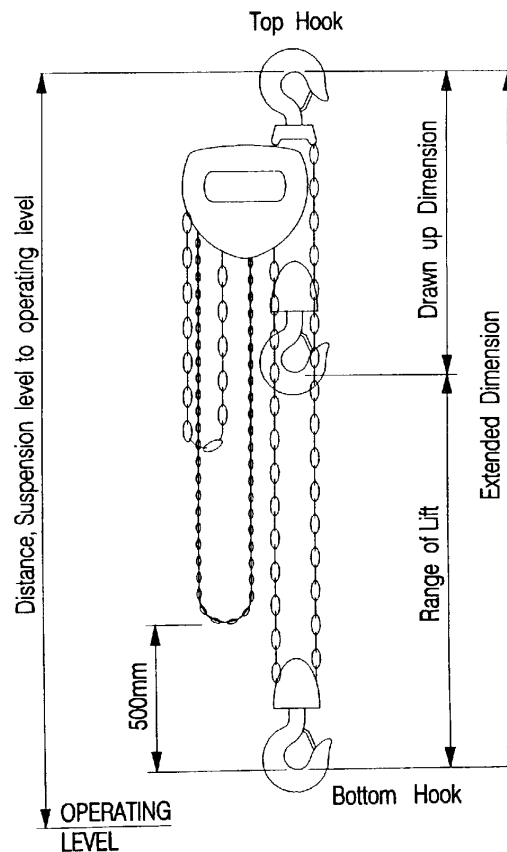


Fig 1

The operating chains (which normally hang 500mm above than the lowest position of the bottom hook) can be made any length to suit the operating position e.g. the operator may be standing on a mezzanine level with the bottom hook required to reach the deck 5 metres below. In this case, the hand chain should be 5 metres shorter than normal to avoid snagging and jamming in the floor grating.

One of the major considerations when selecting a chainblock is the 'drawn up dimension' (closed height) due to the lack of headroom in offshore modules. This dimension varies with the different manufacturers and with the different configurations.

Standard chainblocks are supplied with swiveling top and bottom hooks for ease of attachment to suspension point and load. They can be hooked into a beam trolley for travelling along a runway beam or can be supplied 'built-in' to a trolley to save headroom. Where clearance between runway beam and load is particularly restrictive, Ultra-low headroom trolley hoists can be utilised (see Fig 2).



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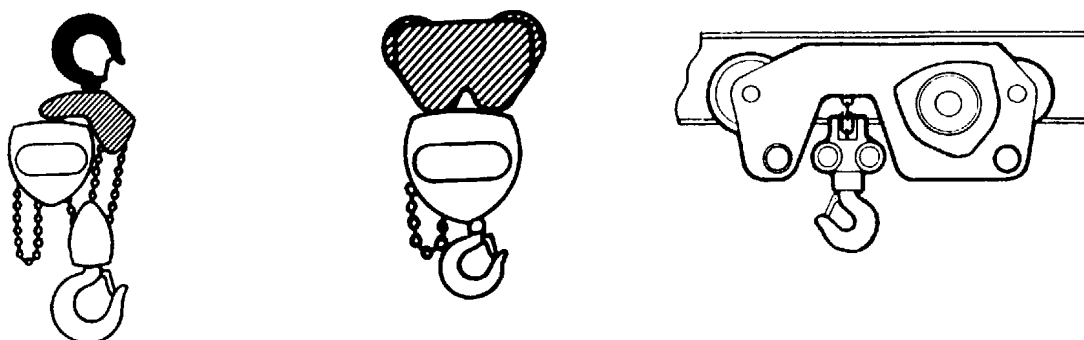


Fig 2

Chainblocks have classifications according to their duties but these are not really relevant in the Oil and Gas Industry where the hoists are only used intermittently for construction / maintenance work and hence are not in constant use. Should further information be required on classification, it can be found in BS 3243.

Chainblocks are primarily designed to be operated vertically but hook suspended blocks will work efficiently up to 45% off the vertical. Chainblocks hooked or built into trolleys (or combined units) should only be used in the vertical position. If a horizontal pull is required, Lever hoists (Pullifts) or Wire rope hoists should be utilised (refer to sections 2.4 and 2.5).

Chainblocks which have un-coated aluminium components in their construction should be avoided (i.e. they should be suitable for use in an offshore environment).

Prior to being received by **QatarEnergy-IS**, each block will be subjected to a Proof Load relative to its working load limit (WLL)

i.e.: Up to 10 tonne - $1.5 \times \text{WLL}$

10 - 20 tonne - $\text{WLL} + 5\text{t}$

Over 20 tonne - $1.25 \times \text{WLL}$

And will be supplied with a proof load test certificate and plated accordingly with the relevant information.

Spark-proof hoists are available for use in hazardous areas but there is a very limited amount of manufacturers. Hoists can also be supplied specially coated for use in a saliniferous environment.

Most chain blocks are suitable for sub-sea work but not all. It is imperative that the manufacturers are contacted for confirmation as to the suitability of their equipment as the brake material of certain models precludes their use in this practice.

Whenever possible, in an effort to minimise the permutation of chain blocks on each installation, preference should be given to the make and model of chain blocks already in use on the installation.

To aid selection of chain block type for special projects etc., specifications (i.e. capacity / dimensions) of the more popular models are contained in the following pages.

Note: Once selection is made, the purchasing department will have to be contacted to ensure that the use of the particular model of chainblock has not been restricted for any reason.

The information to be supplied when raising the requisition will be as follows:

- 1) Quantity
- 2) Manufacturer and model
- 3) SWL
- 4) Height of lift
- 5) Vertical length (drop) of operating chains



- 6) The method of suspension (i.e. with top hook or with trolley)
- 7) If trolley required, indicate if geared or push travel and give beam dimensions
- 8) If an alternative to that mentioned in (2) would be acceptable
- 9) If headroom is restricted, maximum drawn-up dimension allowable

Note: If the hoists are only required for a particular project, they can generally be obtained on a rental basis.

3.6 Lever Hoists (Pullifts)

All lever hoists shall be designed and constructed in accordance with BS 4898 or an equivalent International Standard.

A lever hoist or pullift is a lifting machine which, through the use of a gearbox and brake mechanism, allows the user to raise heavy loads (commonly up to 6 tonne) by applying a force of approximately 40kgf to the operating lever. They can be procured in various capacities i.e. WLL 750kg, 1.5te, 3te and 6te. (Certain manufacturers supply capacities between those stated above and as high as 15 tonne but these are awkward to use and not popular with the users).

The standard range of lift is 1.5 metres but this can be increased to suit any specific requirement. Should this increase be 6 metres or more, the manufacturer should be consulted as regards to the design of the brake mechanism. The same applies with light loads where low operating efforts are necessary.

They are available with two types of chain i.e.: standard short link chain and roller chain (see Fig 1). Link chain models are the most popular as they are flexible and easy to use whereas roller chain models tend to be more rigid and the chains corrode and seize more readily.

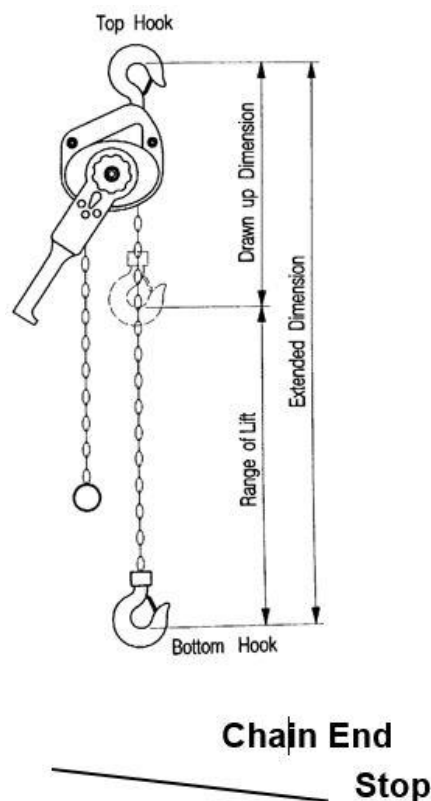


Fig 1

One of the major advantages when using a pullift instead of a chain block is the 'drawn up dimension' (closed height) due to the lack of headroom in offshore modules. This



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dimension varies with the different manufacturers.

One of the disadvantages is that to operate the pullift, the user has to be at the same level as the suspension point which could create a requirement for scaffolding.

Pullifts are supplied with swiveling top and bottom hooks for ease of attachment to suspension point and load. They can also be hooked into a beam trolley for traveling a load along a runway beam.

It is recommended that the WLL should be reduced by 20% if the pullifts are in constant use at full or near full capacity. This is not really relevant in the Oil and Gas Industry as the hoists are only used intermittently for construction / maintenance work. Should further information be required, refer to BS 4898.

Unlike chain blocks, pullifts are designed to be operated both vertically **and** horizontally and can also be hooked into a beam trolley for linear travel.

Pullifts which have un-coated aluminium components in their construction should be avoided (i.e. they should be suitable for use in an offshore environment).

Prior to being received by **QatarEnergy-IS**, each pullift will be subjected to a Proof Load of its working load limit (WLL) x 1.5 and will be supplied with a proof load test certificate and plated / hard stamped accordingly with the relevant information.

Spark-proof pullifts are available for use in hazardous areas but there is a very limited amount of manufacturers. Pullifts can also be supplied specially coated for use in a saline environment.

Most pullifts are suitable for sub-sea work but not all. It is imperative that the manufacturers are contacted for confirmation as to the suitability of their products as the brake material of certain models precludes their use in this practice.

Whenever possible, in an effort to minimise the permutation of pullifts on each installation, preference should be given to the make and model of pullifts already in use on the installation.

To aid the selection of pullift type, specifications (i.e. capacity / dimensions) of the more popular models are contained in the following pages.

Note: Once selection is made, the purchasing department will have to be contacted to ensure that the use of the particular model of pullift has not been restricted for any reason.

The information to be supplied when raising the requisition will be as follows:

- 1) Quantity
- 2) Manufacturer and model
- 3) SWL
- 4) Height of lift
- 5) Type of chain (link or roller)
- 6) If an alternative to that mentioned in (2) would be acceptable
- 7) If headroom is restricted, maximum drawn-up dimension allowable

Note: If the pullifts are only required for a particular project, they can generally be obtained on a rental basis.

3.7 Wire Rope Hoists

Wire rope hoists (Tirfors*) originated in France and consequently no British Standard or indeed any International Standard exists to regulate their manufacture.

*The use of commercial names is avoided throughout this suite of documents however, in this case, the generic name 'Tirfor' is being used in the interest of safety, the reason for which will become apparent in 2.5.7.

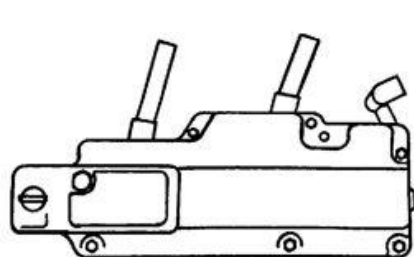
Tirfor rope hoists are available in two ranges i.e.: the lightweight range (T500 series) and the heavy-duty range (**TU series - see Fig 1**)



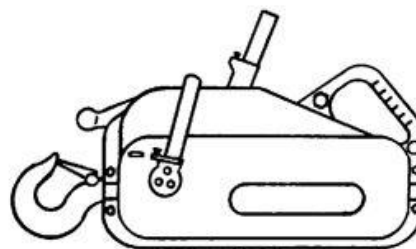
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T500 series



TU series

Fig 1

Each range has three models which are dual rated for lifting (with a factor of safety 5:1) and pulling /cross-hauling (with a factor of safety of approx. 3:1). e.g.

Model (Pulling)	SWL (Lifting)	Capacity
T508/Tu8	800kg	1200kg
T516/Tu16	1600kg	2500kg
T30	3000kg	5000kg
T532/Tu32	3200kg	5000kg

Note: In the interest of safety, all hoists will be rated and marked with their lifting capacity only.

It is important to differentiate between a **lifting application** and a **pulling application** therefore the following philosophy should be used:

- 1) A **lifting application** is one in which the load does not become stationary should either the machine or any of its associated equipment fail.
- 2) A **pulling application** is one in which a load becomes stationary should either the machine or any of its associated equipment fail.

Tirfor rope hoists are normally manually operated but hydraulically powered hoists can be procured to suit specific purposes, e.g. extra long lifts / pulls. **(see Fig 2)**



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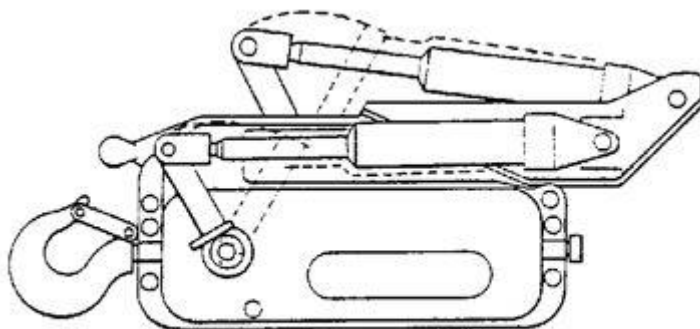


Fig 2

The manual hoists are operated by means of a telescopic handle placed over the operating levers whereas with hydraulic hoists, powered rams operate the lever system.

The hoists are supplied with a specially constructed rope (Maxiflex) to withstand the gripping power of the jaws and to prevent 'Bird-caging'. To use any other rope in the hoist would invalidate its certification. The standard length of rope supplied is 20 metres but shorter or longer ropes are readily available.

It is imperative that the correct rope is used with the hoist as the machines are designed to work with a specific diameter of rope and these vary according to different manufacturers. See table below:

Comparison of Rope Diameters

Manufacturer	SWL 800kg	SWL 1.6t	SWL3.2t
Tirfor	8.2mm	11.3mm	16.3mm
'A' another	8mm	11.2mm	16.0mm
'B' another	8.0mm	11.0mm	16.0mm

Should any of the other manufacturer's ropes be used in a Tirfor hoist, a potentially hazardous situation could develop i.e. it is likely that the rope will slip through the machine under load. For this reason, only Tirfor machines and ropes shall be used on **QatarEnergy-IS** Rigs and worksites. (This appears to be the practice adopted by the majority of operators and contractors throughout the industry possibly due to previous problems).

Although certain models are approved by the manufacturers for man-riding subject to certain safety requirements, they will not be used for this purpose on **QatarEnergy-IS** installations and worksites.

Note: Once selection is made, the purchasing department will have to be contacted to ensure that the use of the particular model of wire rope hoist has not been restricted for any reason.

Hoists and ropes should be supplied with separate certificates as the machine will outlast many ropes which will have to be replaced.

The information to be supplied when raising the requisition will be as follows:

- 1) Quantity
- 2) Manufacturer and model



3) SWL

4) Length of rope

Note: If the Tirfor hoists and ropes are only required for a particular project, they can generally be obtained on a rental basis.

3.8 Pneumatic Chain Hoists

Most of the power-operated hoists used in the Oil and Gas Industry are pneumatically powered for obvious safety reasons. It is also convenient as all installations have a utility air supply in the majority of their modules.

Pneumatic chain hoists are not manufactured in the UK and consequently there is no British Standard to give guidance as to their design. The main safety feature built into this type of hoist however appears to be constant with all manufacturers and that is that the brakes are sprung in the “on” position and require air supply to release them. This in effect means that in the event loss of power (e.g. a burst hose etc.) the brakes are automatically applied to prevent any load falling. In addition, the control buttons must return to neutral on release.

The hoists must also be fitted with overhoist limiters and trip devices to prevent over-hoisting or over-lowering.

Air hoists can be procured in various capacities i.e. WLL 500kg, 1te, 3te, 5te, 10te, 15te, 25te and 50te. (Certain manufacturers supply capacities between those stated above).

The standard range of lift is 3 metres but this can be increased to suit any specific requirement. Should this be a substantial increase, the manufacturer should be consulted as it may be necessary to derate the hoist due to the excess weight of load chain. For long lifts, chain collector boxes can be fitted to the hoist.

The control pendant or chains (which normally hang 1 metre above than the lowest position of the bottom hook) can be made any length to suit the operating position e.g. the operator may be standing on a mezzanine level with the bottom hook required to reach the deck 5 metres below. In this case, the control pendant or chains should be 5 metres shorter than normal.

Pendant controls (with push buttons) are the preferred option but tend to stick if exposed to contaminated air supply. If the hoist is going to be operating in areas exposed to the elements, pull chain or cord controls are often more effective as there are less components to corrode.

One of the major considerations when selecting an airhoist is the 'drawn up dimension' (closed height) due to the lack of headroom in offshore modules. This dimension varies with the different manufacturers and with the different configurations.

Standard airhoists are supplied with swivelling top and bottom hooks for ease of attachment to suspension point and load. They can be hooked into a beam trolley for travelling along a runway beam or can be supplied 'built-in' to a trolley to save headroom. Where clearance between runway beam and load is particularly restrictive, Ultra-low headroom air trolley hoists can be utilised (see Fig 2).

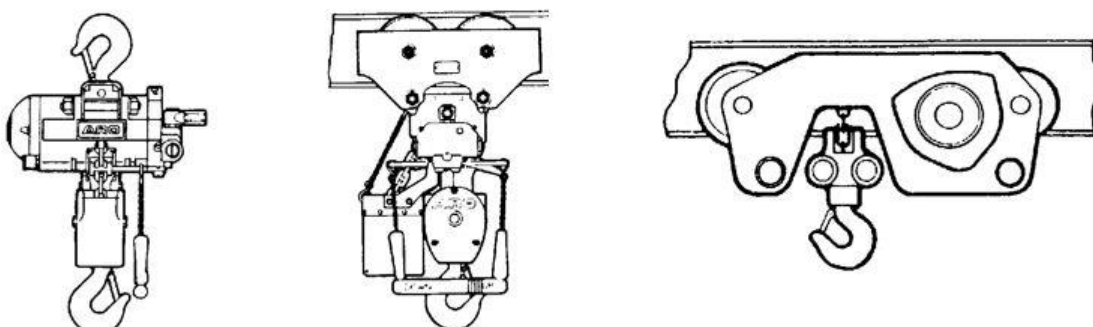


Fig 2

Airhoists are primarily designed to be operated vertically but hook suspended blocks will work efficiently up to 45% off the vertical. Airhoists hooked or built into trolleys (or combined units) should only be used in the vertical position. If a horizontal pull is required, Lever hoists (Pullifts) or Wire rope hoists should be utilised (refer to sections 2.4 and 2.5).

Airhoists which have un-coated aluminium components in their construction should be avoided (i.e. they should be suitable for use in an offshore environment).

Prior to being received, each hoist will be subjected to a Proof Load relative to its working load limit (WLL)

i.e.:

Up to 10 tonne	-	1.5 x WLL
10 - 20 tonne	-	WLL + 5t
over 20 tonne	-	1.25 x WLL

and will be supplied with a proof load test certificate.

Spark-proof and / or explosion proof hoists are available for use in hazardous areas but there is a limited amount of manufacturers. Hoists can also be supplied specially coated for use in a saliniferous environment.

Airhoists are not suitable for subsea work.

Whenever possible, in an effort to minimise the permutation of airhoists on each installation, preference should be given to the make and model of airhoists already in use on the installation.

To aid selection of airhoist type for special projects etc., specifications (i.e. capacity / dimensions) of the more popular models are contained in the following pages.

Note: Once selection is made, the purchasing department will have to be contacted to ensure that the use of the particular model of airhoist has not been restricted for any reason.

The information to be supplied when raising the requisition will be as follows:

- 1) Quantity
- 2) Manufacturer and model
- 3) SWL
- 4) Height of Lift
- 5) The type of controls preferred i.e. pendent or chains and the length
- 6) The method of suspension (i.e. with top hook or with trolley)
- 7) If trolley required, indicate if powered, geared or push travel and give beam dimensions



- 8) If an alternative to that mentioned in (2) would be acceptable
- 9) If headroom is restricted, maximum drawn-up dimension allowable
- 10) The type of air service unit (e.g. filter, regulator and lubricator and the type of fittings
- 11) The maximum permissible noise levels

Note: If the airhoists are only required for a particular project, they can generally be obtained on a rental basis.

3.9 Electric Chain Hoists

Electrically powered chain hoists are not widely used in the Oil and Gas Industry however, on offshore installations, they can be useful in areas such as workshops where 'hot work' is not a problem.

Electric chain hoists are not manufactured in the UK and consequently there is no British Standard to give guidance as to their design. The main safety feature built into this type of hoist however appears to be constant with all manufacturers and that is that the brakes are sprung in the 'on' position and require power supply to release them. This in effect means that in the event loss of power the brakes are automatically applied to prevent any load falling. In addition, the control buttons must return to neutral on release.

The hoists must also be fitted with overhoist limiters and trip devices to prevent over-hoisting or over-lowering.

Electric hoists can be procured in various capacities i.e. WLL 500kg, 1te, 1.5te, 2te and 3te. Above this capacity, electric hoists tend to be fitted with wire rope.

The power supply for the hoists can either be single phase or 3 phase depending on the location.

The standard range of lift (**see Fig 1**) is 3 metres but this can be increased to suit any specific requirement. Should this be a substantial increase, the manufacturer should be consulted as it may be necessary to derate the hoist due to the excess weight of load chain. For long lifts, chain collector boxes can be fitted to the hoist.

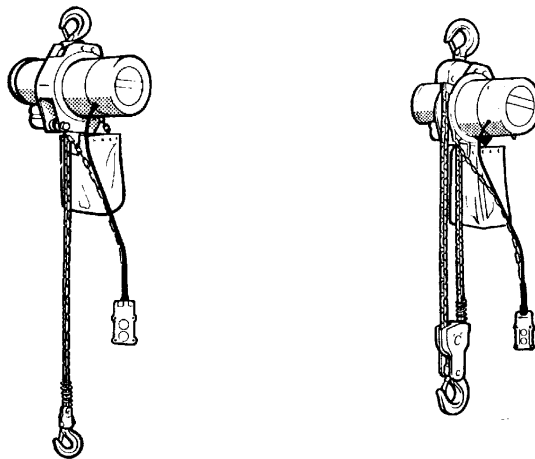


Fig 1

The control pendant (which normally hangs 1 metre above than the lowest position of the bottom hook) can be made any length to suit the working position e.g. the operator may be standing on a mezzanine level with the bottom hook required to reach the deck 5 metres below. In this case, the control pendant or chains should be 5 metres shorter than normal.



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One of the major considerations when selecting an electric hoist is the 'drawn up dimension' (closed height) due to the lack of headroom in offshore modules. This dimension varies with the different manufacturers and with the different configurations. Standard electric hoists are supplied with swivelling top and bottom hooks for ease of attachment to suspension point and load. They can be hooked into a beam trolley for travelling along a runway beam or can be supplied 'built-in' to a trolley to save headroom. This type of hoist however is not available in ultra-low headroom models. Electric hoists are primarily designed to be operated vertically but hook suspended blocks will work efficiently up to 45° off the vertical. Electric hoists hooked or built into trolleys (or combined units) should only be used in the vertical position. If a horizontal pull is required, Lever hoists (Pullifts) or Wire rope hoists should be utilised (refer to sections 2.4 and 2.5).

Electric hoists which have un-coated aluminium components in their construction should be avoided (i.e. they should be suitable for use in an offshore environment).

Prior to being received by **QatarEnergy-IS**, each hoist will be subjected to a Proof Load relative to its working load limit (WLL) i.e.: $WLL \times 1.25$ and will be supplied with a proof load test certificate and plated accordingly with the relevant information.

Spark-proof and / or explosion proof hoists are available for use in hazardous areas but there is a limited amount of manufacturers. Hoists can also be supplied specially coated for use in a saline environment.

Whenever possible, is an effort to minimise the permutation of hoists on each installation, preference should be given to the make and model of hoists already in use on the installation.

To aid selection of electric hoist type for special projects etc., specifications (i.e. capacity / dimensions) of the more popular models are contained in the following pages.

Note: Once selection is made, the purchasing department will have to be contacted to ensure that the use of the particular model of hoist has not been restricted for any reason.

The information to be supplied when raising the requisition will be as follows:

- 1) Quantity
- 2) Manufacturer and model
- 3) SWL
- 4) Height of lift and the length of control pendant
- 5) The power supply
- 6) The method of suspension (i.e. with top hook or with trolley)
- 7) If trolley required, indicate if powered, geared or push travel and give beam dimensions
- 8) If an alternative to that mentioned in (2) would be acceptable
- 9) If headroom is restricted, maximum drawn-up dimension allowable

Note: If the hoists are only required for a particular project, they can generally be obtained on a rental basis.

3.10 Winches

3.10.1 Utility Winches

There is a British Standard covering the design and manufacture of winches (namely BS 3701) but it is only applicable to manual versions whereas the majority of utility winches used offshore are usually pneumatically powered (**see Fig 1**).

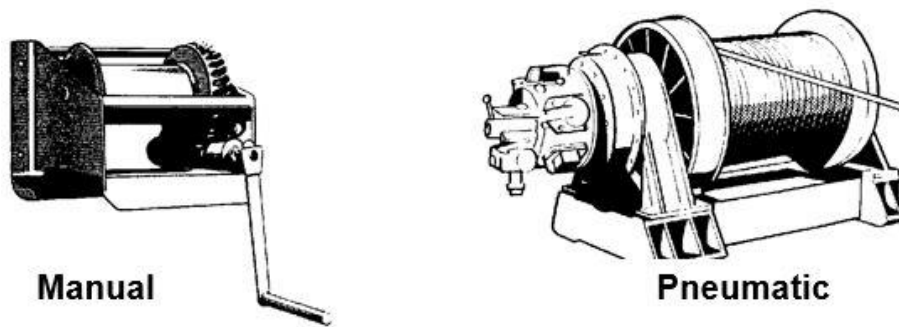


Fig 1

Winches may also be hydraulically or electrically powered but the use of petrol driven or diesel driven units is not permitted.

Winches can be procured in various capacities i.e. WLL 1000lbs, 2000lbs, 3000lbs, 4000lbs, 5000lbs, 7500lbs, 10,000lbs 16,000lbs and 20,000lbs. (Certain manufacturers supply capacities between those stated above).

When selecting a winch, the most important consideration is the required line pull. It must be borne in mind that the majority of manufacturers' line pull ratings are based on half drum performance and this line pull decreases as the drum fills up (**see Fig 2**).

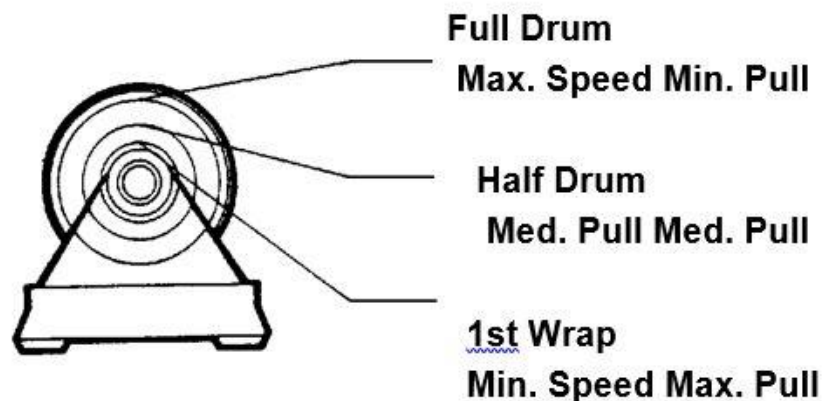


Fig 2

Note: Half drum rating establishes average performance only. This should not be confused with wire rope capacity which is approximately only 36% of full drum storage at this point.

The next consideration should be the drum capacity i.e. how much wire rope is required for the particular operation as this can often lead to a higher capacity winch being required due to loss of line pull. At this stage, check that the platform air supply is adequate to drive the selected winch.

The main control levers shall be arranged to operate in a logical manner and shall be marked clearly and permanently in such a way to demonstrate their purpose and mode of operation. The main control lever will be designed to return to the neutral position upon release, i.e. 'dead man's handle' principle.



A fail-safe braking system, activated when the main control lever is in the neutral position and on failure of the primary power supply, shall be fitted.

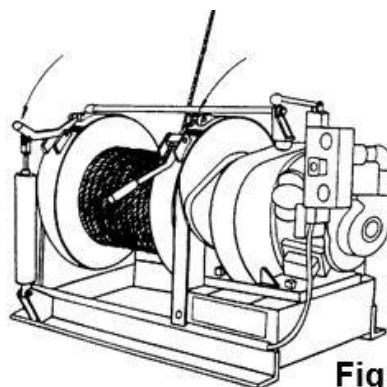
Winches are supplied as a certified unit having been proof loaded to 25% above their SWL but after installation on board, they have to be subjected to a further test to prove the anchorage.

3.10.2 Man-riding Winches

If the winch is to be used for the hoisting and lowering of personnel, the winch design shall be approved by the appropriate Certifying Authority. The winch shall be provided with a secondary, mechanical, means of manual operation which will permit the safe temporary use of the unit in the event of a failure of the primary power supply.

The winch shall incorporate a secondary braking system which is independent of both the automatic brake and the primary power source. This brake should be of a simple design and preferably manual in operation (**see Fig 3**). It must also be fitted with a rope guard.

**Automatic
Brake**



**Manual
Brake**

Fig 3

The clutch must be of a simple design which includes features such that the clutch cannot be disengaged. The design should incorporate additional features which protect critical components from the elements. The winch drum and other moving parts shall be suitably guarded.

The main hoist cable and attachments shall have a minimum factor of safety of 10:1 relative to the hauling load. The main hoist cable and attachments used on casing stabbing boards shall have a minimum factor of safety of 10:1.

The winch should be of adequate capacity to raise and lower the rated load at a speed not in excess of 10m/min only winches which have been originally and specifically designed for man-riding should be used. The rated capacity of the winch based on the line pull on the top layer shall be based on a 10:1 factor of safety.

To facilitate maintenance and inspection procedures, access openings to prime movers, reduction gearing, bearings, brakes and control mechanisms shall be provided.

Man-riding winches will be subjected to much more stringent load testing, the details of which can be found in section 5.1 of this suite of documents.

All winches shall be permanently marked by a rating plate (stating that it is suitable for man-riding) prominently displayed.



Whenever possible, in an effort to minimise the permutation of winches on each installation, preference should be given to the make and model of winches already in use on the installation.

To aid selection of winch type for special projects etc., specifications (i.e. capacity / dimensions) of the more popular models are contained in the following pages.

Note: Once selection is made, the purchasing department will have to be contacted to ensure that the use of the particular model of winch has not been restricted for any reason.

The information to be supplied when raising the requisition will be as follows:

- 1) Quantity
- 2) Manufacturer and model
- 3) SWL
- 4) Whether being used for utility or man-riding
- 5) If a secondary braking system is required (on utility winches)
- 6) Rope diameter, length and type of end fitting
- 7) If an alternative to that mentioned in (2) would be acceptable

Note: If the winches are only required for a particular project, they can generally be obtained on a rental basis.

3.11 Beam Clamps

A beam clamp is a simple portable item of lifting equipment which is used to attach a hoist or similar portable lifting appliance to a structural beam capable of sustaining a load. At present no British Standards exist to govern their design or manufacture but the Industry as a whole tends to use only one specific type and that is the type which actually clamp onto and grip the beam by means of a threaded adjusting bar (**see Fig 1**).

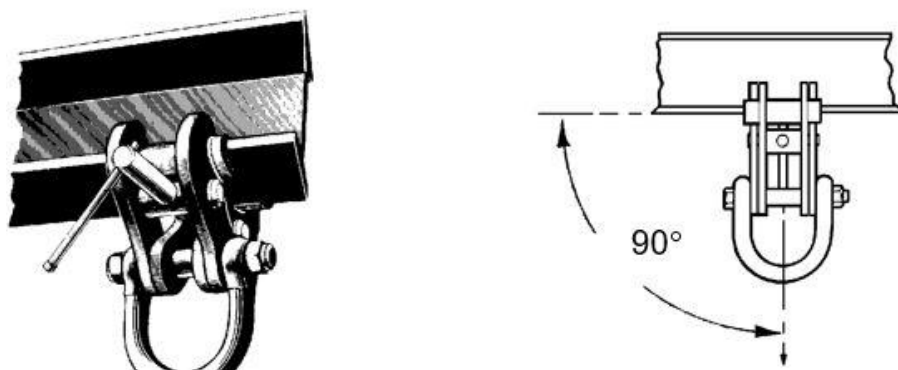


Fig 1

They can be procured in various capacities i.e. WLL 1te, 2te, 3te, 4te, 5te, 6te, 10te, 15te, 20te, 25te and 30te. The clamps become larger as their capacity increases but each one will fit a range of beam sizes (see data sheets at the end of this section).

Beam clamps were primarily designed to operate with a strictly vertical load only but the main manufacturer, Riley Lifting Equipment Ltd who produce the 'Superclamp' range, have created de-rating tables (for certain models only) which allow the beam



clamps to be used for angular loading. This has proved invaluable as lifting beams or support steelwork are rarely directly above the item to be lifted. It must be pointed out however, that the beam may require additional stiffening locally. A design check should be carried out (see Fig 2).

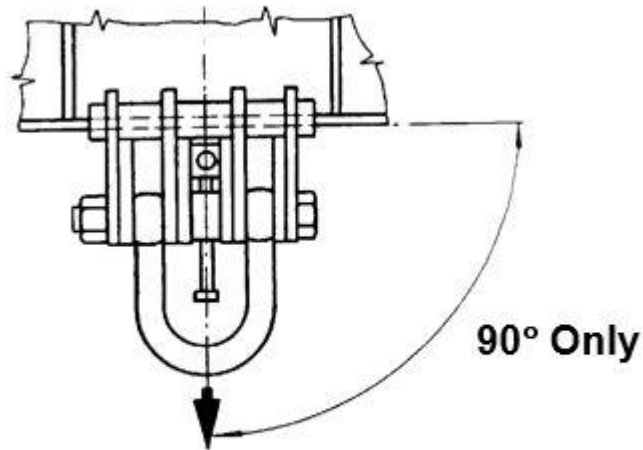
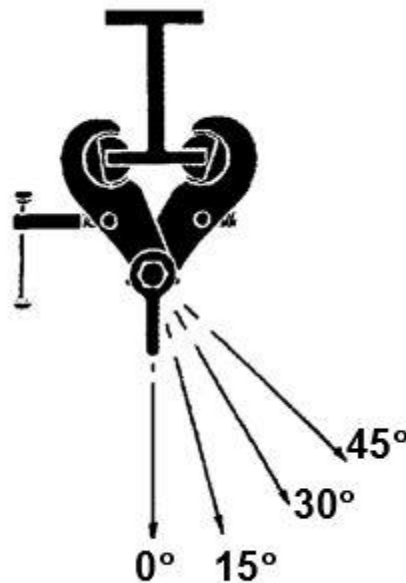


Fig 2

Any load supporting arrangement which cannot be used safely, and is accessible such that casual use could occur, shall be identified and tagged, '**NOT SUITABLE FOR USE**', and any associated equipment removed or isolated to prevent normal usage.





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Tables of allowable angles and loads are contained in the table below.

Reduction in working load limits when side loads applied				
Angle From Vertical	0°	15°	30°	45°
Reduction Factor	Nil	17%	34%	50%
Models	WII	WII	WII	WII
S2a,S5a	3 t	2.5 t	2 t	1.5 t
S3	4 t	3.3 t	2.6 t	2 t
S3x,S3a,S6,S6a	5 t	4.1 t	3.3 t	2.5 t
S4s	6 t	5 t	4 t	3 t
S4a,S11	10 t	8.3 t	6.6 t	5 t
S12,S14	15 t	12.4 t	10 t	7.5 t
S15,S16	20 t	16.6 t	13.2 t	10 t
S17,S18	25 t	20.7 t	16.5 t	12.5 t
S19,S20	30 t	25 t	19.8 t	15 t

The above working load limits and de-rations have been established specifically for most "S" type clamps and only apply in overhead beam attachment i.e. do NOT apply if clamps are to be used for lifting beams. (The side load clamp "SLC range" has been specifically designed for this purpose). The tables apply to Riley clamps only (selected models) and we strongly advise that stress calculations should be carried out by the user's engineering department for any/all support steelwork.

Warning:

All clamps must be correctly applied to the beam by a competent person and fully hand tightened. If in doubt, contact the manufacturer for their recommendations.

Note: Clamp models S1, S2 & S5 are not suitable for any side loading as they are of a lightweight design.

Prior to being received by **QatarEnergy-IS**, each beam clamp will be subjected to a Proof Load relative to its working load limit (WLL) i.e.: WLL x 1.5 or with some manufacturers, WLL x 2 and will be supplied with a proof load test certificate and plated accordingly with the relevant information.

All beam clamps are suitable for sub-sea work.

Whenever possible, is an effort to minimise the permutation of beam clamps on each installation, preference should be given to the make and model of beam clamps already in use on the installation.

To aid selection of beam clamp type for special projects etc., specifications (i.e. capacity / dimensions) of the more popular models are contained in the following pages.

Note: Once selection is made, the purchasing department will have to be contacted to ensure that the use of the particular model of beam clamp has not been restricted for any reason.



The information to be supplied when raising the requisition will be as follows:

- 1) Quantity
- 2) Manufacturer and model
- 3) SWL
- 4) Beam dimensions
- 5) If an alternative to that mentioned in (2) would be acceptable
- 6) If headroom is restricted, maximum dimension allowable between beam flange and bearing point for hoist

Note: If the clamps are only required for a particular project, they can generally be obtained on a rental basis.

3.12 Beam Trolleys

A beam trolley is a portable item of lifting equipment which is used to attach a hoist or similar portable lifting appliance to a structural beam capable of sustaining a load and to allow the load to be moved horizontally. At present no British Standards exist to govern their design or manufacture.

Beam trolleys are in the main, adjustable to suit a specific range of beam flange widths. Adjustment is achieved by means of a screwed adjusting bar or by the positioning of spacer washers on the load bar between the side plates (**see Fig 1**).

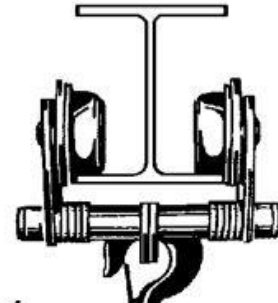
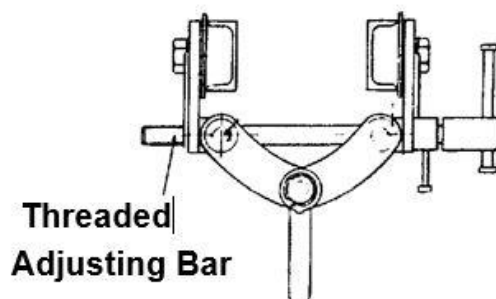


Fig 1

There are two types of beam trolley available, push travel and hand geared travel. Push travel trolleys are normally used for loads of up to 2 tonne where linear movement is achieved by pushing against the load. Geared travel trolleys are preferred when loads in excess of 2 tonne are to be handled or accurate positioning is required. In this case, linear movement is achieved by operating a handchain (**see Fig 2**).

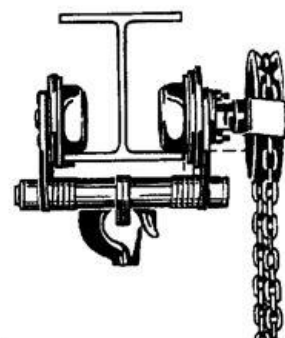
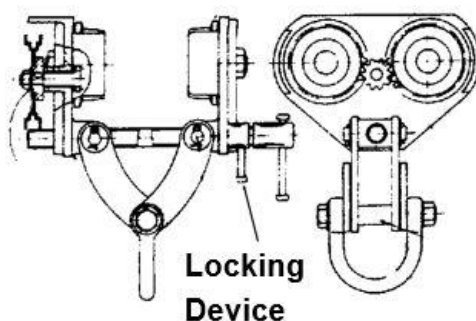


Fig 2



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When selecting the Safe Working Load (SWL) of the beam trolley, the weight of the hoist unit must be taken into account and the operating chains if fitted must be long enough to hang approximately half a metre above the operating level.

They can be procured in various capacities i.e. WLL 1te, 2te, 3te, 4te, 5te, 6te, 10te, 15te, 20te etc. (these can vary from one manufacturer to the next. The trolleys become larger as their capacity increases but each one will fit a range of beam sizes (see data sheets at the end of this section).

Beam trolleys are designed to operate with a vertical load only and are not suitable for side loading i.e. the load should always be picked, traveled and landed directly below the beam.

It must be pointed out that the beam trolleys should only be used on designated runway beams with the exception that they may be used on a beam forming part of a structure where a specific design check for this purpose has been made and end stops or a suitable means of preventing the trolley running off the end of the beam has been fitted. The beam should also be sufficiently level under all loading conditions such that the trolley will not run away under gravity.

Prior to being received by **QatarEnergy-IS**, each beam trolley will be subjected to a Proof Load relative to its working load limit (WLL) i.e.: $WLL \times 1.5$, will be supplied with a proof load test certificate and will be plated accordingly with the relevant information. Beam trolleys are suitable for sub-sea work.

Whenever possible, is an effort to minimise the permutation of beam trolleys on each installation, preference should be given to the make and model of beam trolleys already in use on the installation.

To aid selection of beam trolley type for special projects etc., specifications (i.e. capacity / dimensions) of the more popular models are contained in the following pages.

Note: Once selection is made, the purchasing department will have to be contacted to ensure that the use of the particular model of beam trolley has not been restricted for any reason.

The information to be supplied when raising the requisition will be as follows:

- 1) Quantity
- 2) Manufacturer and model
- 3) SWL
- 4) Beam dimensions
- 5) Length of operating chains
- 6) If an alternative to that mentioned in (2) would be acceptable
- 7) If headroom is restricted, maximum dimension allowable between beam flange and bearing point for hoist

Note: If the trolleys are only required for a particular project, they can generally be obtained on a rental basis.

3.13 Plate Clamps

Plate clamps although manufactured in the UK are not covered by a British Standard. For this reason, the design strength will have to meet **QatarEnergy-IS** requirements as stated in the following paragraphs.

Each plate clamp shall be designed with an ultimate strength equal to five times the Safe Working Load in the specified operating mode.

The manufacturer shall, in the construction of the clamp, use material which is of appropriate strength and dimension for the designed capacity of the plate clamp.



Whenever possible, all components used in the assembly, e.g. bolts, springs etc, shall also conform to recognised standards.

All welds shall be of adequate strength and produced to recognised welding standards. The manufacturer shall ensure that each plate clamp is assembled correctly and to high standards of workmanship. He shall also ensure that each plate clamp is capable of operating at maximum efficiency and is free from defects and flaws.

Each plate clamp shall be subjected to a Proof Load Test equal to twice the Safe Working Load. The load shall be applied in a similar manner to that applied during normal working conditions. It is important that the test should make allowance for permissible off-centre loading if appropriate.

When the plate clamp is capable of turning a plate through a 90° arc the proof load test shall simulate this operation.

Plate clamps are available in two basic designs i.e. Horizontal plate clamps (used in pairs and usually suspended by a two leg sling from a beam) for handling plates in the horizontal position only, and Universal plate clamps for handling plates in the vertical position. (These clamps can also lift from horizontal to vertical and vice-versa, but should NOT be used to transport plates horizontally). **See Fig 1.**

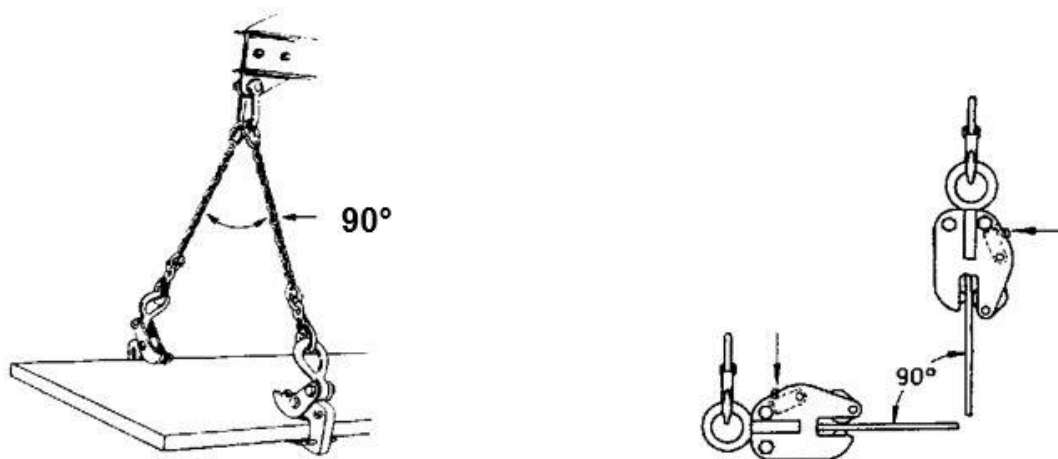


Fig 1

The first consideration when selecting the clamp type is how the plate is to be transported and stacked i.e. horizontally or vertically.

The second consideration is the weight of the plates to be handled which will determine the WLL of the clamp(s).

The third consideration is the thickness of the plate which will determine the correct model of clamp with the appropriate jaw opening.

It is important to select the WLL of the clamp close to (but never below) the weight of the plate as over-specified clamps will not be as efficient.

Whichever type of clamp is selected, it will only be suitable for lifting one plate at a time. For lifting more than one plate (in the horizontal position only) there are specially designed clamps for this purpose known as sheet bundle clamps (**see Fig 2**).

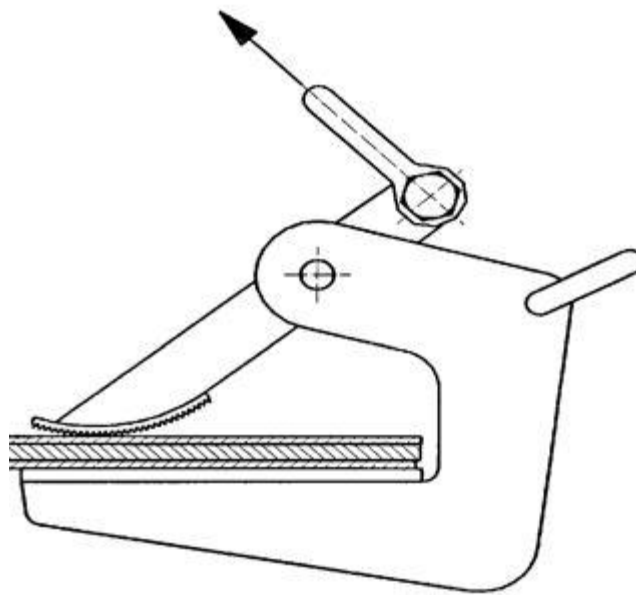


Fig 2

Prior to being received by **QatarEnergy-IS**, each plate clamp will be subjected to a Proof Load relative to its working load limit (WLL) i.e.: $WLL \times 2$ and will be supplied with a proof load test certificate and plated accordingly with the relevant information. Plate clamps are not suitable for sub-sea work.

Whenever possible, in an effort to minimise the permutation of plate clamps on each installation, preference should be given to the make and model of plate clamps already in use on the installation.

Note: Once selection is made, the purchasing department will have to be contacted to ensure that the use of the particular model of plate clamp has not been restricted for any reason.

The information to be supplied when raising the requisition will be as follows:

- 1) Quantity
- 2) Manufacturer and model
- 3) Type i.e. horizontal or vertical
- 4) SWL
- 5) Jaw opening
- 6) If an alternative to that mentioned in (2) would be acceptable

Note: If the plate clamps are only required for a particular project, they can generally be obtained on a rental basis.

3.14 Wire Rope Pulley Blocks

All wire rope pulley blocks used by **QatarEnergy-IS** shall be designed and constructed in accordance with BS 4018 (if under SWL25 tonne), BS 4536 (if over 25 tonne) or the equivalent international standards.

Pulley blocks or sheave blocks are available in the form of single sheave or multi-sheave with a choice of head fittings, the most popular being swivel shackle or swivel oval eye (see Fig 1).

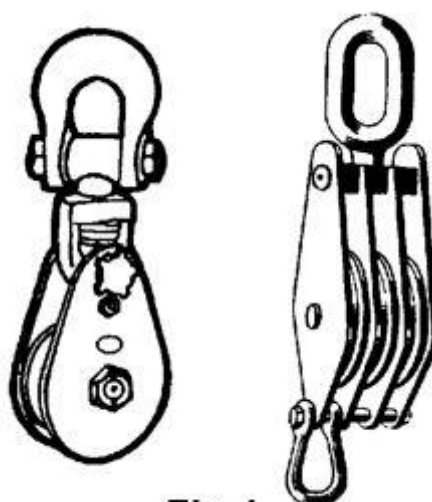


Fig 1

The main function of a single sheave block is to change the direction of the hoisting or pulling rope whereas multi-sheave blocks reduce the necessary pull required on the lead rope to lift the load i.e. the pull required reduces as the number of falls (sheaves) increases.

The two main types of block available are the British Standard blocks (usually in the multi-sheave variety) and the US pattern blocks (usually the single sheave variety). Either type is acceptable to **QatarEnergy-IS**.

The main factor to be aware of is the different methods of rating the blocks i.e. British blocks are rated by 'line pull' whereas US pattern are rated by head load. The BS blocks have a FOS of 5:1 where the US pattern have a FOS of 4:1

e.g. A British Standard single sheave block rated at 8 tonne means that a load of 8 tonne can be raised which, in the worst case, could result in a 16 tonne (plus friction) load on the head fitting. It will be fitted with a 16 tonne head fitting.

In the same scenario, a US pattern single sheave block rated at 8 tonne * means that a load of only 4 tonne can be raised as the head fitting is only capable of supporting slightly over 8 tonne (Head fitting is usually an 8te shackle).

* - Sometimes rated in short tons (2,000lbs per ton).

Pulley blocks or sheave blocks can be procured in various capacities and with or without beackets as required.

The most popular blocks in constant use offshore are 'Snatch blocks' which have a side opening gate or similar to allow the loading of a rope without having to remove the hard eye. Certain modifications have been made to the design of these blocks due to previous minor incidents. These are the fitting of safety shackles as head fittings and the fitting of an 'R' clip to prevent inadvertent dislodgement of the retaining bolt.

Prior to being received by **QatarEnergy-IS**, each sheave block will be subjected to a Proof Load relative to its Working Load Limit (WLL) and will be supplied with a proof load test certificate and plated accordingly with the relevant information.

Sheave blocks are suitable for sub-sea work.

Whenever possible, is an effort to minimise the permutation of sheave blocks on each installation, preference should be given to the make and model of sheave blocks already in use on the installation.



To aid selection of sheave block type for special projects etc., specifications (i.e. capacity / dimensions) of the more popular models are contained in the following pages.

Note: Once selection is made, the purchasing department will have to be contacted to ensure that the use of the particular model of sheave block has not been restricted for any reason.

The information to be supplied when raising the requisition will be as follows:

- 1) Quantity
- 2) Manufacturer and model
- 3) SWL
- 4) Number of sheaves
- 5) Diameter of sheaves
- 6) If an alternative to that mentioned in (2) would be acceptable

Note: If the sheave blocks are only required for a particular project, they can generally be obtained on a rental basis.

3.15 Hydraulic Jacks, Rams and Pumps

The design and manufacture of hydraulic lifting equipment (namely jacks, rams and pumps) is not governed by any British Standard.

Hydraulic lifting equipment is available in two basic formats i.e. a ram / plunger and cylinder with an integral pump unit (**see Fig 1**) and a ram and pump set which consists of a ram / plunger and cylinder with a remote pump unit connected by a high pressure hose (**see Fig 2**).



Fig 1

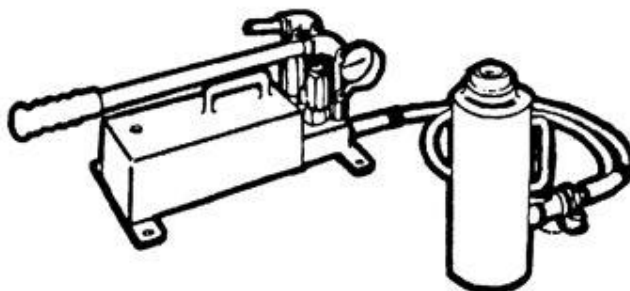


Fig 2

When selecting your hydraulic equipment, the most important consideration is the capacity. (The term SWL does not apply as hydraulic jacks and rams etc. are not subjected to a proof load).

Another major consideration is the number of jacking positions and the distance between them e.g. should the positions be far apart, if jacks were to be used, an operator would be required for each position whereas if rams with remote pump units were utilised, the pumps could be brought together or connected to a manifold and worked by one operator.



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If the jacking operation is in a confined area, it is preferable to use rams and pumps as opposed to jacks to ensure the safety of the operator(s).

Should there be a lack of space to insert a jack, certain models of jack (with extended bases) can be fitted with a claw attachment. It must be borne in mind however that the use of a claw attachment reduces the capacity of the jack to 40%

As an alternative to a jack with a claw there are low-headroom rams known as pancake rams or pad jacks. The major drawback with this type of unit is the very limited stroke. Jacks and rams can be procured in various capacities and dimensions according to the different manufacturers.

Should the load require to be monitored throughout the jacking procedure, pressure / load gauges can be fitted.

Should the load have to be sustained, jacks / rams can be procured with locking collars (see Fig 4).

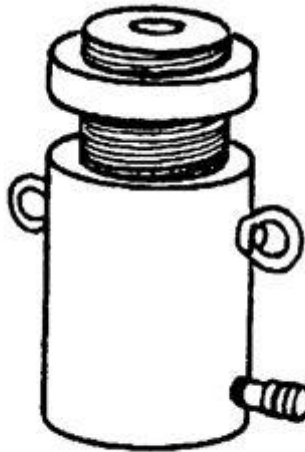


Fig 4

As an additional safety measure, a lock off safety valve can be fitted to the ram to maintain pressure in the event of a hose failure.

Prior to being received by **QatarEnergy-IS**, each ram or jack will be subjected to a test load equal to its capacity and will be supplied with a letter of conformity and hard stamped accordingly with the relevant information.

Standard rams and jacks are not suitable for sub-sea work. The manufacturers should be contacted for specialised equipment for this type of work.

Whenever possible, is an effort to minimise the permutation of hydraulic equipment on each installation, preference should be given to the make and models already in use on the installation.

To aid selection of hydraulic equipment type for special projects etc., specifications (i.e. capacity / dimensions) of the more popular models are contained in the following pages.

Note: Once selection is made, the purchasing department will have to be contacted to ensure that the use of the particular model of jack or ram and pump has not been restricted for any reason.

The information to be supplied when raising the requisition will be as follows:

- 1) Quantity
- 2) Manufacturer and model (of Ram or Jack)



- 3) Capacity
 - 4) Stroke
 - 5) Whether or not gauges are required
 - 1) (If applicable)
 - 6) Manufacturer and model (of pump) and Qty
 - 7) Length of hydraulic hose(s) and Qty
 - 8) Type of manifold (if required) and Qty
 - 9) If an alternative to that mentioned in (2) and (6) would be acceptable
 - 10) If headroom / space is restricted, maximum closed height allowable
- Note:** If the hydraulic equipment is only required for a particular project, it can generally be obtained on a rental basis.

3.16 Wire, Ropes & Fittings

Wire rope for general purposes as used in the engineering industry shall be designed and manufactured in accordance with BS 302 or an equivalent international standard. The construction of wire rope consists of individual steel wires spun into a number of strands, which are in turn laid helically around a central core (See Fig 1).

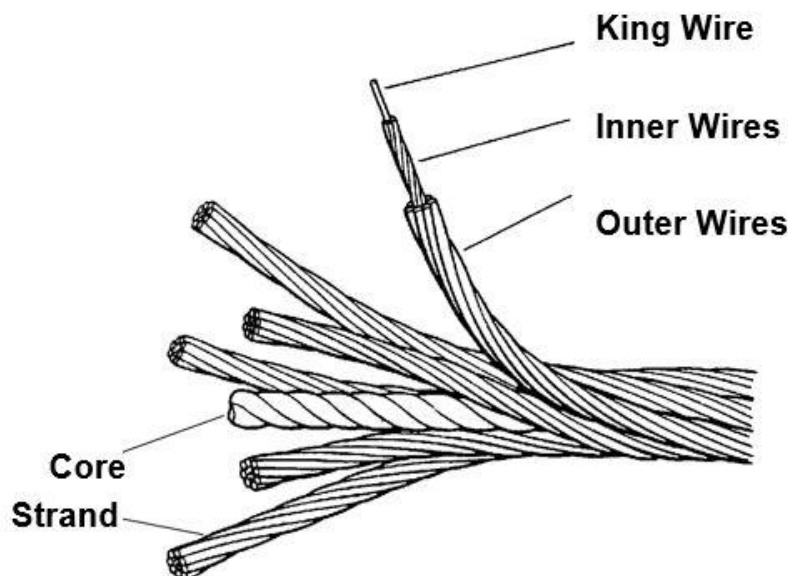


Fig 1

General purpose steel wire ropes are manufactured in three basic types; round strand, multi-strand and triangular strand. The latter type is not commonly found on offshore installations.

There are two types of core, fibre and steel, both serve the same purpose in affording support to the strands laid around it. Fibre cores can be constructed from natural or man-made fibre. Certain man-made fibres, such as polypropylene, have the advantage that they neither absorb nor retain moisture. Steel cores usually comprise, an independent wire rope core (IWRC) around which the outer strands are laid. In smaller size ropes, less than 13mm diameter, a wire strand core might be used. A rope with a steel core is stronger than a rope with a fibre core and is much more resistant to deformation, crushing and stretch.

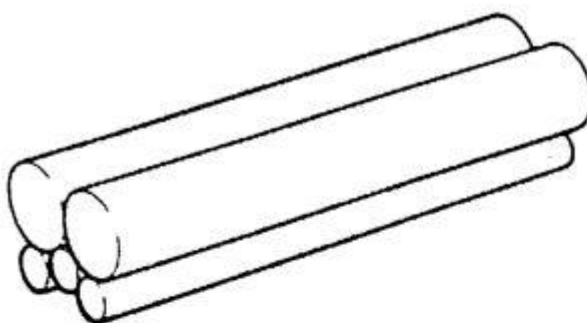
Note: It is **QatarEnergy-IS** policy that only IWRC can be used in the manufacture of their slings. Slings made with fibre core wire rope are NOT acceptable.



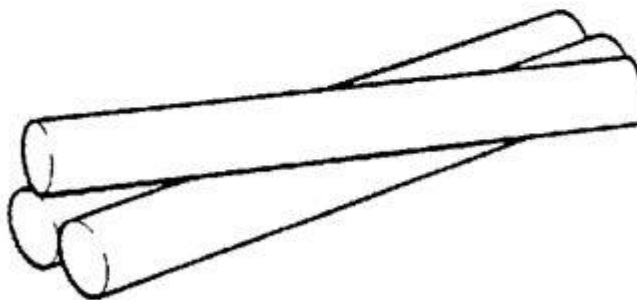
There are two methods of laying wires in a strand.

- 1) Equal lay
- 2) Cross lay

Fig 2 shows the difference between the two methods of laying the wires. In an equal lay rope construction, the wires in each layer in the strands lie either in a bed format or, alternatively, along the crown of an underlying wire. In a cross lay construction the individual wires cross over the crowns of underlying wires.



Equal Lay



Cross Lay

Fig 2

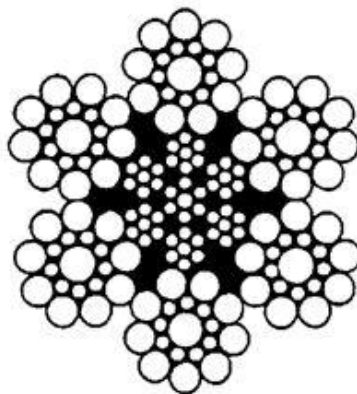
Comparatively, equal lay ropes have a higher braking strength, better bend fatigue performance and better resistance to crushing. For these reasons cross lay constructions are not readily available and are not recommended for general use.



There are three basic equal lay strand constructions:

1) Seale

In this construction the strands have two layers of wire around a king wire. Both layers have the same number of wires and the same length of lay. **(See fig 3)**

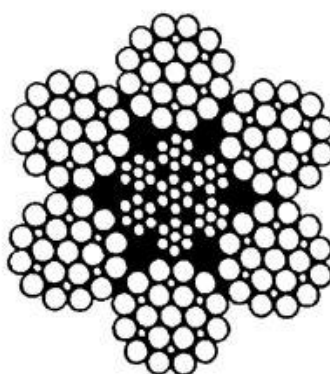


Typical Seale Strand rope 6 x 19 (9/9/1) IWRC

Fig 3

2) Filler

In this construction, the strands have two layers of wires about a king wire. There are twice as many wires used in the outer layer as there are in the inner layer. Both layers have the same length of lay. To position the wires properly in the outer layer, small filler wires are incorporated. **(See Fig 4)**



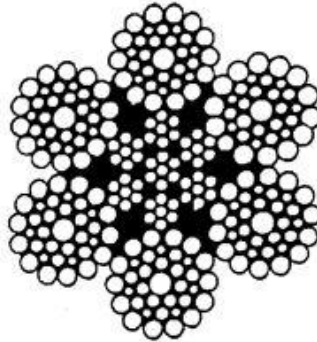
Typical Filler Strand Rope 6 x 19 (12/6+6F/1)

Fig 4



3) Warrington Seale

In this construction the strands have three layers or wire around a king wire. In the 6 x 36 construction, the first layer comprises seven wires of equal diameter. The intermediate layer (Warrington layer) consists of fourteen wires, seven large diameter and seven small diameter, laid large and small alternatively. This layer is covered by fourteen wires of the same diameter (Seale layer). All three layers have the same length of lay. (See Fig 5)



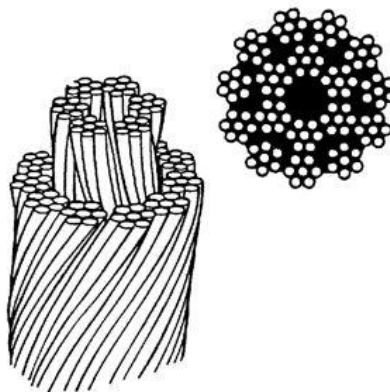
Typical Warrington-Seale Strand Rope

6 x 36 (14/7+7/7/1) IWRC

Fig 5

Multi-strand Rope has two or more layers of strands. The inner and outer strands are laid in opposite directions so that their tendencies to unlay, act in opposition. The type of lay, (ordinary or Lang's) in the outer and inner layers, is at the discretion of the rope manufacturer. These ropes are sometimes referred to as Rotation Resistant.

Fig 6 shows two typical examples of multi-strand rope. However, there are many relatively new ropes available on the market which have exceptionally high fill factors (a term used to relate the area of steel to the total cross sectional area of the rope).



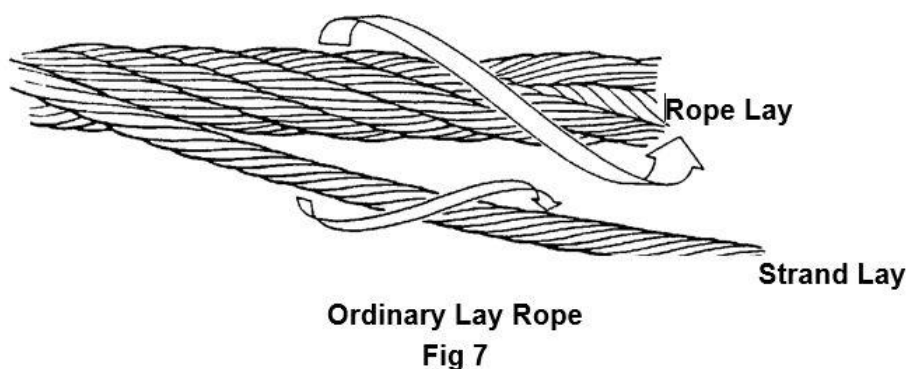
Example of Multi-Strand Rope

Fig 6

There are two types of rope lay:

1) Ordinary Lay (US terminology is 'Regular Lay')

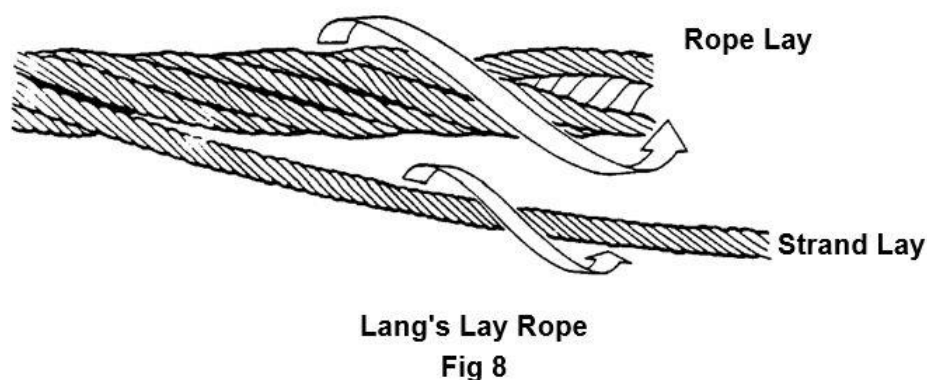
An ordinary lay rope is one in which the strand wires are laid in one direction and the completed strands laid in the opposite direction (See Fig 7).



2) Lang's Lay

This rope is one in which the strand wires are laid in the same direction as the strands in the rope (See Fig 8). The Lang's lay rope exposes the outer wires for a longer length and therefore has better wear properties than ordinary lay rope.

Caution: Lang's lay rope must not be used for slings or when one end is free to rotate.



Of the two types, the ordinary lay rope is easier to handle, more resistant to crushing and distortion and is the more commonly used on offshore installations.

The majority of ropes are of right-hand lay, although occasionally left-hand lay ropes can be supplied. The direction of rope lay is important, to ensure correct coiling and spooling of the rope.

The length of lay, is the distance measured parallel to the longitudinal axis or centre line, of the rope in which the strand makes one complete spiral around the axis (See Fig 9).



Measuring One Rope Lay In a Six-Strand Rope

Fig 9

The largest proportion of ropes supplied are pre-formed, which means, that the finished strands are formed prior to closing, into the helical shape which they will take in the finished rope. This ensures that the strands lie naturally in the rope, thus reducing the internal stresses which could reduce the fatigue life of the rope.

Steel wire ropes are available in different tensile strengths. In Europe the specified tensile strength, or grade, relates to the minimum tensile strength of the individual wires. All the wires in a rope, with the exception of those in the IWRC, must be of the same tensile grade. All wires in the IWRC must be of the same tensile grade, albeit a lower grade than those in the main rope.

All wire ropes must be at least 180 grade. The only exceptions to this are elevator ropes, which may be of dual tensile strength of less than 180 grade.

A number of cranes use ropes that are specified to the US Federal or API Specification. Both of these specifications refer to tensile strength as 'Improved Plow' and 'Extra Improved Plow' steel. These terms correspond approximately to 180 grade and 200 grade respectively. Unlike the European system, the US grading system has a sliding scale of minimum tensile strength relative to the wire diameter. Care must be exercised when substituting European-specified rope for US-specified rope and vice versa.

Note: Problems can also arise when fitting metric ropes to imperial equipment and vice versa.

When selecting a rope for a specific application, the following points should be considered:

- 1) The type of equipment on which the rope will be used
- 2) The maximum load which will be applied to the rope
- 3) The factors which will cause rope deterioration

The minimum factor of safety (for lifting purposes) which must be applied is 5:1, therefore the maximum Safe Working Load permitted is one fifth of the minimum breaking load of the rope. Certain applications will require an increased factor of safety, e.g. rope used on a man-riding winch should have a minimum factor of safety of 10:1. When considering likely causes of deterioration, elements such as fatigue, corrosion, abrasion and mechanical damage must be examined.

The selection of a rope could rely on any one of the points listed in (1), (2) and (3) or, it may be a combination of them all. It is important to select the most suitable construction for the most significant feature, or worst condition affecting the performance of the rope.

Fatigue is a major consideration when selecting the rope construction. The major cause of fatigue is the repeated bending of rope under tensile load when it is passing around pulleys, drums etc. The factors relating to the severity of the effects of fatigue are:

- 1) The load on the rope
- 2) The ratio of the sheave or drum diameter to the wire rope diameter



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- 3) The number of operating cycles
- 4) The construction of the rope

Rope life will decrease as the tensile load increases. The change is non-linear, but at normal working loads, and provided that all other factors remain unchanged, the rope life can be expected to be reduced by 50%, if the average tension is doubled. Conversely, if the minimum ratio of sheave and drum diameters to rope diameter is doubled, the rope life will be expected to increase four times.

If the sheave diameter / rope diameter ratio is small, bending fatigue is likely to be the main problem. The rope chosen for this situation must be flexible, e.g. a 6 x 36 Warrington-Seale construction will be more suitable than a 6 x 19 Seale construction. As the effects of fatigue can be increased if the rope becomes dry, it is important to keep the rope well lubricated.

Corrosion presents another aspect to affect your choice of construction. Contract to the most suitable construction to compensate for the effects of fatigue, corrosion is best compensated for by using a rope which has large outer wires. A suitable construction would be 6 x 19 (9/9/1). If the effects of corrosion are severe, the use of good quality galvanised rope should be considered.

Abrasion occurs in the outer wires and therefore ropes with large outer wires give better rope performance under abrasive conditions. The use of Lang's lay construction, which exposes a longer length of wire on the surface, may also improve resistance to the effects of abrasion. However, care must be exercised to ensure that it is safe to use a Lang's lay rope for the particular application.

Mechanical Damage can be sustained due to rotation. When using ordinary lay, six-strand rope, a swivel should be used to ensure that the load does not rotate when being lifted. The use of a swivel is not recommended when using multi-strand rope or low rotation, six-strand rope. In such situations the guidance of the manufacturer or competent person should be sought.

There is little benefit in using a swivel on a multi-part reeving system. However, if swivels are used on such systems they must be capable of being locked.

There are many designs of high tensile, multi-strand, anti-rotating ropes on the market which have advantages over the standard multi-strand ropes. Care should be exercised when using these high-fill factor ropes, since they can develop internal fatigue problems.

Elongation can occasionally be a problem on applications offshore. The following factors relate to elongation:

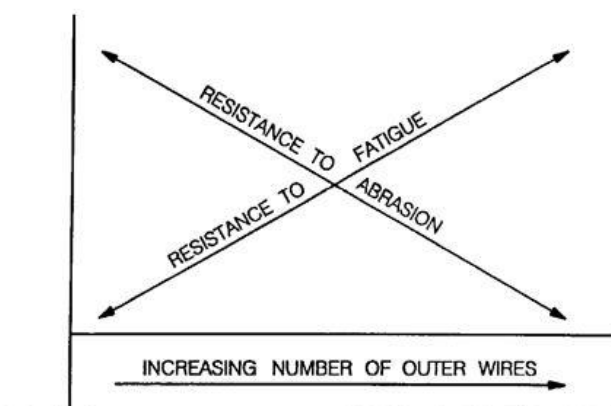
- 1) Constructional stretch, the result of the individual wires and strands settling when the load is applied
- 2) Elastic stretch, the result of the component wire stretching under load
- 3) Extension of rope lay, the result of rotation

Should elongation be a critical consideration, the individual manufacturer selected should be consulted for more specific information.

3.17 Crushing

The effects of crushing should be reduced by using wire ropes with IWRC of ordinary lay and with relatively few wires in the construction.

It is apparent that there has to be a compromise when choosing the most suitable construction. The rope which reduces the effects of corrosion or abrasion will aggravate fatigue problems and vice versa. Figure 10 illustrates the point.



Relative Fatigue And Abrasion Resistance of Six-Strand Rope

Fig 10

Note:

For ordering purposes, measurement of wire rope diameter is made over the highest points of the strands (see Fig 11).

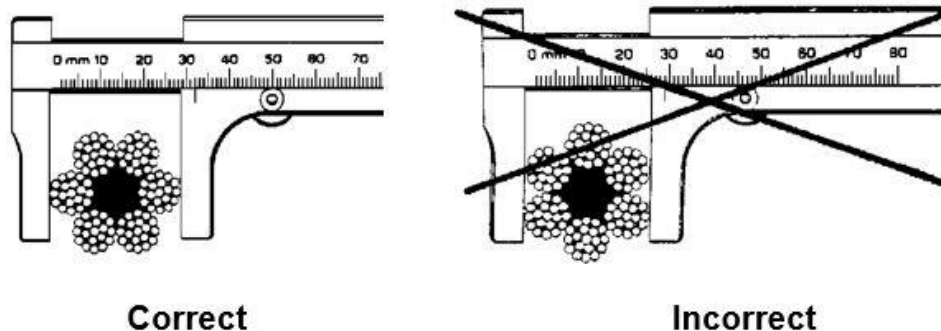
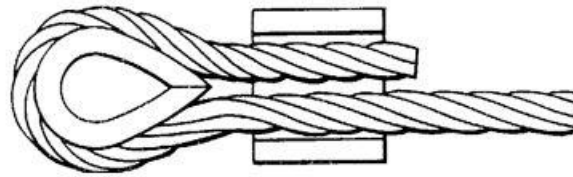


Fig 11

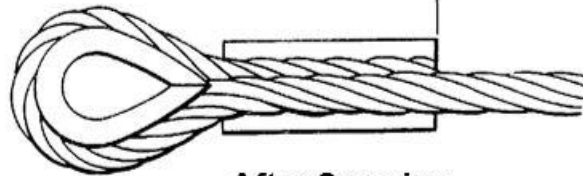
Should you be re-ordering rope, ensure you measure the rope correctly to avoid being supplied with an UNDER-SIZED rope.

There are various methods of terminating a wire rope, the most popular being an aluminium alloy talurit secured eye. This is where a wire rope is passed through the oval shaped talurit, formed into a loop and passed back through the talurit where upon the talurit is compressed to a cylindrical shape. This is a purely mechanical splice (see Fig 12).



Before Swaging

“Dead end” of rope
should be flush or
slightly protruding.



After Swaging

FIG 12

As an alternative method of termination, there is the steel ferrule secured flemish eye also known as a Superloop or Yankee splice. This is a combination of a hand and a mechanical splice (see Fig 13).

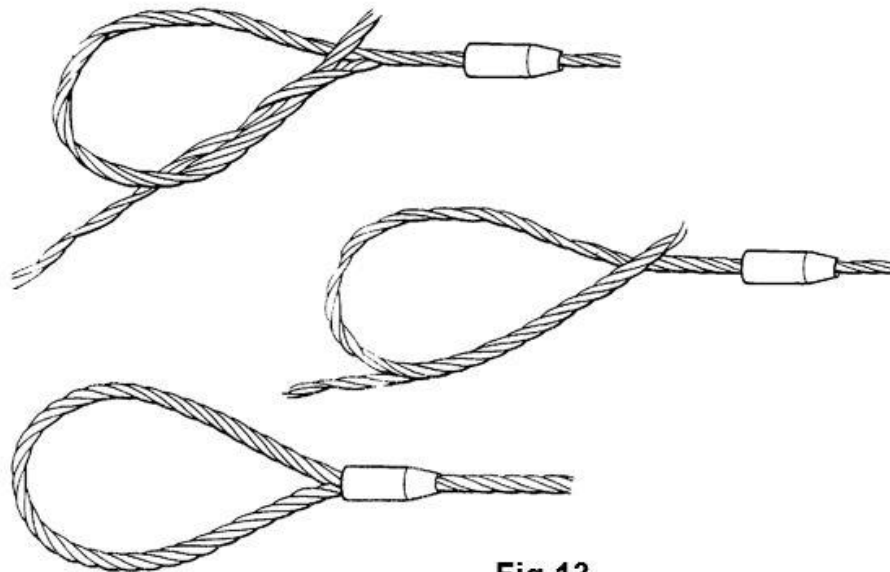


Fig 13

Note 1: Only the first type is acceptable to QatarEnergy-IS. It is QatarEnergy-IS policy NOT to use slings constructed with steel or tapered ferrules.



Note 2: The use of purely hand spliced terminations in lifting applications is forbidden.

In either case, the eye can be formed with wire only (commonly known as a soft eye) or fitted with a protective thimble (commonly known as a hard eye) **see Fig 14.**



Note 3: When the wire rope is being used to make slings, single leg slings for general use are usually fitted with soft eyes whereas multi-leg slings where the eyes are fitted to masterlinks and shackles should always be fitted with hard eyes.

The next most popular type of termination is the spelter socket. There are two basic patterns, those made to British standard (BS 463) and those made to US Federal Specification RR-S-550D Type A and B and both patterns have open and closed types. (Refer to tables 5, 6, 7 & 8 at the end of this section for dimensions).

The occasion will arise where the wire rope will have to be reeved up offshore prior to the termination being fitted. This being the case, the only termination that can be safely and efficiently installed on-site is the open wedge socket (**see Fig 15**). This is an independently certified unit and fitted correctly, does not require re-testing. (Fitting details can be found in the Rigging and Lifting Handbook). Refer to table 9 at the end of this section for dimensions.

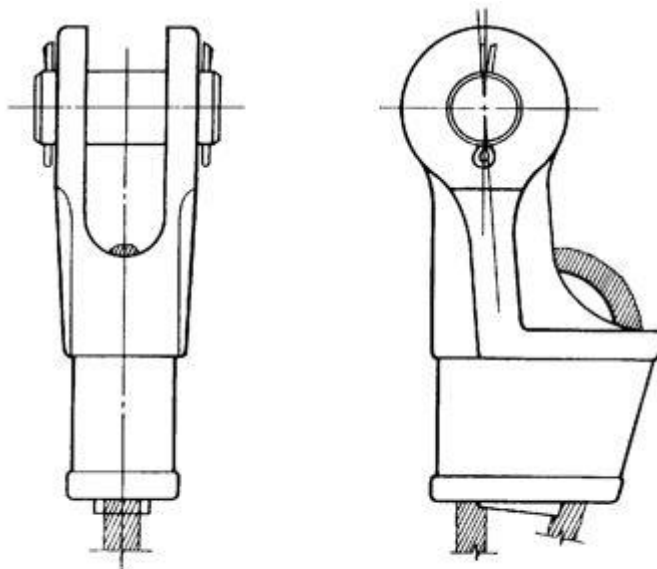


Fig 15



There are alternative temporary methods of terminating a wire rope such as Fist grips, Eureka clamps and Wire rope grips but they are not to be used in a lifting application.

The information to be supplied when raising the requisition will be as follows:

- 1) Quantity
- 2) Length
- 3) Diameter
- 4) Construction
- 5) Tensile Strength
- 6) SWL
- 7) Type of terminations

3.18 Wire Rope Slings

Wire rope slings shall be manufactured in accordance with BS 1290 / BS EN 13414 from wire rope which meets the requirements of BS 302 or an equivalent International Standard. The tensile strength of the wire is normally 180kg/mm² (1770N/mm²). The most common wire constructions are 6x36 or 6x19, IWRC or fibre core. They can be manufactured as single leg slings with hard eyes or soft eyes or endless to suit specific requirements (see Fig 1).

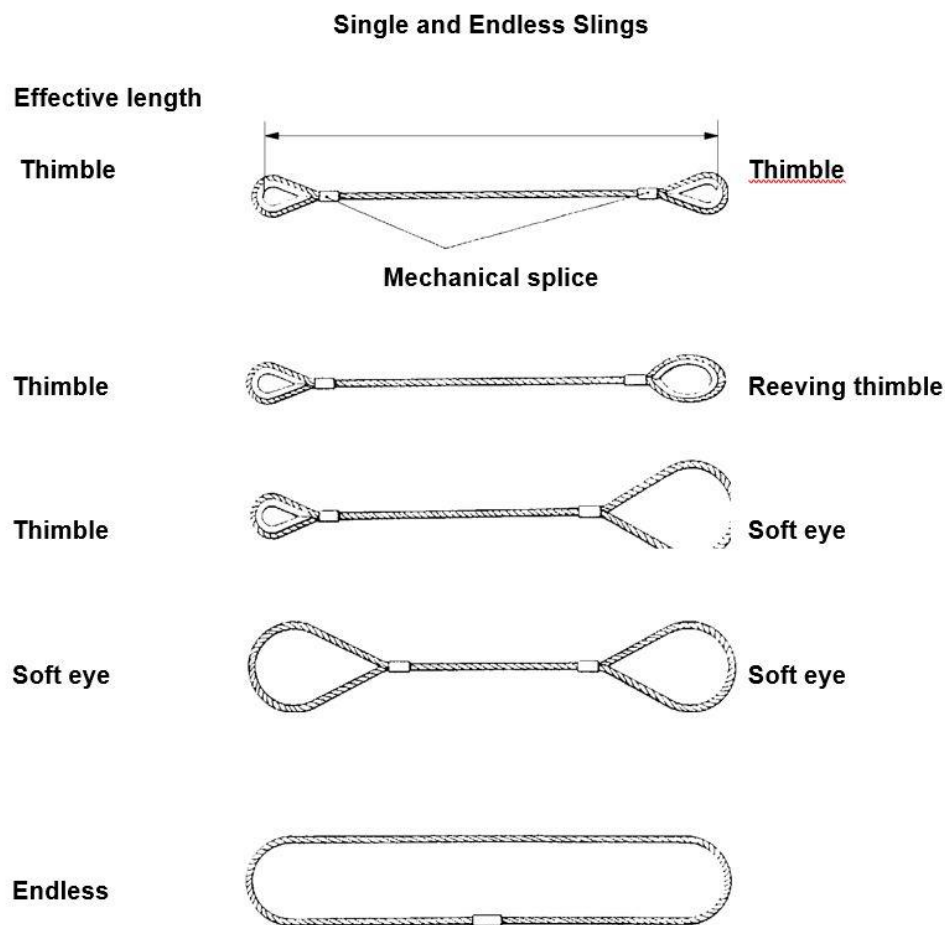
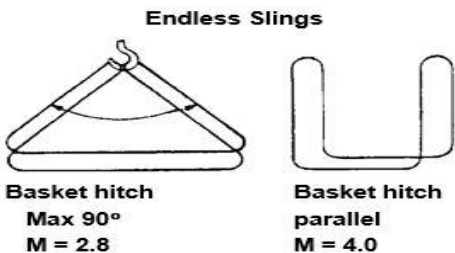
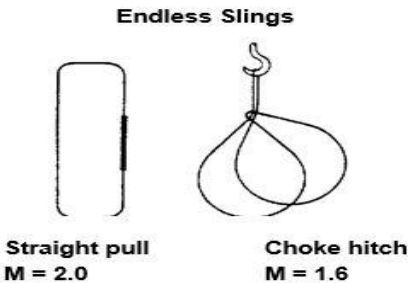
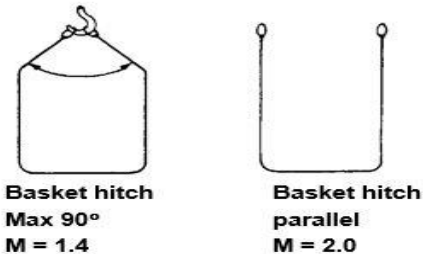
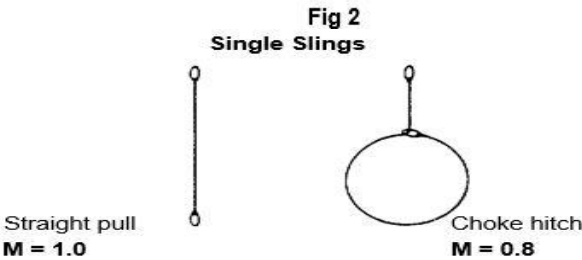
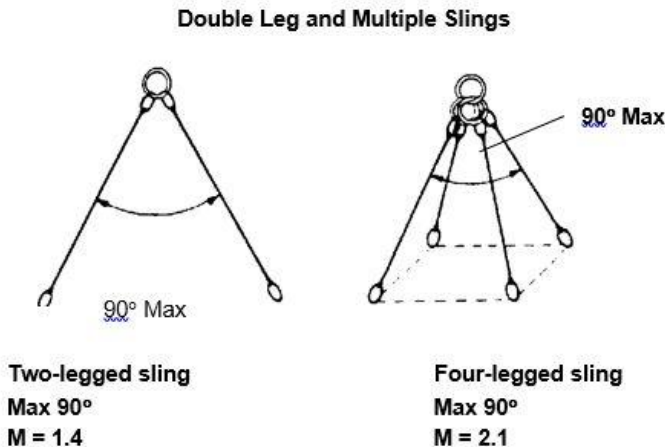


Fig 1



When working with single leg and endless slings, the way in which they are used can affect their overall capacity i.e. it can either increase or decrease (see Fig 2). i.e. multiply the SWL of one leg by the Mode Factor “M” to obtain SWL of the configuration.





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As well as single leg options, they can be supplied as double leg slings fitted to a single link (known as a masterlink) or as multi-leg slings (3 or 4 legs) fitted to a master link assembly (known as a quadruple assembly). **See Fig 3.**



Two Leg



Three Leg



Four Leg

Fig 3

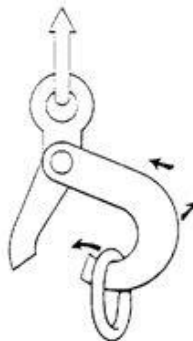
In either case, a top lifting leg can be fitted to ease attachment to crane hook by the user. With double leg and multi-leg slings, all sling legs terminated onto the link or quadruple assembly should be fitted with thimbles to increase the bend radius for the wire and also improve the wear situation. This same rule applies to the other end of the sling leg if compatible sized shackles are being used. Soft eyes would only be efficient if oversized shackles were being used (for example to suit existing padeyes) or if the sling was picking up on lifting trunnions.

For loads with an offset centre of gravity, slings can be constructed with odd-length legs to maintain a level lift.

In addition to having hard eyes, wire slings (such as crane pennants) are often fitted with hooks. The preferred type are the 'BK' pattern which, due to their closed/smooth design, do not tend to snag on protrusions such as handrails etc. These are suitable up to SWL 25 tonne. As an alternative but not so popular there are SF safety hooks (see Fig 4).



BK Type



SF Type

Fig 4



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The SWL of wire rope slings for general use is one fifth of the breaking load and the load rating is based on the uniform load method which give the following results:

Single leg sling (with adjustment for sling mode as depicted in Fig.2)	= 1.0 x SWL of a single leg
Double leg sling from 0° to 90°	= 1.4 x SWL of a single leg
Three and Four leg sling from 0° to 90°	= 2.1 x SWL of a single leg
Double leg sling from 90° to 120°	= 1.0 x SWL of a single leg
Four leg sling from 90° to 120°	= 1.5 x SWL of a single leg

Prior to being received by **QatarEnergy-IS**, each wire rope sling will be subjected to a Proof Load relative to its Safe Working Load (SWL) i.e. SWL x 2 and will be hard stamped accordingly with the relevant information.

To aid selection of sling leg diameters, wire rope breaking loads are to be found in Table 1 (2.14.26).

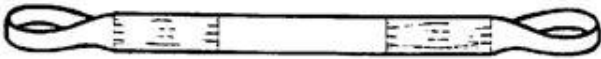
Note: Once selection is made, the purchasing department will have to be contacted to ensure that the use of any of the particular components of the wire rope sling has not been restricted for any reason.

The information to be supplied when raising the requisition shall be as follows:

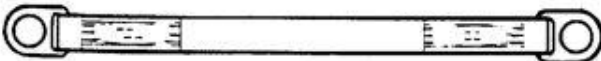
- 1) Quantity of slings
- 2) Wire rope diameter
- 3) Number of legs
- 4) Length of legs
- 5) SWL (If multi-leg, from 0°-90°)
- 6) Type and size of master link
- 7) Type of end fittings
- 8) Details of any additional fittings (e.g. hooks or shackles etc)

3.19 Man-Made Fibre Slings

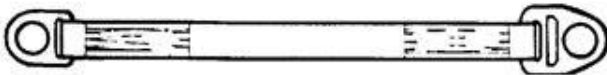
Man-made fibre slings shall be manufactured in accordance with BS EN 1492-1 (Flat web slings) or BS EN 1492-2 (Round slings) or equivalent International Standards. They can be manufactured in various formats e.g.: as flat web slings with soft eyes, hard eyes or endless to suit specific requirements. They can also be supplied as "round" slings (see Fig 1).



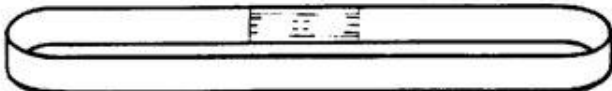
Soft Becketed Eyes



"D" Links



"D" Link and Reeveable Link



Endless (Flat Webbing)



Endless (Round Sling)

Fig 1

Format

- 1) With soft becketed eyes
- 2) With "D" links
- 3) With "D" link and reeveable link
- 4) Endless (Flat webbing)
- 5) Endless (Round sling)

Lifting modes

- Multi-purpose
- Straight or basket lifts
- Straight, basket and choke lifts
- Multi-purpose
- Multi-purpose

Generally man-made fibre slings are used for slinging fragile loads or for suspending of loads from structural steelwork where the coating has to be protected. They are not under any circumstances to be used as the primary lifting slings for the transportation of goods or cargo between the shore base and offshore locations. Flat web slings are manufactured in single thickness (known as simplex) where every 50mm of width equates to 1 tonne of capacity. They are also manufactured in double thickness (known as duplex) where every 50mm of width equates to 2 tonne of capacity.



When describing your requirement for man made fibre slings, the length of flat web slings is always taken as bearing point to bearing point. However, when describing endless sling 'length', you can either quote the circumference (**see Fig 2**) or refer to the bearing point to bearing point measurement as the “Effective Working Length” EWL.

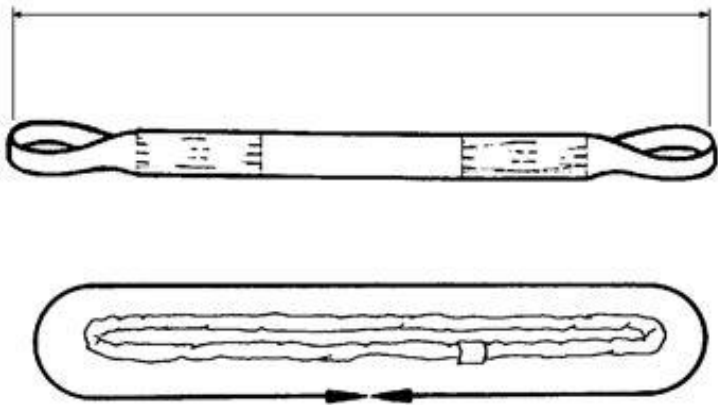


Fig 2

The majority of manmade fiber slings are made from polyester (QatarEnergy-IS do not allow the use of nylon slings for general purpose lifting). Here follows a guide to their resistance to various chemicals and conditions liable to be found in a hostile environment:

Agent Resistance

Acid	good
Alkali	poor
Water / steam	good
Long term exposure to dry heat	excellent
oxidizing agents	excellent
Reducing agents	excellent
Solvents	good

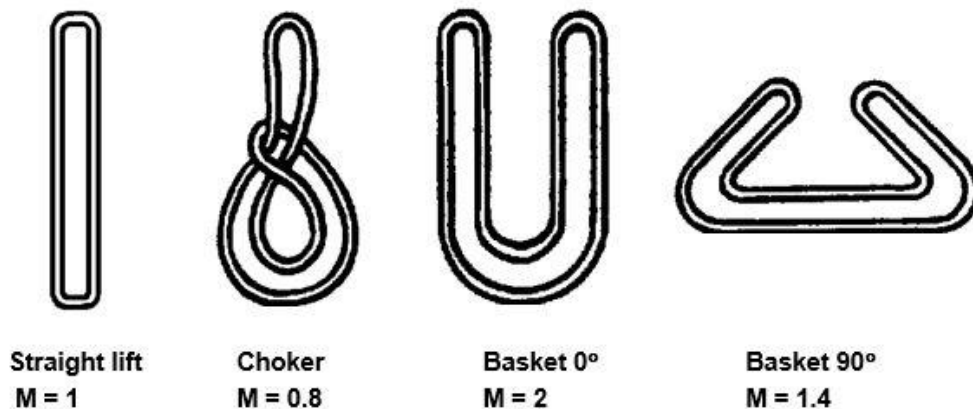
When working with web slings, the way in which they are used can affect their overall capacity i.e. it can either increase or decrease.
To calculate the capacity of the sling, the SWL should be multiplied by the mode factor “M” (**See Fig 3**).



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To aid selection of web or round sling type, specifications are contained in the following pages.

Note: Once selection is made, the purchasing department will have to be contacted to ensure that the use of any of the particular components of the web sling has not been restricted for any reason.

The information to be supplied when raising the requisition will be as follows:

- 1) Quantity of slings
- 2) Type of sling i.e. flat web or round sling
- 3) Width of webbing
- 4) Length of web sling (circumference of endless or round sling)
- 5) SWL
- 6) Type of end fittings

3.20 Chain Slings and Fittings

Chain slings as an assembly, are not covered by a single British Standard but Alloy grade 80 (or grade 'T') chain slings shall be manufactured from chain made in accordance with BS EN 818-7. Alloy fittings shall be designed and manufactured in accordance with ISO 8539. Other fittings such as hooks, shackles, turnbuckles etc. are all covered by their individual standards.

Alloy grade 80 chain slings were developed to replace the older mild steel and high tensile chain slings but unlike these forerunners, they do not work-harden or require periodic annealing / remedial heat treatment. Whereas the mild steel and high tensile slings used to be fabricated by a blacksmith, grade 80 chain slings are constructed from individual components which can be assembled in numerous configurations to suit the task in hand (**see Fig 1**). The only tool required is a hammer.

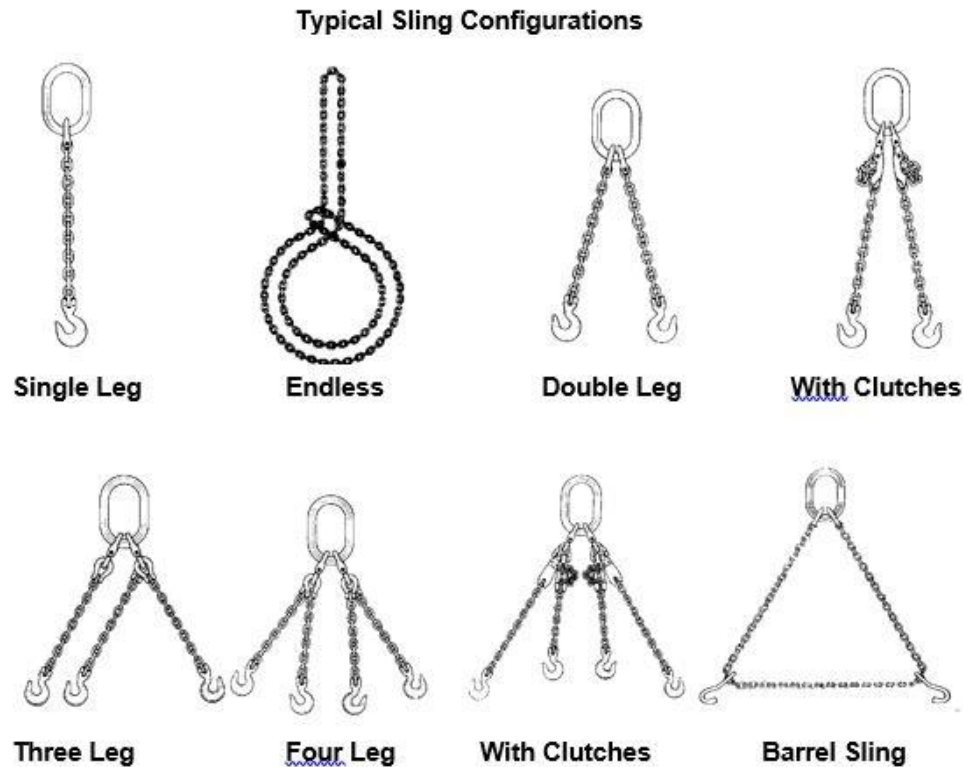


Fig 1

The rating of chain slings is based on the uniform load method which give the following results:

Single leg sling = 1.0 x SWL of a single leg

Double leg sling = 1.4 x SWL of a single leg
from 0° to 90°

Three and Four leg sling = 2.1 x SWL of a single leg
from 0° to 90°

Double leg sling = 1.0 x SWL of a single leg
from 90° to 120°

Four leg sling = 1.5 x SWL of a single leg
from 90° to 120°

Note: When a chain sling or sling assembly is rigged in specific ways, it may be necessary to reduce the safe working load. Further information can be found in the Rigging and Lifting Handbook in section 7.4.



Grade 80 chain slings, although approximately one third of the weight of the high tensile type, are still strength for strength heavier than wire rope slings but have three main advantages namely:

- 1) Greater resistance to corrosion
- 2) More durable
- 3) Adjustable leg lengths (for loads with an offset centre of gravity (see Fig 2)

The only drawback is that they cannot be pushed under the load as is possible with a wire rope sling.

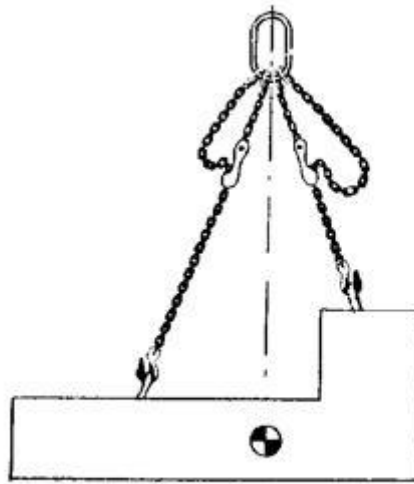


Fig 2

Where the slings are to be fitted with hooks, the desired type for safety is the BK type (see Fig 4 in the wire rope sling section).

Chain slings fitted with shortening clutches are ideal for lifting loads with an offset centre of gravity as the leg length can be adjusted to position the lifting ring directly over the centre of gravity. This allows the load to be lifted level.

Chain slings are not to be used for the transportation of loads to or from an offshore location.

Prior to being received by **QatarEnergy-IS**, each chain sling will be subjected to a Proof Load relative to its Working Load Limit (WLL) i.e. $WLL \times 2$ and will be hard stamped accordingly with the relevant information.

To aid selection of chain sling type, specifications of the chain and fittings are contained in the following pages.

Note: Once selection is made, the purchasing department will have to be contacted to ensure that the use of any of the particular components of the chain sling has not been restricted for any reason.

The information to be supplied when raising the requisition will be as follows:

- 1) Quantity of slings
- 2) Manufacturer and chain diameter
- 3) Number of legs
- 4) Length of legs
- 5) SWL
- 6) Type and size of master link



- 7) Type of end fittings
- 8) Details of any additional fittings (e.g. clutches)
- 9) If an alternative manufacturer would be acceptable

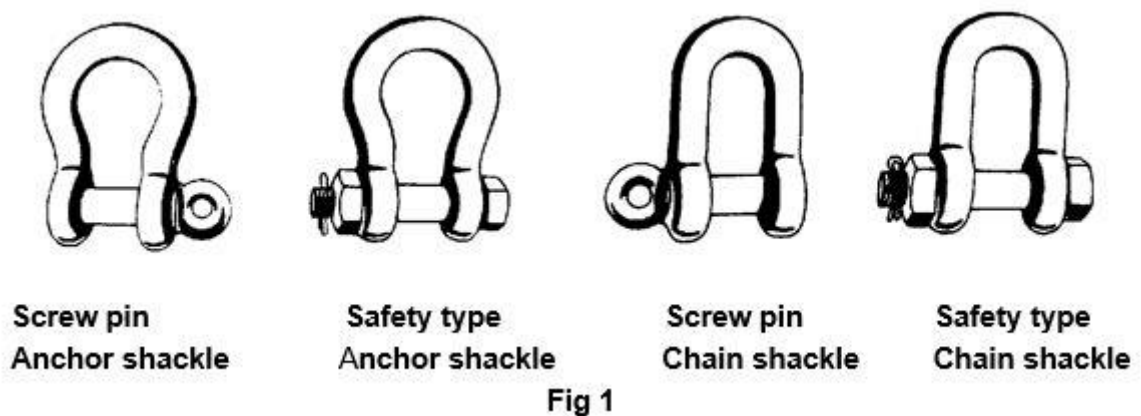
Note: If the chain sling is of a basic format, and only required for a particular project, it may be obtained on a rental basis.

3.21 Shackles

All shackles used by **QatarEnergy-IS** shall preferably be of the alloy variety and designed and manufactured in accordance with the US Federal Specification RR-C-271D or an equivalent international standard.

The use of higher tensile shackles or mild steel shackles is to be avoided for reasons of safety (e.g. A high tensile shackle of similar size to an Alloy shackle is approximately only half the strength - confusing the two could lead to inadvertent overload).

There are two types of shackle commonly used and these are known as anchor shackles (Bow) and chain shackles (Dee). Both are available with screw pin or safety pin (see Fig 1).



In the first instance, shackles should be selected to suit the load being lifted allowing for any increased loadings due to sling angles etc. The dimensions of the shackle will often be governed by the hole diameter and the thickness of the lifting eye.

The selection between bow type and dee type will depend on the number of components being connected (see Fig 2).

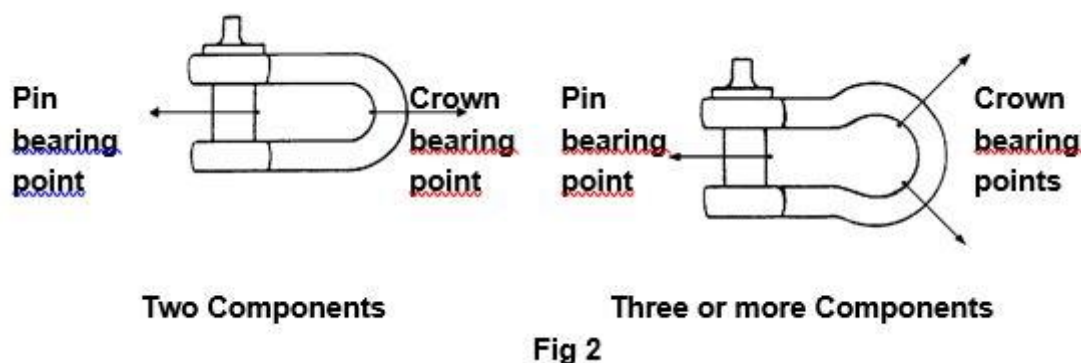


Fig 2

Should headroom be a problem, Dee shackles have a shorter inside length (refer to the tables at the end of this section for actual sizes).

When it comes to selecting between safety pin and screw pin, safety pin are preferred (bow type being the most popular) but in certain situations e.g. diving operations, screw pin are often more practical.

Prior to being received by **QatarEnergy-IS**, each shackle will be subjected to a Proof Load relative to its safe working load

i.e.: Up to 25 tonne - SWL x 2

Over 25 tonne, in accordance with ILO pamphlet 188

It will be supplied with a proof load test certificate and marked accordingly with identification number and SWL.

To aid selection of shackle type for special projects etc., specifications (i.e. capacity / dimensions) of the more popular makes are contained in the following pages.

Note: Once selection is made, the purchasing department will have to be contacted to ensure that the use of the particular type of shackle has not been restricted for any reason.

The information to be supplied when raising the requisition will be as follows:

- 1) Quantity
- 2) SWL
- 3) Manufacturer
- 4) Type

Note: If the shackles are of substantial SWL or quantity and are only required for a particular project, they can generally be obtained on a rental basis. This would only apply to shackles with a SWL of 25 tonne and above.

Note: Screw pin shackle are not acceptable on Cargo Carrying Units.

3.22 Rigging Screws / Turnbuckles

All rigging screws and turnbuckles used by **QatarEnergy-IS** shall be manufactured in accordance with BS 4429, US Federal Spec. FF-T-791-type 1 or an equivalent international standard.

The terms 'rigging screw' and 'turnbuckle' are often applied to a piece of rigging equipment which, through the use of screw threads, provides a means of length adjustment or tensioning. However, there are differences between them.

A rigging screw consists of a tubular body (closed body) sometimes referred to as a box or bottle with a screwed fitting each end (see Fig 1).

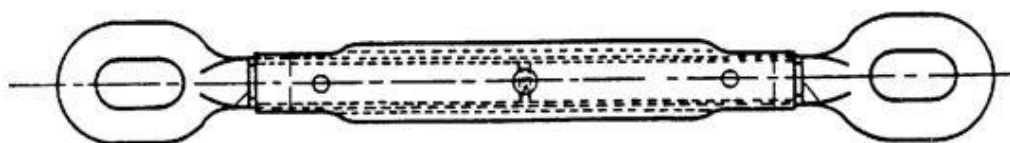


Fig 1

A turnbuckle consists of two reins with a boss at each end (open body) with a screwed fitting each end (see Fig 2).



Fig 2

In either case, they are internally threaded at one end with a right hand thread and

3.23 Containers

All containers manufactured for, bought by or hired to **QatarEnergy-IS** shall be tested and visually examined in accordance with BS 7072.

The majority of containers can be procured on a rental basis. Refer to the table below for dimensions and capacities. The dimensions may vary slightly depending on the supplier.

Typical Container Dimensions							
Type		Width mm	Length mm	Height mm	Tare kg	SWL kg	Gross Wt kg
Mini	(Ext)	1860	1590	2490	1040	4040	5080
	(Int)	1750	1480	2270			
8 ft	(Ext)	1980	2440	2290	1220	5480	6700
	(Int)	1810	2250	1980			
10 ft	(Ext)	2440	3000	2600	1320	8680	10000
	(Int)	2350	2900	2200			
20 ft	(Ext)	2440	6060	2600	2540	12700	15240
	(Int)	2350	5900	2360			
20 ft 1/2 Height	(Ext)	2440	6060	1220	2240	13000	15240
	(Int)	2200	5850	900			



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Apart from the units described in the above table, there are various cargo boxes, cargo baskets, offshore skips, etc. available on a rental basis.

Certain valuable or fragile items of plant may require their own specific containers. This being the case, guidance for their design and manufacture is contained in appendix A of BS 7072. In addition, the following standards should be consulted:

- | | |
|-------------|---|
| BS 449 pt 2 | Specification for the use of structural steel in building |
| BS 5950 | Structural use of steelwork in building |
| BS 2573 | Rules for the design of cranes |

For calculation purposes, the working stress should be based on 2.5 times the maximum gross weight i.e. the tare weight of the container plus its SWL (payload) x 2.5.

Note: Materials used in the construction of structures, containers or other CCUs - **QatarEnergy-IS** will NOT accept Aluminum based units, their use is forbidden.

Containers shall be procured as a fabrication with a full documentation package. This should include the following (where not supplied by QatarEnergy-IS):

- 1) Design calculations
- 2) Fabrication drawings
- 3) Test certificate (plus certificate for any associated rigging)
- 4) QA/QC Release note
- 5) Material Certificates
- 6) Welders Qualifications
- 7) Weld Procedures
- 8) Weld Identification (if relevant)
- 9) N.D.T. Operators Qualifications
- 10) N.D.T. Procedures
- 11) N.D.T. Reports (Pre and Post load test)
- 12) Paint/Coating specification

Important Notes

Containers

- 1) When designing open top containers, half-heights, skips etc., i.e. units which are liable to fill with water, it is imperative that suitable drainage facilities are incorporated in the design.

Sling Sets

- 1) BS 7072 requires that the factor of safety for container slings is increased from the normal 5:1 to 6.5:1 which equates to multiplying the maximum gross weight of the container by 1.3 before applying the rating coefficients contained in BS 1290
- 2) The minimum diameter of wire rope to be used in the sling make up is 13mm
- 3) The main ring of the quadruple assembly (or master link at top of fifth leg if fitted) should have minimum inside dimensions of 270mm x 140mm

(Refer to sections 2.14 and 2.15 for further information on wire rope / slings).



4.0 DATA FOR PROCUREMENT OF LIFTING EQUIPMENT

4.1 Introduction

This 3 part section is designed to assist the Purchasing Department in the procurement of quality lifting equipment for **QatarEnergy-IS** worksites.

The parts are as follows:

Basic Information to be supplied to Manufacturers / Vendors

List of Recommended Manufacturers / Products

Purchasing Restriction Notice

Part 4.1 gives the information on standard lifting equipment that should be passed on to the vendor to ensure the correct equipment is supplied. It also contains details on certification to be supplied with goods. If required, further technical information can be found in Section 2.0 but in all cases, any equipment or accessories supplied must meet with the requirements of **QatarEnergy-IS** LECOP.

Part 4.2 is a list of preferred equipment manufacturers.

Part 4.3 is a "Purchasing Restriction Notice". There are occasions where individual lifting appliances or pieces of lifting gear fail in service or are involved in incidents / accidents and become the focal point of an investigation. Depending on the severity of the incident, until the cause of failure is determined, it is often advisable to restrict the use of and indeed the purchasing of similar equipment. These restriction notices should be continually monitored and updated in line with the results of any investigations.

4.1.1 Overhead Travelling Cranes, Runway Beams and Lifting Beams

When purchasing an overhead travelling crane, it shall be designed and built in accordance with BS 2573 / BS EN 13001. If powered, it must also meet the requirements of BS 466 (An equivalent International Standard would be acceptable). In the first instance to obtain a budget price, the most important document to relay to the prospective manufacturers is a drawing of the building or module in which the crane(s) will be housed. In addition, they would require the following information:

- 1) Quantity
- 2) Type (single or double girder)
- 3) SWL
- 4) Height of lift
- 5) Span (width of building / module)
- 6) Distance of long travel (length of building/module)
- 7) Type of hoist unit i.e. chain / wire rope - rail mounted or trolley mounted
- 8) Single speed or dual speed
- 9) Type of power supply (electrical or pneumatic)
- 10) Hazardous zoning classification
- 11) Mode of operation (i.e. under slung cab / pendant control etc.)
- 12) Paint/coating specification
- 13) The maximum drawn-up dimension allowable (restricted headroom)

The crane manufacturer should also supply the following information with the quotation to allow the engineering department to carry out any stress calculations required:

- 1) Total weight of crane bridge including end carriages and crab / trolley hoist unit.
- 2) Maximum individual wheel loads including horizontal loads
- 3) End carriage trolley wheel centres
- 4) Track width of trolley wheels
- 5) Recommended minimum section of down shop support beams/rails



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The following documents should be supplied as part of the contract:

- 1) Construction dossier c/w proof load test certificate (Proof load normally SWL x 1.25 - refer to Lifting equipment / legislation matrix in Section 1.4). The crane requires to be tested again after installation

Note: The certificate should state the relevant **QatarEnergy-IS** plant No.

- 2) Operating instructions
- 3) Maintenance procedures
- 4) Lubrication details
- 5) List of recommended spares

Runway beams and Lifting beams are not purchased 'off the shelf', but are designed and constructed to suit specific requirements and are procured as a fabrication. Guidance for the design of runway beams is given in BS 2853 and lifting beams should meet the requirements of BS 2573.

Documentation Requirements

- 1) Proof load test certificate (normally SWL x 1.25 for runway beams* but varies with lifting beams - refer to lifting equipment matrix).
* The runway beam requires to be tested again after installation
- 2) In addition to the test certificate, a full documentation package would be required and should include the following (where not supplied by QatarEnergy-IS):
 - a) Design Calculations
 - b) Fabrication Drawings
 - c) Material Certificates
 - d) Welders Qualifications
 - e) Weld Procedures
 - f) Weld Identification (if relevant)
 - g) N.D.T. Operators Qualifications
 - h) N.D.T. Procedures
 - i) N.D.T. Reports (Pre and Post load test)
 - j) Test Certificates for any associated rigging
 - k) Paint / Coating specification
 - l) QA / QC Release Note
 - m) EC declaration of conformity

4.1.2 Lifting Points (Padeyes, Eyebolts, Eyeplates, Eyenuts, etc)

Lifting points can be procured in the form of eyebolts, eyenuts or admiralty eyeplates. When dealing with eyebolts, only collar type are considered suitable for use on **QatarEnergy-IS** Rigs or associated worksites **i.e. the use of dynamo eyebolts is prohibited**. When ordering either eyebolts or eyenuts,

The information to be supplied to the vendor shall be as follows:

- 1) Quantity
- 2) Thread diameter
- 3) Thread type
- 4) Whether standard or long shank (standard shank length = 1.5 to 1.75 x dia of thread)
- 5) With eyebolts or eyenuts, the finish i.e. black or galvanized

Note: Admiralty pattern eyeplates are always supplied black i.e. ready for welding

- 6) Plant / ID numbers
- 7) Color Code



Documentation Requirements

The following documents should be supplied as part of the contract:

- 1) Proof load test certificate (Proof load normally SWL x 2)
- 2) EC declaration of conformity

In the case of Admiralty eyeplates, the pattern number should be given to describe the items required. These would normally be supplied with a certificate of conformity only as they are required to be tested after installation. It is advisable to obtain a material certificate to allow a suitable welding procedure to be selected.

Note: Catalogue details normally quote the proof load figure. Be aware that the SWL is normally **half** this figure.

As an alternative to the above-mentioned lifting points, there is the fabricated padeye. As with the aforementioned Runway beams and Lifting beams, they are not purchased "off the shelf", but are designed and constructed to suit specific requirements and are procured as a fabrication Guidance for the design of padeyes can be found in BS 2573.

Documentation Requirements for Alternatives

- 1) If feasible, Proof load test certificate (normally SWL x 2 * but varies with the capacity). * The padeye is required to be tested again after installation
- 2) In addition to the test certificate, a full documentation package would be required and should include the following (where not supplied by QatarEnergy-IS):
 - a) Design calculations
 - b) Fabrication drawing
 - c) Material certificate(s)
 - d) Weld Procedures
 - e) QA/QC Release Note
 - f) EC declaration of conformity

4.1.3 Manually Hoist (Chainblocks)

Manually operated chain hoists or chainblocks shall be manufactured in accordance with BS 3243 or an equivalent international Standard.

The information to be supplied to the vendor shall be as follows:

- 1) Quantity
- 2) Preferred manufacturer and model
- 3) SWL
- 4) Height of lift
- 5) Vertical length (drop) of operating chains
- 6) The method of suspension (i.e. with top hook or with trolley)
- 7) If trolley required, indicate if geared or push travel and give beam dimensions
- 8) If an alternative to that mentioned in (2) would be acceptable
- 9) If headroom is restricted, maximum drawn-up dimension allowable
- 10) Hazardous zoning classification

Note: If the hoists are only required for a particular project, they can generally be obtained on a rental basis.

- 11) Whether you wish to hire or purchase
- 12) Plant / ID numbers
- 13) Color Code



Documentation Requirements

The following documents should be supplied as part of the contract:

- 1) Proof load test certificate (Proof load = $SWL \times 1.5$ + Light Load Test)
- 2) Operating Instructions
- 3) Maintenance procedures
- 4) Lubrication details
- 5) EC Declaration of Conformity

4.1.4 Lever Hoist (Pullifts)

Lever hoists or pullifts shall be manufactured in accordance with BS 4898 or an equivalent international Standard.

The information to be supplied to the vendor shall be as follows:

- 1) Quantity
- 2) Preferred manufacturer and model
- 3) SWL
- 4) Height of lift
- 5) Type of chain i.e. link or roller (Roller chain not recommended for offshore use)
- 6) If an alternative to that mentioned in (2) would be acceptable
- 7) If headroom is restricted, maximum drawn-up dimension allowable
- 8) Hazardous zoning classification

Note: If the pullifts are only required for a particular project, they can generally be obtained on a rental basis.

- 9) Whether you wish to hire or purchase
- 10) Plant / ID numbers
- 11) Color Code

Documentation Requirements

The following documents should be supplied as part of the contract:

- 1) Proof load test certificate ($SWL \times 1.5$)
- 2) Operating Instructions
- 3) Maintenance procedures
- 4) Lubrication details
- 5) EC declaration of conformity

4.1.5 Wire Rope Hoist (Tirfor)

The original Wire Rope Hoist (Tirfor) was designed and manufactured in France therefore no British Standard exists to cover its design and manufacture.

The information to be supplied to the vendor shall be as follows:

- 1) Quantity
- 2) Preferred manufacturer and model*
- 3) SWL
- 4) Length of rope

Note: If the Tirfor Hoists and ropes are only required for a particular project, they can generally be obtained on a rental basis.

- 5) Whether you wish to hire or purchase
- 6) Color code



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Documentation Requirements

The following documents should be supplied as part of the contract:

- 1) Proof load test certificate (Proof load = SWL x 1.5 for the hoist and SWL x 2 for the rope)
- 2) Operating Instructions
- 3) Maintenance procedures
- 4) Lubrication details
- 5) EC declaration of conformity

* In the interest of safety, it is preferred to use only 'Tirfor' models due to historic problems with other manufacturers' wire ropes i.e. they are of a smaller diameter than Tirfor ropes and if inadvertently used in a Tirfor hoist, they can slip under load thus creating a hazardous situation

4.1.6 Pneumatic Hoists (Air Hosts)

Pneumatic chain hoists or airhoists are governed by European Standard (Din F.E.M. Sect 9) which applies and this is acceptable along with any equivalent International Standards in existence.

The information to be supplied to the vendor shall be as follows:

- 1) Quantity
 - 2) Preferred manufacturer and model
 - 3) SWL
 - 4) Height of lift
 - 5) The type of controls preferred i.e. pendant or chains and the length
 - 6) The method of suspension (i.e. with top hook or with trolley)
 - 7) If trolley required, indicate if powered, geared or push travel and give beam dimensions
 - 8) If an alternative to that mentioned in (2) would be acceptable
 - 9) If headroom is restricted, maximum drawn-up dimension allowable
 - 10) Hazardous zoning classification
- Note:** If the air hoists are only required for a particular project, they can generally be obtained on a rental basis.
- 11) Whether you wish to hire or purchase
 - 12) If an air service unit is required (filter, regulator and lubricator)*
 - 13) Plant / ID numbers
 - 14) Color code

*Air hoists are very susceptible to contaminated air supplies which can cause seizure of the motor. It is recommended that air service units are fitted to prevent this problem arising.

Documentation Requirements

The following documents should be supplied as part of the contract:

- 1) Proof load test certificate (Proof load normally SWL x 1.25)
- 2) Operating Instructions
- 3) Maintenance procedures
- 4) Lubrication details
- 5) List of recommended spares (if purchasing)
- 6) EC declaration of conformity



4.1.7 Electric Chain Hoists

The information to be supplied to the vendor shall be as follows:

- 1) Quantity
 - 2) Preferred manufacturer and model
 - 3) SWL
 - 4) Height of lift and the length of control pendant
 - 5) The power supply
 - 6) Hazardous zoning classification
 - 7) The method of suspension (i.e. with top hook or with trolley)
 - 8) If trolley required, indicate if powered, geared or push travel and give beam dimensions
 - 9) If an alternative to that mentioned in (2) would be acceptable
 - 10) If headroom is restricted, maximum drawn-up dimension allowable
- Note:** If the hoists are only required for a particular project, they can generally be obtained on a rental basis.
- 11) Whether you wish to hire or purchase
 - 12) Plant / ID numbers and Color Code

Documentation Requirements

The following documents should be supplied as part of the contract:

- 1) Proof load test certificate (Proof load normally SWL x 1.25)
- 2) Operating Instructions
- 3) Maintenance procedures and lubrication details
- 4) Classification docs (BASEEFA etc.)
- 5) List of recommended spares (if purchasing)
- 6) EC declaration of conformity

4.1.8 Pneumatic Winches (Air Winches/Tuggers)

The information to be supplied to the vendor shall be as follows:

- 1) Quantity
- 2) Preferred manufacturer and model
- 3) SWL
- 4) Whether being used for utility or man-riding
- 5) If a secondary braking system is required (on utility winches)
- 6) Rope diameter, length and type of end fitting
- 7) If an alternative to that mentioned in (2) would be acceptable

Note: If the winches are only required for a particular project, they can generally be obtained on a rental basis.

- 8) Whether you wish to hire or purchase
- 9) Plant / ID numbers
- 10) Color Code

Documentation Requirements

The following documents should be supplied as part of the contract:

- 1) Proof load test certificate (Proof load normally SWL x 1.25)
- 2) Operating Instructions (including operating pressure, air consumption, fittings, etc.)
- 3) Maintenance procedures
- 4) Lubrication details
- 5) List of recommended spares (if purchasing)
- 6) EC declaration of conformity



4.1.9 Beam Clamps

The information to be supplied to the vendor shall be as follows:

- 1) Quantity
- 2) Preferred manufacturer and model
- 3) SWL
- 4) Beam dimensions
- 5) If an alternative to that mentioned in (2) would be acceptable
- 6) If headroom is restricted, maximum dimension allowable between beam flange and bearing point for hoist

Note: If the clamps are only required for a particular project, they can generally be obtained on a rental basis.

- 7) Whether you wish to hire or purchase
- 8) Plant / ID numbers
- 9) Color code

Documentation Requirements

The following documents should be supplied as part of the contract:

- 1) Proof load test certificate (Proof load normally SWL x 2)
- 2) Instructions for users
- 3) Maintenance and lubrication details
- 4) EC declaration of conformity

4.1.10 Beam Trolleys

The information to be supplied to the vendor shall be as follows:

- 1) Quantity
- 2) Preferred manufacturer and model
- 3) SWL
- 4) Beam dimensions
- 5) Length of operating chains
- 6) If an alternative to that mentioned in (2) would be acceptable
- 7) If headroom is restricted, maximum dimension allowable between beam flange and bearing point for hoist

Note: If the trolleys are only required for a particular project, they can generally be obtained on a rental basis.

- 8) Whether you wish to hire or purchase
- 9) Plant / ID numbers
- 10) Color code

Documentation Requirements

The following documents should be supplied as part of the contract:

- 1) Proof load test certificate (Proof load normally SWL x 1.5 but certain manufacturers test to SWL x 2)
- 2) Instructions for users
- 3) Maintenance and lubrication details
- 4) EC declaration of conformity

4.1.11 Plate Clamps

The information to be supplied to the vendor shall be as follows:

- 1) Quantity
- 2) Preferred manufacturer and model
- 3) Type i.e. horizontal or vertical



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- 4) SWL
 - 5) Jaw opening
 - 6) If an alternative to that mentioned in (2) would be acceptable
- Note:** If the plate clamps are only required for a particular project, they can generally be obtained on a rental basis.
- 7) Whether you wish to hire or purchase
 - 8) Plant/ID numbers
 - 9) Color code

Documentation Requirements

The following documents should be supplied as part of the contract:

- 1) Proof load test certificate (Proof load = SWL x 2)
- 2) Instructions for users
- 3) Maintenance and lubrication details
- 4) EC declaration of conformity

4.1.12 Pulley Blocks (Sheave Blocks, Snatch Block, etc)

Wire rope pulley blocks up to SWL 25 tonne shall be designed and manufactured in accordance with BS 4018 and if over SWL 25 tonne, BS 4536 or equivalent international standards.

The information to be supplied to the vendor shall be as follows:

- 1) Quantity
- 2) Preferred manufacturer and model
- 3) SWL
- 4) Number of sheaves
- 5) Diameter of sheaves
- 6) Diameter of rope being used
- 7) If an alternative to that mentioned in (2) would be acceptable

Note: If the sheave blocks are only required for a particular project, they can generally be obtained on a rental basis.

- 8) Whether you wish to hire or purchase
- 9) Plant / ID numbers
- 10) Color code

Documentation Requirements

The following documents should be supplied as part of the contract:

- 1) Proof load test certificate (Proof load = the maximum load on head fitting x 2)
- 2) Instructions to users
- 3) Maintenance and lubrication details
- 4) EC declaration of conformity

4.1.13 Hydraulic Jacks (Rams, Pumps and Cylinders)

The information to be supplied to the vendor shall be as follows:

- 1) Quantity
 - 2) Preferred manufacturer and model (of Ram or Jack)
 - 3) Capacity
 - 4) Stroke
 - 5) Whether or not gauges are required
- If applicable:
- 6) Preferred manufacturer and model (of pump) and Qty



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- 7) Length of hydraulic hose(s) and Qty
 - 8) Type of manifold (if required) and Qty
 - 9) If an alternative to that mentioned in (2) and 6) would be acceptable
 - 10) If headroom/space is restricted, maximum closed height allowable
- Note:** If the hydraulic equipment is only required for a particular project, it can generally be obtained on a rental basis.
- 11) Whether you wish to hire or purchase
 - 12) Plant / ID numbers
 - 13) Color code

Documentation Requirements

The following documents should be supplied as part of the contract:

- 1) Certificate of conformity or function test (Proof load normally the same as the SWL / Capacity as relief valves are usually fitted to prevent overload)
- 2) Operating Instructions
- 3) Maintenance and lubrication details
- 4) Gauge calibration certificates
- 5) Certification for hydraulic hoses i.e. pressure test certificate
- 6) EC declaration of conformity

4.1.14 Wire Ropes & Fittings

Wire rope for general purposes as used in the engineering industry shall be designed and manufactured in accordance with BS 302 or an equivalent International Standard. The information to be supplied to the vendor shall be as follows:

- 1) Quantity of ropes
- 2) Wire rope diameter
- 3) Length of rope
- 4) Tensile strength
- 5) Construction
- 6) Minimum breaking load
- 7) SWL
- 8) Type of end fittings
- 9) Details of any additional fittings (e.g. hooks or shackles etc.)
- 10) Lubrication type
- 11) Identification numbers

Documentation Requirements

The following documents should be supplied as part of the contract:

- 1) A wire rope certificate fully describing the rope and giving the load at which a sample of the rope destructed
- 2) If the rope has a termination, a proof load test certificate for that termination and, where required
- 3) A confirmation of origin (country of manufacture)
- 4) EC declaration of conformity

4.1.15 Wire Rope Slings

Wire rope slings shall be manufactured in accordance with BS 1290 / BS EN 13414 from wire rope which meets the requirements of BS 302 or equivalent International Standards. Generally the vendor should be advised as to what use the sling will be put



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e.g. If the sling is an assembly to be used on a container, different factors of safety apply under BS 7072.

The information to be supplied to the vendor shall be as follows:

- 1) Quantity of slings
- 2) Wire rope diameter
- 3) Number of legs
- 4) Length of legs
- 5) SWL (If multi-leg, from 0°-90°)
- 6) Type and size of masterlink
- 7) Type of end fittings (e.g. soft eyes or hard eyes)
- 8) Details of any additional fittings (e.g. hooks or shackles etc.)

Note: If the wire rope sling is of a basic format, of large capacity and only required for a particular project, it may be obtained on a rental basis.

- 9) Whether you wish to hire or purchase
- 10) Plant / ID numbers
- 11) Color code

Documentation Requirements

The following document should be supplied as part of the contract:

Proof load test certificate (Proof load = SWL at 0° x 2 and/or the SWL of each leg x 2)

4.1.16 Man Made Fibre Slings (Web Slings and Round Slings)

Manmade fiber slings shall be manufactured in accordance with BS EN 1492-1 (Flat web slings) or BS EN 1492-2 (Round slings) or equivalent International Standards.

The information to be supplied to the vendor shall be as follows:

- 1) Quantity of slings
- 2) Type of sling i.e. flat web or round sling
- 3) Width of webbing
- 4) Length of web sling (circumference of round sling)
- 5) SWL
- 6) Type of end fittings
- 7) Plant/ ID numbers
- 8) Color code
- 9) Preferred material - Polyester

Documentation Requirements

The following document should be supplied as part of the contract:

- 1) Test certificate (Man made fiber slings are not subjected to a proof load but samples are pulled to destruction to verify the type).
i.e. a certificate would normally show the SWL and the breaking load. Metal end fittings however should be tested to twice the SWL.

Note: It is **QatarEnergy-IS** policy that man-made fibre slings are **not used** for the transportation of goods or cargo to or from an offshore Rig or installation.

- 2) EC declaration of conformity

4.1.17 Chain Slings & Fittings

Chain slings shall be made up using chain conforming to BS EN 818-7 and fittings conforming to ISO 8539. The information to be supplied to the vendor shall be as follows:

- 1) Quantity of slings



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- 2) Preferred manufacturer
 - 3) Grade and chain diameter
 - 4) Number of legs
 - 5) Length of legs
 - 6) SWL
 - 7) Type and size of masterlink
 - 8) Type of end fittings
 - 9) Details of any additional fittings (e.g. clutches)
 - 10) If an alternative to that mentioned in (2) would be acceptable
- Note:** If the chain sling is of a basic format, and only required for a particular project, it may be obtained on a rental basis.
- 11) Whether you wish to hire or purchase
 - 12) Plant / ID numbers
 - 13) Color code

Documentation Requirements

The following document should be supplied as part of the contract:

- 1) Proof load test certificate (Proof load = SWL at 0° x 2)
- 2) EC declaration of conformity

4.1.18 Shackles

The preferred shackles for general use in our industry are those manufactured to US Federal Specification RR-C-271 D.

The information to be supplied to the vendor shall be as follows:

- 1) Quantity
- 2) SWL
- 3) Preferred manufacturer
- 4) Type of body (Bow or Dee)
- 5) Type of pin (Safety or screw pin)

Note: If the shackles are of substantial SWL or Quantity and are only required for a particular project, they can generally be obtained on a rental basis. This would only apply to shackles with a SWL of 25 tonne and above.

- 1) Whether you wish to hire or purchase
- 2) Plant / ID numbers
- 3) Color code

Documentation Requirements

The following document should be supplied as part of the contract:

- 1) Proof load test certificate (Proof load normally SWL x 2 - refer to Lifting equipment/legislation matrix)
- 2) EC declaration of conformity

4.1.19 Rigging Screws / Turnbuckles

Rigging screws / turnbuckles shall be manufactured in accordance with BS 4429, US Federal Specification FF-T-791-Type 1 or an equivalent International specification. The information to be supplied to the vendor shall be as follows:

- 1) Quantity
- 2) Preferred manufacturer or Standard
- 3) SWL
- 4) Model number or description



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- 5) Thread diameter
- 6) Type of end fittings
- 7) Whether locknuts are required or not

Note: If the screws are on the large side e.g. 10 tonne and over and are only required for a particular project, they can possibly be obtained on a rental basis.

- 8) Whether you wish to hire or purchase
- 9) Plant / ID numbers
- 10) Color code

Documentation Requirements

The following document should be supplied as part of the contract:

- 1) Proof load test certificate (Proof load normally SWL x 2 - refer to Lifting equipment / legislation matrix)
- 2) EC declaration of conformity

4.1.20 Containers (Cargo Boxes, Skips, etc)

Guidance for the design and manufacture of containers is incorporated in BS 7072 / BS EN 12079. This Standard also lists lifting appliances which although not immediately obvious, are deemed to be containers e.g. bottle racks, fuel tanks, skips, etc. All containers manufactured for, bought by or hired to **QatarEnergy-IS** shall meet with or at least be inspected and tested in accordance with the criteria contained therein. The majority of containers can be procured on a rental basis but certain valuable or fragile items of plant may require containers specially manufactured for them. This being the case, they would be procured as a fabrication with the usual criteria applicable (see below).

Documentation Requirements

If the container is being rented:

- 1) Proof load test certificate (Maximum gross weight x 2)
- 2) Post load test NDT report
- 3) A six-monthly examination form to comply with the requirements of BS 7072 / BS EN 12079 (where applicable)

If the container is being fabricated, in addition to the above, a full documentation package would be required and should include the following (where not supplied by **QatarEnergy-IS**):

- a) Design calculations
- b) Fabrication drawings
- c) Material certificates
- d) Welders Qualifications and Weld Procedures
- e) N.D.T. Operators Qualifications
- f) N.D.T. Procedures
- g) N.D.T. Reports (Pre and Post load test)
- h) Test certificates for any associated rigging
- i) Paint / Coating specification
- j) QA/QC Release Note

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4.2 List of Recommended Manufacturers / Products

Equipment	Recommended Manufacturer
Overhead travelling Cranes	Liebherr, Morris Material Handling Ltd., Stahl Material Handling
Runway beams	Liebherr, Morris Material Handling Ltd., Stahl Material Handling
Lifting beams	Liebherr, Morris Material Handling Ltd., Stahl Material Handling
Lifting Points	Liebherr, Morris Material Handling Ltd., Stahl Material Handling
Chainblocks	Elephant Chain Block Co., Morris Material Handling Ltd, Nitchi Chain Hoists Ltd, Vital
Pullifts	Elephant Chain Block Co., Morris Material Handling Ltd, Nitchi Chain Hoists Ltd
Wire Rope Hoists	Tirfor, Spanset, Crosby
Air Hoists	ARO Corporation, Beebe, Ingersoll Rand Co., JDN Neuhaus, Samiia
Electric Chain Hoists	Stahl Material Handling
Air Winches/Tuggers	Beebe, Ingersoll Rand Co., Samiia
Beam Clamps	Riley Lifting Equipment Ltd
Beam Trolleys	ARO Corporation, Elephant Chain Block Co., Morris Material Handling Ltd, Nitchi Chain Hoists Ltd, Riley Lifting Equipment Ltd,
Plate Clamps	Camlok, Crosby, Tirfor
Pulley Blocks (Sheave blocks, Snatch blocks, etc)	Ansell Jones, Crosby, Morris Material Handling Ltd
Hydraulic jacks (rams, pumps, cylinders)	Enerpac, Tangye
Wire ropes & fittings	Bridon, Brunton Shaw, Crosby
Wire rope slings	Tirfor, Bridon, Crosby
Man-made fiber Slings (Web slings, Round Slings)	Spanset, Crosby, Bridon
Chain Slings & Fittings	Gunnebo, Parsons Chain Company,
Shackles	Crosby, Tirfor

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4.3 Purchasing Restriction Notice Number

<i>Such Notice shall be endorsed by QA/QC department and issued as MEMO to PMI team</i>	
Date:Safety Alert / Notice attached:	
Description of Item Restricted:	
Reason for Restriction:	
By Whom:	
Date Restriction Lifted:	
By Whom:	
Any Specific Instruction for Future Purchase:	
Signature:	



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5.0 CONTROL DOCUMENTS FOR THE HIRING OF HOISTING AND LIFTING EQUIPMENT AND USAGE OF CONTRACTOR'S GEAR

5.1 Instructions to Vendor to guarantee the quality of Long term rented equipment i.e supply of rigging lofts

Vendor Requirement

The selected vendor will supply **QatarEnergy-IS** worksites with Portable Lifting Equipment for use on the site all year round.

Equipment Standards

All equipment must comply with British Standards, International Standards or American Federal specification. It must have track record in the Offshore / Onshore Oil and Gas Industry and be approved by **QatarEnergy-IS**. On no account must prototype equipment be supplied without the express, written approval of **QatarEnergy-IS**.

Note 1: For safety reasons, the practice of supplying "de-rated" lifting appliances e.g. a SWL 3 tonne chain hoist supplied as a SWL 1 tonne hoist is no longer acceptable.

Note 2: **QatarEnergy-IS** may on occasion "restrict" the use of equipment should problems develop / incidents occur. Contractors and vendors will be informed accordingly via safety notices etc.

Shipment and Storage

The equipment will be shipped to and housed in rigging lofts. Refer to Part 4.3 for Rigging Loft specification and operating instructions.

Certification / Changeout

The equipment will be fully certified and suitable for 6 months service . It will be changed out every 6 months.

Marking Tracking

Each and every item will be permanently marked (hard stamped) with an identification or plant number to facilitate traceability to the test certificate and allow for tracking during its service at the worksite. (Web slings can be marked with permanent felt pen). A register of the loft's contents will be supplied with the loft along with the current certification. An issuing and returns register will also be provided along with a "T" card system to assist in tracking the equipment. (Further details of the "T" card tracking system can be found in Part 4.3)

Color Coding

Each and every item will be color coded in accordance with **QatarEnergy-IS** existing system. Lifting appliances should be painted externally with the coded color to ease visual location at the end of the certification period. Wire slings will be color coded at one or both ends but care must be taken to ensure that the identification / plant number remains visible. Other items of loose gear will be colored coded in such a place as to minimise the possibility of it being rubbed off.

Maintenance / Re-certification

The vendor will be responsible for the maintenance and re-certification of the lifting equipment to a standard acceptable to **QatarEnergy-IS**.

All items with enclosed parts will be overhauled (prior to each changeout) to a standard no less than that contained in **QatarEnergy-IS** Portable Equipment Overhaul Procedures contained in section 5.1 of this manual.

All items with no enclosed parts (e.g. loose gear) will be visually examined and where necessary tested in accordance with **QatarEnergy-IS** Offshore Thorough Examination and Re-certification Procedures contained in section 5.2 of this manual.



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Ad Hoc Equipment

During shutdowns or for special projects, the vendor may be required to provide additional lifting equipment (on short term hire) for the duration of the work. This being the case, it will still require to have the current color code but must also have a contrasting band of color through it to ease recognition and gathering at the end of the work. The standard of equipment will be the same as that supplied on the long-term basis.

5.2 Instructions to Subcontractors to guarantee the quality of their Lifting equipment when it is being used on QatarEnergy-IS worksites

Contractor Requirement

Contractors will often be required to supply their own portable lifting equipment for use on / at **QatarEnergy-IS** worksites.

Equipment Standards

All equipment must comply with British standards, International Standards or American Federal specification. It must have track record in the Offshore / Onshore Oil and Gas Industry and be approved by **QatarEnergy-IS**. On no account must prototype equipment be used without the express, written approval of **QatarEnergy-IS**.

Note 1: For safety reasons, the practice of supplying “de-rated” lifting appliances e.g. a SWL 3 tonne chain hoist supplied as a SWL 1 tonne hoist is no longer acceptable.

Storage

The equipment shall be contained and stored on the installation in rigging lofts fitted out with storage facilities e.g. racks, bins etc. Should the lofts have heating and lighting systems, they must be compatible with the worksites power supply, comply with current safety regulations and meet **QatarEnergy-IS** electrical specifications.

Certification / Changeout

The equipment shall have current certification when it arrives at the worksite. Should it remain in situ beyond its certification date, it will be subjected to an Inspection to assess its suitability for further service. As an alternative, it can be changed out completely and replaced with newly re-certified equipment.

Marking / Tracking

Each and every item shall be permanently marked (hard stamped) with an identification or plant number to facilitate traceability to the test certificate and allow for tracking during its service on the worksite. (Web slings can be marked with permanent felt pen). A register of the equipment shall be required along with the current certification. An issuing and returns register should also be provided along with a card system to assist in tracking the equipment.

Color Coding

Each and every item shall be color coded in accordance with **QatarEnergy-IS** existing system, but should also have a contrasting color through it to identify it as contractors' equipment. Items of loose gear shall be colored coded in such a place as to minimise the possibility of it being rubbed off.

Maintenance / Re-certification

The contractor shall be responsible for the maintenance and re-certification of the lifting equipment to a standard acceptable to **QatarEnergy-IS**.

All items with enclosed parts shall be overhauled (prior to each changeout) to a standard no less than that contained in **QatarEnergy-IS** Portable Equipment Overhaul Procedures contained in section 5.1 of this manual.



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All items with no enclosed parts (e.g. loose gear) shall be visually examined and where necessary tested in accordance with **QatarEnergy-IS** Offshore Thorough Examination and Re-certification Procedures contained in section 5.2 of this manual.

Ad Hoc Equipment

During shutdowns or for special projects, the contractor may be required to provide additional lifting equipment (on short term hire) for the duration of the work. This being the case, it shall still require to have the current color code with a contrasting band of color through it to ease recognition and gathering at the end of the work. The standard and quality of equipment shall be the same as that supplied on the long-term basis.

5.3 Rigging Loft Specifications and Operations

The Rigging Loft will consist of a suitably sized lockable container fitted out with the appropriate storage hooks, racks and bin compartments. It will be supplied complete with a register and a procedural document to control issues, receipts etc.

Where required, it shall come complete with adequate heating and lighting compatible with the worksites power supply and to the appropriate explosion proof/hazardous area rating.

Quarantine areas will be designated 'For Inspection' and 'Hold - Do Not Use'.

The container will be equipped with an inventory of portable equipment. The equipment will be color coded and complete with certification. The Rigging Loft shall contain an issuing register and a suitable tracking system e.g. 'T' card.

The Rigging Loft Controller shall have sufficient knowledge and skill to examine and identify defective equipment and shall be responsible for withdrawing it from service, placing it in quarantine to prevent further use and maintaining the records appropriately.

Each individual item of Lifting Equipment will be color coded to indicate the current inspection status. The current color code will be clearly displayed at the entrance to the Rigging Loft.

All items of lifting equipment will be recorded in the Issue Register as follows:

1. Date issued
2. Name and company of personnel using
3. Date returned
4. Return inspection comments

Lifting appliances such as Tirfors, chainblocks, pullifts, snatch blocks etc. will have their status indicated using the 'T' card tracking system operated by the Rigging Loft Controller.

The 'T' card system shall be divided into sections designated 'in use' 'in stock' and 'hold-inspection'.

Upon issue the 'T' card corresponding to the item number shall be removed from the 'in stock' column and placed in the 'in use' column.

Upon return of ANY item, the item shall be located in the 'hold-inspection' area of the Rigging Loft, and where appropriate the corresponding 'T' card placed in the 'hold-inspection' column.

All returned items shall be subjected to inspection prior to acceptance back into the Rigging Loft.

Items awaiting inspection will be held in the designated 'for inspection' area of the Rigging Loft until such time as they can be inspected by the Rigging Loft Controller.

Dependent upon satisfactory inspection of the item, the item will be returned to its allocated storage area, and where appropriate the corresponding 'T' card placed in the 'in stock' column.

Non-conforming items shall be quarantined in the designated hold area of the Rigging



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Loft and tagged accordingly. Where appropriate the corresponding 'T' card shall be placed in the 'hold-inspection' column, and only removed from the system when the item has been disposed of or returned to the Rigging Loft supplier's shore base for repair.

Items of defective lifting equipment shall be quarantined by means of either individual identification (e.g. 'HOLD - DO NOT USE' label) or posted location until returned for repair. The label shall be endorsed on the reverse side with the vendor's non-conformance report number (NCR No.) They should also have any current color code painted over in red. Defective equipment can be identified by anyone but must be reported to the Mechanical Supervisor for processing.

If the defective equipment is in a dangerous condition (and obviously scrap) it should be rendered unusable to prevent any inadvertent use prior to being sent back to the supplier.

Should it be expected that the defective equipment may be scrapped or will take extended time to be returned to the worksite, replacements should be requested from the Rigging Loft Supplier.

Certification records for the equipment which has been scrapped shall be endorsed accordingly and returned to the Rigging Loft Supplier to be retained for 12 months beyond the scrap date to provide traceability.

5.4 Mobile Crane Requirements

Any mobile crane working on a **QatarEnergy-IS** worksite must meet with the minimum requirements specified below.

The crane must have the following documentation available for inspection:

- 1) Current MOT certificate for roadworthiness
- 2) Current proof load test certificate (i.e. tested within the last 4 years)
- 3) Test certificates for all lifting accessories e.g. slings, shackles, hooks etc.
- 4) Latest Examination reports for any of the above whose test certificate is more than 6 months old
- 5) Proof of calibration of the safe load indicator
- 6) Completed and up-to-date daily / weekly maintenance schedules
- 7) Load / duty charts

The crane operator must have the following documentation available for inspection:

- 1) Current and valid Qatari driving license
- 2) Certificate of training to the appropriate standard.
- 3) Proof of competence to operate the particular capacity / type of crane in question. Competence to have been assessed by a qualified Skills Assessor
- 4) Certificate of medical fitness

In addition, the operator must be physically fit and of sufficient stature to be able to operate the crane safely.

Further information on safe crane operations can be found in section 1 part 3 of this manual and in the Crane Operations and Cargo Handling Pocketbook.

5.5 Forklift Truck Requirements

Any Forklift truck working on a **QatarEnergy-IS** worksite must meet with the minimum requirements specified below.

The forklift truck must have the following documentation available for inspection:

- 1) Current certificate for roadworthiness (if applicable)
- 2) Current proof load test certificate (i.e. tested within the last 4 years)



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- 3) Proof load test certificates for the mast lifting roller chains
- 4) Proof load test certificates for any attachments
- 5) Completed and up-to-date daily/weekly maintenance schedules
- 6) MPI certificate / report for the forks

The forklift operator must have the following documentation available for inspection:

- 1) Current and valid driving license
- 2) Certificate of training to the appropriate standard (e.g. RTITB standards)
- 3) Proof of competence to operate the particular capacity / type of forklift in question
- 4) Certificate of medical fitness

In addition, the operator (whether contracted in with the equipment or employed directly by **QatarEnergy-IS**) must be physically fit and of sufficient stature to be able to operate the machine safely.

Further information on forklift operations can be found in the Rigging and Lifting handbook.

6.0 MAINTENANCE OF EQUIPMENT

6.1 Introduction

The information contained in each lifting equipment type procedure is comprehensive so far as to provide sufficient information and when used in conjunction with the manufacturers recommended procedures will allow the competent third party inspector and / or the company person responsible) to understand more readily the various technical aspects of the lifting appliance.

Throughout this document there is a call for certain tests “to be witnessed”. This is a quality assurance measure to prevent the repairer from certifying his own work. This witness could be any person in the workshop not involved in the repair.

In an effort to improve levels of competence and safety standards, it is the intention of the company to continually update existing information and, when necessary, introduce new guidelines for use within operations. To this end, the content of this manual will be reviewed annually to ensure that it reflects current information.

It is the aim of this manual to provide the means and information necessary to control the use of lifting equipment to conform to the Regulations.

The Regulations do not directly refer to appropriate standards associated with the manufacture, use, testing and maintenance of lifting equipment. The onus, to prepare and provide written instructions and guidance, based on the most appropriate recognised and approved standards, therefore, falls on the owner of the worksite. This manual has been prepared on this basis.

Note: Certain British Standards (BS) and International Standards (ISO) use the term Working Load Limit (WLL) in preference to Safe Working Load (SWL); the following definitions are relevant:

Safe Working Load (SWL): the maximum load, as certified by a competent person, that an item of lifting equipment may raise, lower or suspend under particular service conditions.

Working Load Limit (WLL): the maximum load that an item of lifting equipment is designed to raise, lower or suspend.

The SWL may be equal to, or less than, the WLL.

In order to avoid confusion, this document will refer only to Safe Working Load. However, equipment users should be familiar with both terms since they are used by manufacturers and appear in standards and regulations. Although no distinction is



made between the two terms, it is still the responsibility of a competent person to consider all the factors affecting each lifting operation and to make the appropriate adjustments to the rated capacity of the equipment being used.

6.2 Portable Lifting Equipment / Gear Overhaul Procedures

6.2.1 Manual Hoists

Check List

A. Dismantling

- 1) Remove gear case cover, wash gears and check for alignment marks, wear, cracks or broken teeth
- 2) Remove hand chain wheel cover and dismantle brake units, clean metal component parts (bead blast, wire brush or emery) and examine for wear / distortion. Check condition of brake discs and measure thickness to ensure compliance with minimum requirement. Check condition of pawl and pawl spring and ensure it operates freely
- 3) Remove load chain, disconnect bottom hook, tumble chain to remove corrosion etc. and examine and measure load chain for worn, stretched or deformed links
- 4) Examine bottom hook for wear / stretch, ensure it rotates freely and check bearings. Check function of safety catch. In the case of multi-fall units, check load chain sprocket for wear and ensure it rotates freely
- 5) Examine bottom hook load pin for wear / distortion
- 6) Extract drive pinion and examine for wear / distortion
- 7) Examine load chain sprockets for possible wear and ensure satisfactory seating of load chain in pockets, check guide rollers for wear and alignment. Check load chain stripper is in place
- 8) Remove top hook and examine for wear / stretch, ensure it rotates freely and check function of safety catch
- 9) Examine top hook load pin for wear / distortion
- 10) Remove and examine load chain live and dead-end anchor pins for distortion
- 11) Check alignment of internal frame side plates and examine load pinholes for signs of wear/stretch. Where applicable, check top hook yoke for wear / distortion
- 12) Examine all remaining bearings / bushes for wear and check smoothness of operation

B. General

- 1) Ensure all components are clean and free from contamination prior to re-assembly
- 2) Replace any defective components
- 3) Where any doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

C. Re-assembly

- 1) Re-assemble in accordance with the manufacturer's instructions lubricating the various moving components as required

Notes:

- i) Special attention is to be paid to the pawls and pawl springs. Ensure there is adequate clearance between the brake cover and the pawl



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- ii) Where applicable, ensure the welds in the load chain links do not run on the chain sheaves
- 2) Subject the hoist to a S.W.L. + 50% proof load (in such a fashion that all the gearing makes one full revolution under load) and a light load test of between 2% and 5% of SWL to check for efficient operation of the brake. (These tests are to be witnessed). Re-examine hoist to ensure that the proof load has not exposed any defects
- 3) If all satisfactory, hard stamp unit with identification number and S.W.L
- 4) Pass to paint shop for painting and chain coating
- 5) Return to engineer for final check and issue of identification plate. (Ensure that the painting has not interfered with the operation of any moving component)

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an overhaul.

The vendor should have the relevant manufacturer's manuals "in house" and they should be made available to the engineers carrying out the overhauls for further reference etc.



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Job No:			Condition
Model:			A = Satisfactory
S.W.L:			B = Worn but serviceable
Serial No:			C = Condemned
Plant No:			D = Replaced
			E = N.D.T.'D

Description		Condition					Commen
		A	B	C	D	E	
	Top Hook						
	Safety Catch						
	Swivel Unit / Yoke						
	Load Pin						
	Load Chain Sheaves						
	Load Chain Sheave Bearings						
	Load Chain Sheave Pins						
	Gear Cover						
	Gears						
	Drive Shaft / Pinion						
	Load Wheel						
	Load Wheel Bearings						
	Side Plate (L.H.)						
	Side Plate (R.H.)						
	Chain Guide Rollers						
	Chain Stripper						
	Load Chain "Live End" Anchor						
	Load Chain "Dead End" Anchor						
	Ratchet Pawl						
	Pawl Spring						
	Brake Disc Hub						
	Ratchet Wheel						
	Friction Washers						
	Hand Chain Wheel						

Tick relevant boxes / delete where applicable

Signed:	Date:	Test witnessed by:

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Job No:			Condition
Model:			A = Satisfactory
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			E = N.D.T.'D

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Signed:

Date:

Test witnessed by:



6.2.2 Pneumatic Chain Hoists

Check List

A. Dismantling (Type 1 - Rotary Vane Motor)

- 1 Remove air motor and strip down internal parts taking care not to use excessive force (i.e. only wood, rubber, copper or hide mallet to be used)
- 2 Wash down all internal parts and casing
- 3 Check rotor cylinder and blades for wear and scoring. Check condition of blade springs (if fitted)
- 4 Examine shuttle valve making sure the control shuttle runs true and is free from contamination
- 5 Check all "O" rings and seals for damage and ensure they are not perished
- 6 Examine end plates for flatness and wear
- 7 Check ball bearings for wear and smoothness of operation
- 8 Examine brake pads for wear and that they are oil free
- 9 All brake parts should be lightly rubbed with emery paper to remove glazing
- 10 Examine exhaust casing for distortion or cracking
- 11 Remove direction control bar and examine for wear and distortion

B. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Replace any defective components
- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

C. Re-assembly

- 1 Re-assemble in accordance with the manufacturer's instructions lubricating the various moving components as required
- 2 Before connecting brake, run lubricated air through the motor to ensure it runs true and free
- 3 Re-assemble and fit brake to motor and set aside complete unit for final assembly in hoist

D. Dismantling (Type 2 - Piston Type Motor)

- 1 Remove piston air motor from hoist unit
- 2 Remove motor adapter and crank shaft assembly

Note: Undue force when dismantling motor will result in distorted and ill fitting parts at the re-assembly state (use only wood, rubber, copper or hide mallet)

- 3 Remove control handle and casing then pull out rotary valve control
- 4 Begin to dismantle pistons by removing cylinder heads and gaskets
- 5 Remove all internal parts for washing and cleaning
- 6 Examine all bearings for smoothness of operation
- 7 Check all shafts and machined holes for scoring and belling
- 8 Check all gaskets, nuts, screws and rods for stretch or distortion
- 9 Examine pistons for cracking, scoring or wear. All compression rings, oil rings and seals should sit neatly on pistons and should not hamper efficiency of the motor
- 10 Examine rotary valve control for wear, distortion and check that it is free from contamination

E. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Replace any defective components



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- 3 Where doubt exists as to the integrity of any component, N.D.T. techniques should be employed to verify their status

F. Re-Assembly

- 1 Re-assemble in accordance with the manufacturers instructions lubricating the various moving components as required
- 2 Run lubricated air through the motor to ensure it runs true and free
- 3 Set aside complete unit for final assembly in hoist

G. Gear Box

- 1 Remove gearbox from hoist central section
- 2 Remove cover plate or upper ball bearing depending on type / model and inspect internal gearing system
- 3 Check gears for alignment, wear or broken teeth and inspect shaft and bearings for wear and smoothness of operation
- 4 If no apparent damage is observed, avoid dismantling gears
- 5 If damage is apparent, dismantle and replace all necessary parts
- 6 When re-assembling take note of all alignment marks
- 7 Re-grease internal parts and place aside until required for overall assembly of pneumatic hoist

H. Hoist Centre Section

- 1 Remove load chain, clean if required and examine for wear, stretch or distortion
- 2 Check top hook on central section for wear, stretch and ensure it rotates freely. Check function of safety catch
- 3 Dismantle centre section and examine load chain sprockets checking for wear / distortion and ensure load chain seats correctly in pockets
- 4 Examine sprocket bearings and check for wear and smoothness of operation
- 5 Examine centre section housing and check for wear, deformation or signs of cracking
- 6
 - 1) Single fall bottom hook:
Check hook for wear, distortion and ensure it rotates freely. Check function of safety catch. Examine load pin for wear and distortion
 - 2) Multi-fall bottom hook:
Check hook for wear, distortion and ensure it rotates freely. Check function of safety catch. Split casing in half and examine for cracking or wear in the load pin sockets. Examine load pins for wear and check the bearings for smoothness of operation

I. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Replace any defective components
- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

J. Re-assembly

- 1 Re-assemble in accordance with the manufacturer's instructions lubricating the various moving components as required
- 2 Note: When re-reeving load chain, observe the manufacturer's instructions as to the positioning of the weldment on the load chain links in relation to the load sprockets to avoid exerting adverse stresses in the links.
- 3 Subject the hoist to a S.W.L + 25% proof load (in such a fashion that all the gearing makes one full revolution under load) and check for efficient operation



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of the brake. (This test to be witnessed). Re-examine hoist to ensure that the proof load has not exposed any defects

- 4 If all satisfactory, hard stamp unit with identification number and S.W.L
- 5 Pass to paint shop for painting and chain coating
- 6 Return to engineer for final check and issue of identification plate. (Ensure that the painting has not interfered with the operation of any moving component)
- 7 Where a beam trolley is fitted or is an integral part of the hoist, refer to the relevant overhaul procedure. (Should the trolley be air powered, overhaul the air motor as per the aforementioned)

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an overhaul.

The vendor should have the relevant manufacturer's manuals "in house" and they should be made available to the engineers carrying out the overhauls for further reference etc.



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S.W.L:			B = Worn but serviceable
Serial No:			C = Condemned
Plant No:			D = Replaced
			E = N.D.T.'D

	Description	Condition					Comment
		A	B	C	D	E	
	Top Hook						
	Safety Catch						
	Swivel Unit						
	Load Pin (Dead end)						
	Load Chain Sheaves						
	Load Chain Sheave Bearings						
	Load Chain Sheave Pins						
	Bottom Hook						
	Safety Catch						
	Swivel Unit						
	Load Pin (Live End)						
	Load Chain Sheave						
	Load Chain Sheave Bearings						
	Load Chain Sheave Pins						
	Gear Casing						
	Gears						
	Drive Shaft / Drive Connector						
	Bearings and Seals						
	Exhaust Casing						
	Motor Casing						
	Control Shaft						
	Shuttle Valve c/w Housing & "O"						
	End Plate						
	Barrel & Housing						

Tick relevant boxes / delete where applicable

Signed:	Date:	Test witnessed by:

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Serial No:			C = Condemned
Plant No:			D = Replaced
			E = N.D.T.'D

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Signed:

Date:

Test witnessed by:



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6.2.3 Lever Hoists

Checklist

A. Dismantling

- 1) Remove gear case cover, wash gears and check for alignment marks, wear, cracks or broken teeth
- 2) Remove operating lever and dismantle brake unit, clean metal component parts (bead blast, wire brush or emery) and examine for wear / distortion. Check condition of brake discs and measure thickness. Ensure the thickness is above the minimum allowable
- 3) Remove load chain, disconnect bottom hook, tumble chain to remove corrosion etc. and examine and measure load chain for worn, stretched or deformed links
- 4) Examine bottom hook for wear / stretch, ensure it rotates freely and check bearings. Check function of safety catch. In the case of multi-fall units, check load chain sprocket for wear and ensure it rotates freely
- 5) Examine bottom hook load pin for wear / distortion
- 6) Extract pinion shaft and examine for wear / distortion
- 7) Examine load chain sprockets for possible wear and ensure satisfactory seating of load chain in pockets, check chain de-twister for wear and alignment. Check chain stripper (where applicable) for same
- 8) Remove top hook and examine for wear / stretch, ensure it rotates freely and check function of safety catch
- 9) Examine threaded part of hook shank for wear, stretch and cracking
- 10) Remove and examine load chain live and dead end anchor pins for distortion / wear and check chain end ring / slack end pins for distortion
- 11) Check body pressing / casting and where applicable, top hook yoke for wear and distortion
- 12) Examine all remaining bearings / bushes for wear and check smoothness of operation

B. General

- 1) Ensure all components are clean and free from contamination prior to re-assembly
- 2) Replace any defective components
- 3) Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

C. Re-assembly

- 1) Re-assemble in accordance with the manufacturers instructions lubricating the various moving components as required
Note: Special attention is to be paid to the pawls and the pawl spring. Ensure there is adequate clearance between the brake cover and the pawl
- 2) Subject the hoist to a S.W.L. + 50% proof load (in such a fashion that all the gearing makes one full revolution under load), and a S.W.L. x 10% light load test to check for efficient operation of the brake. (These tests to be witnessed). Re-examine hoist to ensure that the proof load has not exposed any defects
- 3) If all satisfactory, hard stamp unit with identification number and S.W.L
- 4) Pass to paint shop for painting and chain coating
- 5) Return to engineer for final check and issue of identification plate. (Ensure that the painting has not interfered with the operation of any moving component)
 This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an overhaul.



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The vendor should have the relevant manufacturer's manuals "in-house" and they should be made available to the engineers carrying out the overhauls for further reference etc.

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			E = N.D.T.'D

	Description	Condition					Comment
		A	B	C	D	E	
	Top Hook						
	Safety Catch						
	Swivel Unit / Yoke						
	Load Pin						
	Load Chain Sheaves						
	Load Chain Sheave Bearings						
	Load Chain Sheave Pins						
	Gear Cover						
	Gears						
	Drive Shaft / Pinion						
	Load Chain Wheel						
	Load Chain Wheel Bearings						
	Body Casting / Casing						
	Chain Guide Rollers						
	Chain De-twister						
	Load Chain "Live End" anchor						
	Load Chain "Dead End" Anchor						
	Reversing Lever						
	Ratchet Pawl						
	Pawl Spring						
	Brake Disc Hub / Shoulder Piece						
	Ratchet Ring						
	Friction Washers						
	Ratchet Wheel						

Tick relevant boxes / delete where applicable

Signed:	Date:	Test witnessed by:

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Job No:			Condition
Model:			A = Satisfactory
S.W.L:			B = Worn but serviceable
Serial No:			C = Condemned
Plant No:			D = Replaced
			E = N.D.T.'D

[illegible]

Signed:

Date:

Test witnessed by:



6.2.4 Wire Rope Hoists (Tirfors)

Check List

A. Dismantling

- 1 Place unit flat on the bench with levers upwards and rope guides (rear of machine) to the right then remove side operating lever (if fitted)
- 2 Unscrew all retaining nuts and part the side casings by using a screwdriver inserted between them
- 3 Disconnect the return spring from the release lever and remove the side case reinforcing plate and rope guide
- 4 Lift out jaw assembly, wash and inspect for wear / cracks on linkage plates between front and rear jaws
- 5 Separate front and rear jaw assemblies and insert respective "wear assessment jigs" to check for wear
- 6 Wash both side casings and examine for distortion, cracks, wear in jaw block guides, misalignment of the spindle bearing and distortion of the anchor pin bearing
- 7 Examine the load pin / load hook and hook seating in the side cases for wear, distortion or cracking

B. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Replace any defective components
- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

C. Re-assembly

- 1 Re-assemble in accordance with manufacturer's instructions, lubricating the various moving components and linkages as required
- 2 Check the free operation of the machine and with rope inserted, check the stroke length in both directions
- 3 Subject the hoist to a S.W.L. + 50% proof load and the rope to a S.W.L. + 100% proof load. (These tests to be witnessed) Sustain the load on the hoist for approximately 4 minutes and check for creep. Re-examine hoist and rope to ensure that the proof load has not exposed any defects
- 4 If all satisfactory, hard stamp unit with identification number and affix name plate with test details
- 5 Replace spare shear pins in handle

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an overhaul.

The following literature must be held "in-house" by the vendor and be made available to the engineers carrying out the overhauls for further reference etc.

- 1) Tirfor T508 - Periodic inspection, overhaul and repair manual
- 2) Tirfor T516 - Periodic inspection, overhaul and repair manual
- 3) Tirfor T532 - Periodic inspection, overhaul and repair manual
- 4) Tirfor operating and maintenance instructions - Models J5/Junior, TU8, TU16 and TU32
- 5) Tirfor operating and maintenance instructions - Models T7, T13, T35, T35S and T30
- 6) Installation, operation and maintenance instructions - Models TU16H and TU32H

Further documents may be required to be held "in-house" as and when additional or alternative equipment is acquired or the particular models are superseded.



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	Description	Condition					Comment
		A	B	C	D	E	
	Operating Handle						
	Shear Pins						
	Casing - Right Hand						
	Casing - Left Hand						
	Side Case Bushes						
	Side Case Bosses						
	Rope Guide						
	Machine Anchoring Hook / Clevis						
	Safety Catch						
	Seating Plate for Anchor Hook						
	Telescopic Operating Handle						
	Front Jaw (Complete)						
	Rear Jaw (Complete)						
	Upper Jaw Assembly (Front)						
	Upper Jaw Assembly (Rear)						
	Lower Jaw Assembly (Front)						
	Lower Jaw Assembly (Rear)						
	Jaw Spring Assembly (Front)						
	Jaw Spring Assembly (Rear)						
	Jaw Side Plate Assembly (Front)						
	Jaw Side Plate Assembly (Rear)						
	Nylon Rollers						
	Crank c/w Nylon Bushes						
	Nylon Bush for Crank						

Tick relevant boxes / delete where applicable

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[illegible]

Signed:	Date:	Test witnessed by:



6.2.5 Beam Trolleys

Check List

A. Dismantling

- 1 Remove nuts from spacer bars / load bars and separate side plates
- 2 Remove spacers from load bar / bars and examine for deformation
- 3 Where applicable, examine suspension plates / yokes / shackle for wear, distortion and cracking
- 4 Remove trolley wheels and examine axles, bearing, wheel tread and flanges
- 5 Examine side plates and check for wear, distortion and cracks
- 6 If geared travel, check gear teeth on wheel flanges and drive sprocket of hand chain wheel shaft for alignment, broken teeth etc.
- 7 Examine hand chain and hand chain wheel and ensure chain seats properly in the chain sprockets
- 8 Check hand chain guides are not deformed and are correctly positioned to ensure smooth feed of chain onto chain wheel
- 9 Where an adjusting screwed bar is fitted, check for straightness and examine threads for stretch / damage and check function of locking device
- 10 Ensure both the female threaded spigots (mounted on the side plates) are not damaged or deformed

B. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Replace any defective components
- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

C. Re-assembly

- 1 Re-Assemble in accordance with the manufacturers instructions (reverse of dismantling procedure), lubricating the various moving components as required
Note: Should the trolley be of the "Super-Clamp" design, check that when the trolley is adjusted to its maximum width (i.e. when the handle butts up against the lock nut and side plate), the threaded bar still protrudes through the opposite spigot.
- 2 Subject beam trolley to a S.W.L. + 50% proof load. ("Riley Superclamp" Trolleys can be subjected to a S.W.L. + 100% proof load). This test should be witnessed
- 3 Re-examine the clamp to ensure that the proof load has not exposed any defects
- 4 If all satisfactory, hard stamp unit with identification number and S.W.L
- 5 Pass to paint shop for painting
- 6 Where applicable, apply a light covering of grease to the threaded adjusting bar and coat any hand chains

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an overhaul.

The vendor should have the relevant manufacturer's manuals "in-house" and they should be made available to the engineers carrying out the overhauls for further reference etc.

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[illegible]

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6.2.6 Beam Clamps

Check List

A. Dismantling

- 1 Remove locknut from suspension shackle pin
- 2 Remove load pin and examine for wear / distortion
- 3 Remove washers, spacers and suspension shackle
- 4 Examine shackle for wear / distortion
- 5 Using a pin punch, knock out roll pin securing handle to screwed adjusting bar and remove handle
- 6 Rotate bar free of both inner and outer clamp halves
- 7 Check screwed bar for straightness and examine threads for stretch / damage
- 8 Check both clamp halves for wear / distortion particularly around the jaws and the suspension shackle bolt holes
- 9 Ensure female threaded spigot in both clamp halves is not deformed and free to swivel
- 10 Where swivel jaws are fitted, check they are free to move

B. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Replace any defective components
- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

C. Re-assembly

- 1 Re-Assemble in accordance with the manufacturers instructions (reverse of dismantling procedure), lubricating the various moving components as required
Note: After assembling the adjusting screwed bar, ensure that when the clamp is fully opened, (i.e. When the tommy bar of the handle butts against the clamp half) the threaded bar should still protrude through the other spigot.
- 2 Subject clamp to a S.W.L. + 50% proof load. ("Super Clamps" can be subjected to a S.W.L. + 100% proof load). This test should be witnessed
- 3 Re-examine the clamp to ensure that the proof load has not exposed any defects
- 4 If all satisfactory, hard stamp unit with identification number and S.W.L
- 5 Pass to paint shop for painting
- 6 Apply a light covering of grease to protect threaded adjusting bar

This procedure is to be followed as a guideline to **QatarEnergy-IS** minimum requirements for an overhaul.

The vendor should have the relevant manufacturer's manuals "in-house" and they should be made available to the engineers carrying out the overhauls for further references etc.

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[illegible]

Signed:

Date:

Test witnessed by:



6.2.7 Horizontal Plate Clamps

Check List

A. Dismantling

- 1 Withdraw the 2 or 3 load bolts to separate hook-ring serrated jaw / rocker arms, main body and swivel toe (if fitted)
- 2 Examine load bolts and check for wear / distortion
- 3 Examine hook ring and look for wear / stretch in the crown and load bolt hole
- 4 Examine the serrated jaw / rocker arms and swivel toe, ensure the serrations are not worn or deformed. (Do not attempt to repair serrations with a file as they are in a hardened condition and filing may remove the hardness). Examine load bolt holes and check for wear / distortion
- 5 Examine main body and check load bolt hole for wear / stretch, check toe plate for deformation and ensure handle is still fitted

B. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Replace any defective components
- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

C. Re-assembly

- 1 Re-assemble in the reverse order of above (paying attention to any specific recommendations of the manufacturer) and lubricate the relevant moving components
- 2 Subject the clamp to a S.W.L. + 100% proof load and re-examine to ensure that proof loading has not exposed any defects
- 3 If all satisfactory, hard stamp unit with identification number and S.W.L
- 4 Pass to paint shop for coating

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an overhaul.

The vendor should have the relevant manufacturer's manuals "in-house" and they should be made available to the engineers carrying out the overhauls for further reference etc.

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Job No:			Condition
Model:			A = Satisfactory
S.W.L:			B = Worn but serviceable
Serial No:			C = Condemned
Plant No:			D = Replaced
			E = N.D.T.'D

[illegible]

Signed:

Date:

Test witnessed by:



6.2.8 Universal Plate Clamps

Check List

A. Dismantling

- 1 Push down on hook ring until hook ring pin is in line with hole in body shell and push out the load pin
- 2 Remove hook ring and examine for wear / distortion in both the suspension hole and pin hole
- 3 Examine the load pin and check for wear / deformation
- 4 Remove jaw pin and nut and examine for wear / deformation
- 5 Disconnect jaw spring and remove internal assembly
- 6 Examine internal links and check for wear / deformation
- 7 Remove pin connecting links to serrated jaw and check for wear / deformation
- 8 Examine serrated jaws and check for wear / deformation. (Do not attempt to repair serrations with a file as they are in a hardened condition and filing may remove the hardness)
- 9 Examine main body shell and check for wear, deformation and cracking
- 10 Ensure locking lever is free to operate and is not deformed

B. General

- 11 Ensure all components are clean and free from contamination prior to re-assembly
- 12 Replace any defective components
- 13 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

C. Re-assembly

- 1 Re-assemble in the reverse order of this procedure (paying attention to any specific recommendations of the manufacturer) and lubricate the relevant moving components
- 2 Subject the clamp to a S.W.L. + 100% proof load and re-examine to ensure proof loading has not exposed any defects
- 3 If all satisfactory, hard stamp with identification number and S.W.L. then pass to paint shop for coating

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an overhaul.

The vendor should have the relevant manufacturer's material "in-house" and they should be made available to the engineers carrying out the overhauls for further references etc.

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[illegible]

Signed:

Date:

Test witnessed by:



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6.2.9 Sheave Blocks / Crane Blocks

Check List

A. Dismantling

- 1 Remove nut from main load pin, withdraw pin and remove sheaves from body plates. (Degrease)
- 2 Examine sheaves for wear in the rope groove, cracks or distortion
- 3 Check bushes for wear and tightness in sheave
- 4 Examine bearings and ensure smoothness of operation
- 5 Examine main load pin checking for wear or scoring. (Ensure all grease ports are clean and unblocked)
- 6 Check function of grease nipples
- 7 Remove swivel head fitting or bottom load hook and examine for wear / stretch
- 8 Examine thrust bearing / washer and ensure smoothness of operation
- 9 Examine upper load pin / spigots and check for wear / distortion
- 10 If head fitting / hook is of the shank type, examine threads of shank and nut checking for stretch / distortion and examine cross-head for wear
- 11 Examine head fitting shackle / eye checking for wear, stretch or cracking
- 12 Ensure any lubrication ports are clean and check function of grease nipples
- 13 Examine side plates / straps and check for distortion, wear or cracking (especially around main load pin hole and top suspension hole)
- 14 Ensure there are no sharp edges or burrs on the side plates which may be detrimental to the wire rope
- 15 If fitted, remove becket and check for wear, stretch or cracking
- 16 If fitted, examine weighted cheek plates (check for cracks)
- 17 Check all spacers and tie bolts and ensure they are not deformed

B. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Replace any defective components
- 3 Where doubt exists as to the integrity of any component, N.D.T. techniques should be employed to verify their status

C. Re-assembly

- 1 Re-assemble in accordance with the manufacturer's instructions. (Where none exist, reverse the dismantling procedure)
- 2 Lubricate the various moving components as required and coat / paint items which will become inaccessible after assembly (i.e. inner cheek plates, sides of sheaves etc.)
- 3 Subject the block to a proof load test in accordance with the I.L.O. Tables (Copy attached). (This test to be witnessed)
- 4 If a becket is fitted, it must be subjected to the relevant proof load dependant on the number of sheaves

No of Sheaves in block	Proportion of total proof load of block to be applied to becket
One (Top Block)	One-Half
One (Btm Block)	One-Third
Two	One-Quarter
Three	One-Sixth



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Four	One-Eighth
Five	One-Tenth
Six	One-Twelfth
Seven	One-Fourteenth
Eight	One-Sixteenth

- 1 After load test, withdraw main load pin and remove sheaves for a further examination to detect if any defect has been exposed by the overload
- 2 If all satisfactory, re-stamp the block with identification number, S.W.L. and new date of test

Note: In the case of single sheave blocks, the "line pull" S.W.L. should also be indicated.

e.g. S.W.L. 8 tonne (Head fitting)
LINEPULL 4 tonne

- 3 Pass to paint shop for final coating

This procedure is to be followed as a guideline to **QatarEnergy-IS** minimum requirements for an overhaul.

The vendor should have the relevant manufacturer's manuals "in-house" and they should be made available to the engineers carrying out the overhauls for further reference etc



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Table as Issued by the International Labour Office
Pamphlet No. 188 - "Safety and Health in Dock Work"

Appendix 1

Safe Working Load W (Resultant on Head fitting)		Excess Load		Proof Load
t	%	t		t
25 or less	100	W		2W
30	85	25		55
35		30		65
40	75	30		70
45		30		75
50	65	35		85
55		35		90
60	60	35		95
65		35		100
70	55	40		110
75		40		115
80	50	40		120
85		40		125
90	46	40		130
95		40		135
100	43	45		145
110	41	45		155
120	38	45		165
130	36	45		175
140	35	50		190
150	34	50		200
160	33	55		215
170	33	60		230
180	33	60		240
Over 180	33	-		-

The proof load is quoted in tonnes for application by weights.



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	Description	Condition					Comment
		A	B	C	D	E	
	Main Load Pin						
	Sheaves						
	Sheave Bushes						
	Sheave Bearings						
	Load Pin Grease Nipple						
	Head Fitting Swivel						
	Head Fitting Swivel Bearing						
	Head Fitting Shackle						
	Cross Head						
	Cross Head Grease Nipple						
	Oval Eye c/w Shank						
	Shank Nut						
	Load Hook						
	Safety Catch						
	Side Plates / Straps						
	Becket						
	Spacers						
	Tie Bolts						
	Weighted Cheek Plates (if fitted)						

Tick relevant boxes / delete where applicable

Signed:	Date:	Test witnessed by:



6.2.10 Pneumatic Winches

Check List

A. Gear Drive End

- 1 Remove gear case cover (and drum guard if fitted)
- 2 Insert packing between winch drum and base (to support drum)
- 3 Disconnect brake band and slide off drum
- 4 Un-bolt gear end support plate and remove to expose ring gear / planet gears
- 5 Withdraw drive shaft
- 6 Where applicable, remove all planet gears from stub shafts and examine needle bearings and ring gear teeth
- 7 Examine winch drum and check for wear, cracking and distortion
- 8 Examine rope anchor and ensure setscrews are free

B. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Replace any defective components
- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

C. Re-assembly

- 1 Re-assemble in the reverse order of above lubricating the relevant moving components as necessary

Lay aside for final assembly

D. Motor End Drive

- 1 Remove motor from motor support end plate
- 2 Remove valve chest
- 3 Remove all cylinder heads
- 4 Withdraw wrist pins from pistons and check piston rings / oil rings for wear.
- 5 Turn motor upside down to expose crank and conrods
- 6 Feed conrods from cylinder apertures and remove the complete crank from its motor case
- 7 Examine all bearings on crank ends and ensure they run smoothly and there is no side play
- 8 Examine crank pin bushings (should wear be found, they have to be replaced, it is important to mark each conrod in numerical sequence to facilitate correct re-assembly in the reverse order)

E. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Replace any defective components
- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

Lay aside for final assembly

F. Valve Chest

Caution: Valve chest bushings are prone to wear. The rotary valve bushing and reverse valve bushing must be clear of any scoring. Any pitting on the rotary or reverse valves will have a drastic effect on the efficiency of the winch performance due to leakage and should be replaced.

- 1 Remove pipe plug on top of valve chest, remove spring, poppet valve and throttle valve ball. Examine all components for wear and check efficiency of spring
- 2 Remove and examine reverse valve for wear (refer to important note above)



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- 3 Examine rotary valve bushing for wear (refer to important note above)
- 4 Dismantle throttle control handle by removing allen screw, withdraw spring and plunger. Examine for wear and check efficiency of spring. If all satisfactory re-grease and re-assemble
- 5 Examine throttle lever spring and check tension

G. Where Winch is Fitted with Automatic Brake

- 1 Remove all brake linkage pins and check for wear on pins (and holes in link arms)
- 2 Examine link arms and check for bending / distortion
- 3 Remove brake springs and check for possible fatigue (by bending to touch both ends together)
- 4 Remove automatic brake cylinder and check for corrosion on chromed piston rod
- 5 Examine supply air line to brake cylinder and check for impact damage. (This is susceptible to forklift damage during handling)
- 6 Examine exhaust valve and function test
- 7 Re-assemble and lay aside for final assembly

H. Winch Base

- 1 Examine steel frame and check all weldments. Checks for corrosion / cracking around tie down bolt holes

Note: If the winch is to be used for the raising and lowering of personnel, refer to procedure 11 which must be read in conjunction with this procedure.

I. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Replace any defective components
- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

J. Re-assembly

- 1 Bolt motor end plate to winch base
- 2 Slide winch drum onto drum shaft
- 3 Replace planet gears on stub shaft
- 4 Bolt gearbox end and plate to winch base
- 5 Install drive gear and motor shaft
- 6 Grease all ring gears, planet gears and drive gears with a good quality gear grease
- 7 Refit reverse valve and throttle lever to valve chest
- 8 Slide valve chest into motor aperture, fit rotary valve through rotary valve bush and engage bush in slot
- 9 Refit valve chest cover taking care to torque the bolts evenly
- 10 Fit manual / automatic brake (adjustment / tension to be set during testing)
- 11 Fill with oil / lubricate in accordance with lubrication schedule

K. Load Testing

- 1 Bolt down winch to suitable test bed and fit a wire of such length that no fewer than five turns remain on drum during any part of the test

1) Test 1 (Dynamic) -

Raise a proof load of the S.W.L. + 25% to the required height to allow one full rotation of drum under load. (This test can be simulated on a hydraulic test bed where a variable pressure relief valve has been fitted.) Apply manual brake to sustain load and check winch drum for creep



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Note: When testing winches with an automatic brake on a hydraulic test bed, there will be a slight drop in load due to the time it takes for the brake to activate.

2) Test 2 (Static) -

Disconnect power source to winch and apply brake. Using an alternative pulling source i.e. pull lift / chainblock on the winch rope, apply a proof load of the S.W.L. + 50% to check the braking efficiency. (The brakes may require adjusting to achieve this). Sustain the load for a short time and check winch drum for creep

3) Test 3 (Function) -

Re-connect power supply to winch and pull up to S.W.L. (To ensure the brake has not been over tightened to such an extent as to affect the winch's performance)

Note: Ensure all levers / controls return to neutral on release

- 2 Re-examine winch to ensure that proof load testing has not exposed any defects the above tests are to be witnessed

Note: Further tests are required if this winch is to be used for the raising or lowering of personnel. **See 5.1.11**

- 3 Fit winch drum guard where applicable
- 4 If all satisfactory, mask / plug all inlets and exhausts and pass to paint shop for coating
- 5 Return to engineer for final plating

Note: Ensure S.W.L. and all directional arrows are clearly visible.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an overhaul.

The vendor should have the relevant manufacturer's manuals "in-house" and they should be made available to the engineers carrying out the overhauls for further references etc.



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	Description	Condition					Comment
		A	B	C	D	E	
	Gear Case Cover						
	Brake Band						
	Gear End Support Plate						
	Ring Gear						
	Planet Gears						
	Drive Shaft						
	Needle Bearing						
	Winch Drum						
	Rope Anchor						
	Motor Support End Plate						
	Cylinder Heads						
	Pistons						
	Piston Rings						
	Oil Rings						
	Crank Shaft						
	Conrods						
	Crank End Bearings						
	Crank Pin Bushings						
	Valve Chest						
	Rotary Valve Bushing						
	Reverse Valve Bushing						
	Pipe Plug						
	Spring & Poppet Valve						
	Throttle Valve Ball						

Tick relevant boxes / delete where applicable

Signed:	Date:	Test witnessed by:

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[illegible]

Signed:	Date:	Test witnessed by:



6.2.11 Pneumatic Winches (Personnel)

Check List

Note: For general overhaul, follow the steps laid out in 5.1.10

A. Additional Instructions

Winches converted for the purpose of raising / lowering personnel

- 1 Winches used for the raising and lowering of personnel will have a **dual** braking system usually consisting of one manual brake and one automatic brake (which will activate should there be a loss of power). These brakes are to be overhauled in accordance with procedure 10
- 2 Where limit switches are fitted to prevent over-travel in both hoisting and lowering, function test to ensure their operation
- 3 Ensure the winch does **not** have a disengaging clutch or pawl and ratchet mechanism
- 4 Ensure that the winch has been adequately de-rated to reflect a factor of safety of no less than 7:1 (Note: Man-riding winch rope to have F.O.S. of 8:1)
- 5 Ensure the winch is clearly marked that it is to be used solely for raising and lowering personnel

B. Testing

- 1 Bolt down winch to suitable test bed and fit a wire of such length that no fewer than five turns remain on drum during any part of the test

Test 1 (Dynamic) -

Raise a proof load of S.W.L. + 25% to the required height to allow one full rotation of drum under load. (This test can be simulated on a hydraulic test bed where a variable pressure relief valve has been fitted.) Apply manual brake to sustain load and check winch drum for creep

Note: When testing winches with an automatic brake on a hydraulic test bed, there will be a slight drop in load due to the time it takes for the brake to activate.

Disconnect automatic brake so full load is taken on manual brake.

Test 2 (Static) -

Disconnect power source to winch and apply brake. Using an alternative pulling source i.e. pull lift/chainblock on the winch rope, apply a proof load of S.W.L. + 50% to check the braking efficiency. (The brakes may require adjusting to achieve this). Sustain the load for a short time and check winch drum for creep

Re-connect automatic brake and with the manual brake in the off position, repeat **test 2** so full load is taken on automatic brake

Note: Should any of the brake drums be exposed to the elements, the aforementioned tests should be carried out with the drums wet, to simulate operating conditions. Make any brake adjustments as required.

Test 3 (Function) -

Re-connect power supply to winch and pull up to S.W.L. (To ensure the brake has not been over tightened to such an extent as to affect the winch's performance)

Note: Ensure all levers / controls return to neutral on release

- 2 Re-examine winch to ensure that proof load testing has not exposed any defects the above tests are to be witnessed
- 3 Fit winch drum guard where applicable
- 4 If all satisfactory, mask / plug all inlets and exhausts and pass to paint shop for coating
- 5 Return to engineer for final plating

Note: Ensure S.W.L. and all directional arrows are clearly visible.

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This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an overhaul. The vendor should have the relevant manufacturer's manuals "in-house" and they should be made available to the engineers carrying out the overhauls for further reference etc. **Refer to procedure 10 for checklist**

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Job No:			Condition
Model:			A = Satisfactory
S.W.L:			B = Worn but serviceable
Serial No:			C = Condemned
Plant No:			D = Replaced
			E = N.D.T.'D

[illegible]

Signed:	Date:	Test witnessed by:



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6.2.12 Manual Winches

Check List

A. Dismantling

- 1 Remove winding handle and examine for distortion
- 2 Withdraw drum shaft and examine for wear / distortion
- 3 Remove drum and examine for wear and distortion and ensure rope anchorage is functional
- 4 Examine teeth on ratchet wheel / worm gear (mounted on side of drum) and check for wear / broken teeth
- 5 Remove ratchet pawl / worm and examine for wear / distortion
- 6 Examine pawl spring (where fitted) and ensure it still has adequate strength
- 7 Examine winch frame and check for distortion, wear and cracking, paying special attention to the holes which locate the drum shaft and the hold down bolts

B. Additional: (Spur Geared Winches)

- 1 Remove and examine friction discs and ensure they are free from contamination
- 2 Examine pressure plates on handle drive shaft and ensure they are flat, un-scored and clean
- 3 Examine all spur gears, checking for alignment, wear on gear teeth, truth of gear shafts and smooth running of bearings (where fitted)

C. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Replace any defective components
- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

D. Re-assembly

- 1 Re-assemble in accordance with the manufacturers instructions, (or in reverse of aforementioned) lubricating all relevant moving parts
- 2 Subject the winch to a S.W.L. + 50% proof load in such a fashion that the drum makes one full revolution under load. (This test to be witnessed). Re-examine the winch to ensure that testing has not exposed any defects
- 3 If all satisfactory, hard stamp with identification number and S.W.L. then pass to paint shop for coating
- 4 Return to engineer for final inspection and plating

This procedure is to be followed as a guideline to **QatarEnergy-IS** minimum requirements for an overhaul.

The vendor should have the relevant manufacturer's manuals "in-house" and they should be made available to the engineers carrying out the overhauls for further reference etc.

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[illegible]

Signed:

Date:

Test witnessed by:



6.2.13 Pallet Stackers (Also applicable to Switchgear Lifters)

Check List

A. Dismantling (Manual Hydraulic)

- 1 Ensure ram is fully closed and drain hydraulic oil from system
- 2 Disconnect pipe work and couplings between pump and cylinder, and examine for impact damage and / or cuts which may affect the pressure capacity
- 3 Disconnect wire rope / chain anchors at:
 - 1) Rear of stacker (positioned or horizontal cross bar) and
 - 2) Bottom of fork support frame
- 4 Examine wire / chain and anchor fittings checking for wear, broken wires, slack chain links, kinks and distortion
- 5 Remove sheave assembly from top of ram and examine sheave, axle pin and bearings, checking for wear / distortion
- 6 Remove cylinder and examine for impact damage, leaking seals and wear at the end connections
- 7 Remove pump and examine, checking for leaks at the piston seal and non-return valve
- 8 Re-connect the hydraulic system (remote from the stacker), re-fill with hydraulic oil and bleed system as required
- 9 Pump up the ram to its maximum stroke and check that the positive stop functions
- 10 Examine the ram, checking for corrosion / scoring
- 11 Release hydraulic pressure and check that ram returns smoothly
- 12 Should the cylinder or pump require further overhaul, proceed as follows:

B. Dismantling (Cylinders / Rams)

- 1 Unscrew inlet coupling and remove
- 2 Disconnect ram return spring (if fitted)
- 3 Undo and remove gland neck ring (re-new "O" ring)
- 4 Withdraw ram c/w return spring and examine for corrosion, wear and scoring (check spring tension)
- 5 Remove packing locking ring and re-new ram packing
- 6 Examine internal bore of cylinder checking for corrosion, wear and scoring

Note: Larger cylinders are slightly easier to strip as they do not usually have a return spring connecting the ram to the cylinder. Access to the internals is usually obtained by removing the cylinder head or gland neck ring.

C. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Renew ALL software i.e. "O" rings, seals and packing and replace any defective components
- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

D. Re-assembly

- 1 Re-assemble in the reverse order of this procedure adhering to any specific instructions by the manufacturer
- 2 When re-fitting the inlet (and outlet) coupling, ensure all old thread tape has been removed. Refit the coupling using thread tape to give a complete seal
- 3 Function test the cylinder, pumping up to full stroke
- 4 Proof load test to S.W.L. + 25% and sustain load for approximately 5-10 minutes. Check for any oil leaks or drop in pressure which will show on the test rig gauge



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- 5 Re-examine unit to ensure proof load testing has not exposed any defects
- 6 If all satisfactory, set aside for final assembly

E. Dismantling (Pumps)

- 1 Remove filler plug and drain off hydraulic oil
- 2 Remove pivot pin fastener and press out pivot pin
- 3 Remove handle base c/w plunger pin
- 4 Withdraw both high pressure and low-pressure plungers / pistons. Unscrew H.P. gland and replace packing
- 5 Unscrew delivery valve plug and remove delivery and suction balls (replace from standard repair kit)
- 6 Undo clamp nut, remove tank end assembly and withdraw oil tube / tank
- 7 This exposes the pressure relief valve - refer to manufacturer's instructions for torque setting

F. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Renew ALL software i.e. "O" rings, seals and packing and replace any defective components
- 3 Where doubt exists as to the integrity of any component, N.D.T. techniques should be employed to verify their status

G. Re-assembly

- 1 Re-assemble in the reverse order of this procedure adhering to any specific instructions of the manufacturer
- 2 Re-fill with clean oil taking care not to over fill (check with dip stick)
- 3 Replace filler plug and function test for entrapped air (bleed as required)
- 4 If all satisfactory, set aside for final assembly.

Note: Should the stacker be purely manual, it will be fitted with a small winch unit which should be removed from the stacker and overhauled as follows:

H. Dismantling (Direct Drive / Worm Geared Winches)

- 1 Remove winding handle and examine for distortion
- 2 Withdraw drum shaft and examine for wear / distortion
- 3 Remove drum and examine for wear and distortion and ensure rope anchorage is functional
- 4 Examine teeth on ratchet wheel / worm gear (mounted on side of drum) and check for wear / broken teeth
- 5 Remove ratchet pawl / worm and examine for wear / distortion
- 6 Examine pawl spring (where fitted) and ensure it still has adequate strength
- 7 Examine winch frame and check for distortion, wear and cracking, paying special attention to the holes which locate the drum shaft and the hold down bolts

I. Additional: (Spur Geared Winches)

- 1 Remove and examine friction discs and ensure they are free from contamination
- 2 Examine pressure plates on handle drive shaft and ensure they are flat, un-scored and clean
- 3 Examine all spur gears, checking for alignment, wear on gear teeth, truth of gear shafts and smooth running of bearings (where fitted)

J. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Replace any defective components



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- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

K. Re-assembly

- 1 Re-assemble in accordance with the manufacturer's instructions (or in reverse of aforementioned), lubricating all relevant moving parts
- 2 Subject the winch to a S.W.L. + 50% proof load in such a fashion that the drum makes one full revolution under load (this test to be witnessed). Re-examine the winch to ensure that testing has not exposed any defects
- 3 If all satisfactory, set aside for final assembly

Note: The manual stacker normally has a sheave mounted on the underside of the horizontal cross bar at the top of the mast. This should be dismantled and examined as per item 1.05 of this procedure.

L. Dismantling (Main Frame Work)

- 1 Remove forks, drum forks, boom or specialized lifting brackets (as per switchgear type stacker) and examine for wear and distortion
- 2 Remove fork support frame and examine for wear / distortion (ensure mast guide rollers are free to rotate)
- 3 Remove and examine wheels / rollers / castors and ensure all bearings are sound
- 4 Where swivel castors are fitted with foot brakes, function test to ensure they are in working order
- 5 Examine remaining mast and base steelwork (depending on the age and condition of the unit, it may be prudent to shotblast and carry out N.D.T. especially around weldments)

Note: Should any remedial work be required i.e. re-welding, replacing members etc., **QatarEnergy-IS** quality assurance department must be contacted to authorise material selection, weld procedures and relevant N.D.T.

M. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Replace any defective components
- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

N. Final Re-assembly

- 1 Re-fit wheels, castors, rollers, etc. lubricating bearings etc. as required
- 2 Re-install fork support frame
- 3 Re-attach forks / drum forks / boom or specialised lifting brackets
- 4 Where applicable, refit sheave assembly to cross beam at top of mast / top of ram
- 5 Re-fit winch unit / hydraulics and re-connect wire / chain accordingly
- 6 Make necessary adjustments to rope / chain anchors to ensure correct range of vertical travel on mast
- 7 With swivel castors locked, subject the stacker to a S.W.L. + 25% proof load and raise to its maximum height. (This will also verify stability)

This test should be witnessed.

- 8 Re-examine unit to ensure proof loading has not exposed any defects.
(A post load test N.D.T. may also be advisable)
 - 9 If all satisfactory, hard stamp unit with identification number and S.W.L then pass to paint shop for coating
 - 10 Return to engineer for final check and issue of identification plate
- This procedure is to be followed as a guideline to **QatarEnergy-IS minimum**

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requirements for an overhaul, etc. The vendor should have the relevant manufacturer's manuals "in house" and they should be made available to the engineers carrying out the overhauls for further reference, etc.



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	Description	Condition					Comment
		A	B	C	D	E	
	Hydraulic Pipe work						
	Hydraulic Couplings						
	Cylinder						
	Ram						
	Gland Neck Ring						
	Ram packing						
	Packing Locking Ring						
	Return Spring (if fitted)						
	Seals						
	"O" Rings						
	Inlet Coupling (if fitted)						
	Outlet Coupling (if fitted)						
	Pump Unit:						
	Handle & Handle Base						
	Pivot Pins						
	H.P & L.P Plungers						
	"O" Rings and Seals for above						
	H.P Plunger Gland						
	L.P Relief Valve Plug						
	L.P Relief Valve						
	Valve Spring						
	Release Valve Assembly						
	Relief Valve						
	Oil Tube / Tank						
	"O" Rings / Seals for above						

Tick relevant boxes / delete where applicable

Signed:	Date:	Test witnessed by:



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	Description	Condition					Comment
		A	B	C	D	E	
	Centre Stud c/w Clamp Nut						
	Tank End Assembly						
	Handle Locking Pin c/w						
	Access & Delivery / Suction						
	Winch Unit:						
	Winding Handle						
	Drum Shaft						
	Drum						
	Ratchet Wheel / Worm						
	Ratchet Pawl / Worm						
	Ratchet Pawl Spring						
	Winch Frame						
	Friction Discs						
	Pressure Plates						
	Spur Gears						
	Spur Gear Shafts						
	Bearings						
	Main Framework						
	Rope / Chain Sheave						
	Forks / Drum Forks / Boom						
	Specialized Lifting Brackets						
	Fork Support Frame						
	Guide Rollers						
	Wheel / Wheel Bearings						
	Castors						

Tick relevant boxes / delete where applicable

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Signed:

Date:

Test witnessed by:



6.2.14 Pallet Trucks

Check List

A. Dismantling

- 1 Lower the pull bar and remove nut and rubber washer then lift the pull bar off
- 2 Remove back axle
- 3 Dismantle pump and remove cylinder (check seals and ram)
- 4 Remove yoke unit and dismantle steering wheels. Check bearings and wheels for smoothness of operation
- 5 Remove front rollers and climbing wheels and check for smoothness of operation
- 6 Examine push rods and ensure they are in good working order

B. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Replace any defective components
- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

C. Re-assembly

- 1 Re-assemble in accordance with the manufacturer's instructions, lubricating the various moving components as required
- 2 Function test the truck with its full S.W.L. and check for hydraulic leaks. Re-examine to ensure that the load test has not exposed any defects
- 3 If all satisfactory, hard stamp unit with identification number and S.W.L
- 4 Pass to paint shop for painting
- 5 Return to engineer for final check and the issue of letter of conformity

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an overhaul.

The vendor should have the relevant manufacturer's manuals "in house" and they should be made available to the engineers carrying out the overhauls for further reference etc.

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[illegible]

Signed:

Date:

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6.2.15 Lifting Beams / Frames

Check List

A. Dismantling

- 1 Remove all ancillary lifting tackle i.e. slings, shackles etc. (These are to be overhauled in accordance with their specific procedures)
- 2 If the beam / frame is of bolted construction, dismantle completely
- 3 Examine all bolts / fixings and check for wear, bending and corrosion
- 4 Shotblast all other steel work to bare metal
- 5 Carry out appropriate N.D.T. checks on all load-bearing weldments i.e. M.P.I. on fillets and U.T. on full pen welds
- 6 Examine all padeyes / lifting points and check for wear, distortion and cracking
- 7 Examine actual beam / frame members and check for bending and distortion
- 8 Should the beam / frame members be constructed from hollow section or tubulars, ensure they have not incurred impact damage which could affect their rigidity

Note: Should any remedial work be required i.e. re-welding, replacing members etc. **QatarEnergy-IS** quality assurance department must be contacted to authorize material selection, weld procedures and relevant N.D.T.

- 9 Replace any worn bolts, connections or damaged components during re-assembly
- 10 Subject the beam / frame to the relevant proof load according to it's S.W.L (See Table 1)
- 11 Re-examine structure to ensure that proof loading has not exposed any defects
- 12 Carry out post load test N.D.T. i.e. M.P.I./U.T
- 13 If all satisfactory, hard stamp identification number and S.W.L
- 14 Pass to paint shop for coating
- 15 Ensure the S.W.L. is prominently stenciled on the unit

This procedure is to be followed as a guideline to **QatarEnergy-IS** minimum requirements for an overhaul.

Table 1

SWL	Proof Load
Up To & Inc. 3t	+ 100%
Over 3 Up To & Inc. 6t	+ 3t
Over 6 Up To & Inc. 20t	+ 50%
Over 20 Up To & Inc. 40t	+ 10t
Over 40t	+ 25%



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6.2.16 Hydraulic Cylinders / Rams / Pumps

Check List

A. Dismantling (Cylinders / Rams)

- 1 Unscrew inlet coupling and remove
- 2 Disconnect ram return spring (if fitted)
- 3 Undo and remove gland neck ring (re-new "O" ring)
- 4 Withdraw ram c/w return spring and examine for corrosion, wear and scoring (check spring tension)
- 5 Remove packing locking ring and re-new ram packing
- 6 Examine internal bore of cylinder checking for corrosion, wear and scoring

Note: Larger cylinders are slightly easier to strip as they do not usually have a return spring connecting the ram to the cylinder. Access to the internals is usually obtained by removing the cylinder head or gland neck ring.

B. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Renew ALL software i.e. "O" rings, seals and packing and replace any defective components
- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

C. Re-assembly

- 1 Re-assemble in the reverse order of this procedure adhering to any specific instructions of the manufacturer
- 2 When re-fitting the inlet (and outlet) coupling, ensure all old thread tape has been removed. Re-fit the coupling using thread tape to give a complete seal
- 3 Function test the cylinder, pumping up to full stroke
- 4 Proof load test to S.W.L. + 25% and sustain load for approximately 5-10 minutes. Check for any oil leaks or drop in pressure which will show on the test rig gauge
- 5 Re-examine unit to ensure proof load testing has not exposed any defects
- 6 If all satisfactory, hard stamp with identification number and S.W.L, fit coupling protection caps and pass to paint shop for coating
- 7 Return to engineer for final inspection

D. Dismantling (Pumps)

- 1 Remove filler plug and drain off hydraulic oil
- 2 Remove pivot pin fastener and press out pivot pin
- 3 Remove handle base c/w plunger pin
- 4 Withdraw both high pressure and low-pressure plungers / pistons. Unscrew H.P. gland and replace packing
- 5 Unscrew delivery valve plug and remove delivery and suction balls. (Replace from standard repair kit)
- 6 Undo clamp nut, remove tank and assembly and withdraw oil tube / tank
- 7 This exposes the pressure relief valve - refer to manufacturers instructions for torque setting



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E. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Renew ALL software i.e. "O" rings, seals and packing and replace any defective components
- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

F. Re-assembly

- 1 Re-assemble in the reverse order of this procedure adhering to any specific instructions of the manufacturer
- 2 Refill with clean oil taking care not to over fill. (Check with dipstick)
- 3 Replace filler plug and function test for entrapped air. (Bleed as required)
- 4 Pressure test pump against master gauge to working pressure + 25% proof pressure and ensure relief valve functions

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an overhaul.

The vendor should have the relevant manufacturer's manuals "in-house" and they should be made available to the engineers carrying out the overhauls for further references etc.



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Plant No:			D = Replaced
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Item	Description	Condition					Comments:
		A	B	C	D	E	
1	Cylinder						
2	Ram						
3	Gland Neck Ring						
4	Ram Packing						
5	Packing Locking Ring						
6	Return Spring (if fitted)						
7	Seals						
8	"O" Rings						
9	Inlet Coupling						
1	Outlet Coupling (if fitted)						
	Pump Unit						
1	Handle and Handle Base						
1	Pivot Pins						
1	H.P. and L.P. Plungers						
1	"O" Rings / Seals for above						
1	H.P. Plunger Gland						
1	L.P. Relief Valve Plug						
1	L.P. Relief Valve						
1	Valve Spring						
1	Release Valve Assembly						
2	Relief Valve						
2	Oil Tube						
2	"O" Seals for above						
2	Centre Stud c/w Clamp Nut						

Tick relevant boxes / delete where applicable

Signed:	Date:	Test witnessed by:

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[illegible]

Signed:

Date:

Test witnessed by:



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6.2.17 Hydraulic Jacks

Check List

A. Dismantling

- 1 With ram in fully lowered position, remove filler screw and drain off all oil from reservoir
- 2 Unscrew and remove base cap screws
- 3 Separate cistern from base, (this may require a reasonable amount of force to break the seal of the gasket) to expose the pump unit and ram assembly
- 4 Unscrew ram assembly
- 5 Separate ram from liner by pushing down from the top
- 6 Examine the bore of the liner checking for wear, corrosion and scoring (Renew software i.e. ram wiper, "O" rings, seals etc.)
- 7 Examine ram checking for wear, corrosion and scoring. (Re-new software i.e. ram packing, seals etc.)
- 8 If all satisfactory, re-assemble and lay aside for final assembly

B. Pump Unit

- 1 Withdraw high pressure and low pressure pistons and examine for wear / scoring (replace "O" rings)
- 2 Replace H.P. & L.P. ball bearing and spring (from standard repair kit)
- 3 Re-assemble pump unit and lay aside for final assembly

C. Cistern

- 1 Examine for impact damage / cracking
- 2 Replace quadrant seal and "O" seal (reservoir to cylinder seal)

D. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Renew ALL software i.e. "O" rings, seals, packing, and gaskets and replace any defective components

Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status.

E. Final Re-assembly

- 1 Fit a new cylinder to base seal and screw cylinder into base plate
- 2 Fit new pump body seal and re-fit pump unit to base
- 3 Re-fit cistern (renewing reservoir gasket) taking care to re-attach the trip mechanism
- 4 Re-fill with clean hydraulic oil and replace filler screw
- 5 Function test the jack by pumping up to full stroke (without load) checking that ram rises a regular amount with each pump
- 6 Examine for oil leaks
- 7 Using a calibrated test rig, pump up the jack to its S.W.L. + 25% (unless fitted with a relief valve) and sustain for approximately 10 minutes, checking for oil leaks or loss of pressure

Note: Should the jack be fitted with a relief valve, it is normally set up to relieve between 105% and 110% of rated load. In this case, pump up to rated capacity and sustain for 10 minutes. (Check as above).

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- 8 Re-examine unit to ensure proof load testing has not exposed any defects
- 9 If all satisfactory, hard stamp with identification number and S.W.L. then pass to paint shop for coating
- 10 Return to engineer for final inspection

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an overhaul.

The vendor should have the relevant manufacturer's manuals "in house" and they should be made available to the engineer carrying out the overhauls for further reference etc.



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Item	Description	Condition					Comment
		A	B	C	D	E	
	Liner						
	Ram						
	Base						
	Cistern						
	Liner Seal						
	Cistern Joint (Gasket)						
	Base Plug						
	Base Plug Ball						
	Cistern Seal						
	Cistern (Base) Screws						
	Filler Screws						
	Dip Stick						
	Pump Quadrant Shaft						
	Quadrant Shaft Seal						
	Pump Plunger						
	Pump Plunger Seal						
	Pump Barrel						
	Pump Barrel Joint						
	Valve Balls						
	Valve Springs						
	Suction Valve Screw						
	Delivery Valve Screw						
	Valve Screw Joint						
	Suction Filter						

Tick relevant boxes / delete where applicable

Signed:	Date:	Test witnessed by:



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	Description	Condition					Comment
		A	B	C	D	E	
	Release Valve (Complete)						
	Cistern Breather Pipe						
	Operating Lever						
	Operating Lever Key						
	Ram Wiper						
	Ram Wiper Retaining Ring						
	Ram Cap						
	Retaining Pins						
	Control Ring C/W Spring						
	Trip Bar & Pin						
	Valve Seating						
	Trip Valve Cage						
	Trip Valve Ball & Spring						
	Plug Screw						
	Seal Washer						
	Pump Packing Anti-Extrusion						
	Pump Packing Retaining Sleeve						
	Relief Valve Sub Assembly						
	Breather Valve Assembly						
	Rubber Gaiter						
	Ram Packing Anti Extrusion Ring						
	Ram Line Seal						
	Trip Valve Gauge						
	Valve Support Wire						

Tick relevant boxes / delete where applicable

Signed:	Date:	Test witnessed by:

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Serial No:			C = Condemned
Plant No:			D = Replaced
			E = N.D.T.'D

[illegible]

Signed:

Date:

Test witnessed by:



6.2.18 Mobile (Access) Platforms

Check List

A. Dismantling (Manual Hydraulic)

- 1 Drain all hydraulic oil from system
- 2 Disconnect pipe work between pump and cylinder
- 3 Remove cylinder and examine for impact damage, leaking seals and wear on the two end connections
- 4 Remove pump and examine, checking for leaks at the piston seal and non-return valve
- 5 Examine the pipe work and couplings and check for impact damage and / or cuts which may affect the pressure capacity
- 6 Re-connect the hydraulic system (remote from the mobile platform), re-fill with hydraulic oil and bleed the system
- 7 Pump up the ram to its maximum stroke and check that the positive stop functions
- 8 Examine the ram checking for corrosion / scoring
- 9 Release hydraulic pressure and check that the ram returns smoothly
- 10 Should the cylinder or pump have to be stripped further, proceed as follows:

B. Dismantling (Cylinders / Rams)

- 1 Unscrew inlet coupling and remove
- 2 Disconnect ram return spring (if fitted)
- 3 Undo and remove gland neck ring (re-new "O" rings)
- 4 Withdraw ram c/w return spring and examine for corrosion, wear and scoring (check spring tension)
- 5 Remove packing locking ring and re-new ram packing
- 6 Examine internal bore of cylinder checking for corrosion, wear and scoring

Note: Larger cylinders are slightly easier to strip as they do not usually have a return spring connecting the ram to the cylinder. Access to the internals is usually obtained by removing the cylinder head or gland neck ring.

C. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Renew ALL software, i.e. "O" rings, seals and packing and replace any defective components
- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

D. Re-assembly

- 1 Re-assemble in the reverse order of this procedure adhering to any specific instructions of the manufacturer
- 2 When re-fitting the inlet (and outlet) coupling, ensure all old thread tape has been removed. Re-fit the coupling using thread tape to give a complete seal
- 3 Function test the cylinder, pumping up to full stroke
- 4 Proof load test to S.W.L. + 25% and sustain load for approximately 5-10 minutes. Check for any oil leaks or drop in pressure which will show on the test rig gauge
- 5 Re-examine unit to ensure proof load testing has not exposed any defects
- 6 If all satisfactory, set aside for final assembly

E. Dismantling (Pumps)

- 1 Remove filler plug and drain off hydraulic oil
- 2 Remove pivot pin fastener and press out pivot pin
- 3 Remove handle base c/w plunger pin



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- 4 Withdraw both high pressure and low-pressure plungers / pistons. Unscrew H.P. gland and replace packing
- 5 Unscrew delivery valve plug and remove delivery and suction balls. (Replace from standard repair kit)
- 6 Undo clamp nut, remove tank end assembly and withdraw oil tube / tank
- 7 This exposes the pressure relief valve - refer to manufacturers instructions for torque setting

F. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly (renew ALL software i.e. "O" rings, seals and packing and replace any defective components)
- 2 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

G. Re-assembly

- 1 Re-assemble in the reverse order of this procedure adhering to any specific instructions of the manufacturer
- 2 Refill with clean oil taking care not to over fill (check with dip stick)
- 3 Replace filler plug and function test for entrapped air (bleed as required)
- 4 If all satisfactory, set aside for final assembly
- 5 Should the hydraulic system be air driven, proceed as follows:

H. Dismantling (Pneumatic Rotary Vane Motor)

- 1 Remove air motor and strip down internal parts taking care not to use excessive force (i.e. only wood, rubber, copper or hide mallet to be used)
- 2 Wash down all internal parts and casing
- 3 Check rotor cylinder and blades for wear and scoring. Check condition of blade springs (if fitted)
- 4 Examine control valve making sure it functions correctly and is free from contamination
- 5 Check all "O" rings and seals for damage and ensure they are not perished
- 6 Examine end plates for flatness and wear
- 7 Check ball bearings for wear and smoothness of operation
- 8 Examine exhaust casing for distortion or cracking
- 9 Remove direction control lever and examine for wear and distortion

I. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Replace any defective components
- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

J. Re-assembly

- 1 Re-assemble in accordance with the manufacturers instructions lubricating the various components as required
- 2 Run lubricated air through the motor to ensure it runs true and free
- 3 If all satisfactory, set aside complete unit for final assembly

K. Dismantling (Main Steelwork)

- 1 Remove wheels and check axles for wear
- 2 Examine wheels and where fitted, check bearings for smoothness in operation
- 3 If wheels are fitted with pneumatic tires, check condition of tires and that the pressure conforms to the manufacturer's recommendations
- 4 Dismantle the outriggers, screw down jacks and foot pads and examine for wear / distortion



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- 5 Dismantle main frame work and examine all hinge pins checking for wear / bending
- 6 Examine main frame steelwork. (Depending on the age and condition of the unit, it may be prudent to shotblast and carry out N.D.T. especially around weldments)
- 7 Examine work platform and ensure handrails are secure
- 8 Ensure safety bar and chain is fitted

Note: Should any remedial work be required i.e. re-welding, replacing members etc., **QatarEnergy-IS** quality assurance department must be contacted to authorize material selection, weld procedures and relevant N.D.T.

L. General

- 1 Ensure all components are clean and free from contamination prior to re-assembly
- 2 Replace any defective components
- 3 Where doubt exists as to the integrity of any components, N.D.T. techniques should be employed to verify their status

M. Final Re-assembly

- 1 Re-assemble the main steel framework in the reverse order of part 11 of this procedure, lubricating the relevant moving parts, i.e. all hinge pins, screw down jack threads etc.
- 2 Re-fit hydraulic cylinders and pump unit
- 3 Where relevant, re-install air motor
- 4 With the outriggers in the operating position, subject the mobile access platform to S.W.L. + 25% proof load and operate the unit to the extremes of its range. (This will also verify stability). This test should be witnessed
- 5 Re-examine unit to ensure proof loading has not exposed any defects. (A post load test N.D.T. may also be advisable)
- 6 If all satisfactory, hard stamp unit with identification number and S.W.L. then pass to paint shop for coating
- 7 Return to engineer for final check and issue of identification plate

This procedure is to be followed as a guideline to **QatarEnergy-IS** minimum requirements for an overhaul.

The vendor should have the relevant manufacturer's manuals "in house" and they should be made available to the engineers carrying out the overhauls for further reference etc.



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Item	Description	Condition					Comment
	Hydraulics	A	B	C	D	E	
	Hydraulic Pipe work						
	Hydraulic Couplings						
	Cylinder						
	Ram						
	Gland Neck Ring						
	Ram Packing						
	Packing Locking Ring						
	Return Spring (if fitted)						
	Seals						
	"O" Rings						
	Inlet Coupling						
	Outlet Coupling (if fitted)						
	Pump Unit:						
	Handle & Handle Base						
	Pivot Pins						
	H.P. & L.P. Plungers						
	"O" Rings & Seals for above						
	H.P. Plunger						
	L.P. Relief Valve Plug						
	L.P. Relief Valve						
	Valve Spring						
	Release Valve Assembly						
	Relief Valve						

Tick relevant boxes / delete where applicable

Signed:	Date:	Test witnessed by:



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	Description	Condition					Comment
		A	B	C	D	E	
	Oil Tube / Tank						
	"O" Rings / Seals for above						
	Centre Stud c/w Clamp Nut						
	Tank End Assembly						
	Handle Locking Pin c/w Chain						
	Access & Delivery Suction Balls						
	Air Motor (If Fitted)						
	Motor Housing						
	Cylinder Sleeve						
	Rotor						
	Rotor Valve						
	Vane Springs (if fitted)						
	Bearing Disc (end plate) 1						
	Bearing Disc (end plate) 2						
	Bearing 1						
	Bearing 2						
	Cylinder Covers						
	Control Valve						
	Control Levers						
	Seals						
	Exhaust / Silencer						
	Air Inlet Coupling						

Tick relevant boxes / delete where applicable

Signed:	Date:	Test witnessed by:

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[illegible]

Signed:

Date:

Test witnessed by:



6.2.19 Wire Rope Slings

Check List

A. Examination Procedure

1 Examine each individual leg along its entire length and check for:

- a) Wear
- b) Corrosion
- c) Abrasion
- d) Mechanical Damage
- e) Broken Wires
- f) Internal Deterioration*

*Internal deterioration should be checked for by using two specially designed clamps to open up the rope to expose the wire / fibre core

- 2 Measure the rope diameter as any reduction from original size could indicate either amount of wear on outer wires, partial collapse of the core or stretch due to overload
- 3 Examine each ferrule and ensure the correct size of ferrule has been fitted
- 4 Check that the end of the loop does not terminate inside the ferrule (i.e. the rope end should protrude slightly but not more than 1/3rd of the dia.) unless the ferrule is of the longer tapered design which has an internal step
- 5 The ferrule should be free from cracks or other deformities
- 6 Examine each thimble and check for correct fitting, snagging damage and elongation. (Stretched thimbles / eyes could indicate possible overload)
- 7 Examine wire rope around thimbles as it is often abraded due to sling being dragged over rough surfaces
- 8 If fitted, examine master link / quadruple assembly and check for wear, corrosion and cracking
- 9 Any damaged components / rope should be replaced
- 10 Subject the sling to a S.W.L. + 100% proof load and re-examine to ensure testing has not exposed any defects
- 11 If all satisfactory, hard stamp identification number and S.W.L. (if not already on) and test date
- 12 Re-lubricate as required

Note: The original test certificate relating to the manufacturer's certification issued for component parts of multiple leg slings i.e. Masterlinks, Quadruple Assemblies and Shackles, **must** be retained by the owner / operator after any repairs and re-testing. The new load test certificate should be cross-referenced with the original certificate as it does **not** provide traceability for individual component parts remaining on that assembly.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an overhaul.



6.2.20 Chain Slings

Check List

A. Examination Procedure

- 1 Lay out chain sling on floor and remove all twists from legs
- 2 Match up legs to check for obvious signs of stretch
- 3 Visually examine the chain sling along its entire length and check for:
 - a) Distortion in chain links (nicks, bends, twists, elongation)
 - b) Wear between chain links
 - c) Wear between chain links and load pins
 - d) Heat damage
 - e) Chemical damage
 - f) Wear, stretch, twisting on end terminations
 - g) Operation of safety catches (where fitted)
 - h) Secure load pins
 - i) Correct identification number and S.W.L
- 4 Replace any damaged components. If chain is damaged replace complete leg or legs
- 5 Subject the chain sling to a S.W.L. + 100% proof load and re-examine to ensure testing has not exposed any defects
- 6 If all is satisfactory, hard stamp identification number and S.W.L. (if not already marked) and test date

Note: The original test certificate relating to the manufacturer's certification issued for component parts of multiple leg slings i.e. Masterlinks, Quadruple Assemblies and Shackles, **must** be retained by the owner / operator after any repairs and re-testing. The new load test certificate should be cross-referenced with the original certificate as it does **not** provide traceability for individual component parts remaining on that assembly.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an overhaul.

The vendor should have the relevant manufacturer's manuals "in house" and they should be made available to the engineers carrying out the overhauls for further references etc.



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6.2.21 Shackles

Check List

A. Examination Procedure

- 1 Remove shackle pin and examine for wear, deformation and cracking
- 2 Ensure it is the correct pin for the shackle
(i.e. **NOT** a higher tensile pin in an alloy shackle)
- 3 Check pin threads for wear / deformation
- 4 Examine shackle body and check for wear in the crown and pinholes, deformation and cracking
- 5 Subject shackle to a proof load test in accordance with the loads specified in the International Labor Office pamphlet 188 (See procedure 9 - Appendix 1) and re-examine to ensure proof loading has not exposed any defects
- 6 If all satisfactory, hard stamp with identification number and S.W.L. (if not already marked). Hard stamp only the side of the shackle and not the crown
- 7 In the case of safety pin shackles, ensure split pins are fitted

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an overhaul.

6.2.22 Eyebolts / Eyenuts

Check List

A. Examination Procedure

- 1 Clean up threads and wire brush if necessary
(if eyebolts / eyenuts are unused, remove protective tape / plug to allow inspection of threads)
- 2 Examine threads and check for wear, stretch or impact damage. The threads must be complete (no broken threads) and full (no flats on top)
(The thread should be concentric and fit neatly in a standard nut on a standard bolt. If stretch is suspected, a thread gauge should be used to confirm condition)
- 3 Examine eye of bolt / nut and check for wear / stretch / distortion and look for hairline cracks at the crown of the ring
- 4 Check squareness of shank / screwed hole against shoulder
- 5 If all satisfactory, subject to S.W.L. + 100% proof load and re-examine to ensure proof loading has not exposed any defects
- 6 Hard stamp identification number and S.W.L. around shoulder of eyebolt / nut

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an overhaul.



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6.2.23 Rigging Screws

Check List

A. Examination Procedure

1. Unscrew both ends of rigging screw to separate the three main components
2. Clean and examine the threads and check for wear, stretch or impact damage
3. Examine the eyes and hooks and check for wear / stretch
(Jaw type - Remove clevis pin and examine pin and pin holes for wear / stretch and distortion)
4. Re-grease the threads and re-assemble
5. Prior to load testing, ensure the male threads are fully through the female threads
6. Load test to a S.W.L. + 100% Proof Load and re-examine to ensure the testing has not exposed any defects
7. If all satisfactory, hard stamp with identification number and S.W.L.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an overhaul.

6.2.24 Open Wedge Sockets

Check List

A. Examination Procedure

1. Remove main load pin and wedge
2. Examine load pin and check for wear / distortion
3. Examine wedge and check for wear / distortion / cracking
(Ensure it is the correct wedge for the rope dia. marked on the socket)
4. Remove any coating on the socket to expose bare metal
5. Examine socket and check for wear / elongation of pin holes and cracking
6. If all satisfactory, re-assemble and subject the unit to a S.W.L. + 100% proof load*
7. Re-examine and carry out M.P.I. to ensure proof loading has not exposed any defects
8. Hard stamp identification number, S.W.L. and test date and pass to paint shop

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an overhaul.

* SWL will vary according to actual diameter and type of rope fitted. Efficiency of termination = 80%.



7.0 INSTRUCTIONS AND PROCEDURES FOR INSPECTIONS

7.1 Introduction

The main purpose of these inspection procedures is to indicate to prospective lifting equipment inspection companies, the minimum amount of examination required on any specific piece of equipment in order to qualify as an acceptable visual examination i.e. it is a guideline to **QatarEnergy-IS** minimum requirements. These procedures can be issued to the tendering companies at the bid stage of the contract to give an indication of the scope of work and can also be used as an audit document once the contract has been awarded and is in place.

The information contained in each lifting equipment type procedure is comprehensive so far as to provide sufficient information which will allow the competent third party inspector (and / or the company person responsible) to understand more readily the various technical aspects of the lifting appliance.

It is important to understand the difference between the requirement for testing and the requirement for periodic inspection / examination. Generally, the proof load test is valid for the lifetime of the equipment unless it is repaired, modified or otherwise altered, to such an extent as to affect its integrity. This being the case, the equipment would have to be retested.

The visual inspection for accessories i.e. slings, shackles, eyebolts etc. is valid for 6 months only, thereafter the equipment has to be re-inspected. The visual inspection on appliances is valid for a maximum of twelve months, **however, it is QatarEnergy-IS policy that this period is reduced to 6 months** except for certain items listed below:

- 1) Lifting support steel work with no moving parts such as runway beams, padeyes etc
- 2) Permanently attached or integral padeyes on installed plant and equipment
As long as the following criteria is met:
 - a) They are not used for supporting personnel
 - b) Their use is infrequent to the extent that wear is not an adverse factor
 - c) They are adequately coated / protected so that corrosion is not an adverse factor
- 3) Permanently installed hoists on runway beams

For further info. refer to Lifting Equipment Matrix contained in the written scheme of examination

Throughout these procedures there are itemised checklists designed as an aide memoir for examiners. The checklists should only be completed in the event of the equipment under examination being defective. This is to assist the onshore repairers in identifying the faults and carrying out remedial work.

In an effort to improve levels of competence and safety standards, it is the intention of the company to continually update existing information and, when necessary, introduce new guidelines for use within operations. To this end, the content of this manual will be reviewed annually to ensure that it reflects current information.



7.2 Portable / Fixed Lifting Equipment and Gear Thorough Examination & Testing Procedure

7.2.1 Pedestal / Mobile Cranes

The legal responsibility for ensuring that Pedestal / Mobile Cranes are properly designed, installed and maintained rests upon either the operator or the owner of the installation on which the appliances are located.

It is important that the competent person engaged on this work should be familiar with the contents of the standards and be able to apply his knowledge.

The Pedestal Crane must be load tested before being put into use after installation, or after substantial alteration / repair and subject thereto, at regular intervals (12 months maximum*) for thorough examination.

The purpose of thorough examination and testing is to determine whether the condition of the crane after installation or repairs is such that it can continue to be used safely.

* It is **QatarEnergy-IS** policy that this period between examinations will remain at 6 months.

Introduction

This procedure has been produced for cranes situated on **QatarEnergy-IS** offshore & onshore worksites.

This procedure is to be used for guidance by the testing company when preparing the crane for examination and / or proof load test. It should also be referred to in the following circumstances:

1. Initial testing after installation / commissioning of crane
2. Statutory quadrennial testing
3. Testing after completion of any major boom repair or replacement of major components
4. Retesting after an existing and certified boom arrangement is either shortened or lengthened

i.e. the crane must be re-tested and certified prior to use if:

- a) A boom is shortened (i.e. a section removed) on a tested and certified crane boom
- b) A tested and certified crane requires a new, repaired or reconditioned section fitted to the existing boom
- c) The crane undergoes any substantial structural or mechanical repair or modification

Note: If the boom referred to in (a) is later returned to its original length it is **recommended** that a maximum proof load test be carried out to ensure that it has been refitted correctly.

5. At the request of the certifying authority, where there is clear justification
This procedure is to be followed as a guideline to **QatarEnergy-IS** minimum requirements for an examination and / or proof load test.

Preparation of Equipment

- 1) Ensure that test equipment is available on the installation / at the worksite prior to commencement of testing, and is placed in a convenient "pick up" position within the crane's working radius
- 2) Ensure that each test weight is clearly identified and the exact weight is marked thereon
- 3) Ensure each test weight is fitted with a suitable certified lifting set secured in such a manner that it cannot be inadvertently dislodged



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- 4) Ensure the lifting cradle complete with slings and shackles has current certification and is capable of carrying the total test load
- 5) If water bags are being used, ensure they are free from any defects and a suitable water supply is available

Note: Special permission is required to connect into the firewater mains

- 6) The certified and calibrated test clock/dynamometer or load cell to be used in conjunction with the water bags should be stored safely until the test commences
- 7) Where possible ensure any obstructions on the deck of the platform / ground which are within the working radius of the crane (during testing) are cleared

Preparation of crane

Prior to any testing, a thorough visual examination of the crane must be carried out. This and all subsequent actions required for the test must be performed by a test engineer who is:

1. Trained, qualified and fully experienced with the type of crane to be tested and
2. Fully conversant with all related operating requirements and safety rules / regulations

Note: As part of the visual examination all clutch and brake linings must be examined for signs of wear and checked for ingress of oil or grease on the linings and drums.

Wire ropes are to be inspected in accordance with manufacturer's laid down specification and procedures.

In the case of mobile cranes, ensure they are set up and correctly on hard, flat ground, clear of any subterranean services which may collapse under load and clear of any overhead power cables.

The out riggers must be fully extended, locked and pinned in position using mud mats / spreaders as required. The crane must be truly levelled using gauges on crane's chassis.

Function Tests

A thorough function test of the crane must be carried out **without load** and any defects or adjustments corrected in accordance with the manufacturer's recommendations. When all defects have been rectified proceed with the safe working load and proof load tests.

Function Test as follows:

Boom Hoist System

Raising the Boom - prior to raising the boom where applicable, **engage** the pawl.

Raise the boom and test the efficiency of both hoisting mechanism, brake and pawl.

Lowering The Boom - Prior to lowering the boom where applicable, **disengage** the pawl. Lower the boom and test the efficiency of lowering mechanism and brake.

Ensure the boom angle indicator is functioning correctly.

Extend and retract telescopic booms if applicable.

Check maximum and minimum radius limits are functioning and are correctly set up.

Main Hoist Line

Raise and test the efficiency of hoisting mechanism and brake.

Lower and test the efficiency of hoisting / lowering mechanism and brake.

Check overhoist limit.

Check lower limit. (Use auto-spooler).

Whip Line (if fitted)

Raise and test the efficiency of hoisting mechanism and brake.

Lower and test the efficiency of hoisting/lowering mechanism and brake.

Check overhoist limit.



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Check lower limit. (Use auto-spooler).

Slewing

Slew left and test the efficiency of the brake (and clutch where applicable).

Slew right and test the efficiency of the brake (and clutch where applicable).

If fitted, ensure any slew limits function correctly.

Testing Procedures

Safe Working Load Test

- 1) Position and stack test weights at appropriate 'measured' radius. (All radius measurements must be taken from the centre of the pedestal to the hoist line)
- 2) Raise the appropriate test load weights just clear of the deck, operate the crane in all modes and slew (as far as possible) through 360°
- 3) Check the accuracy of the safe load and radius indicators

Proof Load Test

The proof loads applied will vary with the crane radius but shall not exceed the percentage overload as indicated below.

- | | |
|-------------------------------------|---------------------|
| a) Up to 20 tonnes: | Capacity + 25% |
| b) Between 20 tonnes and 50 tonnes: | Capacity + 5 tonnes |
| c) Above 50 tonnes: | Capacity + 10% |

The proof load is to be applied at minimum radius, maximum radius and at **least** one radius in between. The proof load shall include the weight of the test cradle and rigging

- 4) Raise the appropriate proof load test weights just clear of the deck, slew the crane (as far as possible) through 360°
- 5) At the commencement of overload testing, correct function of the safe load alarm systems should be verified
- 6) Automatic safe load indicators may then be disconnected for the duration of the overload tests but they must be re-connected immediately afterwards
- 7) After re-connection, the safe load indicators must be tested to establish that they are functioning

Warning:

Operators are reminded that it is absolutely forbidden for any alarm, warning or safety device to be disconnected unless specifically authorised by a suitably qualified person and only then under strictly controlled conditions.

Action Upon Completion of Tests

- 1) Carry out thorough examination of the crane to establish that all parts have withstood the test programme without defect
- 2) All test weights, waterbags, test clocks, loadcells and associated equipment must be returned to a suitable storage position until it can be back loaded
- 3) The Company must be informed immediately of any defects found which would prevent the crane becoming certified
- 4) A copy of the certification and / or report form is to be left onboard the installation and official typed copy is to be sent to the relevant onshore department
- 5) Reconnect any alarms / warning devices which were disconnected during proof load testing



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Crane Inspection Form & Check List		
1	Owner	
2	Location	
3	Crane Model	
4	Inspection Engineer	
5	Date of Inspection	
6	Date of Test	
7	Current Cert No.	
8	Name & Address of Maker of Crane	
9	Type of Crane and Nature of Power	
10	Date of Manufacture of Crane	
11	Identification	
	a) Makers Serial Number	
	b) Owners Distinguishing Mark or Number	
12	Make & Type of Automatic Safe Load Indicator	
13	Make and Type of Derricking Interlock	
14	Date of Last Previous Test of Crane	
15	Date of Last Previous Thorough Examination of Crane	
16	In the case of a crane with a derricking jib or jibs, the maximum radius and load at which the jib may be worked	

The following checklist must be completed before and after load test and at the time of any Periodic Visual Examination:



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Please INITIAL the applicable columns to the crane under test (Do not tick)

Components	Satisfactory	Requires Attention	W.O. Number	Requires Renewal
Boom / Structure				
Telescopic Boom				
Hydraulic Rams				
Tires (condition / pressure)				
Outriggers				
Chassis				
Operating cab				
Head Section				
Root Section				
Intermediate Sections				
Joining Pins Or Bolts				
Foot Pins & Bushes				
Sheaves And Pins				
Keep Plates				
Whip line Extension				
Structure & Fittings				
10% UT On Sheaves				
10% UT On Chords				
10% MPI On Boom Foot Pins				
10% MPI On Boom Foot Clevises				
G.O.P. Fittings				
Boom Hoist System				
Boom Hoist Rope				
Rope Termination & Anchor				
Bridle To Include Sheaves				
Pennants & Pins				
'A' Frame Sheaves				
'A' Frame				
Max & Min Limit Switches				
Pawl				
Brakes & Clutch				
Control Condition				



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Please INITIAL the appropriate column (do not tick)

Components	Satisfactory	Requires Attention	W.O. Number	Requires Renewal
Hydraulics				
Motors				
Pump				
Tank And Oil				
Hydraulic Cylinders				
Power Unit				
Engine				
Mountings				
Couplings				
Splitter Gear Box				
Slew Unit				
Ring And Pinion				
Control System				
Clutch (If Fitted)				
Brake				
Slew Ring Bolts				
Slew Limit Switches				
Main Hoist				
Rope And Fittings				
Control System				
Clutches				
Brakes				
Bottom Block				
Hook				
Overhoist Limit Switch				
Lower Limit				



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Please INITIAL the appropriate column (do not tick)

Components	Satisfactory	Requires Attention	W.O. Number	Requires Renewal
Aux. Hoist				
Rope And Fittings				
Control Gear				
Clutches				
Brakes				
Hook				
Overhoist Limit				
Lower Limit				
Safe Load Indicator				
Load Indicator				
Radius Indicator				
Audible Warnings				
Visible Warnings				
Winches:				
1) Boom Hoist				
Hold Down Bolts				
Spooling				
Rope Termination				
Boom Pawl				
2) Main Hoist				
Hold Down Bolts				
Spooling				
Rope Termination				
3) Whip Hoist				
Hold Down Bolts				
Spooling				
Rope Termination				
Ancillary Equip.				
Helicopter Warning				
Boom Lights				
'A' Frame Light				



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Auxiliary Equipment**Condition of Crane****Remarks**

Safe Working Load or Loads	(i) Length of Jib	(ii) Radius	(iii) Test Load	(iv) SWL
In the case of a crane with a variable operating radius (incl. a crane with a derricking jib or with interchangeable jibs of different lengths) the safe working load at various radii of the jib, jibs trolley or cab must be given. Test loads at various radii should be given in column (iii) and in the case of a safe working load which has been calculated without the application of a test load "NIL" should be entered in that column.				

Inspection carried out by:**Print Name:****Signature:****Date:**



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Testing Company: _____

Certificate of Test and Thorough Examination of Crane

1.Name and address of owner of crane				
2.Name and address of maker of crane				
3.Type of crane and nature of power (e.g. Scotch derrick - manual, Rail mounted tower - electric)				
4.Date of manufacture of crane				
5.Identification No				
a) Makers Serial No				
b) Owners distinguishing mark or No.				
6.Make and type of automatic safe load indicator, if required (See Regulation 30)				
7.Make and type of derricking interlock, if required (See Regulation 22)				
8.Date of last previous test of crane				
9.Date of last previous thorough examination of crane				
	1) Length of Jib	2) Radius	3) Test Load	4) SWL
10.Safe Working Load or loads in the case of a crane with variable operating radius (including a crane with a derricking jib or with inter-changeable jibs of different lengths) the safe working load at various radii of the jib, trolley or crab must be given in column (3) and in the case of a safe working load, which has been calculated without the application of a test load "NIL" should be entered in that column				
11.In the case of a crane with a derricking jib or jibs the maximum radius at which the jib or jibs may be worked				
12.Defects noted and alterations or repairs required before crane is put into service. (if none enter "NONE")				

I hereby certify that on (date) the crane described in this certificate was tested and thoroughly examined and the above particulars are correct.

Signature:

Position:

Date of Certificate:

Cert. No.:



7.2.2 Overhead Travelling Cranes

The legal responsibility for ensuring that overhead travelling cranes are properly designed and installed rests upon either the operator or the owner of the worksite at which the appliances are located.

It is important that the competent person engaged on this work should be familiar with the contents of the Standards and be able to apply his knowledge.

The overhead travelling crane must be load tested before being put into use after installation, or after substantial alteration / repair and subject thereto, at regular intervals (12 months maximum*) for thorough examination.

The purpose of thorough examination and testing is to determine whether the condition of the overhead travelling crane after installation or repairs is such that it can continue to be used safely.

* It is **QatarEnergy-IS** policy that this period between examinations will remain at 6 months.

Pre-examination Procedure

Prior to any maintenance or inspection the engineer should satisfy himself with the following:

1. Has the owner's representative been advised of your presence on site?
2. Is the owners representative satisfied with the performance of equipment?
3. Does the owner's representative have any complaints regarding the equipment?
4. Are all lubrication points adequately lubricated and clean?
5. Are all oil reservoirs clean and filled to correct level?
6. Is equipment functioning as designed?
7. Is equipment adequately clean and work areas adequately illuminated?
8. Do all safety features operate satisfactorily?
9. Are all parts of equipment correctly adjusted and operational?
10. Is equipment safe in operation?
11. Is the crane isolated and have the fuses been removed?
12. Are "under repair" signs displayed?
13. Are all tools being used secure from falling off apparatus?

Note: Great care must be taken whilst manoeuvring on overhead beams.

Functional Controls Examination

1. Check isolating switch for damage (i.e.) handle or cover broken, making it possible to expose live parts
2. In the case of a pendant operated overhead travelling crane, the rope, cable and control panel must be in good condition with good mobility in all directions of travel
3. All directional displays on control panel must be legible

Crab Examination

1) Structure

Check frame for loose rivets, bolts, broken welds and damage.

2) Crab Hoist Motor

- a) Examine motor frame and fixings for possible defects



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- b) Check the slip rings, brushes and brush holders for wear and ensure that they are secure with bearings in good condition and sufficiently lubricated

3) Crab Coupling Bushes

- a) Ensure that coupling is correctly aligned with bolts and bushes in good condition and keys tight

4) Crab Electro-Magnetic Brake

- a) Examine coils and armature. If wear has taken place on the armature and laminations, it is important to ascertain that the air gap has not been reduced to less than 0.005" or in the case of the plunger type solenoid that no distortion of the plunger or coil formers has taken place
- b) Check that all fixing bolts are secure, that the spring is correctly tensioned and the brake properly adjusted, shoes correctly balanced and linings not unduly. All pivots should be free from excessive wear and properly lubricated

5) Crab Gear Box

- a) Examine for wear or damage. Where the gears are in an enclosed gearbox and an unusual noise is being emitted indicating a possible defect then the box should be opened up for an in-depth examination
- b) Check oil level

6) Crab Bearings

- a) Check for wear, damage and proper lubrication

7) Crab Drum

- a) Examine for wear, damage or cracks with particular attention to the grooves. If the tips of the grooves have been damaged but can be considered safe for use the broken surfaces should be dressed to remove any sharp edges

8) Crab Limit Switch

- a) Remove the cover to ensure that the contacts are free from excessive wear, burning or pitting
- b) Lubricate all moving parts and check they are free from undue wear
- c) Levers and return springs should be in good condition

9) Crab Load rope

- a) Check for wear or damage. Ropes should be considered unusable when broken wires exceed 10% over a length of eight times the diameter of the rope, if the reduction in diameter of the outside wires is 50% over the whole of the working length or any sign of necking indicating internal damage. Ensure adequate lubrication

10) Crab Bottom Block

- a) Examine for wear in the grooves, bore of the sheave and on the crosshead. Ensure that these parts and the guard plates are free from damage or defects
- b) Examine for wear in the saddle of the hook, this should not exceed 10%. Check throat opening against standard dimension to ensure hook has not stretched or deformed
- c) The top nut should be securely locked



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Cross Travel Examination

1) Cross Travel Motor

- a) Examine the frame and fixings for any defects
- b) Check the slip rings, brushes and holders for wear and ensure that they are secure. Bearings should be in good condition and properly lubricated

2) Cross Travel Coupling bushes

- a) Ensure alignment is correct, all bolts and bushes are in good condition and keys tight

3) Cross Travel Electro-magnetic Brake

- a) Check frame, fixings, solenoid and shoes for excessive wear
- b) Adjustment should give smooth stopping. Ensure all moving parts are adequately lubricated

4) Cross Travel Gearbox

- a) Examine gears for excessive wear or damage. Gearbox linkage should not have too much play. Check oil level

5) Cross Travel Bearings and Runners

- a) Check bearings for wear, damage and proper lubrication
- b) Check flanges of runners for wear

6) Cross Travel Hand Chain Wheel and Hand Chain (if fitted)

- a) If worn and require replacement, give particulars which will help to identify i.e. for:
 - Wheel Pattern No
 - Wheel Diameter
 - Chain Size
 - Length

7) Cross Travel Collectors

- a) Check that they are tight
- b) Check for wear

Bridge Long Travel Examination

1) Bridge Long Travel Motor

- a) Examine frame and fixings for possible defects
- b) Check the slip rings, brushes and holders for wear and ensure that they are secure. Bearings should be in good condition and properly lubricated

2) Bridge Coupling Bushes

- a) Ensure correct alignment, all bolts and bushes are in good condition and keys tight

3) Bridge Electro-Magnetic Brake

- a) Check all parts for excessive wear. Adjustment should give smooth stopping. Ensure all moving parts are adequately lubricated

4) Bridge Gears

- a) Check for wear or damage. Check oil level

5) Bridge Bearings

- a) Check for wear, damage and lubrication



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6) Bridge Tram Wheels and Axles

- a) Check for wear on tread and flanges; if evident, state degree of wear and whether on inside or outside flange. Note degree of wear on axles and bore of wheel and ensure that they are adequately lubricated

7) Bridge Platform etc. (if fitted)

- a) Examine for defective welds, loose bolts and rivets and damaged structure. Make certain that the platform and handrail is secure. Check the cross-wires for wear or damage and the strainer bolts and brackets for possible defects

Load Testing

1) Application of Loads

- a) A test load equivalent to the maximum Safe Working Load shall be raised and traversed along the whole operational length of the long and cross travel. The load shall be halted during its passage along the system and kept at rest at suitable positions, in order that stable deflection readings may be taken and recorded. The deflections are to be taken at the weakest points i.e. mid-span / mid support and should not exceed one seven hundred and fiftieth of the total span
- b) The above procedure shall then be repeated except that a proof load shall be applied which will be equal to SWL plus 25%

2) Inspection and Thorough Examination

- a) During the application of the test load (SWL) and the proof load as specified, the overhead travelling crane shall be kept under such observation as will ensure the ready detection of any obvious defect in the appliance
- b) Following the application of the proof load the crane shall be thoroughly visually examined in order to ascertain that it has withstood the proof load without injury or permanent deformation

3) Certification

- a) It is the duty of the certifier to satisfy himself that the maximum allowable stresses and calculated deflections specified in BS 466/5744 are not exceeded and that all other requirements of this standard are complied with before issuing the relevant certificate

4) Accuracy of Weights

- a) Where practical the load used shall be made up of certified weights such that the sum of inaccuracies of the load do not exceed plus or minus 4%

5) Markings

- a) The safe working load, identification number and any limiting conditions shall be plainly and permanently marked on the overhead travelling crane so as to be clearly visible to the operator and identifiable against its certification

If all satisfactory, record relevant details and issue report and / or certificate.

This procedure is to be followed as a guideline to **QatarEnergy-IS** minimum requirements for an inspection and / or proof load test.

Upon Completion



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Make everything safe and tidy (remembering any loose parts that may be lying around, particularly if you have been working overhead); check all tools are cleared from site and hand over completed crane with report to owner's representative.

Overhead Travelling Crane - Check List

1. Owner:
2. Location:
3. Crane Model:
4. Inspection Engineer:
5. Date of Inspection:
6. Date of Test:
7. Current Cert. No:
8. Name and Address of Maker of Crane:
9. Type of Crane and Nature of Power:
10. Date of Manufacture of Crane:
11. Identification:
12. Maker's Serial Number:
13. Owner's Distinguishing Mark Or Number:
14. Date of Last Previous Test of Crane:
15. Date of Last Previous Thorough Examination of Crane:



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Components	Satisfactory	Requires Attention	Requires Renewal
Pendant Control			
Casing			
Contacts			
Adjustment			
Lubrication			
Push Button Control			
Casing			
Coils			
Contacts			
Buttons			
Lubrication			
Structure			
Fastenings			
Welds			
Crab Unit			
Hoist Motor			
Frame			
Fastenings			
Motor			
Bearings			
Slip Rings			
Brushes			
Lubrication			



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Components	Satisfactory	Requires Attention	Requires Renewal
Coupling Bushes			
Alignment			
Fastenings			
Bushes			
Keys			
Elec. Magnetic Brake			
Coils			
Armature			
Laminations			
Solenoid			
Fastenings			
Springs			
Shoes			
Linings			
Pivots			
Lubrication			
Gear Box			
Gears			
Lubrication			
Crab Bearings			
Rope Drum			
Limit Switches			
Adjustment			
Contacts			
Lubrication			



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Components	Satisfactory	Requires Attention	Requires Renewal
Load Rope			
Bottom Block			
Cross Travel Motor			
Frame			
Fastenings			
Motor			
Bearings			
Slip Rings			
Brushes			
Lubrication			
Coupling Bushes			
Alignment			
Fastenings			
Bushes			
Keys			
Elec. Magnetic Brake			
Coils			
Armature			
Laminations			
Solenoid			
Fastenings			
Springs			
Shoes			
Linings			
Pivots			



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Components	Satisfactory	Requires Attention	Requires Renewal
Elec. Magnetic Brake			
Lubrication			
Gear Box			
Gears			
Lubrication			
Cross travel Bearings			
Hand Chain			
Hand Chain Wheel			
Chain Guides			
Bridge Long Travel			
Motor			
Frame			
Fastenings			
Bearings			
Slip Rings			
Brushes			
Lubrication			
Coupling Bushes			
Alignment			
Fastenings			



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Components	Satisfactory	Requires Attention	Requires Renewal
Coupling Bushes			
Bushes			
Keys			
Electro Magnetic Brake			
Coils			
Armature			
Laminations			
Solenoid			
Fastenings			
Springs			
Shoes			
Linings			
Pivots			
Lubrication			
Gear Box			
Gears			
Lubrication			
Long Travel Bearings			
Tram Wheels			
Axles			
Platform			
Fastenings			
Welds			
Handrails			



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7.2.3 Forklifts

The legal responsibility for ensuring that forklift trucks and their accessories are properly maintained and certified rests upon the owner of the equipment but the operator of the worksite also has a duty to ensure that the forklift trucks are suitable for working under conditions which may be specific to that site.

It is important that the competent person engaged on this work should be familiar with the contents of the Standards and be able to apply his knowledge.

The forklift must be load tested before being put into use or after substantial alteration / repair and subject thereto, at regular intervals (12 months maximum) for thorough examination.

The purpose of thorough examination and testing is to determine whether the condition of the appliance is such that it can continue to be used safely.

Visual Examination

During the thorough examination, the following points should be addressed:

Component	Check for:
Smoke Emission	Density, Sparks
Wheels & Hubs	Fracture, Security, Distortion
Tires	Damage, Tread, Walls
Bodywork	Security, Fouling
Cab Load Guard	Locking Devices, Security
Driver's Seat	Adjustment, Condition
Windscreen	Visibility, Cracks
Wipers	Function
Horn	Function
Driving Controls	Completeness, Condition, Position, Obstruction
Play at Steering Wheel	Not more than 1/5 of diameter
Steering Column	End Float, Side play, Flex coupling
Mech. Brake Hand Lever	Condition, Travel, Obstruction, Hold on
Service Brake Operation	Leaks, Servo, Operation
Parking Brake	Quality Of Drums, Performance, Pressure
Electrical Wiring	Insulation, Battery, Security, Chaffing
Engine Mountings	Condition
Oil Leaks	Quantity, Severity
Fuel Tank System	Condition, Leakage, Filling Cap
Stub Axle	Wear, Condition
Wheel Bearings	Wear, Condition
Steering Linkage	Alignment, Fractures, Locks, Wear
Steering Box	Wear, Sector Shaft, Security
Power Steering	Function, Leakage, Fouling
Transmission	Bearing Wear, Mountings
Mechanical Brake	Cables, Movement, Adjustment Components
Lifting chains	Wear, Anchor Pins, Split pins etc.
Mast Bearings	Bearings (Roller Side)
Rams	Leaks etc.
Bushes	Tilt and Pivot
Forks	Condition, Parallel, Wear, Cracks
Extension	Condition, Parallel, Wear
Carriage	Locating Bar, Pins, Bearings etc.

Note: Forklifts powered by diesel engines must as a minimum, meet with the EEMUA recommendations for protection of diesel engines operating in hazardous areas. (Publication No. 107).



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Load Testing

- 1) Stability load tests are to be carried out in accordance with the manufacturer's ratings + 10% overload, using known certified weights placed securely between forks. The appliance should be on a level normal operating surface and manoeuvred throughout its working ability
- 2) After load testing, re-examine forklift to ensure that load testing has not exposed any defects
- 3) Every 6 months the mast hoist chains should be removed and carefully inspected for broken or cracked links, links with worn pins and elongation. The chain should be carefully measured and if there is over 2% extension the chain should be replaced. On replacement of the chains the fork lift should be proof load tested to SWL + 25% in the stationary mode with the load being lifted just clear of the ground
- 4) The forks should be MPI'd every 6 months to check for cracking.

If all satisfactory, record relevant details and issue report and/or certificate.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection and / or proof load test.

Forklift - Check List

1. Owner:
2. Location:
3. Model:
4. Inspection Engineer:
5. Date of Inspection:
6. Date of Test:
7. Current Cert. No:
8. Name and Address of Maker:
9. Type of Forklift and Nature of Power:
10. Date of Manufacture:
11. Identification:
 - a) Maker's Serial Number:
 - b) Owner's Distinguishing Mark or Number:
12. Make and Type of Automatic Safe Load Indicator:



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Components	Satisfactory	Requires Attention	Requires Renewal
Smoke Emission			
Wheels & Hubs			
Tires			
Bodywork			
Cab Load Guard			
Driver's Seat			
Windscreen			
Wipers			
Horn			
Driving Controls			
Steering Wheel			
Steering Column			
Mech. Hand Brake			
Service Brake			
Parking Brake			
Electrical Wiring			
Engine Mountings			
Oil Leaks			
Fuel Tank System			
Stub Axle			
Wheel Bearings			
Steering Linkage			
Steering Box			
Power Steering			
Transmission			
Lifting Chains			
Mast Bearings			
Rams			
Bushes			
Forks			
Extensions			
Carriage			



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7.2.4 Pallet Stackers (Also Applicable to Switchgear Lifters)

Check List

Examination Procedure

- 1) Examine all hydraulic pipe work and check for impact damage or cuts etc. which may affect the pressure capacity
- 2) Examine all couplings and check for oil leaks
- 3) Examine hydraulic pump, check oil level and operate to check function of non-return valve
- 4) Examine pump piston and check for leaks at the seal
- 5) Examine hydraulic cylinder and check for impact damage
- 6) Examine wheels / rollers / castors and ensure they run smoothly
- 7) Function test castor foot brakes
- 8) Pump up ram to its full stroke and inspect ram checking for corrosion and scoring
- 9) Examine wire / chain and check operation of sheave

Note: Should the stacker be purely manual i.e. no hydraulics, it will be fitted with a winch. This being the case, proceed as follows:

Direct Drive / Worm Geared Winches

- 1) Examine winding handle and check for distortion
- 2) Examine winch drum and check for distortion, wear or cracking (if accessible, examine rope anchorage and ensure it is secure)
- 3) Where applicable, examine ratchet or worm gear (mounted on side of drum) and check for wear / broken teeth
- 4) Examine pawl and pawl spring / worm and check for wear / distortion. Ensure the pawl functions correctly / ensure the worm meshes correctly with the worm wheel
- 5) Examine winch frame and check for distortion, wear and cracking
- 6) Ensure the hold down bolts / welding / clamping is adequate and free from corrosion/distortion

Additional (Spur Geared Winches)

- 1) Visually examine the brake assembly and operate the winch handle to ensure it functions correctly
- 2) Examine all spur gears checking for alignment, wear on gear teeth, truth of gear shafts and smooth running of bearings (where fitted)
- 3) Examine forks / drum forks / boom or specialised lifting brackets (as per switchgear type stacker) and check for wear / distortion
- 4) Examine fork support frame and ensure the mast guide rollers are free to rotate
- 5) Examine mast and base steelwork and check for impact damage / distortion
- 6) Function test complete unit through its complete range

If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection.



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Plant No:			Conditions
Serial No:			A = Satisfactory
Model:			B = Worn but serviceable
SWL:			C = Condemned

Item	Description	Condition		
	Hydraulic Pipe work			
	Couplings			
	Hydraulic Pump			
	Hydraulic Cylinder			
Winch Unit				
	Handle			
	Drum			
	Ratchet Wheel / Worm Wheel			
	Ratchet Pawl/Worm			
	Ratchet Pawl Spring			
	Winch Frame			
	Friction Discs			
	Spur Gears			
	Spur Gear Shafts			
	Bearings			
	Hold Down Bolts / Clamps / Welding			
Main Framework				
	Rope / Chain Sheave			
	Forks / Drum Forks / Boom			

Tick relevant boxes / delete where applicable

Signed:	Date:	Engineer:

Satisfactory	Return for repair



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Serial No:			A = Satisfactory
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SWL:			C = Condemned

Item	Description	Condition		
	Specialized Lifting Brackets			
	Fork Support Frame			
	Guide Rollers			
	Wheels / wheel bearings			
	Castors			
	Castor Brakes			
	Rollers			
	Base Frame			
	Mast / Guide Rails			
	Wire Rope			
	Wire Rope Anchors			
	Roller Chain			
	Roller Chain Anchors			

Tick relevant boxes / delete where applicable

Signed:	Date:	Engineer:

Satisfactory	Return for repair



7.2.5 Pallet Trucks

Check List

Examination Procedure

- 1) Examine general condition of body, making sure it is free from contamination and that it is not bent or twisted in any way as to hamper the safe performance of the machine
 - 2) Check all wheels and bearings for smoothness of operation
 - 3) Raise truck and examine cylinder. There should be a thin film of oil in order to lubricate the piston
 - 4) If leaking occurs a new repair set for the pump piston and the lift piston should be fitted
 - 5) Check function of control lever and raise and lower the toes
- If all satisfactory, record relevant details and issue report.
This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection.



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Plant No:			Conditions
Serial No:			A = Satisfactory
Model:			B = Worn but serviceable
SWL:			C = Condemned

Item	Description	Condition		
	Wheels			
	Structure			
	Pump Piston			
	Lift Piston			
	Control Lever			
	Pump Arm			

Tick relevant boxes / delete where applicable

Signed:	Date:	Engineer:

Satisfactory	Return for repair



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7.2.6 Suspended Cradles

“AS USED FOR THE SUPPORT AND TRANSPORTATION OF PERSONNEL”(ALSO APPLICABLE TO FABRICATED MAN-RIDING CRADLES / BASKETS)

Whether the equipment being examined is of the modular type (as used by scaffolders / painters) or of the fabricated rigid design (man-riding cradles / baskets), the following criteria must be met and checked during the statutory examinations (maximum intervals = 6 months).

BS 2830 - design criteria and check list

- 1) The clear working space (within the toe boards) shall not be less than 440mm in width or greater than 3200mm in length
- 2) Toe boards shall extend around the perimeter of the platform and shall not be less than 155mm in height above the floor
- 3) Rigid guardrails shall be fixed to the stirrups and so locked as to prevent removal or distortion under working conditions
- 4) The underside of the guardrail shall be at a height of 920mm to 1150mm from floor level. (Where the guardrail interferes with working, this height may be reduced to not less than 690mm on the working face)
- 5) The vertical distance between the top of the toe board and the underside of the guard rail shall not exceed 760mm (unless the cradle is of the permanently fabricated design with expandamet / mesh sides)
- 6) The end frames of the cradle (stirrups) shall be so constructed that the suspension rope, bottom pulley block, winch or similar devices can be securely fastened to it at the suspension point
- 7) The suspension points shall be centrally situated in the width of the platform and at a height above the floor level of not less than 1.67 times the clear internal width of the cradle: (in the case of fabricated cradles / baskets with a protection roof, there may be four suspension points, situated one on each corner)
- 8) The safe working distributed load of cradles of 2M clear working length or less shall be 225KG. For cradles with a clear working length between 2M and 3.2M, the safe working distributed load shall be 295KG (with fabricated baskets, the SWL may well be higher depending on construction materials)
- 9) Except for necessary drainage, the platform shall be closely boarded, planked or plated and in the case of timber platform used as a structural member, shall overhang the end frames (stirrups) by more than 100mm but not more than 200mm
- 10) Any gaps in the platform shall not exceed 6mm in width
- 11) The floor should be of sound material and adequately supported
- 12) The floor surface shall be as non-slip as is practicable
- 13) Where applicable, suitable provision shall be made in the design of the cradle for anchorage of safety belts / harnesses etc.
- 14) In the case of fabricated cradles / baskets, they shall be fitted with an internal handrail and access facility (i.e. integral ladder/hinged gate etc.).
- 15) Where a hinged gate is fitted it should be so designed as to open inwards and automatically lock in the closed position
- 16) The locking device will be of such a design that it cannot be disengaged accidentally
- 17) All ancillary equipment i.e. pulleys, sheaves, suspension ropes, slings etc., should be inspected in accordance with the relevant examination procedure
- 18) Where possible, where winch units are fitted, they should be function tested



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If all satisfactory, record relevant details and issue report.

Load Testing

The cradle should be subjected to an evenly distributed proof load of S.W.L. + 25%. On removal of the load, the cradle should be free from permanent deformation, cracks, flaws and other defects.

1) Impact tests

The design of the cradle shall be such that when the following weights are dropped on any part of the platform, it shall not fracture:

- a) A 4.5 kg weight dropped on to an area 25mm square from a height of 1.2M
- b) A 25 kg weight of sand in a bag dropped from a height of 1.2M

2) Guard rail tests

No guard rail shall fracture or show any permanent deformation when a 50 kg weight is hung on its centre point and the platform of the cradle is tilted at an angle of 30° from the horizontal.

Note: Fabricated cradles may be subjected to a higher proof load test depending on construction.

If all satisfactory, record relevant details and issue test certificate.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection / proof load test.



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Model:			B = Worn but serviceable
SWL:			C = Condemned

Item	Description	Condition		
		A	B	C
	Clear working Space			
	Toe boards			
	Guard Rails			
	Height of Guard Rails			
	Distance between Rail & Floor			
	End Frames/Stirrups			
	Suspension Points / Padeyes			
	Load Rating			
	Floor Construction			
	Gaps / Drainage			
	Floor Condition			
	Non-slip Surface			
	Safety Belt Anchorages			
	Internal Hand Rails			
	Integral Ladder			
	Hinged Gate			
	Locking Devices			
	Winch Units			
	Suspension Ropes			
	Lifting Bridle & Shackles			
	Function Test			

Tick relevant boxes / delete where applicable

Signed:	Date:	Engineer:

Satisfactory	Return for repair



7.2.7 Electrically Powered Chain or Wire Rope Hoists

The legal responsibility for ensuring that powered hoists are properly designed and installed rests upon either the operator or the owner of the worksite on which the appliances are located.

It is important that the competent person engaged on this work should be familiar with the contents of the Standards and be able to apply his knowledge.

The hoist must be load tested before being put into use after installation, or after substantial alteration / repair and subject thereto, at regular intervals (12 months maximum) for thorough examination.

The purpose of thorough examination and testing is to determine whether the condition of the powered hoist after installation or repairs is such that it can continue to be used safely.

Pre-examination Procedure

Prior to any maintenance, the engineer should satisfy himself with the following:

- 1) Has the owner's representative been advised of your presence on site?
- 2) Is the owner's representative satisfied with the performance of equipment?
- 3) Does the owner's representative have any complaints regarding equipment?
- 4) Are all lubrication points adequately lubricated and clean?
- 5) Are all oil reservoirs clean and filled to correct level?
- 6) Is equipment functioning as designed?
- 7) Is equipment adequately clean and work areas adequately illuminated?
- 8) Do all safety features operate satisfactorily?
- 9) Are all parts of equipment correctly adjusted and operational?
- 10) Is equipment safe in operation?
- 11) Is the hoist isolated and have the fuses been removed?
- 12) Are "under repair" signs displayed?
- 13) Are all tools being used secure from falling off work platform?

Note: Great care must be taken whilst manoeuvring on overhead beams.

Visual Examination

1) Chain Hoist Motor Examination

- a) The frame should be free from damage or defects
- b) All bolts and screws should be tight
- c) Examine for signs of overheating
- d) Bearings should be in good condition and adequately lubricated

2) Pendant Control

- a) Examine case for damage
- b) Check contacts for burning and wear and also for correct adjustment
- c) Lubricate where necessary

3) Push Button Control (if fitted)

- a) Examine case for damage
- b) Check coils and contacts for burning and wear
- c) Lubricate where necessary
- d) Push buttons should be free and making good contact

4) Limit Switch

- a) Check for wear and damage
- b) Ensure correct adjustment and good contact surfaces
- c) All moving parts are to be lubricated

5) External Brake (for chain hoist brakes with cam shoes)

- a) Ensure free movement of all moving parts
- b) Check for oil leaks from gearbox



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- c) Brake linings to be checked for wear and adjusted if necessary
- 6) External Brake (push button units) (for chain hoist brakes)**
 - a) Check for wear on shoes, levers etc.
 - b) Ensure correct adjustment and that coil is not overheating
 - c) Lubricate all moving parts
- 7) Electric Hoist Brake**
 - Plunger type**
 - a) Check solenoid coil for undue wear or distortion caused by overheating
 - b) Ensure plunger is in correct alignment and quite free in the guides
 - c) Check laminations for deformation
 - d) Check lever for wear and alignment
 - Clapper type**
 - a) Check for excess wear on laminations
 - b) Check air gap at heel of armatures
 - All types**

Check wear on linings, correct spring pressure, correct adjustment and proper lubrication of all moving parts.
- 8) Gears**
 - a) If gears sound unduly noisy, remove cover for checking and examination
 - b) Check for wear, damage or cracks and alignment
 - c) Make sure all keys are tight and all covers secure to give adequate guarding
 - d) Ensure adequate lubrication
- 9) Internal Brake**

If ratchet can be heard, case should be opened up to determine cause of defect.
- 10) Load sheave**

Check for wear and correct seating of load chain in sprockets.
- 11) Load chain**
 - a) Check for wear and stretch
 - b) Check for faulty welds or cracks in the links
- 12) Top Hook**

Examine for wear and distortion.
- 13) Bottom Hook**
 - a) Examine for wear and distortion
 - b) Wear in saddle of hook should not exceed 10%. The throat-opening if found opened more than 5% should be considered unsafe
- 14) Rope Drum**
 - a) Check for cracks or damage, particularly to the grooves
 - b) Dress any damaged grooves and remove sharp edges
- 15) Load Rope (electric wire hoist block)**
 - a) Check for wear or damage
 - b) Ropes should be considered unusable when broken wires exceed 10% over a length of eight times the diameter of the rope, or if reduction in diameter of the rope, or if reduction in diameter of outside wires is 50% over the whole of the working length, or any sign of necking indicating internal damage
 - c) Ensure adequate lubrication
- 16) Bottom Block (electric hoist block)**
 - a) Examine for wear in the sheave, grooves and bore
 - b) Examine for wear in the crosshead
 - c) Ensure above parts and the guard plate are free from damage or defects
 - d) Check that wear in saddle of hook does not exceed 10%
 - e) Ensure top nut is locked



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17) Safe Working Load

Should be clearly marked.

18) Insulation Resistance Test

Megger reading should not be less than one megohm.

19) Supply Cable

Check for damage, worn or perished insulation.

Load Testing

- a) Subject the hoist to a proof load of S.W.L. + 25% (in such a fashion that all the gearing makes one full revolution under load) and check for efficient operation of the brakes. Re-examine hoist to ensure that the proof load has not exposed any defects
- b) Where a beam trolley is fitted or is an integral part of the hoist, refer to the relevant overhaul procedure
If all satisfactory, record relevant details and issue report and/or certificate.
This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection and / or proof load test.

Upon completion

Make everything safe and tidy (remembering any loose parts that may be lying around, particularly if you have been working overhead); check all tools are cleared from site.

Electrically Powered Chain Or Wire Rope Hoists - Check List

- 1) Owner:
- 2) Location:
- 3) Model:
- 4) Inspection Engineer:
- 5) Date of Inspection:
- 6) Date of Test:
- 7) Current Cert. No:
- 8) Name And Address of Maker of Hoist:
- 9) Type of Hoist and Nature of Power:
- 10) Date of Manufacture of Hoist:
- 11) Identification:
 - A. Maker's serial number:
 - B. Owner's distinguishing mark or number:
- 12) Date of Last Previous Test of Hoist:
- Date of Last Previous Thorough Examination of Hoist:



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Components	Satisfactory	Requires Attention	Requires Renewal
Chain Hoist Motor			
Frame			
Fastenings			
Motor			
Bearings			
Pendant Control			
Casing			
Contacts			
Adjustment			
Lubrication			
Push Button Control			
Casing			
Coils			
Contacts			
Buttons			
Lubrication			
Limit Switches			
Adjustment			
Contacts			
Lubrication			
External Brakes			
Gear Box			
Linings			
Levers			
Shoes			
Coil			
Lubrication			



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Components	Satisfactory	Requires Attention	Requires Renewal
Electric Hoist Brake			
Solenoid Coil			
Plunger			
Guides			
Laminations			
Lever			
Armatures			
Linings			
Spring Pressure			
Adjustment			
Lubrication			
Gears			
Alignment			
Keys			
Covers			
Lubrication			
Internal Brakes			
Case			
Ratchet			
Load Sheave			
Sprocket			
Load Chain			
Top Hook			
Bottom Hook			
Rope Drum			
Load Rope			
Bottom Block			
Swl			
Markings			
Supply Cable			
Trolley (If Fitted)			



7.2.8 Pneumatic Chain Hoists

Check List

Examination Procedure

- 1) Examine top and bottom hooks and check for stretch / distortion
- 2) Check function of safety catches
- 3) Ensure hooks swivel freely in yokes and are secure in their mountings
- 4) Where accessible, examine load pin for wear / distortion.
- 5) Examine load chain and sheaves, check for wear / distortion and ensure satisfactory seating of load chain in pockets
- 6) Examine body casing and check for cracks, distortion, missing bolts, screws or any other defect that may interfere with operation or effect the safety of the hoist
- 7) Examine dead end load chain anchor pins and check for distortion or wear
- 8) Check direction controls are free from damage and that they function correctly
- 9) Suspend hoist and run air through the system to check the integrity of the machine. Listen to the sound of the motor as it should sound smooth. (A sluggish sounding motor could indicate an ingress of contamination or lack of lubricant)
- 10) Examine the complete length of load chain and check for stretch, deformed / bent links, nicks, wear and excessive corrosion
- 11) Where a beam trolley is fitted or is an integral part of the hoist, refer to the relevant examination procedure

If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection.



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Serial No:			A = Satisfactory
Model:			B = Worn but serviceable
SWL:			C = Condemned

Item	Description	Condition		
		A	B	C
	Top Hook			
	Safety Catch			
	Bottom Hook			
	Safety Catch			
	Top Hook Swivel / Yoke			
	Load Chain Sheaves			
	Body Casing			
	Load Chain			
	Function Test			
	Bottom Hook Load Pin			
	Bottom Hook Swivel / Yoke			
	Direction Controls (pendant/pull cord)			

Tick relevant boxes / delete where applicable

Signed:	Date:	Engineer:

Satisfactory	Return for repair



7.2.9 Manual Hoists

Check List

Examination Procedure

1. Examine top and bottom hooks and check for stretch / distortion
2. Check function of safety catches
3. Ensure hooks swivel freely in yokes and are secure in their mountings
4. Where accessible, examine load pin for wear/distortion
5. Examine load chain sheaves, check for wear / distortion and ensure satisfactory seating of load chain
6. Examine body casing and check for cracks, distortion, missing bolts / screws or any other defect that may interfere with operation or affect the safety of the hoist
7. Examine (as far as possible) internal frame side plates
8. Ensure chain guide rollers and chain stripper are in position and functioning
9. Examine live end and dead end load chain anchor pins and check for distortion / wear
10. Suspend hoist and spin hand chain wheel (in the hoist mode) to check function of ratchet pawl. There should be a healthy clicking sound but if this is not the case, the pawl may be seized or the spring may be broken
11. Examine hand chain wheel and chain and ensure it seats correctly in the pockets
12. Examine the complete length of load chain and check for stretch, deformed / bent links, nicks, wear and excessive corrosion

If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS** minimum requirements for an inspection.



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Serial No:			A = Satisfactory
Model:			B = Worn but serviceable
SWL:			C = Condemned

Item	Description	Condition
	Top Hook	
	Safety Catch	
	Bottom Hook	
	Safety Catch	
	Top Hook Swivel / Yoke	
	Top Hook Load Pin	
	Load Chain Sheaves	
	Body Casing	
	Side Plate L.H.	
	Side Plate R.H.	
	Chain Guide Rollers	
	Chain Stripper	
	Load Chain Live End Anchor	
	Load Chain Dead End Anchor	
	Ratchet Assembly	
	Hand Chain Wheel	
	Hand Chain	
	Load Chain	

Tick relevant boxes / delete where applicable

Signed:	Date:	Engineer:

Satisfactory	Return for repair



7.2.10 Lever Hoists

Check List

Examination Procedure

1. Examine top and bottom hooks and check for stretch / distortion
 2. Check function of safety catches
 3. Ensure hooks swivel freely in yokes and are secure in their mountings
 4. Where accessible, examine load pins for wear / distortion
 5. Examine load chain sheaves, check for wear / distortion and ensure satisfactory seating of load chain
 6. Examine body casing and check for cracks, distortion, missing bolts / screws or any other defect that may interfere with operation or affect the safety of the hoist
 7. Examine (as far as possible) internal frame side plates
 8. Ensure chain guide rollers and chain stripper / de-twister are in position and functioning
 9. Examine load chain "live" end anchor and "dead" end load chain stop
- Note:** If the dead end chain stop is deformed, it can be pulled through the hoist.
10. Examine the operating lever and ensure it is not deformed (a bent lever could be an indicator of overload). Operate the lever in both the hoisting and lowering modes to check operation of pawls and direction changing mechanism. (There should be a healthy clicking sound but if this is not the case, the pawls may be seized or the spring may be broken)
 11. Examine the complete length of load chain and check for stretch, deformed / bent links, nicks, wear and excessive corrosion

If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS** minimum requirements for an inspection.



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Serial No:			A = Satisfactory
Model:			B = Worn but serviceable
SWL:			C = Condemned

Item	Description	Condition
1	Top Hook	
2	Safety Catch	
3	Bottom Hook	
4	Safety Catch	
5	Top Hook Swivel / Yoke	
6	Top Hook Load Pin	
7	Load Chain Sheaves	
8	Body Casing	
9	Side Plate L.H.	
10	Side Plate R.H.	
11	Chain Guide Rollers	
12	Chain Stripper/De-twister	
13	Load Chain Live End Anchor	
14	Load Chain Dead End Anchor / Stop	
15	Ratchet Mechanism	
16	Operating Lever	
17	Direction Changing Mechanism	
18	Load Chain	

Tick relevant boxes / delete where applicable

Signed:	Date:	Engineer:

Satisfactory	Return for repair



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7.2.11 Tirfor Rope Hoists

Check List

Examination Procedure

- 1) Examine the load pin/hook and hook seating in the side casings for wear, distortion or cracking
- 2) Where hooks are fitted, check operation of safety catch and ensure the hook is free to swivel
- 3) Check operation of both forward and reverse operating levers and ensure the shear pins are intact. (Sheared or cracked pins indicate overloading)
- 4) Check operation of the release lever/handle
- 5) Insert a rope and check the stroke of both the forward and reversing lever
 - i.e. T508 - 55mm approx.
 - T516 - 55mm approx.
 - T532 - 32mm approx.

Note: (a) A longer stroke could indicate internal wear

(b) A shorter stroke could indicate contamination/damage

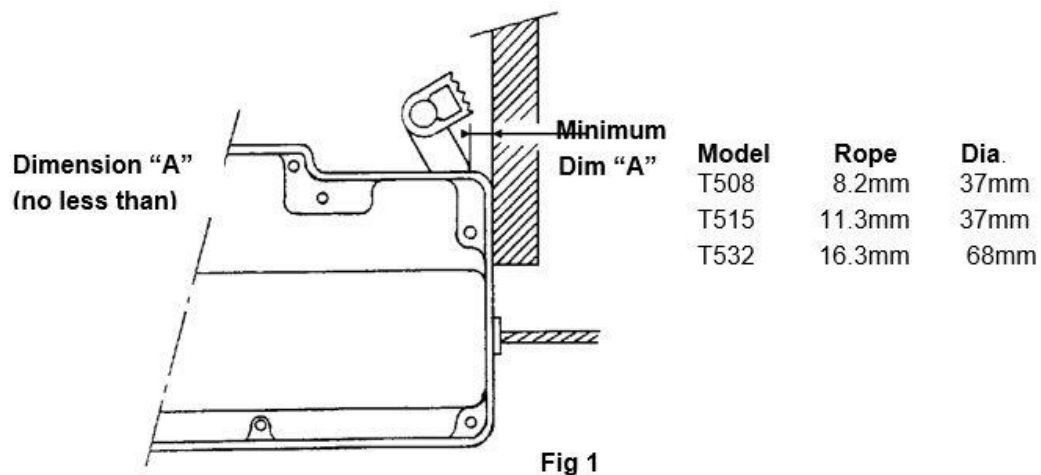
- 6) Examine casing and check for cracks or dents which may affect the workings of the internal mechanism
- 7) Using a torch, examine the internals through the opening at the top of the hoist and ensure it is adequately lubricated

Note: (a) With the T500 series of Tirfor, it is possible to gauge the extent of internal wear without dismantling the hoist. To do this, you require a new or unworn rope and proceed as follows:

- Release the unit and insert the piece of rope
 - Re-engage the machine on the rope
 - Push the release lever towards the anchor point until resistance is felt
 - Measure the distance between the rear of the release lever and the edge of the casing
 - This dimension must not be less than the figures tabled-**See Fig 1**
- (b) Should the dimension be less, the machine is to be returned for repair

If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection.





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Item	Description	Condition		
	Load Pin / Hook			
	Safety Catch			
	Operating Levers			
	Shear Pins			
	Release Lever			
	Forward Stroke			
	Reverse Stroke			
	Casing R.H.			
	Casing L.H.			
	Internals (Condition)			
	Internal Wear Check			

Tick relevant boxes / delete where applicable

Signed:	Date:	Engineer:

Satisfactory	Return for repair



7.2.12 Beam Trolleys

Check List

Examination Procedure

- 1) Examine side plates and check for wear, distortion and cracks
- 2) Examine wheels, axles, bearings, wheel tread and flanges
- 3) If geared travel, check gear teeth on wheel flanges and drive sprocket of hand chain wheel shaft for alignment, broken teeth etc.
- 4) Examine hand chain and hand chain wheel and ensure chain seats properly in the chain sprockets
- 5) Check the chain guides are not deformed and are correctly positioned to ensure smooth feed of chain onto chain wheel
- 6) Examine load bar / bars and check for deformation
- 7) Examine suspension plates / yokes / shackle for wear, distortion and cracking
- 8) Where an adjusting screwed bar is fitted, check for straightness and examine threads for stretch / damage and check function of locking device
- 9) Ensure both the female threaded spigots (mounted on the side plates) are not damaged or deformed

If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection.



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SWL:			C = Condemned

Item	Description	Condition		
		A	B	C
	Side Plate (R.H.)			
	Side Plate (L.H.)			
	Wheels			
	Geared Wheels			
	Wheel Axles			
	Wheel Bearings			
	Drive Pinion			
	Hand Chain Wheel			
	Hand Chain			
	Screwed Adjuster Bar			
	Locking Device			
	Load Bar / Suspension Yoke			
	Suspension Shackle (if applicable)			
	Spacers			
	Hand Chain Guides			

Tick relevant boxes / delete where applicable

Signed:	Date:	Engineer:

Satisfactory	Return for repair



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7.2.13 Beam Clamps

Check List

Examination Procedure

- 1) Examine suspension shackle and check for wear, stretch or distortion
- 2) Examine suspension load bar and check for wear, bending or distortion
- 3) Examine inner clamp half and check for wear, distortion and cracking - check jaws for deformation
- 4) Examine outer clamp half and check for wear, distortion and cracking - check jaws for deformation

Note: Where swivel jaws are fitted, ensure they are free to rotate.

- 5) Operate adjusting bar and check straightness and function. Check threads for wear and stretch
- 6) Examine female screwed spigots (in each clamp half) and ensure they are not deformed due to over/under tightening of clamp on beam
- 7) Check tommy bar handle and ensure it is not bent or has any damage which may be injurious to the operator's hands

If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection.

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Plant No:			Conditions
Serial No:			A = Satisfactory
Model:			B = Worn but serviceable
SWL:			C = Condemned

[illegible]

Signed:

Date:

Engineer:

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Satisfactory

Return for repair

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7.2.14 Horizontal Plate Clamps

Check List

Examination Procedure

- 1) Examine hook ring and check for wear / stretch in crown of link
- 2) Check lateral movement (if any) of hook ring on load bolt connection to serrated jaws / rocker arms as excessive movement could indicate wear / distortion
- 3) Check lateral movement (if any) of serrated jaws/rocker arms on load bolt connection to main body as excessive movement could indicate wear / distortion
- 4) Examine the serrations on the jaws or swivel toe and check for wear/deformity
- 5) Where a toe is fitted, check for lateral movement (if any) of swivel toe on load bolt connection to rocker arms as excessive movement could indicate wear/distortion

If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS** minimum requirements for an inspection.



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Serial No:			A = Satisfactory
Model:			B = Worn but serviceable
SWL:			C = Condemned

Item	Description	Condition		
		A	B	C
	Hook Ring			
	Serrated Jaw - 1			
	Serrated Jaw - 2			
	Rocker Arm - 1			
	Rocker Arm - 2			
	Load Bolt - 1			
	Load Bolt - 2			
	Load Bolt - 3			
	Swivel Toe			
	Main Body			

Tick relevant boxes / delete where applicable

Signed:	Date:	Engineer:

Satisfactory	Return for repair



7.2.15 Universal Plate Clamps

Check List

Examination Procedure

- 1) Examine hook ring and check for wear / distortion in the crown of the ring
- 2) Ensure the hook ring is not too slack in the clamp as this could indicate internal wear
- 3) Examine jaw pin and nut and ensure it is secure and not deformed
- 4) Check operation of cam-assembly locking lever/jaw spring
- 5) Examine serrated jaw and serrated pad and check for wear/deformation
- 6) Examine main body shell and check for wear, cracks or deformation which may affect the operation of internal components

If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection.



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SWL:			C = Condemned

Item	Description	Condition		
		A	B	C
	Hook Ring			
	Jaw Pin & Nut			
	Jaw Spring			
	Serrated Jaw			
	Serrated Pad			
	Main Body Shell			
	Cam Assembly Locking Lever			

Tick relevant boxes / delete where applicable

Signed:	Date:	Engineer:

Satisfactory	Return for repair



7.2.16 Sheave Blocks / Crane Blocks

Check List

Examination Procedure

- 1) Examine sheaves for wear in the rope groove, cracks or distortion
- 2) Try to lift sheaves to check bushes for wear
- 3) Spin the sheaves to check bearings / bushes and ensure smoothness of operation
- 4) Ensure all grease ports are clean and unblocked and that the block is well lubricated
- 5) Examine swivel head fitting / bottom load hook and check for wear / stretch
- 6) Examine (if possible) thrust bearing / washer and ensure smoothness of operation
- 7) Examine upper load pin / spigots and check for wear / distortion
- 8) If head fitting/hook is of the shank type, check security of shank and nut and examine for stretch / distortion. Examine crosshead for wear
- 9) Examine head fitting shackle/eye checking for wear, stretch or cracking
- 10) Examine side plates / straps and check for distortion, wear or cracking (especially around main load pin hole and top suspension hole)
- 11) Ensure there are no sharp edges or burrs on the side plates which may be detrimental to the wire rope
- 12) If fitted, examine becket and check for wear, stretch or cracking
- 13) If fitted, examine weighted cheek plates (check for cracks)
- 14) Check all spacers and tie bolts and ensure they are not deformed

If all satisfactory, record relevant details and issue reports.

This procedure is to be followed as a guideline to **QatarEnergy-IS** minimum requirements for an inspection.



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Serial No:			A = Satisfactory
Model:			B = Worn but serviceable
SWL:			C = Condemned

Item	Description	Condition		
		A	B	C
	Sheaves			
	Sheave Bearings / Bushes			
	Grease Ports / Nipples			
	Head Fitting Swivel			
	Head Fitting Swivel Bearing			
	Head Fitting Shackle OR			
	Head Fitting Hook			
	Cross Head / Hook Yoke			
	Oval Eye c/w Shank			
	Shank Nut			
	Safety Catch			
	Side Plates/Straps			
	Becket			
	Spacers			
	Tie Bolts			
	Weighted Cheek Plates (if fitted)			

Tick relevant boxes / delete where applicable

Signed:	Date:	Engineer:

Satisfactory	Return for repair



7.2.17 Pneumatic Winches

Check List

Examination Procedure

- 1) Examine rope guard (where fitted) and ensure there is no damage / distortion which may obstruct and / or abrade the winch rope
- 2) Examine winch drum and check for wear, distortion or cracks
- 3) Examine brake bands and brake drums and check for wear
- 4) Ensure the brake bands and brake drums are clean and free from contamination
- 5) Where fitted, examine the automatic brake checking springs, link arms and pins
- 6) Examine automatic brake cylinder and check for leaks
- 7) Examine chromed piston rod and check for corrosion
- 8) With power disconnected, check that all operating levers return to neutral when released
- 9) Ensure directional arrows / markings are in place and clearly visible
- 10) Check oil level
- 11) Examine winch base and check for cracked welds, cracks around bolt holes, distortion or impact damage
- 12) **Anchorage:** Ensure the hold down bolts / welding / clamping is adequate and as far as possible, satisfy yourself that the support steel work is capable of sustaining the winch under load
- 13) Check that the power supply is adequate for that particular model of winch
- 14) Connect power supply and function test the winch
- 15) Should wire rope be fitted, it must be examined in accordance with procedure 31

If all satisfactory, record relevant details and issue report.

Note: Further checks are required if this winch is to be used for the raising or lowering of personnel. See procedure Section 21.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection / “pre-load test” examination.

Load Test

Note: Although the winch should be fully certified prior to reaching the worksite, once it has been sited and secured, the anchorage of the winch has to be tested. This is normally achieved by applying a proof load to the anchorage via the winch.

- 1) Using test weights or a test clock with suitable anchor point, apply a proof load of S.W.L. + 25%
- 2) Sustain load and examine anchorage checking for deflection / distortion
- 3) Release load and re-examine to ensure proof loading has not exposed any defects
- 4) Any welding should be N.D.T.'d after load test

If all satisfactory, record all details including position of winch base anchorage, mark with identification and issue test certificate.



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Serial No:			A = Satisfactory
Model:			B = Worn but serviceable
SWL:			C = Condemned

Item	Description	Condition		
		A	B	C
	Rope Guard			
	Winch Drum			
	Brake Bands & Drums			
	Automatic Brake Springs			
	Automatic Brake Cylinder			
	Automatic Brake Link Arms & Pins			
	Operating Levers (Function)			
	Directional / SWL Markings			
	Oil Level			
	Winch Base			
	Winch Anchorage			
	Power Supply			
	Function Test			

Tick relevant boxes / delete where applicable

Signed:	Date:	Engineer:

Satisfactory	Return for repair

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7.2.18 Pneumatic Winches (Personnel)

Note: This procedure is to be read in conjunction with Procedure 17.

Additional Instructions

(Winches converted for the purpose of raising / lowering personnel)

- 1) Winches of this type will be fitted with a dual braking system usually consisting of one manual brake and one automatic brake. These brakes are to be examined in accordance with procedure 17
- 2) Where limit switches are fitted to prevent over travel in both hoisting and lowering, function test to ensure their operation
- 3) Examine the winch and ensure the winch does **not** have a disengaging clutch or pawl and ratchet mechanism
- 4) Check that the winch has been adequately de-rated to reflect a factor of safety of no less than 7:1 (Note: Man-riding winch rope to have F.O.S. of 8:1)
- 5) Ensure the winch is clearly marked stating it can be used for raising and lowering personnel

Note: It is recommended that this type of winch is not used for general duties.

If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection / “pre load test” examination (6 monthly intervals maximum).

Load Test

Note: Although the winch should be fully certified prior to reaching the worksite, once it has been sited and secured, the anchorage of the winch has to be tested. This is normally achieved by applying a proof load to the anchorage via the winch.

- 1) Using test weights or a test clock with suitable anchor point, apply a proof load of S.W.L. + 25%
- 2) Sustain load and examine anchorage checking for deflection/distortion
- 3) Release load and re-examine to ensure proof loading has not exposed any defects
- 4) Any welding should be N.D.T.'D after load test

If all satisfactory, record all details including position of winch base anchorage, mark with identification and issue test certificate.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for a proof load test.

Refer to 6.2.17 for checklist.



7.2.19 Manual Winches

Check List

Direct Drive / Worm Geared Winches

- 1) Examine winding handle and check for distortion
- 2) Examine winch drum and check for distortion, wear or cracking (if accessible, examine rope anchorage and ensure it is secure)
- 3) Where applicable, examine ratchet or worm gear (mounted on side of drum) and check for wear / broken teeth
- 4) Examine pawl and pawl spring /worm and check for wear / distortion. Ensure the pawl functions correctly and ensure the worm meshes correctly with the worm wheel
- 5) Examine winch frame and check for distortion, wear and cracking
- 6) Ensure the hold down bolts / welding / clamping is adequate and as far as possible, satisfy yourself that the support steel work is capable of sustaining the winch under load
- 7) Should wire rope be fitted, it must be examined in accordance with procedure 31

Additional (Spur Geared Winches)

- 1) Visually examine the brake assembly and operate the winch handle to ensure it functions correctly
- 2) Examine all spur gears checking for alignment, wear on gear teeth, truth of gear shafts and smooth running of bearings (where fitted)
- 3) Function check the winch through one complete revolution of the drum
If all satisfactory, record relevant details and issue report.
This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection and / or a “pre-load test” examination.

Load Test

Note: Although the winch should be fully certified prior to reaching the worksite, once it has been sited and secured, the anchorage of the winch has to be tested. This is normally achieved by applying a proof load to the anchorage via the winch.

- 1) Using test weights or a test clock with suitable anchor point, apply a proof load of S.W.L. + 50%
- 2) Sustain load and examine anchorage checking for deflection / distortion
- 3) Release load and re-examine to ensure proof loading has not exposed any defects
- 4) Any welding should be N.D.T.'D after load test

If all satisfactory, record all details including position of winch base anchorage, and mark with identification and issue test certification.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for a proof load test.



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Item	Description	Condition		
		A	B	C
	Handle			
	Drum			
	Ratchet Wheel / Worm Wheel			
	Ratchet Pawl/Worm			
	Ratchet Pawl Spring			
	Winch Frame			
	Friction Discs			
	Spur Gears			
	Spur Gear Shafts			
	Bearings			
	Hold Down Bolts / Clamps / Welding			
	Support Steel Work			

Tick relevant boxes / delete where applicable

Signed:	Date:	Engineer:

Satisfactory	Return for repair



7.2.20 Overhead Runway Beams

The legal responsibility for ensuring that overhead runway beams are properly designed and installed rests upon either the operator or the owner of the worksite in which the overhead runway beams are located.

It is important that the competent person engaged on this work should be familiar with the full contents of the Standard and be able to apply his knowledge.

The overhead runway beam must be load tested before being put into use after installation or after substantial alteration / repair and subject thereto, at regular intervals (12 months maximum) for thorough examination.

The purpose of thorough examination and testing is to determine whether the condition of the overhead runway beam after installation or repairs is such that it can continue to be used safely.

Checking the strength

- 1) Most overhead runway beams are constructed from Universal Beams. As the flanges of these beams are comparatively wide and slender it is necessary to check the strength transversely (i.e.) against the possibility of the flanges bending; as well as the longitudinal bending of the beam
- 2) For simply supported beam spans, the maximum longitudinal stress acts as tension on the bottom flange and compression on the top flange. On cantilevers the maximum tensile stress is in the top flange and the maximum compression stress is in the bottom flange
- 3) For the purpose of checking and design, the load on the runway beam is to be taken as the S.W.L. of the block on the runway beam plus the weight of the lifting appliances i.e. (block and trolley) plus any other superimposed load (the runway beam might be used to support other machinery) plus an allowance for dynamic loading i.e. 25% for power operated or 10% for hand operated appliances
- 4) The self-weight of the beam is often ignored in calculations on the strength of the beam itself, but where it is not, half the weight of the beam in the span under consideration should be added to the above loads

Deflection

- 1) The permissible deflection of a simply supported beam under test is $1/500^{\text{th}}$ of the largest span and $1/250^{\text{th}}$ for cantilevers
- 2) The deflections allowed by the Standard apply to the actual runway beam only and do not include deflections in the supporting structures. Such deflections should be deducted from the total measured deflection in order to determine the actual deflection for comparison with the Standard

Testing and Inspection Procedure

Inspection before application of loads

- a) All joints, connections and supports shall be carefully inspected, special attention being paid to the security of nuts on bolts. The nuts shall not be tightened to such an extent as to over stress the bolts
- b) Attention shall be paid to the condition of walls / steelwork surrounding and adjacent to the end fastening of runways fixed therein
- c) End stops (or stops) must be fitted to prevent the lifting appliance running off the end of the beam or colliding with the support structure / walls and must be so positioned as to prevent the load being imposed within a flange width from the beam end



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Application of loads

- a) The preliminary inspection having been carried out, a test load equivalent to the maximum safe working load, together with the weight of the appropriate heaviest lifting appliance supported by the runway, shall be applied and traversed along the whole length of the runway. The load shall be halted during its passage along the runway and kept at rest at suitable positions, in order that stable deflection readings may be taken and recorded. Deflections are to be taken at weakest points i.e. mid span and at end of cantilevers
- b) The above procedure shall then be repeated except that a proof load shall be applied at an increase of 25% above the S.W.L. calculated

Inspection and thorough examination

- a) During the application of the test load (S.W.L.) and the proof load as specified, the runway shall be kept under such observation as will ensure the ready detection of any obvious defect in the runway beam
- b) Following the application of the proof load the runway shall be thoroughly visually examined in order to ascertain that the runway beam has withstood the proof load without injury or permanent deformation

Certification

It is the duty of the certifier to satisfy himself that the maximum allowable deflections specified in BS 2853 are not exceeded and that all other requirements of this Standard are complied with before issuing the relevant certificate.

Accuracy of weights

Where practical the load used shall be made up of certified weights such that the sum of inaccuracies of the load do not exceed plus or minus 4%.

Markings

The Safe Working Load, Identification Number and any limiting conditions shall be plainly and permanently marked on the runway beam so as to be clearly visible to the operator and identifiable against its certification.

If all satisfactory, record relevant details and issue report and / or certificate.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an Inspection and / or proof load test.

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7.2.21 Lifting Beams / Frames

Check List

Examination Procedure

- 1) Examine all lifting lugs and check for wear, distortion and cracking (around weldments). Ensure the holes in the lugs are not too large for the shackle pins as this could cause problems due to point loading
- 2) If the lifting beam / frame is of bolted construction, check security of all bolts, nuts and split pins etc.
- 3) Examine actual beam / frame and check for bending and distortion
- 4) Should the beam / frame be constructed of hollow section or tubulars, ensure they have not incurred impact damage which could affect their rigidity
- 5) Any ancillary fittings i.e. wire ropes, shackles etc. should be inspected in accordance with their specific examination procedure

If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection.

7.2.22 Hydraulic Cylinders / Jacks / Pumps

Check List

Examination Procedure

- 1) Examine body of cylinder / jack and check for impact damage, cracks and oil leaks. (With cylinders, examine inlet / outlet couplings and check for leakage)
- 2) Operate cylinder / jack, pumping the ram to full stroke
- 3) Examine the ram and check for bellling, scoring and corrosion
- 4) If the ram is threaded externally and fitted with a locking collar, examine threads and check for stretch
- 5) Examine round seal and check for oil leakage
- 6) Turn valve to lower / release and ensure the ram goes down smoothly.
 - a. (Jerkiness could indicate distortion to the ram not visible to the eye and sponginess indicates air in the system which will have to be bled)
- 7) Examine all hoses and fittings and ensure they are not perished, cut or in any way damaged
- 8) With the ram in the lowered position check the oil level of the jack / pump unit
- 9) Function test the pump and ensure the valve does not leak when closed and under pressure
- 10) Where claw attachments are fitted examine for distortion / cracks
- 11) Where gauges are fitted, check for leaks, function test and ensure needle returns to zero

If all satisfactory record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection.



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Item	Description	Condition		
		A	B	C
	Cylinder / Body			
	Inlet / Outlet Couplings			
	Ram			
	Ram Threads			
	Ram Seals			
	Release Valve			
	Hoses			
	Couplings			
	Pump Unit			
	Oil Levels			
	Claw Attachment			
	Gauges			

Tick relevant boxes / delete where applicable

Signed:	Date:	Engineer:

Satisfactory	Return for repair



7.2.23 Transit Containers

It is important that the competent person engaged on this work should be familiar with the full contents of the Standard and be able to apply his knowledge.

The transit containers must be load tested before being put into use, after repairs and subject thereto, intervals of 6 months for thorough examination.

BS 7072 requires the inclusion of a Non Destructive Examination of the lifting points before being used for the first time, after repair or alteration and subject thereto intervals of 12 months. (It may also be necessary to N.D.E. after 6 months, this being at the discretion of the competent person). Further proof load testing must be carried out every 24 months.

The purpose of thorough examination and testing is to determine whether the condition of the container is such that it can continue to be used safely.

Introduction and Plating Instructions

- 1) A container is an item of equipment lifted by slings, whose primary purpose is the transportation of goods, materials or equipment, which may be permanently installed in the container
- 2) Each container should be fitted with a data plate made of non-corrosive material, securely attached externally. The plate should have the following information permanently and legibly marked in characters not less than 4mm in height:
 - Owners name
 - Container identification (unit number)
 - Tare weight (in tonnes)
 - SWL / Payload (in tonnes)
 - Maximum gross weight (in tonnes)
 - Date of last test / examination using the following:
 - Suffix T** for proof load test, NDE and visual examination
 - Suffix VN** for NDE and visual examination
 - Suffix V** for visual examination only

Visual Examination

1) Welds

All accessible load-bearing welds should be visually examined to ensure freedom from defects.

2) Structure

The structure should be visually examined for corrosion, mechanical damage or injurious deformation.

3) Lifting Points

The lifting points should be visually examined for distortion, mechanical damage or any other sign of distress or overload.

4) Door closures

Doorframes, seals, hinges, locks etc. should be visually examined and functionally checked to ensure that they operate in a satisfactory manner without undue force being required.

5) Floor

The floor should be visually examined to check that it is substantially flat with no sign of distress or overload. Drainage facilities where fitted, should be examined for debris etc.

6) Markings



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The paint markings and plates should be checked to see that they meet the recommendations of the standard.

7) Lifting Set

If the container is fitted with a lifting sling set this should be examined.

Note: All lifting sets used in the handling of containers should be manufactured in accordance with BS 1290. Sling assemblies should be selected on the basis of complying with BS 1290 at a rating 1.3 times the specified maximum gross weight of the container to which it is to be fitted.

Non-destructive examination of lifting points

Unless the competent person decides an alternative method is more appropriate, ferromagnetic materials should be examined by magnetic particle inspection (MPI) - (See BS 6072).

Proof Load Testing

- 1) Place a uniformly distributed test load over the floor of the container in such a way that the combined weight of the container and test load is equal to two times the maximum gross weight and lift the container carefully by the lifting points in such a way that no significant acceleration or deceleration forces are applied. If the container cannot be directly loaded because of its construction, the competent person should determine a suitable test such as suspending weights from the frame
- 2) Maintain the proof load on the container for a minimum of 5 minutes then examine the container against the criteria given in the thorough visual examination procedure during and after application of the proof load

If all satisfactory, record relevant details and issue report and / or certificate.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection and / or proof load test.

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7.2.24 Mobile (Access) Platforms

Check List

Examination Procedure

- 1) Examine all hydraulic pipe work and check for impact damage or cuts etc. which may affect the pressure capacity
- 2) Examine all couplings and check for oil leaks
- 3) Examine hydraulic pump, check oil level and operate to check function of non-return valve
- 4) Examine pump piston and check for leaks at the seal
- 5) Examine hydraulic cylinder and check for impact damage
- 6) Examine wheels and tyres and ensure they run smoothly
- 7) Examine out riggers, screw down jacks and footpads. Ensure the locking mechanism functions correctly
- 8) With the out riggers in the operating position, pump up ram to its full stroke and inspect ram checking for corrosion and scoring. (If fitted with an air motor, ensure all operating levers return to neutral on release)
- 9) Examine main framework and all hinge pins ensuring they are secure
- 10) Examine the work platform and ensure handrails are secure
- 11) Examine safety bar / chain

If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection (maximum intervals = 6 months).



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Item	Description	Condition		
		A	B	C
	Hydraulic Pipe work			
	Couplings			
	Hydraulic Pump			
	Hydraulic Cylinder			
	Air Motor (if fitted)			
	Pneumatic Control Levers			
	Wheels			
	Tires			
	Out Riggers			
	Screw Down Jacks			
	Foot Pads			
	Main Framework			
	Work Platform			
	Handrails			
	Safety Bar / Chain			

Tick relevant boxes / delete where applicable

Signed:	Date:	Engineer:

Satisfactory	Return for repair



7.2.25 Fabricated Pad Eyes

It is important that the competent person engaged on this work should be familiar with the full contents of the Standard and be able to apply his knowledge.

The Fabricated Padeyes / Clamps must be load tested before being put into use after installation, or after substantial alteration/repair and subject thereto, at regular intervals (12 months maximum) for thorough examination.

The purpose of thorough examination and testing is to determine whether the condition of the Padeyes / Clamps after installation or repairs is such that they can continue to be used safely.

Visual Examination

- 1) Examine the profile of the padeye (or padeye section of clamp) and ensure all flame cutting marks have been ground smooth to avoid creating possible stress areas
- 2) Ensure the lifting / suspension hole has been either drilled or machined. (Flame cut holes are not acceptable)
- 3) Check that the hole dia. and padeye width are compatible with the appropriate SWL shackle. (Ideally, hole should be Pin Dia. + 10%, width should be jaw gap less 10%)

Note (a) If the hole is too large, the shackle pin could possibly over stress the padeye due to point loading

(b) If the padeye is too narrow, the shackles could sit askew and consequently be subjected to stresses other than those for which it was designed

- 4) Ensure padeye is not bent or distorted in any way
- 5) Where applicable, examine welding of padeye to main steel work. (It may be prudent to N.D.T. prior to load testing)
- 6) In the case of bolt-on padeyes, check the attachment steel work for any local distortion, ensure the bolts are adequate in quantity, diameter and length, and are not distorted
- 7) Check the security of the bolting. (Padeyes designed for an angular pull often have torque values for bolting to prevent slippage along the beam)

If all satisfactory, record the relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection and / or a "pre-load test" examination.

Proof Load Testing

It is recommended that the padeye should be subjected to a proof load of no less than SWL + 50% in the same direction as the forces it would be subjected to under normal working conditions.

1) The three acceptable methods of testing are as follows:

- a) Using "Live" test weights / water weights
- b) Using a test clock / load cell against a suitable anchor point
- c) Using calibrated rams / jacks with a suitable test jig

After load testing, re-examine Padeye / Clamp and support steel work to ensure that load testing has not exposed any defects.

All weldments on the Padeye / Clamp should be subjected to N.D.T.

If all satisfactory, hard stamp with Identification Number and S.W.L. and issue certificate.

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SWL:			C = Condemned

[illegible]

Signed:	Date:	Engineer:

Satisfactory	Return for repair



7.2.26 Load Cells

Check List

Load Cell Shackles

- 1) Examine body of shackle and check for wear / stretch
- 2) Examine pin and bobbin and check for bending
- 3) Ensure electrical plugs and sockets are clean, free from contamination and dry
- 4) Examine supply cable to remote readout indicator
- 5) Examine indicator and function check making sure the unit tares at zero

Test Clocks

- 1) Where shackles are fitted, remove and examine pin and body checking for wear / stretch
- 2) Examine upper and lower lifting / suspension eyes checking for wear / distortion
- 3) Ensure swivel functions correctly and check for play on thrust bearing
- 4) Examine the body of the clock checking for impact damage
- 5) Examine seals and ensure they are not perished or otherwise damaged
- 6) Examine clock face and ensure the needle returns to zero when not under load

General

Ensure the calibration certificate is current i.e. no more than 12 months old.

Where possible, function test the load cell / test clock by suspending from it, a known weight.

If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection.

7.2.27 Wire Rope

Check List

Examination Procedure

- 1) Where possible, sight the original test certificate and obtain the date the rope went into service.
- 2) Visually examine the wire along its entire length and check for:
 - a) Wear
 - b) Corrosion
 - c) Abrasion
 - d) Mechanical Damage
 - e) Broken Wires
- 3) Measure the rope dia. at regular intervals along its length. Any reduction from the nominal diameter could be an indicator to:
 - a) The amount of wear on the outer wires
 - b) Stretch due to overload; or
 - c) Partial collapse of the core

Should reduced diameters be discovered, the positions are to be tagged / marked for further internal inspection.
- 4) Using two specially designed clamps, open up the lay of the rope (at the locations mentioned above) to expose the core of the rope. This can only be carried out when there is no tension on the rope
- 5) Examine and check for the state of internal lubrication, the degree of corrosion, internal fretting and broken wires



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- 6) Replace any lubricant removed during examination and reverse the contra rotation of the clamps to re-bed the outer strands onto the core
- 7) Examine rope terminations and pay special attention to the rope adjacent to the terminations as dissimilar metals in a saliniferous environment can deteriorate due to the electrolytic process
(This is more critical with standing ropes such as crane pendants etc. and ropes immersed in salt water or used Sub-Sea).

Note: The **Department of Energy** recommends that the running ropes on a crane be renewed / changed out on an annual basis. Should the rope have been in service for some time, it may be advisable to have a sample of the rope destructed to ensure there is no loss of strength.

- 8) When examining rotation resistant wire such as whip lines, it is not possible to examine the core as previously stated. It is therefore advisable to obtain a sample from the "used" section of the rope and unwrap the alternative layers to expose the core for examination. It is to be borne in mind that with ropes continually running over pulleys (i.e. flexing), should the internal lubrication be inadequate, deterioration of the core can be rapid and it shows no exterior evidence. If the rope is suspect, once again, a test sample can be destructed to verify the ropes "integrity"

Important

- 9) Replace any lubricant removed during inspection.

If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection.

7.2.28 Wire Rope Slings

Examination Procedure

- 1) Examine each individual leg along its entire length and check for:
 - a) Wear
 - b) Corrosion
 - c) Abrasion
 - d) Mechanical Damage
 - e) Broken Wires
 - f) Internal Deterioration*

*Internal deterioration should be checked for by using two specially designed clamps to open up the rope to expose the wire / fibre core.

- 2) Measure the rope diameter as any reduction from original size could indicate either amount of wear on outer wires, partial collapse of the core or stretch due to overload
- 3) Examine each ferrule and ensure the correct size of ferrule has been fitted
- 4) Check that the end of the loop does not terminate inside the ferrule (i.e. the rope end should protrude slightly but no more than 1/3rd of the dia.) unless the ferrule is of the longer tapered design which has an internal step
- 5) The ferrule should be free from cracks or other deformities
- 6) Examine each thimble and check for correct fitting, snagging damage and elongation. (Stretched thimbles / eyes could indicate possible overload)
- 7) Examine wire rope around thimbles as it is often abraded due to sling being dragged over rough surfaces
- 8) If fitted, examine master link / quadruple assembly and check for wear, corrosion and cracking



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If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection.

7.2.29 Chain Slings

Examination Procedure

- 1) Sight the original test certificate and obtain the date the sling went into service, chain diameter and length from bearing to bearing.
(measure chain length to check for stretch)
- 2) Lay out chain sling on floor, remove all twists from legs.
- 3) Match up legs to check for signs of stretch / deformation in individual legs.
- 4) Visually examine the chain sling along its entire length and check for:
 - a) Distortion in chain links (nicks, bends, twists, elongated links)
 - b) Wear between chain links
 - c) Wear between chain links and load pins
 - d) Heat damage
 - e) Chemical damage
 - f) Wear, stretch, twisting on end terminations
 - g) Function of safety catches (where fitted)
 - h) Security of Load pins
 - i) Correct identification number and SWL

If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection.

7.2.30 Man-made Fibre Slings

Check List

Examination Procedure

- 1) Sight the original test certificate and obtain the date the sling went into service and its original length. (Measure the length to check for stretch).
- 2) Visually examine web sling along its entire length and check for:
 - a) Cuts or tears
 - b) Burst stitching (especially around the eyes)
 - c) Chemical damage
 - d) Heat damage
 - e) Ingress of foreign bodies into fibres
 - f) Distortion/wear in metal eyes (where fitted)
- 3) Ensure S.W.L. is clearly marked and identification tag is still attached
- 4) If any faults are found the web sling should be condemned and rendered unusable

If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection.

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7.2.31 Shackles

Check List

Examination Procedure

- 1) Remove shackle pin and examine for wear, deformation and cracking
- 2) Ensure it is the correct pin for the shackle. (i.e. NOT a higher tensile pin in an alloy shackle)
- 3) Check pin threads for wear / deformation
- 4) Examine shackle body and check for wear in the crown and pin holes, deformation and cracking
- 5) In the case of safety pin shackles, ensure split pins are fitted

If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection.

7.2.32 Eyebolts / Eyenuts

Check List

Examination Procedure

- 1) Clean up threads and wire brush if necessary
(If eyebolts / eyenuts are unused, remove protective tape / plug to allow inspection of threads)
- 2) Examine threads and check for wear, stretch or impact damage. The threads must be complete (no broken threads) and full (no flats on top)
The thread should be concentric and fit neatly in a standard nut/on a standard bolt. If stretch is suspected, a thread gauge should be used to confirm condition
- 3) Examine eye of bolt / nut and check for wear / stretch / distortion and look for hairline cracks at the crown of the ring
- 4) Check squareness of shank / screwed hole against shoulder

If all satisfactory, record relevant details and issue report.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection.

7.2.33 Rigging Screws

Check List

Examination Procedure

- 1) Unscrew both ends of rigging screw to separate the three main components
- 2) Clean and examine the threads and check for wear, stretch or impact damage
- 3) Examine the eyes / hooks and check for wear / stretch
- 4) (Jaw type - Remove clevis pin and examine pin and pinholes for wear/stretch and distortion)
- 5) Re-grease the threads and re-assemble
- 6) Ensure the male threads are fully through the female threads

If all satisfactory, record relevant details and issue report.



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This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection.

7.2.34 Open Wedge Sockets

Check List

Examination Procedure

- 1) Remove main load pin and wedge
- 2) Examine load pin and check for wear / distortion
- 3) Examine wedge and check for wear / distortion / cracking
(Ensure it is the correct wedge for the rope dia. marked on the socket)
- 4) Remove any coating on the socket to expose bare metal
- 5) Examine socket and check for wear /elongation of pin holes and cracking
- 6) Carry out M.P.I. and re-assemble
If all satisfactory, record relevant details and issue reports
- 7) Repaint

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection.

7.2.35 Personnel and Goods Lift

The legal responsibility for ensuring that lifts are properly maintained rests upon either the operator or the owner of the installation in which the appliances are installed.

This procedure is intended to remind persons responsible for making arrangements for the thorough examination and competent persons carrying out the thorough examination of the need to be aware of the criteria of the current standards.

It is important that the competent person engaged on this work should be familiar with the contents of the standards and be able to apply his knowledge.

The appliance must be load tested before being put into use after installation or after substantial alteration/repair and subject thereto, at regular intervals (6 months maximum) for thorough examination.

The purpose of thorough examination and testing is to determine whether the condition of the lift after installation or repairs is such that it can continue to be used safely.

Inspection Criteria

1) Landing and Car Door Interlocks

It should be verified that these devices are in efficient working order and in full compliance with statutory requirements. If the lock is provided with a transparent cover it may be possible to check that it functions correctly without removing the cover. Where the cover is not transparent it will be necessary to remove it or partially dismantle the lock, to confirm its satisfactory condition. The competent person should decide on the frequency at which the interior of the lock should be exposed; however it is recommended that this should be done at intervals not exceeding 12 months.

2) Worm and other Gearing

These mechanisms should be checked for backlash, end float, gear noise, etc. and it may be possible to carry this out without the need for dismantling. If there is evidence of excessive wear it may be necessary to remove gear case cover or otherwise expose the working parts, to permit further examination. It is recommended that such exposure be carried out at intervals not exceeding 10 years and perhaps most conveniently when re-roping is being carried out. Unless considered essential, more frequent dismantling should be avoided.



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3) Main Drive System components

a) Shafts, Plain Bearings and Pulleys

Unless there is evidence to indicate that attention is required at more frequent intervals, it is recommended that when these components are not readily accessible, they be exposed for examination at intervals not exceeding 10 years and also perhaps when re-roping is being carried out. Particular attention should be given to shouldered shafts which are likely to have points of high stress concentration where the change of diameter occurs.

(b) Roller, Ball and Needle Bearings

Exposure of this type of bearings for examination should be undertaken only when evidence obtained from other methods of examination indicates that a more detailed assessment is required.

4) Governors

The over-speed governor, if fitted, should be checked at least every 5 years to ensure correct operation at the appropriate tripping speeds and on every occasion when the over-speed governor has been subject to any repairs which may affect its operation. It should also be checked during renewal of the governor rope.

5) Safety Gear

a) If safety gear has been fully tested and certified before hand-over as specified in British Standard 2655: Part 7: 1970, then a full load speed test should be required only after a major overhaul involving a change of rated load or rated speed, or the disturbance of the car sling and/or safety gear assembly. In the absence of a proof of test, consideration should be given for a full load speed test to be carried out. If the lift has been installed in accordance with British Standard 5655: Part 1: 1979 then the tests should be carried out as specified therein.

b) Linkages and all moving parts of safety gear should be checked at every thorough examination for free and effective operation and for any signs of deterioration and wear. At 5 yearly intervals, or whenever the safety gear rope (if fitted) is renewed, there should be a test in which the safety gears is engaged by hand traction of the tripping rope or lever, or by manual tripping of the governor (if fitted) with empty car descending slowly, until it has been demonstrated that the safety gear has engaged correctly on the guides. The test should be carried out with the car at a convenient height above pit floor level.

c) Where the safety gear complies with British Standard 2655: Part 1: 1970 or British Standard 5655: Part 1: 1979, the car should be moved upwards after the test, sufficiently to demonstrate that the safety gear disengages and so that the pattern of engagement may be examined. The means of moving the car should be as described in the above standards for emergency operation.

6) Suspension Ropes

Typical factors creating visible change in rope condition are the number of broken wires and their position, surface wear, excessive stretching, inequality in rope tension, inequality in diameter and internal conditions such as internal corrosion. Severely corroded ropes cannot be restored to serviceable condition by a belated application of lubricant. All factors affecting rope condition must be taken into account and experience and competence should decide the degree



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of wear that can be tolerated and the stage at which replacement becomes necessary.

7) Broken Wires

The following Table is intended as a guide to the number and position of broken wires (splintering) which, together with other influencing factors, may be used to determine when ropes may be considered to be fit for further service or when immediate replacement is necessary. It indicates the maximum number of broken wires in the worst section of the rope that can be allowed, taken over a length of one complete turn or lay of six diameters.

Condition of Rope and Action Required

	Wearing but Serviceable		Report for ordering and replacement		Replace Immediately	
Rope Construction	6 x 19	8 x 19	6 x 19	8 x 19	6 x 19	8 x 19
Broken wires equally distributed around the strands	Less than 12	Less than 15	12 or more	15 or more	24 or more	30 or more
Broken wires predominate in 1 or 2 strands	Less than 6	Less than 8	6 or more	8 or more	8 or more	10 or more

8) Wear and Age

There are factors which should be given particular attention dependent upon the knowledge of site conditions and type of drive

- In the case of traction drives with "U" or undercut grooves, a reduction of more than 6%, and in case of "V" grooves more than 15%, on the nominal diameter of the rope, is a guide for rejection.
- The life of a suspension rope is dependent on many factors after it has been correctly installed but it can be generally stated that the greatest single factor affecting rope life is the number of bends made by the rope and also the radius of such bending. It will be appreciated that no definite guide as to the life of a suspension rope can be given, but particular care should be exercised where the ropes are more than 10 years old.
- If unusual features are present indicating the possibility of failure, replacement must be considered. Some examples are:
 - ropes exuding a red or rouge dust
 - a local reduction in diameter or similar deformation

9) Lubrication

Ropes are lubricated internally during manufacture and this is usually sufficient to prevent corrosion. The presence of the correct degree of lubrication can best be judged by a slightly oily deposit on the traction sheave grooves, but if the machine should be operating in a contaminated atmosphere, this may prove inadequate and it may be necessary to dress the rope with a non-acid lubricant to inhibit corrosion. Excessive lubrication should be avoided, otherwise slip may occur between the ropes and the traction sheave.



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10) Rope Termination

Rope terminations or anchorages should be fitted in accordance with the manufacturer's instructions and should comply where applicable with the appropriate British Standard. At every thorough examination they should be closely examined for signs of deterioration or wear. The fitment of detachable terminations such as wedge sockets and grips should be checked for security and it should be noted that broken wires can occur at these terminations.

Pre-examination Procedure

Prior to any maintenance the engineer should satisfy himself with the following:

- 1) Has the owner's representative been advised of your presence on site?
- 2) Is the owners representative satisfied with the performance of equipment?
- 3) Does the owner's representative have any complaints regarding equipment?
- 4) Are all lubrication points adequately lubricated and clean?
- 5) Are all oil reservoirs clean and filled to correct level?
- 6) Is equipment functioning as designed?
- 7) Is equipment adequately clean and work areas adequately illuminated?
- 8) Do all safety features operate satisfactorily?
- 9) Are all parts of equipment correctly adjusted and operational?
- 10) Is equipment safe in operation?
- 11) Are main suspension ropes, governor ropes, compensating ropes serviceable?
- 12) Is balance weight overrun clearance adequate?
- 13) Does fire control switch function correctly?
- 14) Have all floor levels been checked for accuracy?
- 15) Does emergency alarm operate?
- 16) Do all indicators operate correctly?
- 17) Do all push buttons operate correctly?
- 18) Do all mechanical door locks function correctly?
- 19) Do all interlocks, limits and slack rope switches operate correctly?

Visual Examination (Quarterly)

- 1) As per section 2
- 2) Strip down lift brake and check brake linings for wear etc.
- 3) Strip down door operator brake and mechanical linkages and check for wear on bearings etc.
- 4) Manually operate safety gear mechanism under lift car and adjust if necessary
- 5) Check condition of all motor bush gear and brushes and adjust or replace if necessary
- 6) Inspect interiors of all door locks and shaft limit switches and adjust if necessary
- 7) Inspect entire length of suspension ropes and anchorage points and report any defects
- 8) Check all overload devices operate as designed. Adjust if necessary
- 9) Inspect condition of all control panel equipment - i.e.
 - a) Relays
 - b) Contactors
 - c) Solenoids
 - d) Electrical Terminations
 - e) Phase Failure and Reversal Unit
 - f) Protective overload equipment
- 10) Check gearbox for wear / or overheating of bearing, bushings etc.



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- 11) Overrun lift and check distance travelled is adequate to prevent lift running onto buffers before emergency stop is induced by final limit switches at terminal floors
- 12) Inspect all traction and deflector sheaves for signs of wear
- 13) Inspect roller guide shoes and / or guide shoe slides for wear
- 14) Inspect electric travelling cables in lift shaft for signs of damage or worn insulation

Visual Examination (Six Monthly)

- 1) As per section 2 and 3
- 2) Carry out actual over-speed governor test
- 3) Single phase lift motor to be checked for operation of protection equipment
- 4) Drain gearbox oil sump and replenish oil after flushing
- 5) Check all motor / generator electrical terminations for tightness
- 6) Carry out stall test on door operator motor
- 7) Overhaul floor selector and adjust if necessary

Visual Examination (Yearly)

- 1) As per section 2,3, & 4
- 2) Check tightness of guide rail clips and fixing brackets
- 3) Check guide rails are parallel. Adjust if necessary
- 4) Check lift car is secure within frame. Tighten if necessary
- 5) Check distances between car sill and landing sills are equal and consistent at every floor level
- 6) Check all guide shoe fixing brackets are tight. Tighten if necessary
- 7) Check all landing doorframes are tight. Adjust as necessary
- 8) Carry out earth continuity tests throughout installation

Load Testing

1) Application of Loads

- a) After the thorough examination, a test load equivalent to the maximum SWL shall be applied and evenly distributed on car floor. Car with weight should be travelled throughout hoisting distance, stopping at each floor / deck level whilst hoist and structure are kept under observation
- b) The above procedure shall then be repeated on the descent
- c) Repeat test procedure 6.1(a) above except that a proof load shall be applied equal to SWL plus 25%

2) Inspection and Examination

- a) Following the application of the proof load the lift shall be thoroughly examined in order to ascertain that it has withstood the proof load without injury or permanent deformation.

3) Certification

It is the duty of the certifier to satisfy himself that the criteria specified in BS 2655/5655/ PD 6500 are in order and that all other requirements of this standards are complied with before issuing the relevant report and/or certificate.

If all satisfactory, record relevant details and issue report and / or certificate.

This procedure is to be followed as a guideline to **QatarEnergy-IS minimum** requirements for an inspection and / or proof load test.

Upon Completion

Make everything safe and tidy. Check all tools are cleared from the site.



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Personnel And Goods Lifts - Check List

- 1) Owner:
- 2) Location:
- 3) Model:
- 4) Inspection Engineer:
- 5) Date of Inspection:
- 6) Date of Test:
- 7) Current Cert. No:
- 8) Name and Address of Maker of Lift:
- 9) Type of Lift And Nature of Power:
- 10) Date of Manufacture of Lift:
- 11) Identification:
 - a) Maker's Serial Number:
 - b) Owner's Distinguishing Mark or Number:
- 12) Date of Last Previous Test of Lift:

Date of Last Previous Thorough Examination of Lift:



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Components	Satisfactory	Requires Attention	Requires Renewal
Interlocks			
Landing			
Car Door			
Gearbox			
Gears			
Lubrication			
Main Drive System			
Shafts			
Bearings			
Pulleys			
Governors			
Limits			
Safety Gear			
Suspension Ropes			
Terminations			
Anchorage			
Sockets			
Suspension Ropes			
Chain			
Chain Wheel			
Sprockets			
Overload Devices			
Brakes			
Linings			
Control Panel			
Relays			
Contactors			
Solenoids			
Elec. Terminations			
Failure Unit			
Limit Switches			
Traveling Cables			



7.3 Drilling Hoisting Equipment Visual Inspection and Thorough Examination

Introduction

This section of the LECOP is for the use of personnel employed directly or indirectly in drilling activities.

The Contractor has the responsibility of ensuring that the range of hoisting tools, as specified in the scope of this document, are supplied, maintained, examined, certified and operated in accordance with **QatarEnergy-IS** LECOP. This responsibility relates to all such drilling hoisting equipment when used on **QatarEnergy-IS** contracted Drilling Rigs.

These controls and procedures are based principally on the requirements of the appropriate standards and recommendations of API (e.g. API Spec 4E, 7, 7K and 8C and API RP 7G (ISO 1047) 8B and 9B, which are considered to be the most authoritative and useful.

All the drilling hoisting equipment currently in service shall comply strictly with the requirements of these documents (and API RP-8B) concerning Visual Inspection, Thorough Examination and Maintenance.

This section sets out the requirements for visual inspection and thorough examination of drilling and production hoisting tools in **QatarEnergy-IS** operations. The inspection terminology used throughout these documents shall mean the following:

Visual Inspection (of hoisting equipment): Full check that relevant documentation is current plus a visual check to ascertain whether the item shows any obvious signs of unsuitability for continued use in service (e.g. obvious signs of cracks, loose fits, elongation of parts, deformation of parts, etc.), plus appropriate confirmatory documentation of pass or failure of the Visual Inspection. Frequency of Visual Inspection: six-monthly; except where a Thorough Examination of an item is scheduled.

Thorough Examination (of hoisting equipment): Full check that relevant documentation is current plus full dismantling of the item (i.e. to permit 100% examination of all load bearing parts and parts which are critical to the operation of the equipment) and requisite examination and testing for suitability to continue in service, plus appropriate confirmatory documentation of pass or failure of the Thorough Examination.

Frequency of Thorough Examination: - as stated in Table 1.

Scope

The requirement for visual inspection and thorough examination will apply to the following range of hoisting equipment:

- Elevator Links
- Tension Members of Subsea Handling Equipment
- BOP Handling Equipment
- Riser and Conductor Handling Tools
- Tubing Hooks
- Spiders Mechanical and Air Powered Slips
- Kelly Bushings
- Casing, Tubing and Drill Pipe Elevators
- Single Joint Elevators



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Circulating Elevators
 Swivel Bail Adapters
 Rotary Slips
 Manual Tongs (including Lines and Posts)
 Power Tongs (including Iron Roughneck and Top Drive Wrenches)
 Safety Clamps
 Travelling Block Dollies
 Deadline Tie-Downs (Anchors)
 Drawworks (Blockline Clamps and Brake Linings)
 Kelly Assembly Connections (including Kelly Driver)
 Top Drive Rotary Connections
 Lifting Caps, Plugs, clamps, hooks and Subs
 Rotary Swivels (including Hoses)
 Kelly Spinners (used as tension member)
 Connectors and Link Adapters
 Power Swivels and Power Subs
 Overhead Drilling Systems (Top Drive System)
 Master Bushings
 Derrick Structure (including Dolly Track)
 Block to Hook Adapters
 Drilling Hooks
 Crown Blocks
 Travelling Blocks
 Rotary Tables
 Drawworks (Clutches and Brakes)
 Hoisting equipment not included in the list (e.g. Racking Arms, Manipulator Arms, Finger Boards, etc.) which have a primary purpose of connecting a load to a lifting appliance shall also meet the requirements of these controls and procedures.

Guidelines for Visual Inspection

- 1) Each item of hoisting equipment shall be checked for its unique identification number and capacity / load rating. The item's test certificate and if appropriate, proof load certificate or most recent certificate of thorough examination, shall be scrutinised and the information checked against the information stamped on the equipment.
- 2) The contractor shall acknowledge that existing equipment may not all have the appropriate certification required by **QatarEnergy-IS** lecop, but that this shall not necessarily be justification to reject equipment.
- 3) Each item of hoisting equipment which has been visual inspected and considered suitable for continuing in service, and has the appropriate current test certificates, shall be colour coded.
- 4) A certificate of visual inspection, shall be issued by the inspection contractor for each item of hoisting equipment inspected on the drilling rig.
- 5) Each certificate will state in detail, the method of inspection used.
- 6) When an item of hoisting equipment is found to be defective, it shall be clearly labelled as unserviceable and have a report attached to the item.
- 7) The inspection contractor shall, as part of this contract, supply suitable labels to clearly identify unserviceable equipment.
- 8) A copy of each report relating to hoisting equipment which cannot continue in service until repaired, shall be submitted to the contractors drilling supervisor



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who shall sign each report and make the appropriate entry in the drilling hoisting register, list of rejected equipment.

- 9) The contractor shall prepare a list of all items of hoisting equipment which have been found defective or for which certification documentation was incomplete. This list shall identify the equipment which has been removed from service and that which has to be removed from service by the platform / modu personnel at the earliest opportunity and shipped onshore.
- 10) A copy of the defect list shall be given to the drilling contractor prior to the inspector leaving the worksite.

Guidelines for thorough examination

- 1) Each item of hoisting equipment shall be checked for a unique identification number and capacity / load rating. The item's test certificate and if appropriate, proof load certificate or most recent certificate of thorough examination (and where applicable, the most recent certificate of visual inspection), shall be examined and the information checked against the information stamped on the equipment.
- 2) The contractor shall acknowledge that existing equipment may not all have the appropriate certification required by API RP-8B , but that this shall not necessarily be justification to reject equipment.
- 3) Each item of equipment shall be disassembled as appropriate and subjected to MPI (see Magnetic Particle Inspection) of the high and medium stressed areas and given a Thorough Examination.
- 4) Each item of hoisting equipment which has been Thoroughly Examined and considered suitable for continuing in service, and has the appropriate current test certificates, shall be colour coded.
- 5) A Certificate of Thorough Examination shall be issued by the Examination Contractor for each item of hoisting equipment Thoroughly Examined on the drilling rig.
- 6) Each Certificate of Thorough Examination shall be accompanied by a detailed MPI report and diagram indicating the areas which have been Thoroughly Examined.
- 7) When an item of hoisting equipment is found to be defective it shall either be placed in a quarantine area and tagged as defective.
- 8) A copy of each report on hoisting equipment which cannot be safely used until repaired or which requires to be repaired within a specific period (not exceeding 28 days) shall be submitted to the Contractors Drilling Department who shall make an appropriate entry in the Drilling Hoisting Register, List of Rejected Equipment. Typed and signed certificates of items passing the Thorough Examination shall be submitted to the Contractors Drilling Department.
- 9) The Inspection Contractor shall also prepare a list of defective equipment and of equipment for which appropriate documentation was incomplete. The Inspection Contractor shall, as part of this contract, supply suitable labels to clearly identify unserviceable equipment.
- 10) The list raised for the above shall identify equipment which has been held in a quarantine area or removed from service.

NOTE: Refer to sketches on following pages for guidance on areas requiring NDT.



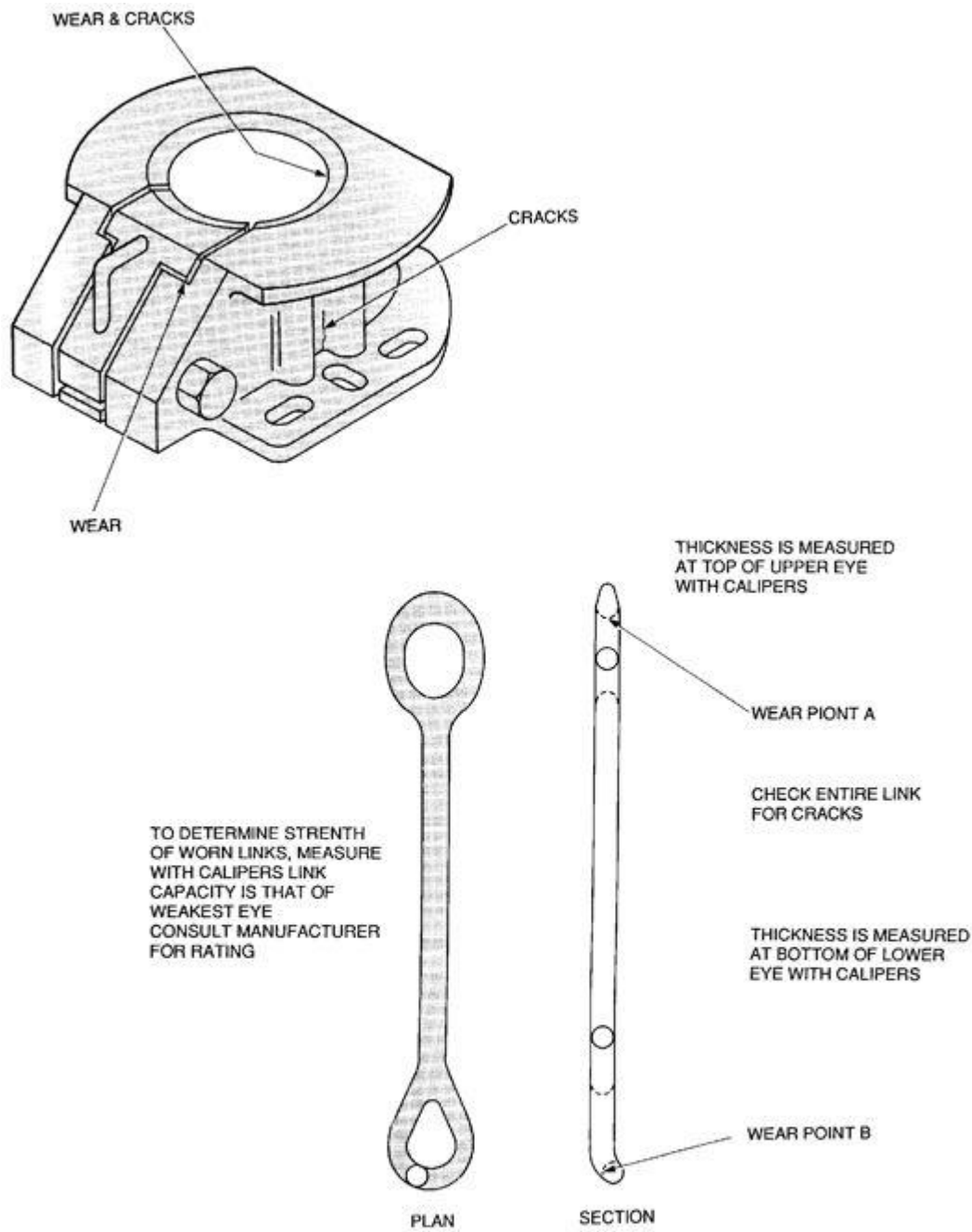
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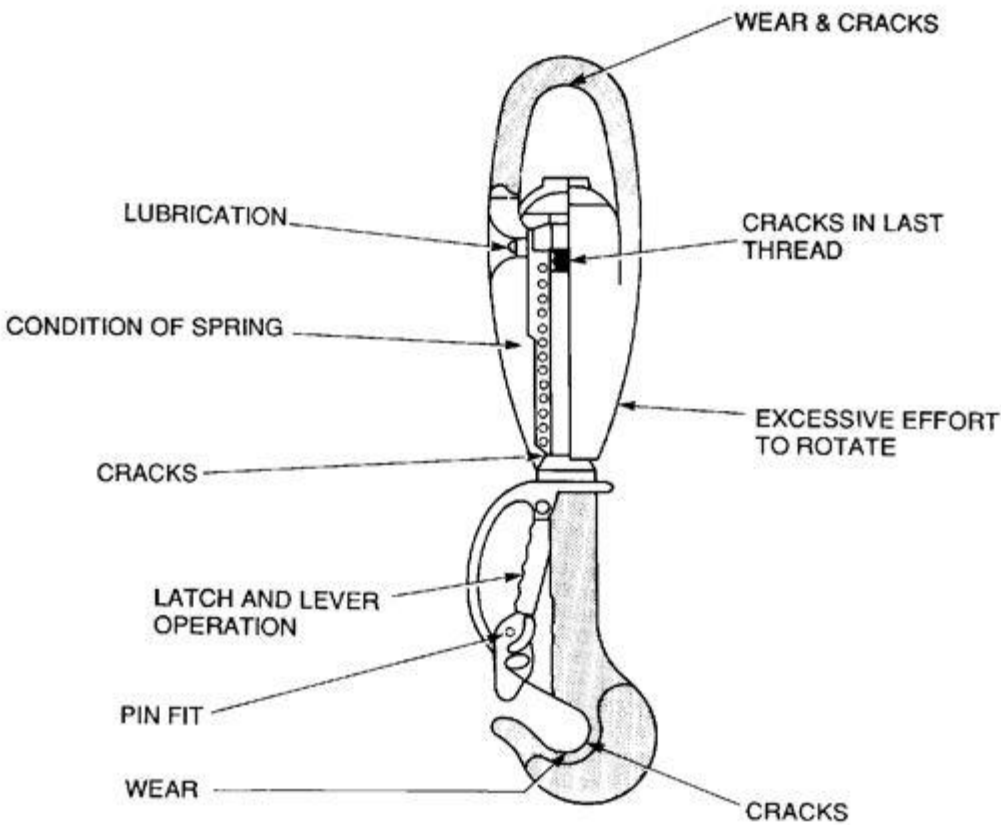
Table 1 Hoisting Equipment Frequency of Dismantling for Thorough Examination

Equipment	Thorough Examination			
	6 M	2 M	24 M	60 M
Elevator Link	X			
BOP Handling Equipment	X			
Riser and Conductor Handling Tools	X			
Tubing Hooks	X			
Spiders	X			
Mechanical and Air Powered Slips	X			
Kelly Bushings	X			
Casing, Tubing and Drill Pipe Elevators	X			
Single Joint Elevators	X			
Circulating Elevators	X			
Swivel Bail Adapters	X			
Rotary Slips	X			
Manual Tongs (incl. Lines and Posts)	X			
Power Tongs (incl. IR and Top Drive Wrenches)	X			
Safety Clamps	X			
Travelling Block Dollies	X			
Deadline Tie-Downs (Anchors)	X			
Drawworks (Blockline Clamps & Brake Linings)	X			
Kelly Assy Connections (incl. Kelly Drives)	X			
Top Drive Rotary Connections	X			
Lifting Caps, Plugs and Subs				
		X		
Rotary Swivels (incl. Hoses)		X		
Kelly Spinners (used as tension member)		X		
Connectors and Link Adapters		X		
Power Swivels and Power Subs		X		
Overhead Drilling Systems (Top Drive System)		X		
Master Bushings		X		
Derrick Structure (incl. Dolly Track)		X		
Block to Hook Adapters			X	
Drilling Hooks			X	
Crown Blocks				X
Travelling Blocks				X
Rotary Tables				X
Drawworks (Clutches and Brakes)				X



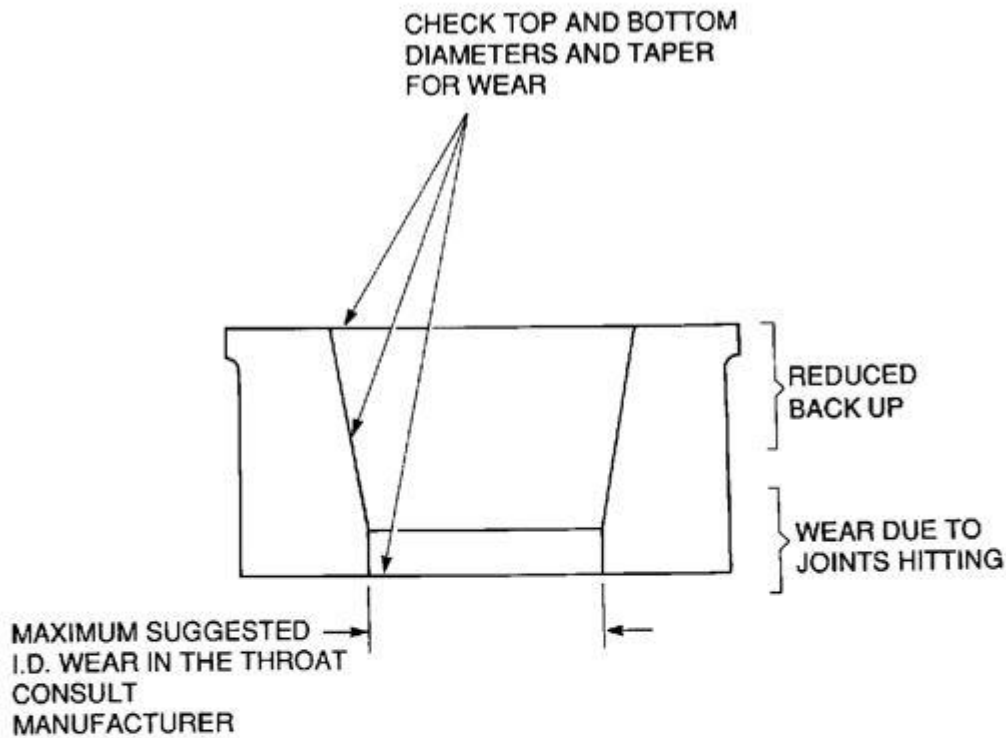
ALL PARTS TO BE EXAMINED BY
M.P.I.

FIG. 1 - Elevator Links



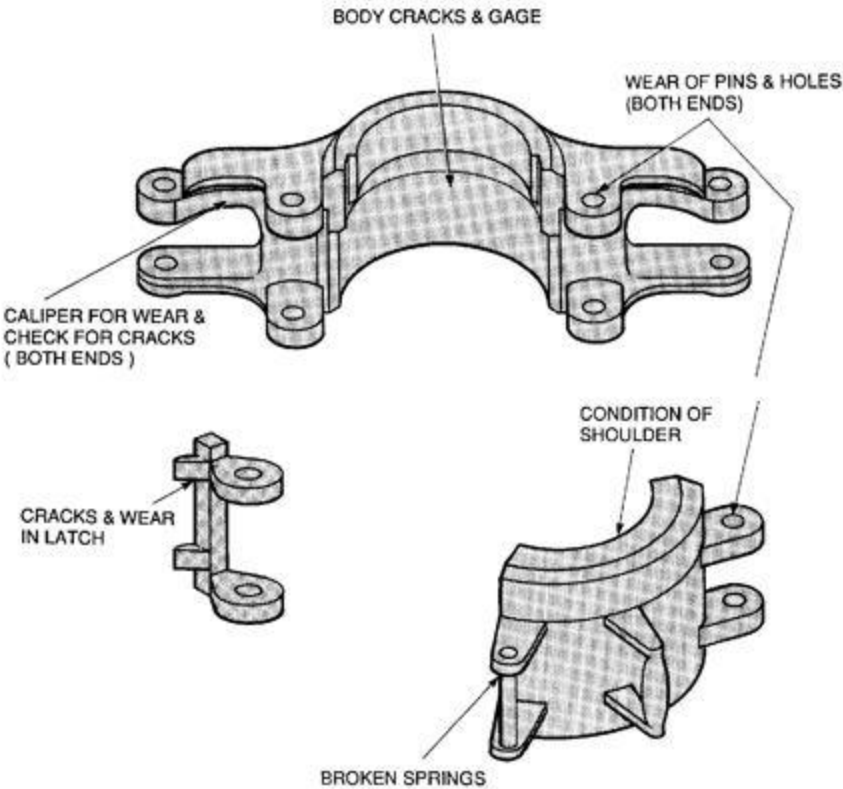
AREAS TO BE EXAMINED BY M.P.I.

FIG. 2 - Tubing Hook



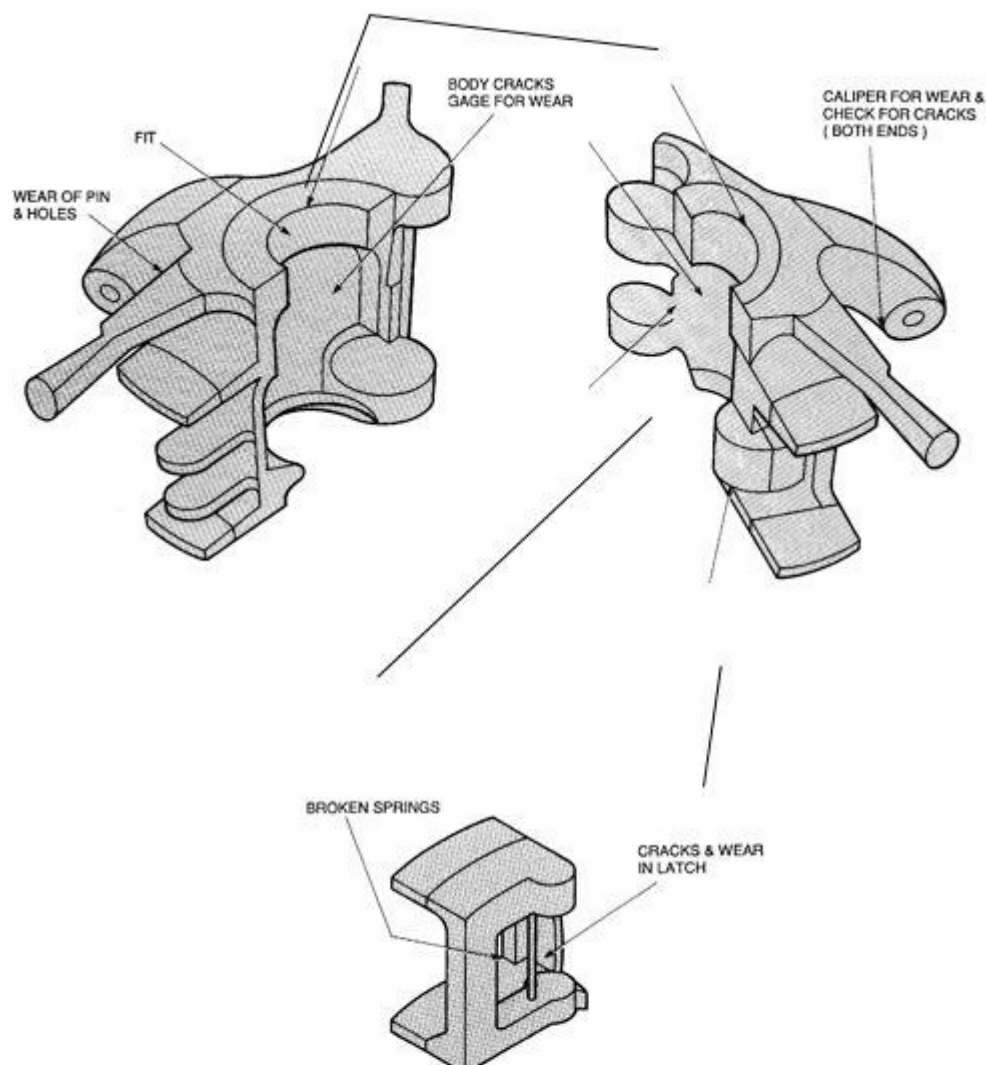
ALL PARTS TO BE EXAMINED BY
M.P.I.

FIG. 3 - Spider



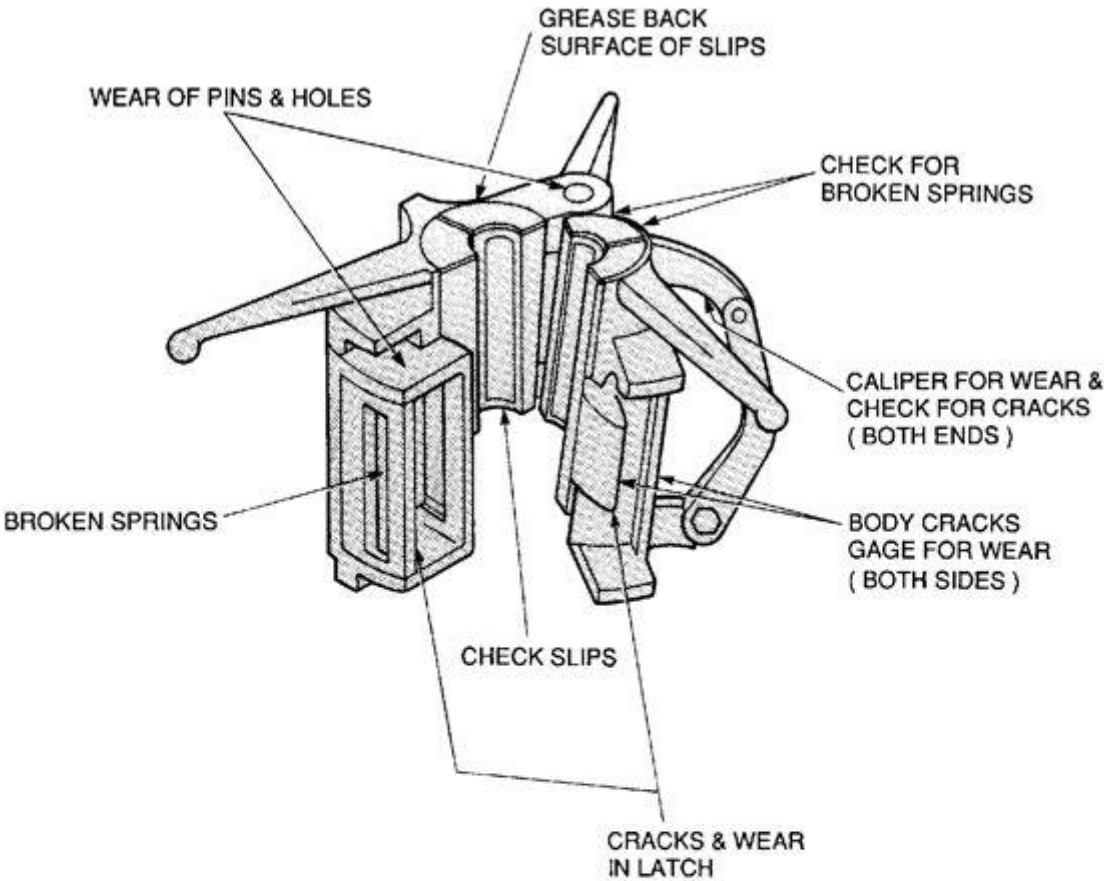
ALL PARTS TO BE EXAMINED BY M.P.I.

FIG. 4 - Side Door Elevator



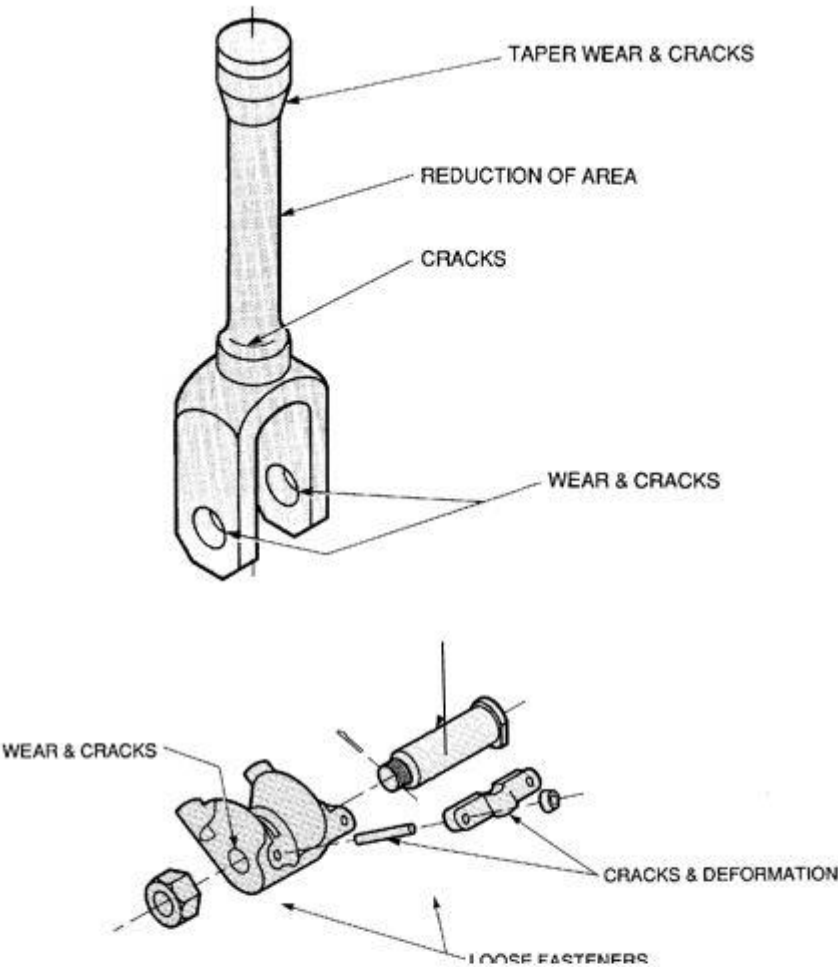
ALL PARTS TO BE EXAMINED BY M.P.I.

FIG. 5 - Centre Latch Elevator



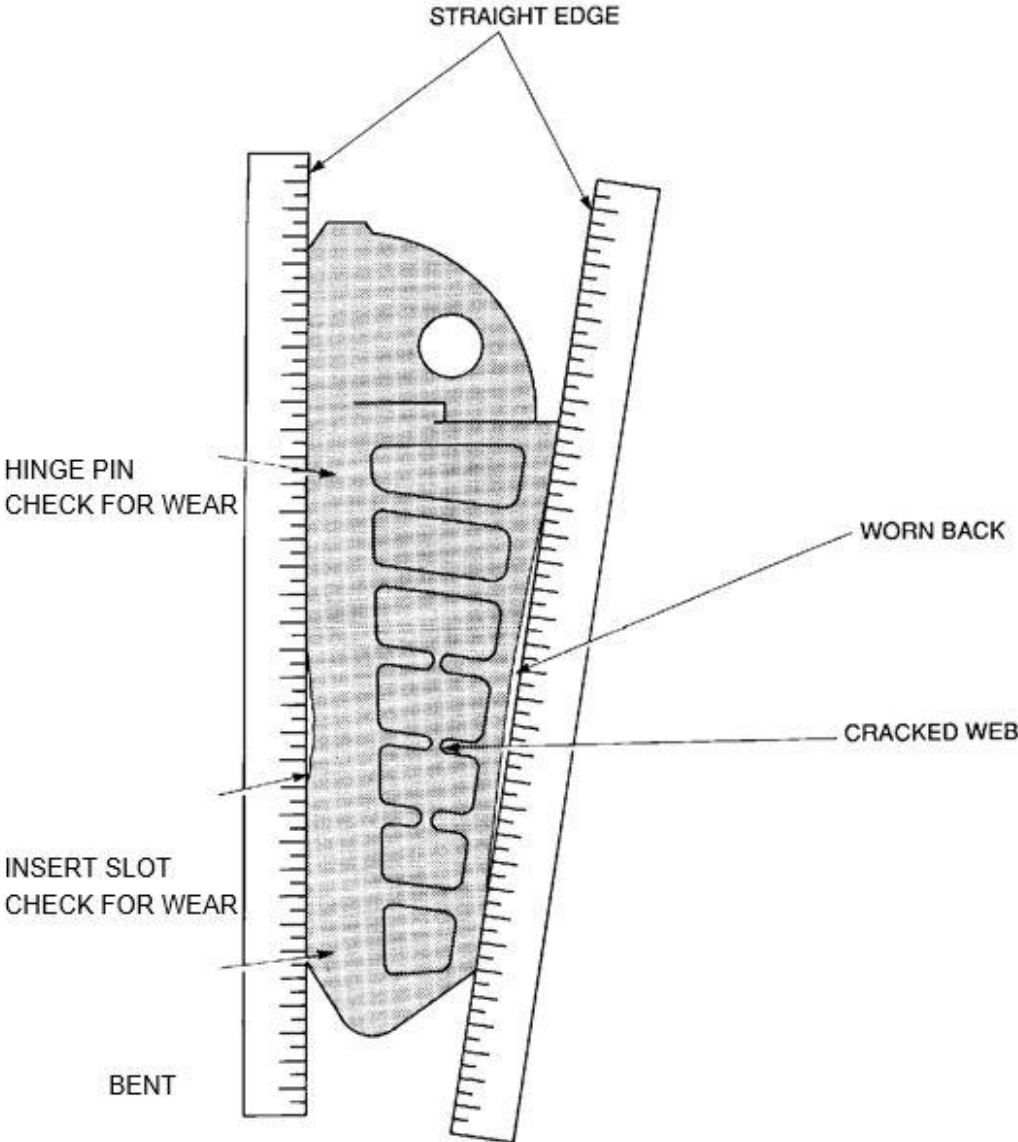
ALL PARTS TO BE EXAMINED BY M.P.I.

FIG. 6 - Slip Type Elevator



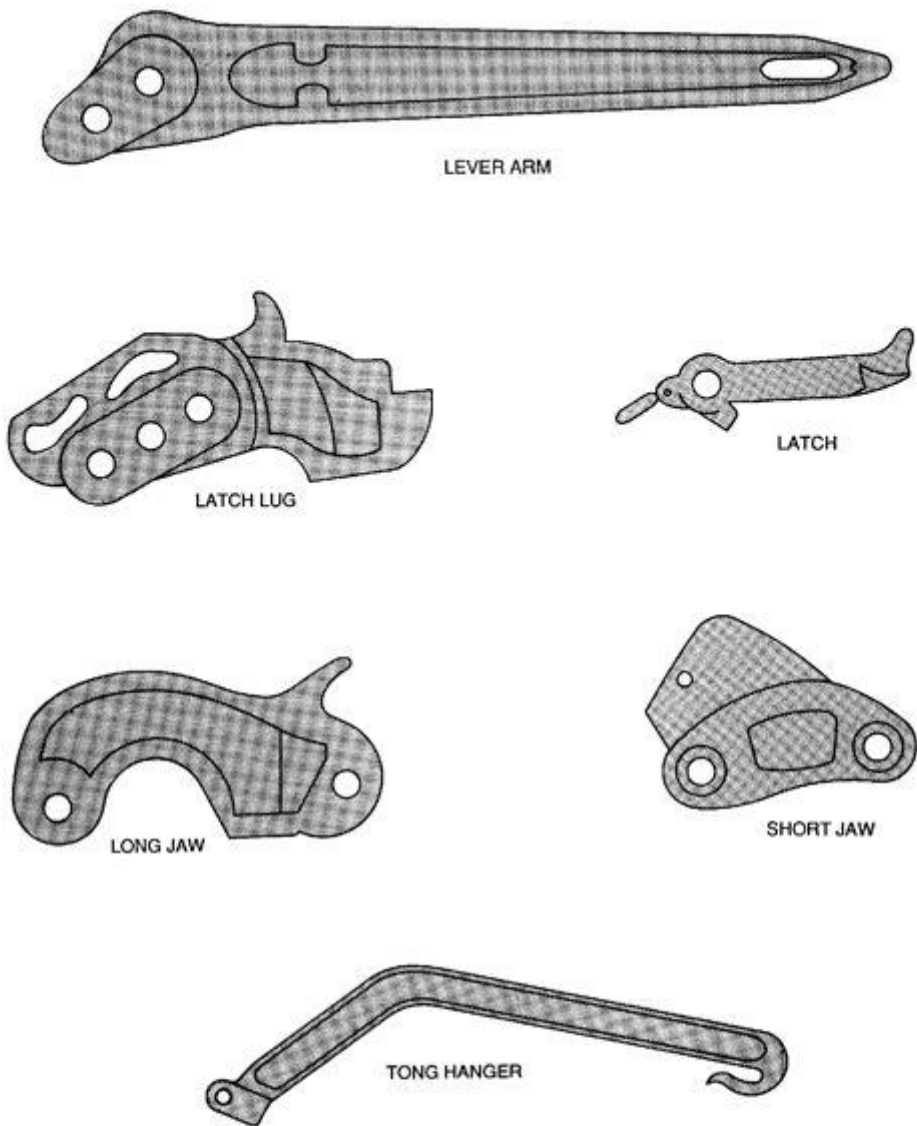
ALL PARTS TO BE EXAMINED BY M.P.I.

FIG. 7 - Swivel Bail Adapter



ALL PARTS TO BE EXAMINED BY M.P.I.

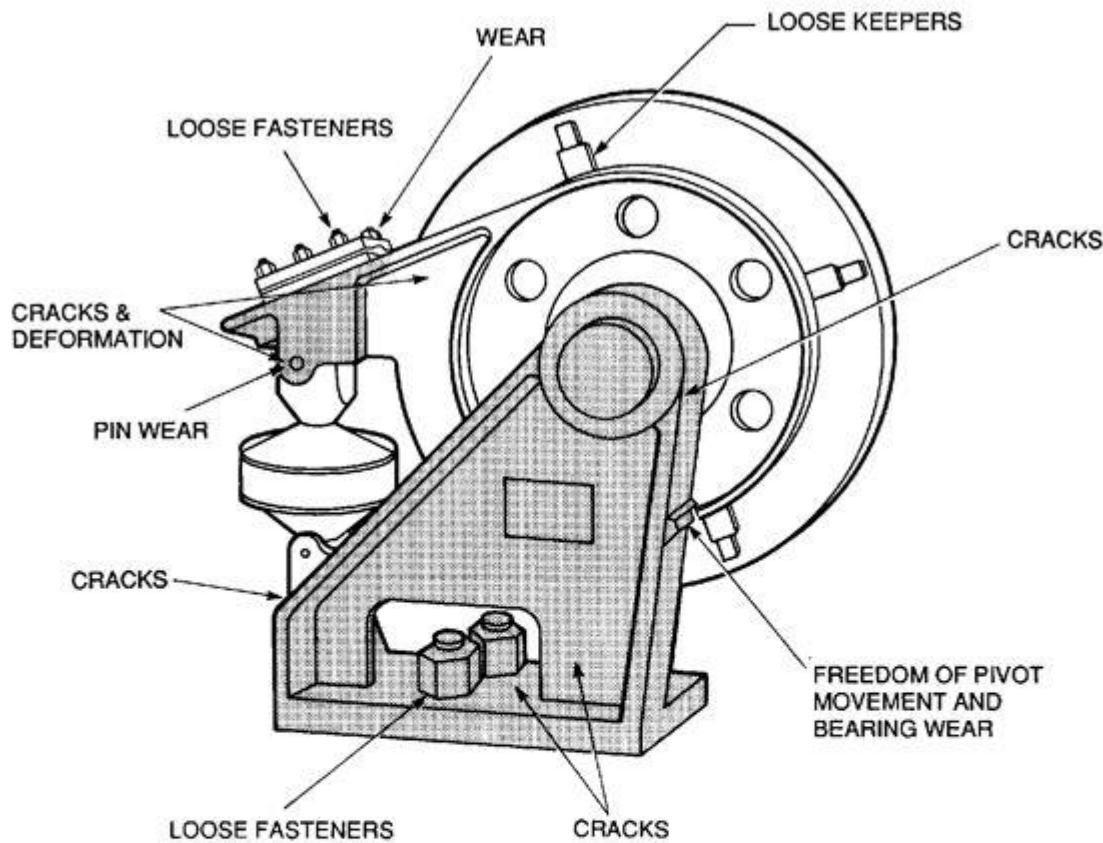
FIG. 8 - Rotary Slips



ALL PARTS TO BE EXAMINED BY M.P.I.

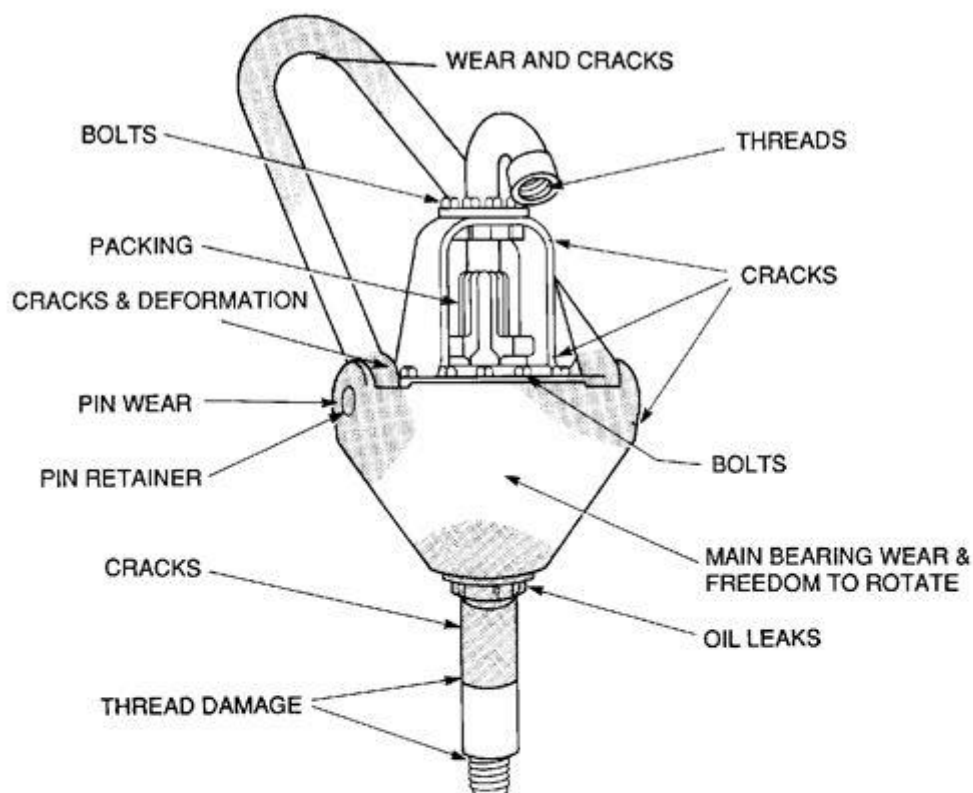
NOTE: Back up posts and lines shall be visually inspected.

FIG. 9 - Manual Tongs



AREAS TO BE EXAMINED BY M.P.I.

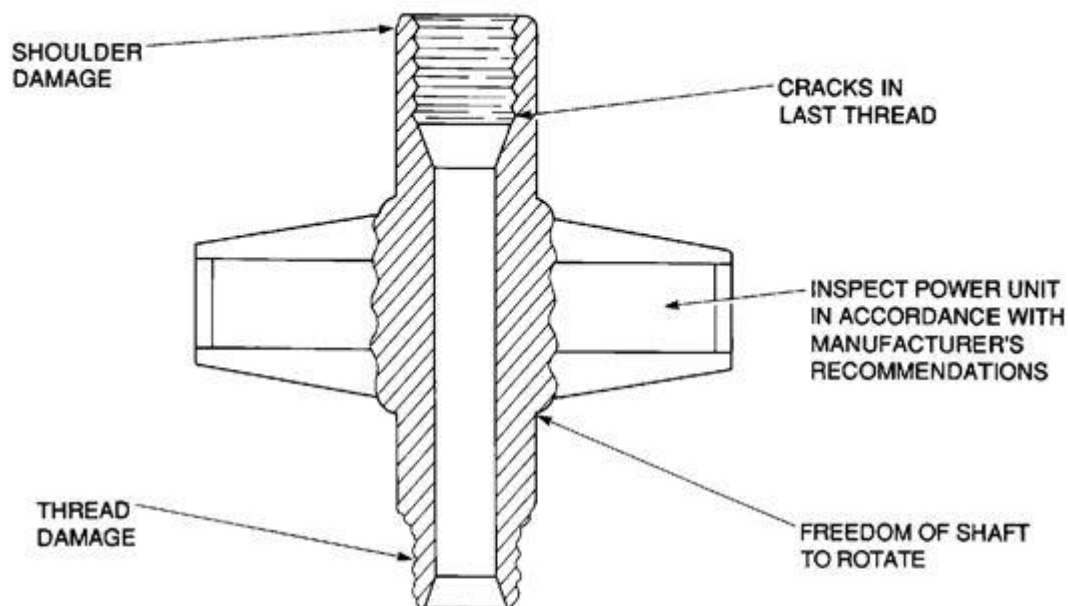
FIG. 10 - Deadline Tie-down (Anchor)



NOTE: Rotary hoses and associated couplings shall be Visually Inspected for damage and wear at least every 12 months.

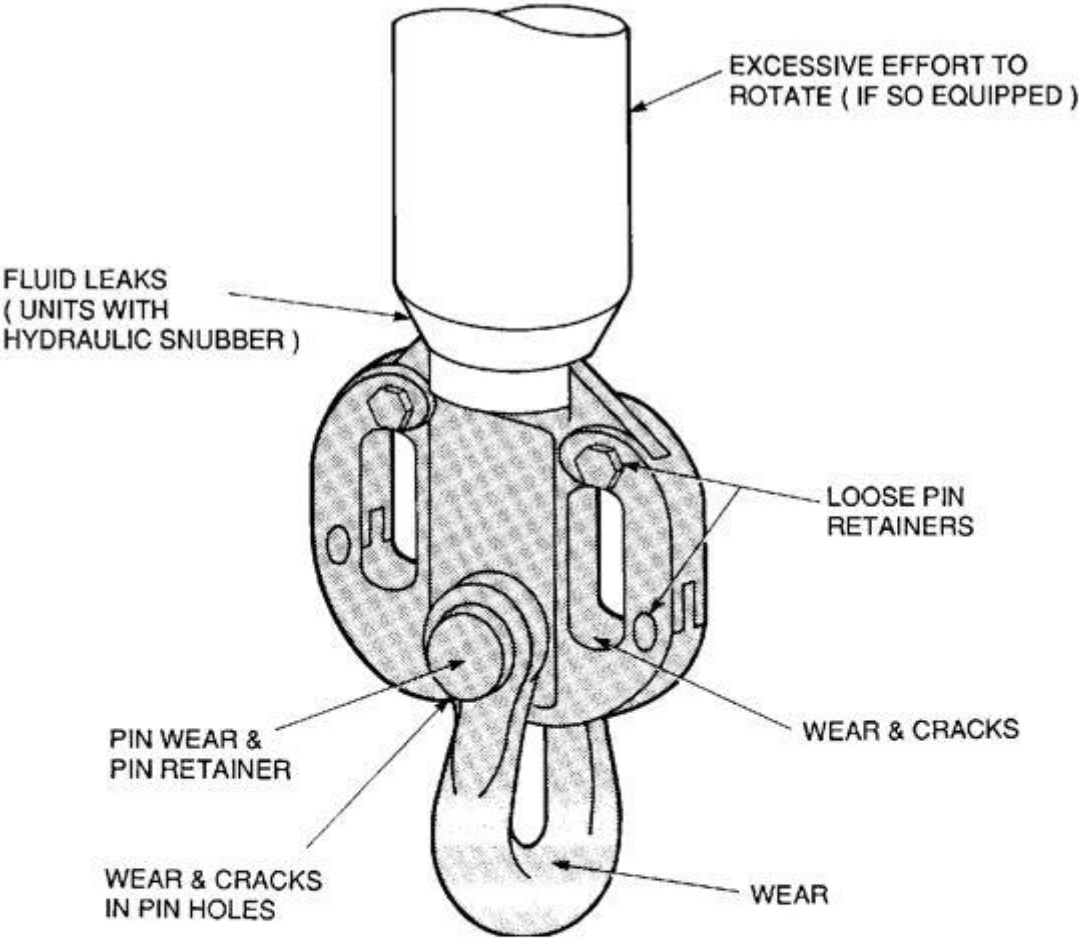
NOTE: Rotary connections shall be Thoroughly Examined every six months in accordance with API RP - 7G for tool joints.

FIG. 11 - Rotary Swivel



NOTE: Inspect pin and box in accordance with API RP - 7G for tool joints

FIG. 12 - Kelly Spinner



AREAS TO BE EXAMINED BY M.P.I.

FIG. 13 - Link Adapter



FIG. 14 - Overhead Drilling System (Top Drive System) Top Drive Systems and/or Kelly/Swivel Assemblies

General

Users of Top Drive Systems shall make themselves aware of the load carrying components. Generally all components of a TDS with rotary connections are load carrying, as are such components of a conventional kelly/swivel assembly.

Inspection

Top Drive designs vary among manufacturers and it is important that inspections be carried out following the recommendations laid down in the individual manufacturer's Service Manuals, together with the manufacturer's Technical or Service Bulletins.

Load carrying members should be visually checked for wear, cracks, flaws and deformation - at least every three (3) months.

In line with current practice for checking rotary connections of kelly assemblies every six (6) months, all top drive connections shall be inspected every six (6) months, generally in accordance with API RP7G. Furthermore, DOE Safety Alert 1/89 requires all dollies to be inspected every six (6) months.

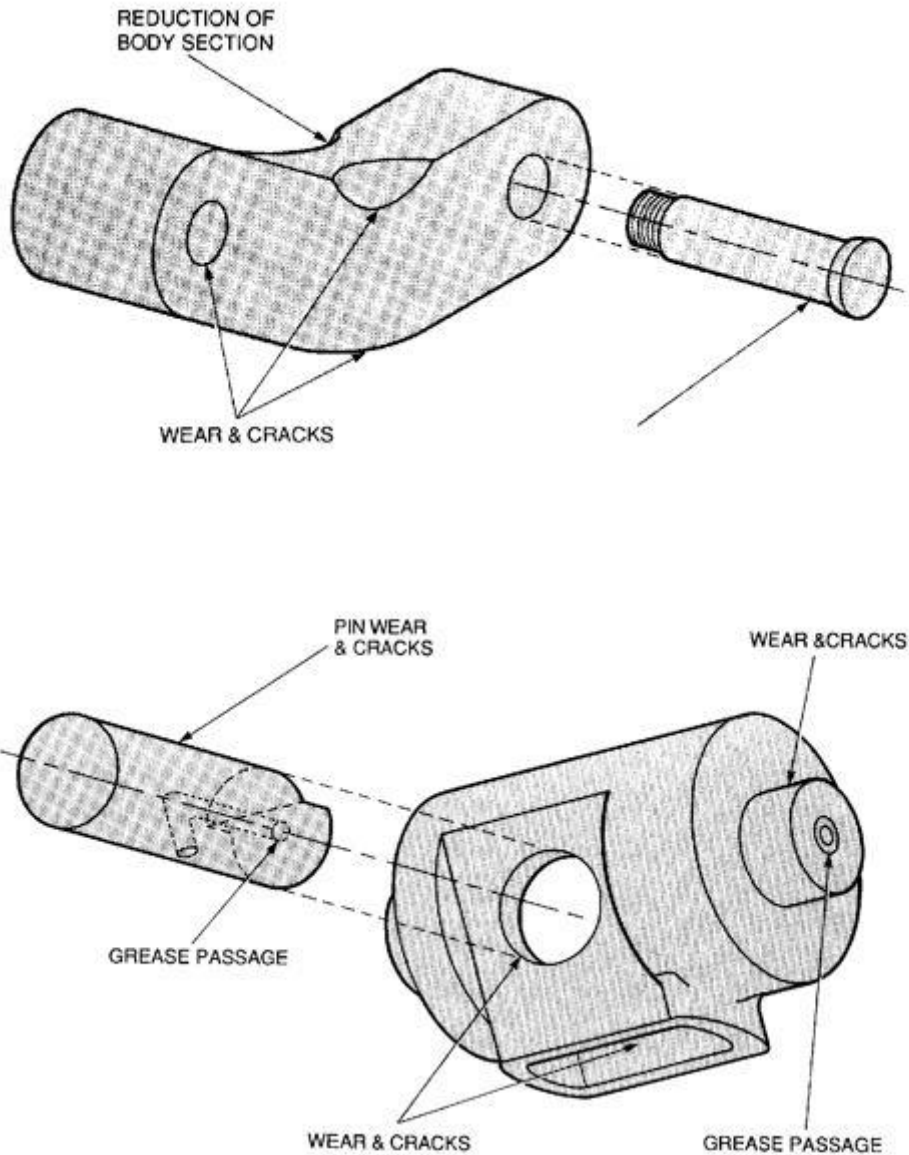
Maintenance

The manufacturer's recommendations shall be followed, especially on requirements for dismantling after a stated number of running hours.

FIG. 15 - Derrick Systems

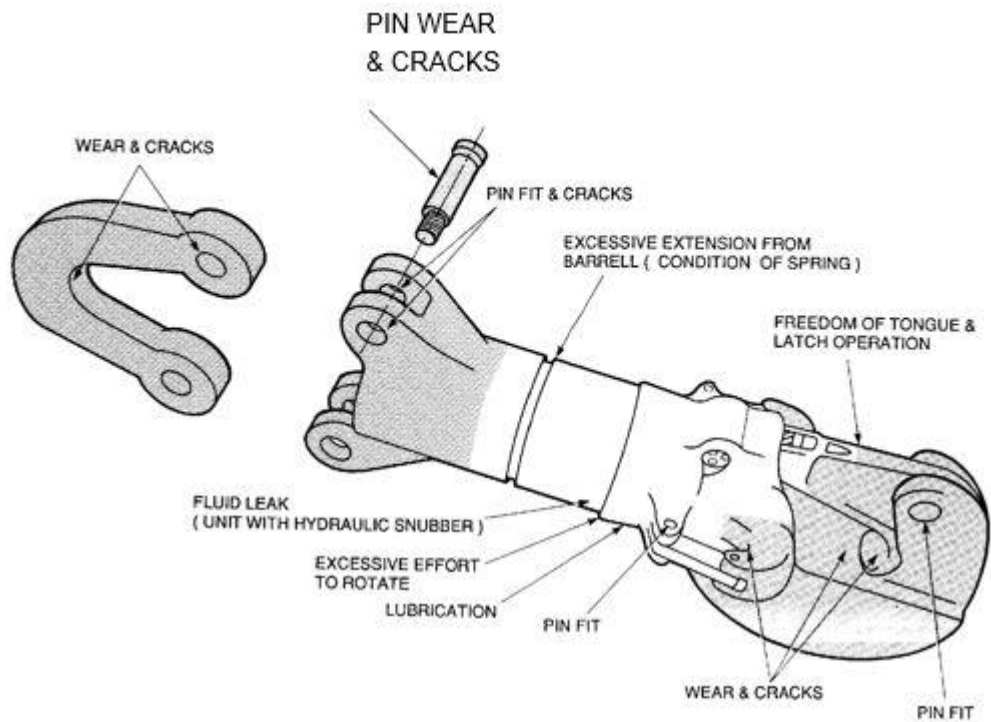
Inspection

Derrick structures including Dolly Tracks, finger boards etc., should be independently inspected every 12 months for damaged beams, slack bolts, etc., using the recommendations of API Spec 4E. This inspection shall be recorded in the Register of Lifting Equipment, together with any remedial action taken, or defects corrected.



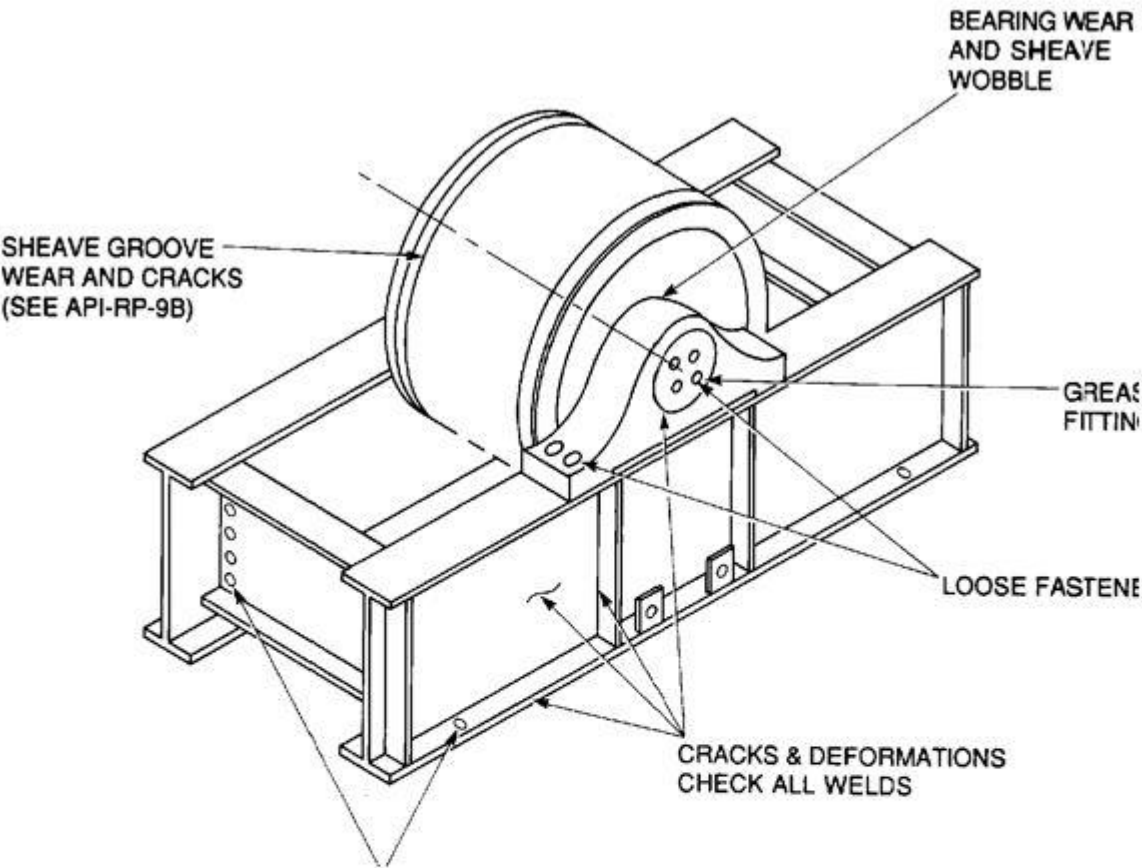
ALL PARTS TO BE EXAMINED BY M.P.I.

FIG. 16 - Block to Hook Adapter



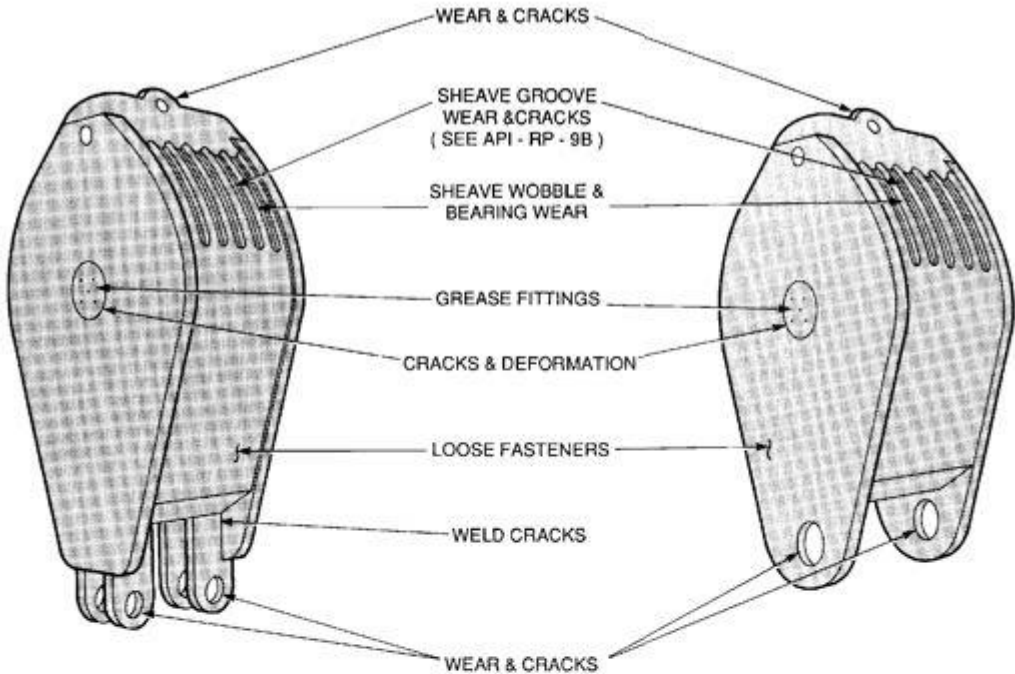
AREAS TO BE EXAMINED BY M.P.I.

FIG. 17 - Drilling Hook



NOTE: Sheaves, shaft, and all beams to be examined by MPI

FIG. 18 - Crown Block



ALL PARTS TO BE EXAMINED BY M.P.I.

FIG. 19 - Travelling Block

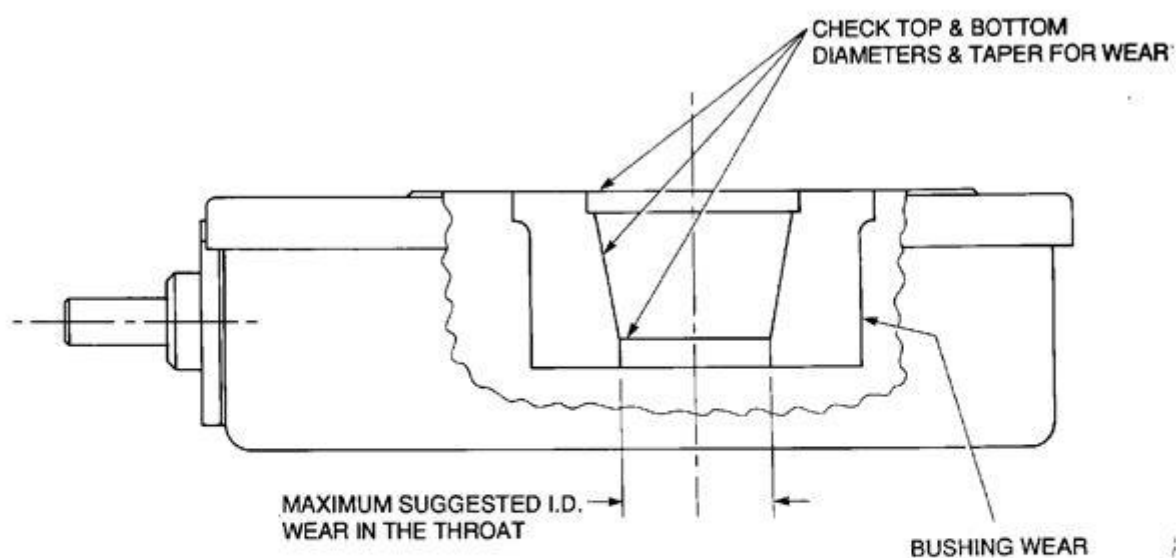


FIG. 20 - Rotary Table



8.0 GUIDELINES FOR PREVENTIVE MAINTENANCE ROUTINES/ OFFSHORE PREVENTATIVE MAINTENANCE ROUTINES

8.1 Introduction

For lifting equipment and accessories on **QatarEnergy-IS** worksites, planned preventative maintenance routines are generally managed within **QatarEnergy-IS** maintenance management system unless otherwise specified, such as the case of Contractor owned Drilling Rigs. The following guidelines identify the minimum requirements for maintaining integrity.

When dealing with lifting equipment in general, due to the certification criteria, any preventative maintenance carried out must be of such a nature, as not to affect any load bearing components or the integrity of the equipment as this would normally negate the certification.

For this reason, preventative maintenance routines are restricted to the following:

- 1) Visual examination of the equipment including functional / operational checks to ensure satisfactory performance and
- 2) Lubrication / re-coating of parts liable to seize or corrode

The criteria for visual examination (in most cases) is contained in Part 5.2 of this manual.

To prevent duplication of instructions, where applicable, the individual procedures will be referred to throughout these PMRs

Safety Precautions

The following precautions are a general guide to safe working practices and should always be observed.

- 1) Ensure that all relevant "permit to work" and "isolation" procedures are complied with
- 2) Adhere to **QatarEnergy-IS** safe operating procedures
- 3) Observe all safety notices, warning signs, safety barriers, PA announcements and area status lights appertaining to plant / module access
- 4) Be aware of the location of firefighting equipment and emergency escape routes
- 5) Wear a safety helmet, safety glasses and safety boots / shoes. Do not wear loose fitting or torn clothing. Remove rings / chains etc. before working on machinery
- 6) Wear ear protection devices when in machinery spaces. High frequencies can damage your hearing
- 7) Whilst carrying out any work, ensure worksite is kept tidy. Upon completion of work ensure the worksite area is left in a safe and clean condition

List of Equipment

Pedestal Cranes*

- 1) Overhead Traveling Cranes (electric)
- 2) Overhead Traveling Cranes (manual)
- 3) Traveling Beams
- 4) Runway Beams
- 5) Swing Jibs
- 6) Mobile Gantry Cranes
- 7) Pneumatic Winches
- 8) Hand Winches
- 9) Forklifts



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***Note:** Due to the complex nature of this type of equipment, generic PMR's are not available in this document.

8.2 Fixed Lifting Equipment

8.2.1 Overhead Travelling Cranes (Electric)

3 Monthly Maintenance Routine by Mechanical Technician

- 1) Isolate crane to make safe and post relevant notices
 - 2) Carry out cursory inspection of cross beam, end carriages and long travel beams / rails looking for obvious defects
 - 3) Check down shop beams / rails are clear of obstructions
 - 4) Re-grease carriage wheels as required
 - 5) Apply grease to hoist via grease nipples and top up oil in gearbox as required
 - 6) Apply grease to rope guide and stop
 - 7) Apply grease to limit switch bar. (Refer to Lubrication Schedule)
 - 8) Check brake wear by position of motor shaft
i.e. if flush with fan hood - adjust brake
if protruding beyond fan hood - replace brake disc
 - 9) Reconnect power supply and raise / lower hoist checking function of limit switches
 - 10) Lubricate wire rope as required. (Refer to Lubrication Schedule)
 - 11) Operate hoist cross-travel and check brake efficiency
 - 12) Operate hoist in long travel direction and check brake efficiency
- Check for crabbing

8.2.2 Overhead Travelling Cranes (Manual)

3 Monthly Maintenance Routine by Mechanical Technician

- 1) Carry out cursory inspection of gantry crane looking for obvious defects
- 2) Wind the long travel hand chain to drive the crane forwards and backwards to check smoothness of operation
- 3) Operate the trolley hoist along the cross beam to check for smooth operation
- 4) Raise / lower hoist through one complete rotation of load wheels. (Raise or lower approximately 300mm)
- 5) Listen for healthy "clicking" of ratchet pawl
- 6) If required, lubricate hand chain wheel gearing on long travel and hoist travel and lightly oil hand chains and load chains. (Refer to Lubrication Schedule)

Note: Avoid any lubrication on or near the brake unit.

6 Monthly Maintenance Routine by Mechanical Technician

- 1) Visually examine complete gantry crane for damage / corrosion / wear etc.
- 2) Check paintwork covering welds for cracking - should any be found, remove paint and carry out N.D.T. (If all satisfactory or after repairs, repaint)
- 3) Check security of components i.e. end carriage wheels, hand chain wheels etc.
- 4) Operate gantry crane forwards and backwards to check for smoothness of operation
- 5) Lubricate where necessary. (Refer to Lubrication Schedule)
- 6) Visually inspect the hoist and check for security of components (particularly the chain guide rollers)
- 7) Examine bottom hook and check for stretch / distortion
- 8) Check function of safety catch
- 9) Ensure hook swivels freely in yoke and is secure in its crosshead



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- 10) Examine load chain sheaves, check for wear / distortion and ensure satisfactory seating of load chain
- 11) Examine body casing and check for cracks, distortion, missing bolts / screws or any other defect that may interfere with operation or affect the safety of the hoist
- 12) Examine (as far as possible) internal frame side plates
- 13) Examine live end and dead end load chain anchor pins and check for distortion / wear
- 14) Examine hand chain wheel and chain and ensure it seats correctly in the pockets
- 15) Examine the complete length of load chain and check for stretch, deformed / bent links, nicks, wear and excessive corrosion
- 16) Lightly oil hand chain and load chain. (Refer to Lubrication Schedule)

8.2.3 Manual Travelling Beam

3 Monthly Maintenance Routine by Mechanical Technician

- 1) Carry out cursory inspection of beam and carriages looking for obvious defects.
- 2) Check rails are clear of any obstructions.
- 3) Wind the long travel hand chain to drive the crane forwards and backwards to check smoothness of operation.
- 4) If required, grease gear wheels and carriage wheels.
- 5) Lightly oil hand chain. (Refer to Lubrication Schedule).

6 Monthly Maintenance Routine by Mechanical Technician

- 1) Visually inspect cross beam and long travel beams for damage / corrosion / wear etc.
- 2) Check paintwork covering welds for cracking - should any be found, remove paint and carry out N.D.T. (If all satisfactory or after repairs, repaint)
- 3) Check security of components i.e. bolted end stops, carriage wheels, hand chain wheel, chain guide and taught wire
- 4) Check rails are clear of any obstructions
- 5) Wind the long travel hand chain to drive the crane forwards and backwards to check smoothness of operation
- 6) If required, grease gear wheels and carriage wheels
- 7) Lightly oil hand chain. (Refer to lubrication schedule)

8.2.4 Runway Beams

Maintenance Routine by Mechanical Technician (Periodicity in accordance with the Written scheme of Examination)

- 1) Visually inspect complete length of runway beam for damage / corrosion / wear etc.
- 2) Check paintwork covering welds for cracking - should any be found, remove paint and carry out N.D.T. (If all satisfactory or after repairs, repaint)
- 3) Check security of fittings i.e. bolted end stops etc.

8.2.5 Geared Swing Jib Cranes

Maintenance Routine by Mechanical Technician (Periodicity in accordance with the Written scheme of Examination)

- 1) Visually inspect jib and column for damage / corrosion / wear etc.
- 2) Check security of any fittings i.e. bolted top bracket and end stop



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- 3) Check paintwork covering welds for cracking - should any be found, remove paint and carry out N.D.T. (if all satisfactory or after repairs, repaint)
- 4) Check jib is clear to slew through complete working sector
- 5) Operate winding handle on worm gear box to slew through full extent
- 6) Apply lubrication to slewing shaft, slew bearings and top up grease in gearbox (Refer to lubrication schedule)

8.2.6 Mobile Gantry Cranes

3 Monthly Maintenance Routine by Mechanical Technician

- 1) Carry out cursory inspection of gantry looking for obvious defects
- 2) Check tires for splits / cuts, wear etc.
- 3) Operate both winding handles for dog clutches to ensure they turn freely and lubricate as required. (Refer to Lubrication Schedule)
- 4) Carry out cursory inspection of hoist looking for obvious defects. (Pay particular attention to chain guide rollers)
- 5) Travel hoist forwards and backwards along the beam to check for smooth operation
- 6) Operate hoist through one complete rotation of load wheel. (Raise or lower approximately 300mm)
- 7) Listen for health "clicking" of ratchet pawl
- 8) Lubricate chains as required but **avoid** any lubrication on or near brake unit. (Refer to Lubrication Schedule)

8.2.7 Pneumatic Winches

Weekly Maintenance Routine by Mechanical Technician

- 1) Carry out cursory inspection of equipment looking for obvious defects
- 2) Ensure the brake bands and brake drums are clean and free from contamination
- 3) Where fitted, examine the automatic brake checking springs, link arms and pins
- 4) Examine automatic brake cylinder and check for leaks
- 5) Examine chromed piston rod and check for corrosion
- 6) Check oil levels and top up as required. (Refer to Lubrication Schedule)

3 Monthly Maintenance Check by Mechanical Technician and Operator Inspection

- 1) Power unit checks are satisfactory
- 2) Coolant system is satisfactory
- 3) Engine lube oil is correct
- 4) Battery is satisfactory (power)
- 5) Gas bottles are satisfactory
- 6) Hydraulic hoses and fittings are secure
- 7) Brake system is mechanically correct
- 8) Brake shoes and drums are satisfactory
- 9) Mast inspection is acceptable
- 10) Lift chain and anchorages are correctly installed
- 11) Main frame and structure is safe and operate correctly
- 12) Electrical systems are safe and operate correctly
- 13) Safety cut outs and limit switches are correctly set
- 14) Steering system, tires and exhaust system are in good condition
 - i. SWL and test dates are displayed

Lubrication

- 1) Lubrication of chain and mast



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- 2) Wheel bearings

3 Monthly Maintenance Routine by Mechanical Technician

- 1) Carry out cursory inspection of equipment looking for obvious defects
- 2) Ensure the brake bands and brake drums are clean and free from contamination
- 3) Where fitted, examine the automatic brake checking springs, link arms and pins
- 4) Examine automatic brake cylinder and check for leaks
- 5) Examine chromed piston rod and check for corrosion
- 6) Check oil levels and top up as required. (Refer to Lubrication Schedule)
- 7) Examine brake bands and brake drums and check for wear
- 8) Check security of anchor bolts / welding

8.2.8 Hand Winches

3 Monthly Maintenance Routines by Mechanical Technician

- 1) Carry out cursory inspection of hand winch looking for obvious defects
- 2) Operate winch through one complete revolution of winch drum
- 3) Grease worm gear as required. (Refer to Lubrication Schedule)

6 Monthly Maintenance Routines by Mechanical Technician

- 1) Examine winding handle and check for distortion
- 2) Examine winch drum and check for distortion, wear or cracking (if accessible, examine rope anchorage and ensure it is secure)
- 3) Where applicable, examine ratchet or worm gear (mounted on side of drum) and check for wear / broken teeth
- 4) Examine pawl and pawl spring / worm and check for wear / distortion. Ensure the pawl functions correctly and ensure the worm meshes correctly with the worm wheel
- 5) Examine winch frame and check for distortion, wear and cracking
- 6) Ensure the hold down bolts / welding / clamping is adequate and as far as possible, satisfy yourself that the support steel work is capable of sustaining the winch under load
- 7) Operate winch through one complete revolution of winch drum
- 8) Grease worm gear as required. (Refer to Lubrication Schedule)

8.2.9 Forklifts

Pre-use Examination by Operator

- 1) Power / fuel system checks:
 - a) Battery fully charged
 - b) Diesel fuel at adequate level to carry out shift workload
 - c) Gas tank pressure sufficient to carry out shift workload
 - d) Check all fuel systems for leaks
- 2) Coolant system is at correct level. Engine lubricating oil is at correct level
- 3) Hydraulic system is at correct level to allow all hydraulic systems to function correctly
- 4) Start-up power unit. Check for leaks (diesel fuel, gas or coolant). Check no oil or grease is present on the control pedals or the floor area. Pedal control rubbers to be inspected for wear
- 5) Function check the parking brake, hand brake and foot brake, also the hydraulic control systems i.e. hoist, mast tilt, lower and side shift if installed



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- 6) Check the steering, and the condition of the wheels and tires. Carry out lubricating schedule
- 7) Function check all safety devices, windscreen wipers and horn, check condition of doors, cab glass and driver's seat
- 8) Check the exhaust emissions for sparks if Diesel / Gas engine is the power source

Note: Before the unit is put into service it is the responsibility of the operator to ensure it is in a serviceable condition.

Weekly Maintenance Routine by Mechanical Technician

The following checks are in addition to the daily checks carried out by the operator:

- 1) Check the external condition of all hoses for chaffing, wear and cuts, check the security of the end fittings and couplings. Check the hose reeling unit if an extension mast is fitted
- 2) Check the brake fluid, linkages and cables
- 3) Check the headlights, brake lights and indicator lights. (if fitted)
- 4) Check the hoist chain anchorages are correctly fitted and adjusted. Ensure no links are broken or damaged
- 5) Check the condition of forks, support frame, and the locking pins
- 6) Check mast slide ways and stops
- 7) Check overall condition of the Forklift. i.e. safety cage, fixing pins, security of fixings and fastenings

Note: Diesel / Gas powered Forklifts should meet the minimum EEMUA standards and Electric Forklifts should meet BASEEFA minimum standards if they are to work in hazardous areas on the Rig.

6 Monthly Maintenance Routine by Mechanical, Electrical and Test Engineers

The following inspection and checks are in addition to the daily / weekly inspection.

- 1) Obtain work permit
- 2) Check the battery cells fluid level, check the specific gravity by using a hydrometer, replenish fluid with distilled water. Check all terminals for corrosion and tightness, and if required clean, adjust and cover the terminals with petroleum jelly. Check condition of battery cables and plug
- 3) Engine coolant system, check the anti-freeze % to coolant, if low top up as required
- 4) Gas / Diesel engines lubricating oil change, use oil as recommended in the lubrication schedule
- 5) Remove drive wheels and brake drums, check the condition of the brake system. If required replace and adjust parts, refit brake drums and drive wheels
- 6) Inspect hoist mast slide ways for wear and deformation, check hydraulic rams for scores and damage, inspect the condition of the ram seals
- 7) Check the condition of hoist chain guide rollers
- 8) Every 6 months the main hoist chain should be removed and carefully inspected for broken or cracked links, links with worn pins and elongation. The chain should be carefully measured and if there is over 2% extension the chain should be replaced
- 9) On replacement of the chains the forklift should be proof load tested to SWL + 25%
- 10) Carry out a full visual inspection of all structural members for cracks and deformation
- 11) Ensure SWL is displayed, also check that the test plate is stamped correctly i.e. Identification Number and the latest test date
- 12) Return permit to control room. Leave equipment in a safe condition

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13) Check list should be completed and returned to the supervisor.

9.0 REFERENCES TO OTHER DOCUMENTS

TABLE 1: REFERENCES TO OTHER DOCUMENTS

Document Title	Document Number	Internal/External Document (if any)	Document Referencing
IS Lifting Equipment Control And Operations Procedure	IS-HSE-PRC-027	Internal	Sideward
QatarEnergy Standard For Lifting Equipment and Operations	QP-PAI-STD-005	Internal	Upward

10.0 DEFINITIONS

Other definitions are contained within this Lifting Operations Guidance Document.

Term	Definition
Alpha / Numeric	A combination of letters and/or numbers used for identification.
Calculated Deflections	The deflections due to the safe load and proof loads, computed by using the formula applicable to the particular circumstances.
Cantilever Beam	Section of beam supported at one end only.
Cargo	Any liquid, solid or gaseous matter transported in a container. Tubulars such as drill pipe, casing, etc.
Certificate of Examination	A written declaration that an examination has been carried out to the satisfaction of the independent competent person
Color Code	A method of marking equipment (normally with plastic tie-wraps or paint) to give a visual indication of its certification status. This color is normally changed every six months.
Competent Person	Someone who is considered, by either the Rig Manager or his employer, to be experienced in the use of lifting equipment and who is familiar with, and has experience in, its operation. This person must also have the requisite knowledge and experience, both theoretical and practical, of the type of material under examination to certify with confidence whether it is free from patent defect and suitable in every way for the duty for which the material is required.
Container	Any form of unit or device used for the transportation of cargo as defined in BS 7072 / BS EN 12079. Also known as a cargo carrying unit (CCU)
Dangerous Goods	Substances defined as such but not limited to those in the CSG Code (e.g. CAA, COSHH, IMDG etc).



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Term	Definition
Drawn-up Dimension	The minimum distance between the suspension level and the bottom hook saddle (also known as closed-height).
Effective Span	The distance between the centres of the adjacent supports for overhead crane beams, runway beams and cantilevers.
Effort	The pull on the hand chain or lever required to lift a specific load (chain blocks and pullifts). The maximum effort allowable is relative to the maximum working load limit of the block.
Examination	A visual examination by a competent person, carried out carefully and critically, and supplemented by other means, such as measurement and where necessary non-destructive testing, in order to arrive at a reliable conclusion as to the condition and safety of the appliance. If necessary for the examination, part of the appliance shall be dismantled.
Extended Dimension	The distance between the suspension level and the bottom hook saddle of a chain block, when the bottom hook is in the extended position. It equals the sum of the drawn up dimension and the range of lift (or height of lift).
Factor of Safety	The ratio of the load that would cause failure of an item of lifting equipment to the load that is imposed upon it in service i.e. SWL (This is to allow for detrimental criteria such as wear and tear, dynamic loadings etc.).
Ferrule	A compression fitting used to secure the eye of a wire rope.
Frame	The primary load bearing elements of the container.
Gross Weight	The maximum allowable weight of a loaded appliance which is equal to the Tare weight (weight of empty appliance) plus the Safe Working Load (weight of cargo which may be carried by the appliance).
Hand Operated Chain Hoist	A block reeved with a load chain and operated by a hand chain so as to give a mechanical advantage (also known as a chain block).
Headroom	Mechanical advantage (also known as a chain block). The maximum vertical distance between the item to be lifted and the point of suspension of the hoisting machine. e.g. between the lifting padeyes and the underside of runway beam.
Height of Lift	The amount of possible travel between the top and bottom connection points (e.g. hooks) of a hoisting machine.
Hook Load	The total weight suspended from the hook.
Inertia Forces	The forces produced by a change of velocity.
Inspection	A visual inspection by a competent person, looking carefully and critically for anything which may impair the safe and efficient working of the lifting equipment.
Lifting Appliance	Appliances or machinery capable of raising, suspending or lowering a load, including any attachments for anchoring, fixing or supporting the appliance.



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Term	Definition
Lifting Accessory	Components for attaching loads(s) to lifting appliances or machinery e.g. slings and shackles.
Lifting Set	Loose gear used to connect the container to the lifting appliance.
Live Loads	Any load except wind load that gives rise to a variation of stress in a member. Such variation may be due to any change of position or magnitude of an externally applied load, or to the movement of the crane structure itself.
Load	Encompasses the concept of either mass or force, and is expressed in the appropriate units.
Load Sensor	A piece of equipment used for detecting the actual loadings at a rigging point e.g. load cell shackle, test clock or compression load cell etc.
Measured Deflections	The deflections measured in such a manner that they relate to precisely the same conditions as those covering the calculated deflection.
Mode Factor	A factor which takes into account the geometry of the sling assembly, the number of parts and other constants as specified in the appropriate British Standard.
Operating Level	The level on which the operator stands.
Permissible Working Stress	The stress numerically equal to the basic stress, multiplied by the relevant duty factor corresponding to the load.
Plant Identification Number /	A unique number given to an item of lifting equipment for registration purposes and to facilitate traceability.
Proof Load Test	The deliberate application of a predetermined overload to a piece of lifting equipment.
Range of Lift	The vertical distance which the bottom hook travels between the extended and highest positions.
Responsible Person	The person (or persons) appointed within an organization to have responsibility for the use and certification of containers or lifting equipment.
Rigging Loft	A container modified specifically to suit the storage of lifting equipment (quite often fitted with lighting and heating).
Runway Beam	An overhead track for carrying loads which are moved along it.
Safe Working Load (SWL)	The maximum load which an item of lifting equipment may raise, lower or suspend safely under particular service conditions. This is normally determined by a competent person and takes into consideration, operational factors which may be detrimental, or affect the integrity of the equipment, e.g. extreme heat, extreme cold, constant / intense use, etc. This will often result in the equipment being down-rated i.e. the SWL can be lower than the WLL set by the manufacturer.



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Term	Definition
Sling Assembly	A sling comprising various components e.g. sling legs, master links, etc
Structural Integrity	The strength of materials and their arrangement to provide reliability of the load bearing structure.
Tank Container	A container which consists of the tank or tanks, and the load bearing structure.
Thorough Examination	A thorough examination by a competent person, looking carefully and critically for anything which may impair the safe and efficient working of the lifting equipment. This examination is to include testing by the competent person where appropriate and may also include disassembly.
Tare	The weight of the container without cargo. Tare weight shall include all fixtures normally fixed to the container in service.
Thimble	A fitting used to protect the eye of the wire rope termination.
Trigonometrical Method	A method traditionally used to calculate the Working Load Limit (WLL) of slings at various angles of the legs.
Uniform Load Method	A method of rating multi-legged slings for use at any included angle between the sling legs of up to 90°, and in special cases for two-legged and four-legged slings at any included angle between 90° and 120°. This is the preferred method for rating general purpose slings in accordance with BS 1290, 1983 and BS 6166, 1981.
Webbing	A part of a flat lifting sling, comprising a woven narrow fabric, generally of a coarse weave and multiple plies, the prime function of which is load bearing.
Wind Load	The forces produced by the velocity of the wind, which is assumed to act horizontally.
Working Load Limit (WLL)	The maximum load which an item of lifting equipment is designed to raise, lower or suspend. This is normally specified by the manufacturer and does NOT take into consideration operational factors which may be detrimental, or affect the integrity of the equipment, e.g. extreme heat, extreme cold, constant / intense use, etc.

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APPENDIX**APPENDIX A - REVISION HISTORY LOG****Revision Number: 01****Date: 07/12/2021**

Item Revised	Revision Description	Page No.
Common	The document was updated based on the new format for Management system documents	
Remarks :		

**APPENDIX B - LIST OF ABBREVIATIONS**

Term	Definition
API	American petroleum Institute
ASME V	American Standards for non-destructive testing
BS	British Standards
FC	Fibre Core
IMDG Code	International Maritime Dangerous Goods Code
IWRC	Independent Wire Rope Core
LOLER	Lifting Operations and Lifting Equipment Regulations SI 1998 No 2307
MBL	Minimum Breaking Load. This is the load specified in the relevant British Standard from which are derived other values such as the working load limit.
MPI	Magnetic Particle Inspection. A technique used to detect surface breaking discontinuities in ferro-magnetic materials.
NDE	Non-destructive Examination. A visual inspection to detect any defects in a material.
NDT	Non-destructive Test. Any technique used to detect defects in a material without subjecting it to physical stress. These techniques include ultrasonics, dye penetrants, magnetic particle inspection and radiography
SWL	Safe Working Load. The maximum load as certified by a competent person that an item of lifting equipment may raise, lower or suspend under particular service conditions.
SWL / Payload	Safe Working Load / Payload. This is the maximum permissible weight of cargo which may be safely be transported by container.
WLL	Working Load Limit. The maximum load that an item of lifting equipment is designed to raise, lower or suspend.